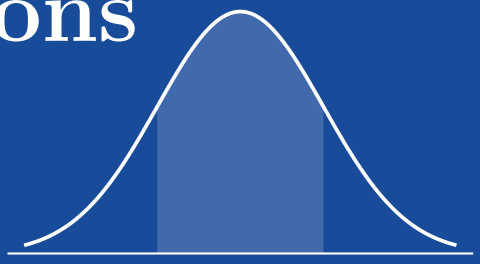
Fundamental Quant Interview Questions

Probability, Statistics, Games & Puzzles



72 concepts in 10 categories

I Conditional Probability and Counting
II Expected Value and Key Distributions
III Coin Flips, Sequences and Markov Chains
IV Optimal Stopping and Dynamic Programming
V Betting, Martingales and Market Making
VI Game Theory
VII Order Statistics and Geometric Probability
VIII Estimation, Logic and Brain Teasers
IX Number Theory and Combinatorial Puzzles
X Geometry, Regions and Graphs

Each concept: a representative problem with a step-by-step solution, a concept summary and applications, and additional practice problems with detailed solutions.

October 2026

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Part I — Conditional Probability and Counting

1 Conditional Probability

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A).$$

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}, \qquad P(B) = \sum_i P(B \mid A_i)\, P(A_i).$$

Independence

A and B are independent iff $P(A \mid B) = P(A)$, equivalently $P(A \cap B) = P(A)\, P(B)$.

Pitfall: $P(A|B) \neq P(B|A)$ in general

$P(\text{positive test} \mid \text{disease}) = 0.90$	High sensitivity
$P(\text{disease} \mid \text{positive test}) = 0.15$	Disease is rare

Use Bayes' theorem to convert: $P(A|B) = P(B|A) \cdot P(A)/P(B)$.

Worked Example

Two fair dice. Given $D_1 + D_2 \leq 5$, find $P(D_1 = 2)$.

Numerator — outcomes where $D_1 = 2$ *and* $D_1 + D_2 \le 5$:

$$D_1 = 2 \;\Rightarrow\; D_2 \le 3, \quad \text{so } (2,1),\,(2,2),\,(2,3) \;\Longrightarrow\; 3 \text{ outcomes.}$$

Denominator — all outcomes where $D_1 + D_2 \le 5$:

$$\begin{aligned} D_1 = 1: &\quad (1,1),(1,2),(1,3),(1,4) \;\to\; 4 \text{ outcomes} \\ D_1 = 2: &\quad (2,1),(2,2),(2,3) \;\to\; 3 \text{ outcomes} \\ D_1 = 3: &\quad (3,1),(3,2) \;\to\; 2 \text{ outcomes} \\ D_1 = 4: &\quad (4,1) \;\to\; 1 \text{ outcome} \end{aligned} \qquad \text{Total: } 10 \text{ outcomes.}$$

Result (the sample space has $6 \times 6 = 36$ equally likely outcomes):

$$P(D_1 = 2 \mid D_1{+}D_2 \le 5) = \frac{P(D_1 = 2 \text{ and } D_1{+}D_2 \le 5)}{P(D_1{+}D_2 \le 5)} = \frac{3/36}{10/36} = \frac{3}{10}.$$

Additional Problem 1 — Bayesian Updating

(a) Two-headed vs. fair coin. A pocket holds one fair coin and one two-headed coin. You pick one at random and flip it: heads. What is the probability the *other* side is tails (i.e. the coin is fair)?

$$P(F \mid H) = \frac{P(H \mid F)P(F)}{P(H)} = \frac{\frac{1}{2}\cdot\frac{1}{2}}{\frac{1}{2}\cdot\frac{1}{2} + 1\cdot\frac{1}{2}} = \frac{1/4}{3/4} = \boxed{\tfrac{1}{3}}.$$

A head is evidence *for* the two-headed coin, so the fair coin drops from a $\frac{1}{2}$ prior to a $\frac{1}{3}$ posterior.

(b) Four coins, given at least one head. Flip four fair coins. Given that at least one is heads, what is the probability exactly one is heads?

$$P(\text{exactly one} \mid \ge 1) = \frac{\binom{4}{1}/16}{15/16} = \frac{4/16}{15/16} = \boxed{\tfrac{4}{15}}.$$

(c) Which coin is the double-headed one? Five coins: one double-headed, four fair. Pick one at random, flip it five times — all heads. Probability it is the double-headed coin?

$$P(D \mid 5H) = \frac{1\cdot\frac{1}{5}}{1\cdot\frac{1}{5} + \left(\frac{1}{2}\right)^5\cdot\frac{4}{5}} = \frac{1/5}{1/5 + 1/40} = \frac{8/40}{9/40} = \boxed{\tfrac{8}{9}}.$$

Five straight heads is strong (but not conclusive) evidence for the certain-heads coin.

(d) $d4$ **vs.** $d6$**.** A fair 4-sided die and a fair 6-sided die; pick one at random and roll it, getting a 2. Probability it was the $d4$?

$$P(d4 \mid X{=}2) = \frac{\frac{1}{4}\cdot\frac{1}{2}}{\frac{1}{4}\cdot\frac{1}{2} + \frac{1}{6}\cdot\frac{1}{2}} = \frac{1/4}{1/4 + 1/6} = \frac{3/12}{5/12} = \boxed{\tfrac{3}{5}}.$$

The $d4$ assigns higher likelihood to any low roll it can produce, so a 2 tilts the posterior toward it (60%).

Additional Problem 2 — Knew the Answer or Guessed? (Q544)

Problem: A test question has 5 possible answers, exactly one correct. You observe that someone answered it correctly. What is the probability that they actually knew the answer rather than guessed?

Key observation — a prior is needed. The answer depends on how likely the person was to know the answer *before* you saw it was correct. Call that prior $p = P(\text{knew})$. Without a stated p, give the formula and then a number for a reasonable choice (usually $p = \frac{1}{2}$).

Step 1 — two ways to be right. Let K = knew, R = right.

- Knew it (probability p): always right, $P(R \mid K) = 1$.
- Guessed (probability $1 - p$): right 1 time in 5, $P(R \mid K^c) = \frac{1}{5}$.

So $P(R) = p + \frac{1-p}{5} = \frac{4p+1}{5}$.

Step 2 — Bayes.

$$P(K \mid R) = \frac{P(R \mid K)P(K)}{P(R)} = \frac{p}{(4p+1)/5} = \frac{5p}{4p+1}.$$

Step 3 — plug in a prior.

prior p	0.1	0.25	0.5	0.75	0.9
$P(\text{knew} \mid \text{correct})$	0.36	0.625	**0.833**	0.94	0.98

Intuition with 100 test-takers ($p = \frac{1}{2}$). 50 know and all 50 are right; 50 guess and about 10 are right. Of the 60 correct answers, 50 come from people who knew: $\frac{50}{60} = \frac{5}{6}$.

Odds view. Someone who knows is 5 times as likely to be right as a guesser (1 vs. $\frac{1}{5}$), so a correct answer multiplies the odds of "knew" by exactly 5: prior odds 1:1 become 5:1, i.e. $\frac{5}{6}$.

$P(\text{knew} \mid \text{correct}) = \dfrac{5p}{4p+1}$, which is $\dfrac{5}{6} \approx 83\%$ for a $50/50$ prior. A correct answer multiplies the odds of "knew" by the likelihood ratio 5.

Additional Problem 3 — Rain on Both Days (Fréchet Bounds) (Q404)

Problem: There is a $p\%$ chance of rain on Saturday and a $q\%$ chance of rain on Sunday. What are the *maximum* and *minimum* possible probabilities that it rains on *both* days?

Key observation. We are told only the two *marginal* probabilities $P(A) = p/100$ (rain Saturday) and $P(B) = q/100$ (rain Sunday). Nothing is said about how the two days are related — they could be independent, strongly positively related (rainy weekends), or negatively related. So $P(A \cap B)$ is not a single number; we must find the *range* of values consistent with the marginals.

Step 1 — Upper bound. "Rain on both days" is a sub-event of "rain on Saturday" and also of "rain on Sunday":

$$A \cap B \subseteq A \Rightarrow P(A \cap B) \leq P(A) = \tfrac{p}{100}, \qquad A \cap B \subseteq B \Rightarrow P(A \cap B) \leq P(B) = \tfrac{q}{100}.$$

Hence $P(A \cap B) \le \min(p, q)\%$. *Attained* when the less likely rainy day only happens together with the other one (e.g. with $p \le q$: whenever it rains Saturday it also rains Sunday, $A \subseteq B$).

Step 2 — Lower bound. Use inclusion–exclusion,

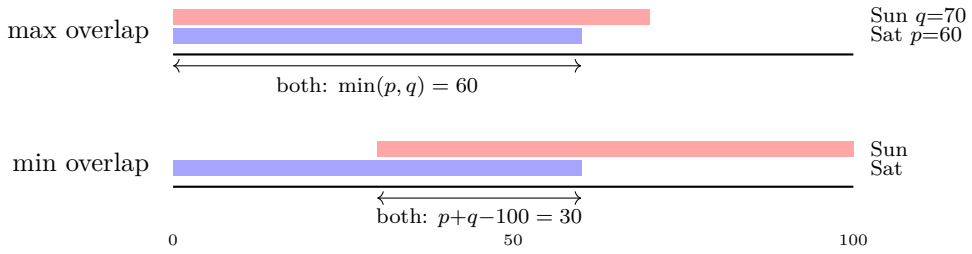
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \Longrightarrow \quad P(A \cap B) = P(A) + P(B) - P(A \cup B).$$

A probability can be at most 1, so $P(A \cup B) \le 1$ and

$$P(A \cap B) \;\ge\; \frac{p}{100} + \frac{q}{100} - 1 = \frac{p + q - 100}{100}.$$

Also $P(A \cap B) \ge 0$. So $P(A \cap B) \ge \max(0,\, p + q - 100)\%$. *Attained* by making the two rainy events overlap as little as possible: if $p + q \le 100$ they can be disjoint (overlap 0); if $p + q > 100$ they must overlap by at least the excess $p + q - 100$.

Picture. Think of the 100 equally likely "weather scenarios" laid out on a line; Saturday's rain occupies p of them, Sunday's q. Slide the two intervals to overlap as much or as little as possible:



$$\max(0,\, p + q - 100)\% \;\le\; P(\textbf{rain both days}) \;\le\; \min(p, q)\%.$$

Example: $p = 60,\ q = 70$ gives a range of 30% to 60%. (If the days were independent the answer would be $pq/100 = 42\%$, which always lies inside the range.)

Takeaway. These are the **Fréchet bounds**: marginals alone pin a joint probability only to an interval. The upper end is "maximal positive dependence" (one event inside the other); the lower end is "maximal negative dependence" (the events overlap only as much as forced). In risk management the same bounds give best/worst cases for the chance two positions default together when only the individual default probabilities are known.

Follow-up — a dry weekend and the effect of correlation (Q533). Saturday rain 50%, Sunday rain 60%, dependence unknown. What is the chance of *no* rain all weekend, and does positive correlation raise or lower it?

Step 1 — write "dry" in terms of "both".

$$P(\text{dry}) = 1 - P(A \cup B) = 1 - \big[P(A) + P(B) - P(A \cap B)\big] = P(A \cap B) - 0.1.$$

So the dry-weekend probability moves *one-for-one* with the probability of rain on both days.

Step 2 — apply the bounds above. With $p = 50$, $q = 60$: $0.1 \le P(A \cap B) \le 0.5$, hence

$$0 \;\le\; P(\text{dry}) \;\le\; 0.4.$$

0 when the rain is spread out as much as possible (every possible weekend has rain on at least one day); 0.4 when rain on Saturday always comes with rain on Sunday.

Step 3 — independence. $P(\text{dry}) = P(A^c)P(B^c) = 0.5 \times 0.4 = 0.2$ (equivalently $P(A \cap B) = 0.3$, and $0.3 - 0.1 = 0.2$).

Step 4 — positive correlation. Positive correlation means rain on one day makes rain on the other more likely:

$$P(A \cap B) > P(A)P(B) = 0.3 \quad \Longrightarrow \quad P(\text{dry}) = P(A \cap B) - 0.1 > 0.2.$$

So the chance of a dry weekend **increases**. (In general, for two events $\text{Cov}(\mathbf{1}_A, \mathbf{1}_B) = P(A \cap B) - P(A)P(B) = P(A^c \cap B^c) - P(A^c)P(B^c)$, so the complements are correlated in exactly the same way: if rainy days cluster, dry days cluster too.)

Unknown dependence: $P(\text{dry weekend}) \in [0, 0.4]$; independent: 0.2; positively correlated: above 0.2. Positive correlation raises $P(\text{both rainy})$, which lowers $P(\text{at least one rainy day})$ and so raises $P(\text{dry})$ — even though each day's own chance of rain is unchanged.

Additional Problem 4 — Stay or Switch Urns After Drawing $10? (Q551)

Problem: Two urns look identical from the outside. Urn A holds 3 one-dollar coins and 7 ten-dollar coins; urn B holds 5 one-dollar and 5 ten-dollar coins. You pick an urn at random and draw a coin: it is a \$10. You may now draw again (the first coin is *not* replaced) from the same urn or from the other urn. Which maximises the chance of another \$10?

Step 1 — what the first \$10 tells you (Bayes).

$$P(\$10) = \tfrac{1}{2} \cdot \tfrac{7}{10} + \tfrac{1}{2} \cdot \tfrac{5}{10} = \tfrac{6}{10}, \qquad P(A \mid \$10) = \frac{\frac{1}{2} \cdot \frac{7}{10}}{6/10} = \frac{7}{12}, \qquad P(B \mid \$10) = \frac{5}{12}.$$

You are now more likely to be holding the richer urn.

Step 2 — stay. Your urn has lost one \$10, leaving 9 coins: A has 6 \$10s left, B has 4.

$$P(\$10 \mid \text{stay}) = \tfrac{7}{12} \cdot \tfrac{6}{9} + \tfrac{5}{12} \cdot \tfrac{4}{9} = \frac{42 + 20}{108} = \frac{62}{108} = \frac{31}{54} \approx 0.574.$$

Step 3 — switch. The other urn is untouched (10 coins). If yours was A, the other is B ($\frac{5}{10}$); if yours was B, the other is A ($\frac{7}{10}$):

$$P(\$10 \mid \text{switch}) = \tfrac{7}{12} \cdot \tfrac{5}{10} + \tfrac{5}{12} \cdot \tfrac{7}{10} = \frac{35 + 35}{120} = \frac{7}{12} = \frac{63}{108} \approx 0.583.$$

Switch, narrowly: $\frac{7}{12} = \frac{63}{108}$ versus $\frac{31}{54} = \frac{62}{108}$ for staying.

Why — two competing effects.

1. *Evidence for staying:* the \$10 makes it more likely you hold the richer urn ($\frac{7}{12}$).
2. *Depletion against staying:* you just removed a \$10 from that urn. With only 10 coins this matters: A's \$10 share falls from 70% to 67%, B's from 50% to 44%, while the other urn is untouched.

Here the evidence is weak (7 vs. 5 \$10s, so the odds move only from 1:1 to 7:5), and depletion wins by a hair.

Contrast (Q574). If the urns hold 3 and 1 \$10s out of 10, a \$10 is *strong* evidence ($P(\text{richer urn} \mid \$10) = \frac{3}{4}$), which outweighs the depletion: *staying* is better (expected draw \$2.50 vs. \$2.35). **Lesson:** neither "always stay" nor "always switch" is right — update with Bayes, account for what you removed, and compute both options.

Additional Problem 5 — Predict the Next Flip of a Random Coin (Q559)

Problem: A bag holds three coins whose heads probabilities are 50%, 60% and 70%. You pull out a coin at random and flip it: heads. If you flip the same coin again, what is the probability it lands heads?

Step 1 — update which coin you hold (Bayes). The first head has probability $P(H_1) = \frac{1}{3}(0.5 + 0.6 + 0.7) = 0.6$. A head is more likely from a more biased coin, so each posterior weight is proportional to that coin's p:

coin	$p = 0.5$	$p = 0.6$	$p = 0.7$
prior	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$P(\text{coin} \mid H_1) = \frac{p \cdot \frac{1}{3}}{0.6}$	$\frac{5}{18}$	$\frac{6}{18}$	$\frac{7}{18}$

Step 2 — predict the next flip. Average the coins' heads probabilities with the posterior weights:

$$P(H_2 \mid H_1) = \tfrac{5}{18}(0.5) + \tfrac{6}{18}(0.6) + \tfrac{7}{18}(0.7) = \frac{2.5 + 3.6 + 4.9}{18} = \frac{11}{18} \approx 0.611.$$

Shortcut. $P(H_2 \mid H_1) = \dfrac{P(H_1 H_2)}{P(H_1)}$, and with a uniformly chosen coin

$$P(H_2 \mid H_1) = \frac{0.5^2 + 0.6^2 + 0.7^2}{0.5 + 0.6 + 0.7} = \frac{1.10}{1.80} = \frac{11}{18} = \frac{\overline{p^2}}{\bar{p}} = \bar{p} + \frac{\mathrm{Var}(p)}{\bar{p}}.$$

$P(\textbf{heads again}) = \frac{11}{18} \approx 61.1\%$**, slightly above the** 60% **average coin.**

Intuition. Without the first flip the answer would be the average $\bar{p} = 0.6$. Seeing a head shifts weight toward the more biased coins, adding $\mathrm{Var}(p)/\bar{p} = 0.00\overline{6}/0.6 \approx 0.011$. The bump is small because the three coins are similar; with very different coins (say one fair and one two-headed, as in Additional Problem 1(c)) the first head tells you much more. The same "posterior $\propto$ prior $\times$ likelihood, then predict" pattern underlies Laplace's rule of succession.

2 Equally Likely Outcomes: Counting Favourable Cases

Many interview probability questions need no clever trick at all — only careful counting. If an experiment has a finite number of outcomes that are all *equally likely* (two fair dice, a shuffled deck, a random arrangement), then

$$P(\text{event}) = \frac{\#\text{favourable outcomes}}{\#\text{total outcomes}}.$$

The whole skill is (1) choosing a description of outcomes in which they really *are* equally likely, (2) counting the total, and (3) counting the favourable cases systematically, using shortcuts such as the complement, symmetry and independence.

Representative Problem — Perfect-Square Sum of Two Dice (Q565, Q493)

Problem: (a) Roll two fair six-sided dice. What is the probability that their sum is a perfect square? (b) Same question for two fair eight-sided dice (faces 1–8).

Step 1: Choose equally likely outcomes — ordered pairs

Record the result as an *ordered* pair (a, b): first die a, second die b. Each of the 6 values of a combines with each of the 6 values of b, giving $6 \times 6 = 36$ ordered pairs, each with probability $\frac{1}{36}$.

Why ordered? If you list only unordered results, "a 1 and a 3" and "two 2s" look like one outcome each, but the first happens in two ways, $(1, 3)$ and $(3, 1)$, while the second happens in only one way, $(2, 2)$. Unordered outcomes are *not* equally likely; ordered ones are. (Imagine the dice are different colours.)

Step 2: Which sums are perfect squares?

Two six-sided dice give sums from 2 to 12. The perfect squares in that range are 4 and 9 (1 is too small, 16 too big).

Step 3: Count each favourable sum

The grid below shows the sum for every ordered pair; the perfect squares are highlighted.

$a\backslash b$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

- Sum 4: $(1, 3), (2, 2), (3, 1)$ — 3 outcomes.
- Sum 9: $(3, 6), (4, 5), (5, 4), (6, 3)$ — 4 outcomes.

Each sum s lies on an anti-diagonal of the grid, so for $s = 2, \ldots, 12$ the counts (worth memorising) are

$$1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1.$$

Step 4: Divide

$$P(\text{square sum, two d6}) = \frac{3+4}{36} = \frac{7}{36} \approx 19.4\%.$$

Step 5: Two eight-sided dice

Now there are $8 \times 8 = 64$ equally likely ordered pairs, and sums run from 2 to 16. The squares are 4, 9 and 16. For two n-sided dice the number of ways to make sum s is $s-1$ for $s \le n+1$ and $2n+1-s$ for $s \ge n+1$ (the grid's anti-diagonals). So

- sum 4: $(1,3),(2,2),(3,1)$ — 3;
- sum 9 ($= n+1$, the peak): $(1,8),(2,7),\ldots,(8,1)$ — 8;
- sum 16: only $(8,8)$ — 1.

$$P(\text{square sum, two d8}) = \frac{3+8+1}{64} = \frac{12}{64} = \frac{3}{16} = 18.75\%.$$

(a) $\frac{7}{36}$ (sums 4 and 9: $3+4$ of 36 ordered pairs). (b) $\frac{3}{16}$ (sums $4, 9, 16$: $3+8+1$ of 64).

Concept Summary and Applications

Equally likely outcomes: $P(E) = \dfrac{\#\text{favourable}}{\#\text{total}}$, provided every outcome has the same probability.

Counting tools.

- **Multiplication rule:** k independent stages with $n_1, \ldots, n_k$ options give $n_1 n_2 \cdots n_k$ outcomes (36 for two dice, 4^3 for three dots on 4 edges).
- **Choosing positions:** arranging a items of one kind and b of another in a row: $\binom{a+b}{a}$ (lattice paths, tic-tac-toe boards).
- **Complement:** $P(E) = 1 - P(\text{not } E)$ when "not E" is easier.
- **Symmetry:** equal-looking cases have equal probability (e.g. $P(X > Y) = P(Y > X)$).

Two dice: sums $2, \ldots, 12$ occur in $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ of the 36 ways.

- **Make outcomes equally likely first:** use *ordered* pairs for dice; for card draws without replacement, multiply step-by-step probabilities; for random processes, look for a description (such as "final board") in which all cases have the same weight.
- **List systematically:** fix the first coordinate and count the second (a table or grid avoids double counting and omissions).
- **Check with a quick sanity test:** counts must add up to the total ($1+2+\cdots+1 = 36$), and symmetric cases must match.
- **Applications:** dice and card problems, digit problems, lattice paths, random arrangements, and any discrete model where you can enumerate cases.

Additional Problem 1 — Perfect-Square Product of Two Dice (Q448)

Problem: Roll two fair dice and multiply the results. What is the probability that the product is a perfect square?

Step 1: Total

As before, 36 equally likely ordered pairs (a, b).

Step 2: Go through the first die

For each a, ask which $b \in \{1, \ldots, 6\}$ make ab a perfect square $(1, 4, 9, 16, 25, 36)$:

- $a = 1$: b itself must be a square: $b = 1$ or 4 (products 1, 4).
- $a = 2$: $2b$ square: $b = 2$ (product 4); $b = 8$ would also work but is not on a die.
- $a = 3$: $3b$ square: $b = 3$ (product 9).
- $a = 4$: $4b$ square $\iff$ b square: $b = 1$ or 4 (products 4, 16).
- $a = 5$: $b = 5$ (product 25).
- $a = 6$: $b = 6$ (product 36).

Favourable pairs: $(1,1), (1,4), (2,2), (3,3), (4,1), (4,4), (5,5), (6,6)$ — 8 of them.

Step 3: Why exactly these? The square-free part

Write each number as (square)×(square-free part): $1 = 1$, $2 = 2$, $3 = 3$, $4 = 4 \cdot 1$, $5 = 5$, $6 = 6$. A product ab is a perfect square exactly when a and b have the *same* square-free part (the leftover primes must pair up). The square-free parts of $1, \ldots, 6$ are $1, 2, 3, 1, 5, 6$: only 1 and 4 share one. So the squares are the 6 doubles plus $(1, 4)$ and $(4, 1)$.

$$P = \frac{8}{36} = \frac{2}{9} \approx 22.2\%.$$ **(Same question: Q623, Q658.)**

Additional Problem 2 — Counting Two-Dice Sums and Comparisons (Q417, Q605, Q52, Q602)

Problems: Roll two fair six-sided dice.

(a) (Q417) What is the probability that the sum is 7?
(b) (Q605) What is the probability that the sum is 10?
(c) (Q52) What is the probability that the second die shows a higher number than the first?
(d) (Q602) What is the probability that the sum is odd?

All four use the same 36 equally likely ordered pairs and the sum grid from the representative problem.

(a) Sum $= 7$

$(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$: 6 outcomes — the longest anti-diagonal of the grid. For *every* first die there is exactly one second value that makes 7, which is why 7 is the most likely sum.

$$P = \frac{6}{36} = \frac{1}{6}.$$

(b) Sum $= 10$

$(4,6),(5,5),(6,4)$: 3 outcomes. (Note $(5,5)$ counts once, the others twice as ordered pairs.)

$$P = \frac{3}{36} = \frac{1}{12}.$$

(c) Second die higher than the first

Direct count: if the first die is 1, the second can be 2–6 (5 ways); first 2: 4 ways; ...; first 6: 0 ways. Total $5+4+3+2+1+0 = 15$, so $P = \frac{15}{36} = \frac{5}{12}$.

Symmetry shortcut: the three cases "second higher", "first higher", "equal" cover everything. "Equal" has 6 outcomes (the doubles), probability $\frac{1}{6}$. The other two cases are mirror images (swap the dice), so they are equally likely:

$$P(\text{second} > \text{first}) = \frac{1 - \frac{1}{6}}{2} = \frac{5}{12}.$$

(d) Odd sum

A sum is odd exactly when one die is odd and the other even. Each die has 3 odd and 3 even faces:

$$\underbrace{3 \cdot 3}_{\text{odd, even}} + \underbrace{3 \cdot 3}_{\text{even, odd}} = 18, \qquad P = \frac{18}{36} = \frac{1}{2}.$$

Shortcut: whatever the first die shows, exactly 3 of the 6 values of the second die make the sum odd.

(a) $\frac{1}{6}$ **(b)** $\frac{1}{12}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{1}{2}$**.**

Additional Problem 3 — Die, Card and Roulette Show the Same Number (Q592)

Problem: You roll a die, draw a card from a standard 52-card deck (Ace = 1, Jack = 11, Queen = 12, King = 13) and spin a roulette wheel. What is the probability that all three show the same number?

Step 1: Which numbers can possibly match?

The die shows 1–6, the card 1–13, the European wheel 0–36 (37 pockets; an American wheel adds 00, 38 pockets). A common value must be on all three, so only $k \in \{1,\dots,6\}$ — the die is the bottleneck.

Step 2: Probability of matching one particular value

For a fixed $k \in \{1,\dots,6\}$:

$$P(\text{die} = k) = \tfrac{1}{6}, \qquad P(\text{card} = k) = \tfrac{4}{52} = \tfrac{1}{13}, \qquad P(\text{wheel} = k) = \tfrac{1}{37}.$$

The three devices are independent, so multiply: $\frac{1}{6} \cdot \frac{1}{13} \cdot \frac{1}{37}$.

Step 3: Add over the six possible values

The events "all show 1", ..., "all show 6" are disjoint, so

$$P = 6 \cdot \frac{1}{6} \cdot \frac{1}{13} \cdot \frac{1}{37} = \frac{1}{13 \cdot 37} = \frac{1}{481} \approx 0.21\%.$$

Simpler view: whatever the die shows, the card must match it ($\frac{1}{13}$) and the wheel must match it ($\frac{1}{37}$). American wheel: $\frac{1}{13 \cdot 38} = \frac{1}{494}$.

Common slip: stopping at $\frac{1}{6 \cdot 13 \cdot 37}$ — that is the chance all three show one *particular* number; there are 6 possible matching values.

$\frac{1}{481}$ (European wheel), $\frac{1}{494}$ (American). (Same question: Q541.)

Additional Problem 4 — Three Random Dots on a Square's Edges (Q333)

Problem: Three dots are placed independently and uniformly at random on the perimeter of a square. (1) What is the probability that the three dots do *not* form a triangle? (2) What is the probability that the dots lie on three different edges?

Step 1: Reduce to which edge each dot lands on

All four edges have the same length, so each dot lands on each edge with probability $\frac{1}{4}$, independently. Record only the edges: there are $4^3 = 64$ equally likely edge assignments (edge of dot 1, of dot 2, of dot 3). Landing exactly on a corner has probability 0 and can be ignored.

Step 2: When do three dots fail to form a triangle?

Three points fail to form a triangle exactly when they are *collinear* (on one straight line). A straight line meets the boundary of a convex shape such as a square in at most two points — unless it runs along an edge. So three boundary dots are collinear only if all three lie on the *same edge*.

Step 3: Count

- **All on one edge:** choose the edge (4 ways); all three dots on it: 1 assignment each. 4 favourable assignments, so $P = \frac{4}{64} = \frac{1}{16}$.
- **Three different edges:** first dot 4 choices, second 3 (different), third 2: $4 \cdot 3 \cdot 2 = 24$, so $P = \frac{24}{64} = \frac{3}{8}$.
- For completeness, exactly two on one edge: $64 - 4 - 24 = 36$, i.e. $\frac{9}{16}$.

(1) No triangle (all on one edge): $\frac{1}{16}$. (2) Three distinct edges: $\frac{3}{8}$.

Additional Problem 5 — Two Red Cards: Small Deck or Big Deck? (Q638)

Problem: You may play with a 52-card deck (26 red, 26 black) or a 104-card deck (52 red, 52 black). You draw two cards at random without replacement and win if both are red. Which deck should you choose?

Step 1: Two draws without replacement

The first card is red with probability $\frac{1}{2}$ in either deck. Given that, one red card is gone, so the second card is red with probability (reds left)/(cards left).

$$P_{52} = \frac{26}{52} \cdot \frac{25}{51} = \frac{25}{102} \approx 0.2451, \qquad P_{104} = \frac{52}{104} \cdot \frac{51}{103} = \frac{51}{206} \approx 0.2476.$$

(Equivalently, by counting: $\binom{26}{2}/\binom{52}{2} = \frac{325}{1326} = \frac{25}{102}$.)

Step 2: Why the bigger deck wins

With r red and r black cards,

$$P(\text{two reds}) = \frac{1}{2} \cdot \frac{r-1}{2r-1},$$

and $\frac{r-1}{2r-1} = \frac{1}{2} - \frac{1}{2(2r-1)}$ increases toward $\frac{1}{2}$ as r grows. Removing the first red card hurts the second draw more in a small deck. A huge deck behaves like drawing *with* replacement, $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Choose the 104-card deck: $\frac{51}{206} \approx 24.8\%$ vs. $\frac{25}{102} \approx 24.5\%$. Related: Q670 (same), Q378, Q382 (full vs. half-size deck).

Additional Problem 6 — Shortest Paths Across a Chessboard (Q609, part 1)

Problem: Moving only along the grid lines of an 8×8 chessboard, how many shortest paths are there from one corner of the board to the opposite corner?

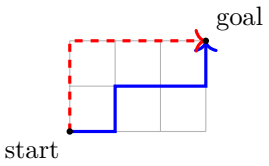
Step 1: What a shortest path looks like

Put the start at the bottom-left corner $(0, 0)$ and the goal at the top-right corner $(8, 8)$. Any path must gain 8 units to the right and 8 units up. A *shortest* path never goes left or down, so it consists of exactly 8 right steps (R) and 8 up steps (U) — 16 steps in total.

Step 2: Paths $\leftrightarrow$ words of R's and U's

Every shortest path is a sequence of 16 letters with 8 R's and 8 U's, and every such sequence is a valid path. Choosing the path = choosing *which* 8 *of the* 16 *positions* are R's:

$$\binom{16}{8} = \frac{16!}{8!\,8!} = 12{,}870.$$



Small example on a 3×2 grid: blue = RURRU, red = UURRR. There are $\binom{5}{3} = 10$ paths.

Step 3: The other reading

If a piece instead moves square to square (right or up) from the corner *square* a1 to h8, it needs only 7 right and 7 up moves: $\binom{14}{7} = 3432$. State which reading you use. (Part 2 of Q609 — the smallest number with 28 divisors, 960 — is a prime-factorisation problem; see Concept CA in the deck.)

Along grid lines, corner to opposite corner: $\binom{16}{8} = 12{,}870$ (square to square: $\binom{14}{7} = 3432$). In general an $a \times b$ grid has $\binom{a+b}{a}$ shortest paths.

Additional Problem 7 — Random Tic-Tac-Toe Ends in a Draw (Q529)

Problem: Two players play tic-tac-toe, each choosing uniformly at random among the empty squares on every move. What is the probability that the game ends in a draw?

Step 1: Every fill-order is equally likely

X moves on turns $1, 3, 5, 7, 9$ and O on turns $2, 4, 6, 8$. Under random play, what matters is the *order* in which the 9 squares are filled. Imagine that, even after someone wins, the players keep placing marks at random until the board is full. Then every one of the 9! fill-orders has the same probability $\frac{1}{9} \cdot \frac{1}{8} \cdots \frac{1}{1} = \frac{1}{9!}$.

Step 2: The stopping rule does not matter for draws

A draw means no line is ever completed. Along any fill-order that ends in a drawn full board, no three-in-a-row appears at any stage, so the real game never stops early. Hence

$$P(\text{draw}) = P(\text{the final full board has no line for either player}),$$

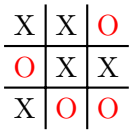
and we can just count final boards.

Step 3: 126 equally likely final boards

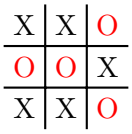
A full board is determined by which 5 squares are X's (the other 4 are O's): $\binom{9}{5} = 126$ boards. Each board arises from exactly $5! \cdot 4! = 2880$ fill-orders (X's squares in any order on X's turns, O's in any order on O's turns), and $126 \times 2880 = 362{,}880 = 9!$. So all 126 boards are equally likely.

Step 4: Count the drawn boards — 16

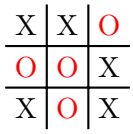
O's four squares must touch all 8 lines (a line with no O would be all X's), yet O must not complete a line. Up to rotations and reflections there are exactly 3 drawn patterns. The board has 8 symmetries (4 rotations $\times$ 2 reflections); a pattern with no symmetry of its own produces 8 distinct boards, a pattern with one mirror symmetry produces 4.



Pattern 1: no symmetry
$\Rightarrow$ 8 boards



Pattern 2: top–bottom mirror
$\Rightarrow$ 4 boards



Pattern 3: anti-diagonal mirror
$\Rightarrow$ 4 boards

Check Pattern 1: its rows are XXO, OXX, XOO and columns XOX, XXO, OXO — no three X's or three O's; both diagonals contain an O and an X. Total drawn boards: $8+4+4=16$ (confirmed by computer enumeration of all 126).

Step 5: The answer

$$P(\text{draw}) = \frac{16}{126} = \frac{8}{63} \approx 12.7\%.$$

For reference, full game-tree enumeration (where wins end the game early) gives X wins $\frac{737}{1260} \approx 58.5\%$, O wins $\frac{121}{420} \approx 28.8\%$. Wins cannot be found by counting final boards, because a winning game can stop early and a full board can contain lines for both players — only the draw reduces to a clean count.

$P(\textbf{draw}) = \frac{16}{126} = \frac{8}{63} \approx 12.7\%$**: all** $\binom{9}{5} = 126$ **final boards are equally likely, and** 16 **of them** $(8+4+4$ **from three patterns) are draws.**

3 Arrangements with Identical Items

Multinomial Coefficient

Arrange n items where type i appears n_i times ($\sum n_i = n$):

$$\text{Distinct arrangements} = \frac{n!}{n_1!\, n_2!\, \cdots\, n_k!}$$

Why: Start with $n!$ permutations of all items as if distinct. Divide by $n_i!$ for each group to cancel identical rearrangements.

Special Case: Binomial Coefficient

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Classic Example: MISSISSIPPI

11 letters: 1 M, 4 I, 4 S, 2 P.

$$\frac{11!}{1!\cdot 4!\cdot 4!\cdot 2!} = \frac{39{,}916{,}800}{1\times 24\times 24\times 2} = 34{,}650.$$

Multinomial Theorem

$$(x_1 + \cdots + x_k)^n = \sum_{\substack{n_1+\cdots+n_k=n \\ n_i\geq 0}} \frac{n!}{n_1!\cdots n_k!}\, x_1^{n_1}\cdots x_k^{n_k}.$$

4 Inclusion–Exclusion and Derangements

Inclusion–exclusion counts the outcomes in which *at least one* of several events happens, by adding the single events, subtracting the overlaps counted twice, adding back the triple overlaps, and so on:

$$|A_1 \cup \cdots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots .$$

Its most famous use is counting *derangements* — arrangements in which nothing lands in its "own" place (no letter in its own envelope, no cup on its own saucer). The answer for large problems is close to $1/e$ of all arrangements.

Representative Problem — Cups and Saucers With No Match (Q608)

Problem: Six cups and six saucers come in three patterns: two red, two white and two starred of each. The cups are placed on the saucers at random (one cup per saucer). What is the probability that no cup sits on a saucer of its own pattern?

Step 1: Total arrangements

Treat the cups and saucers as distinguishable. A placement is a matching of 6 cups to 6 saucers: $6! = 720$ equally likely placements.

Step 2 (Method 1): Count by pattern table

Let a_{ij} be the number of cups of pattern i on saucers of pattern j. A no-match placement needs $a_{ii} = 0$ and every row and column summing to 2:

	R	W	S
R	0	a	$2-a$
W	$2-a$	0	a
S	a	$2-a$	0

$a \in \{0, 1, 2\}$.

(Solve the row/column equations: everything is determined by a.) Now count individual placements for each table:

- $a = 0$ or $a = 2$: each pattern's two cups go together onto the two saucers of one other pattern, $2!$ ways each: $(2!)^3 = 8$ placements.
- $a = 1$: every pattern's two cups are split, one onto each of the other two patterns. Place the red cups first: choose which red cup goes on a white saucer (2), which white saucer (2), and which starred saucer takes the other red cup (2): 8 ways. Now one white and one starred saucer are left, plus both red saucers. White cups: choose which goes on a red saucer (2) and which red saucer (2); the other white cup must take the remaining starred saucer: 4 ways. The two starred cups fill the last red and white saucers: 2 ways. Total $8 \cdot 4 \cdot 2 = 64$ placements.

Total: $8 + 64 + 8 = 80$ placements.

Step 3 (Method 2): Inclusion–exclusion

Let r_k be the number of ways to choose k cups and put each on a *distinct* saucer of its own pattern (the "bad" placements we must exclude). Within one pattern (two cups, two saucers): 0 matches

in 1 way, 1 match in $2 \cdot 2 = 4$ ways, 2 matches in 2 ways. The three patterns are independent, so the r_k are the coefficients of

$$(1 + 4x + 2x^2)^3 = 1 + 12x + 54x^2 + 112x^3 + 108x^4 + 48x^5 + 8x^6.$$

Each such choice can be completed by placing the remaining $6-k$ cups in $(6-k)!$ ways. Inclusion–exclusion then gives the number of placements with no match at all:

$$\sum_{k=0}^{6} (-1)^k r_k \, (6-k)! = 720 - 12 \cdot 120 + 54 \cdot 24 - 112 \cdot 6 + 108 \cdot 2 - 48 + 8 = 80. \checkmark$$

Step 4: Probability

$$P(\text{no cup on its own pattern}) = \frac{80}{720} = \frac{1}{9}.$$

Shortcut: only colours matter, so count the $\frac{6!}{2!\,2!\,2!} = 90$ equally likely colour patterns; $1+8+1 = 10$ of them have no match: $\frac{10}{90} = \frac{1}{9}$.

Step 5: Compare with a plain derangement

If all six cups and saucers were different, the chance of no cup on its own saucer would be $\frac{D_6}{6!} = \frac{265}{720} \approx 0.37 \approx \frac{1}{e}$ (Additional Problem 1). Here each cup has *two* forbidden saucers, so the chance drops to $\frac{1}{9} \approx 0.11$.

80 of the 720 placements have no matching pattern (by the pattern table, or by inclusion–exclusion with $(1 + 4x + 2x^2)^3$), so $P = \boxed{\frac{1}{9}}$.

Concept Summary and Applications

Inclusion–exclusion: #(none of the bad events) $= \sum_{k \geq 0} (-1)^k \, S_k$, where S_k sums the sizes of all k-fold intersections.
Derangements (permutations of n with no fixed point):

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} \approx \frac{n!}{e}, \qquad D_n = (n-1)(D_{n-1} + D_{n-2}), \quad D_1 = 0, \; D_2 = 1.$$

$D_n = 0, 1, 2, 9, 44, 265, \ldots$; $P(\text{no fixed point}) \to \frac{1}{e} \approx 0.368$, already 0.368 for $n = 6$.
Forbidden positions in blocks: with "rook numbers" r_k (ways to place k forbidden matches), the count of good arrangements is $\sum_k (-1)^k r_k (n-k)!$.

- **Recipe:** define A_i = "item i is bad"; compute intersections by symmetry; alternate signs.
- **Complement:** $P(\text{at least one}) = 1 - P(\text{none})$; inclusion–exclusion is often easiest for "none".
- **More forbidden places ⇒ fewer good arrangements:** one forbidden saucer per cup gives $\approx \frac{1}{e}$; two give $\frac{1}{9}$ here.

- **Applications:** secret Santa (no one draws themselves), matching problems, surjections (every type appears), coupon-collector tail probabilities.

Additional Problem 1 — The Classic Hat-Check Problem

Problem: n people check their hats; the hats are returned at random. What is the probability that nobody gets their own hat? Evaluate for $n = 4$ and large n.

Step 1: Bad events

Let $A_i =$ "person i gets their own hat". If a particular set of k people all get their own hats, the other $n-k$ hats can be arranged in $(n-k)!$ ways, and there are $\binom{n}{k}$ such sets:

$$S_k = \binom{n}{k}(n-k)! = \frac{n!}{k!}.$$

Step 2: Inclusion–exclusion

$$D_n = \sum_{k=0}^{n}(-1)^k S_k = n!\Big(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\Big), \qquad P(\text{no match}) = \sum_{k=0}^{n}\frac{(-1)^k}{k!}.$$

Step 3: $n = 4$ and the limit

$n = 4$: $D_4 = 24(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}) = 12 - 4 + 1 = 9$, so $P = \frac{9}{24} = \frac{3}{8} = 0.375$. (Check by listing: the 9 derangements of 1234 are $2143, 2341, 2413, 3142, 3412, 3421, 4123, 4312, 4321$.) For large n the series is the expansion of e^{-1}, so $P \to \frac{1}{e} \approx 0.368$ — and it barely depends on n beyond $n \approx 5$.

Step 4: The recurrence (a second derivation)

Person 1 gets hat $j \neq 1$ ($n-1$ choices). Either person j gets hat 1 (the two swap, leaving a derangement of $n-2$ people: D_{n-2}), or not (then person j behaves as if hat 1 were their "own" forbidden hat: D_{n-1}). So $D_n = (n-1)(D_{n-1} + D_{n-2})$: $D_3 = 2(1+0) = 2$, $D_4 = 3(2+1) = 9$, $D_5 = 4(9+2) = 44$.

$P($nobody gets their own hat$) = \sum_{k=0}^{n}\frac{(-1)^k}{k!}$: $\frac{3}{8}$ for $n = 4$, $\to \frac{1}{e} \approx 36.8\%$ for large n.

5 Birthday and Collision Problems

When n items each land independently and at random in one of d boxes (birthdays, sock colours, hash values), how likely is it that two land in the same box — and how many items until the first repeat? The answer is surprising: repeats become likely after only about $\sqrt{d}$ items, because what matters is the number of *pairs*.

Representative Problem — The Birthday Problem (Q434)

Problem: In a group of n people, what is the probability that at least two share a birthday? How large must n be for this probability to exceed 50%?

The standard assumptions are:

- There are 365 possible birthdays.
- Each birthday is equally likely.
- Birthdays are independent.
- Leap day is ignored.

Step 1: Use the complement

It is much easier to compute the probability that **no two people share a birthday**, and then subtract from 1.

Let

- $A =$ "at least two people share a birthday,"
- $A^c =$ "all n birthdays are different."

Then

$$P(A) = 1 - P(A^c).$$

So the main task is to find $P(A^c)$.

Step 2: Compute the probability that all birthdays are different

Assign birthdays one person at a time.

- The first person can have any birthday, so the probability is 1.
- The second person must avoid the first person's birthday, so the probability is $\frac{364}{365}$.
- The third person must avoid the first two birthdays, so the probability is $\frac{363}{365}$.
- Continuing in this way, the kth person must avoid the previous $k-1$ birthdays, so the probability is

$$\frac{365 - (k-1)}{365}.$$

Thus,

$$P(A^c) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - n + 1}{365}.$$

Equivalently,

$$P(A^c) = \prod_{k=0}^{n-1} \frac{365-k}{365}.$$

Therefore the probability of at least one shared birthday is

$$P(A) = 1 - \prod_{k=0}^{n-1} \frac{365-k}{365}.$$

This is the exact formula.

Step 3: Example with $n = 23$

The famous case is $n = 23$.

Then

$$P(\text{no match}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{343}{365},$$

so

$$P(\text{at least one match}) = 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{343}{365}.$$

Numerically,

$$P(\text{no match}) \approx 0.4927,$$

and hence

$$P(\text{at least one match}) \approx 1 - 0.4927 = 0.5073.$$

So with just 23 people, the probability of a shared birthday is about

$$50.7\%.$$

This is why the result feels surprising.

Step 4: Why the probability grows so quickly

The key point is that we are not comparing one person to everyone else only once. Among n people, there are many possible pairs.

The number of pairs is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

For $n = 23$,

$$\binom{23}{2} = \frac{23 \cdot 22}{2} = 253.$$

So there are 253 different pairs of people who could potentially match, which makes a shared birthday much more likely than intuition suggests.

Step 5: The exponential approximation — where it comes from

(a) The key fact. From the series $e^{-x} = 1 - x + \frac{x^2}{2} - \cdots$, for small x the x^2 term is tiny, so

$$1 - x \approx e^{-x} \qquad \text{(e.g. } e^{-0.01} = 0.99005 \approx 0.99\text{)}.$$

(b) Apply it to every factor. Each factor is $\frac{365-k}{365} = 1 - \frac{k}{365}$, and for $k \le 22$ we have $\frac{k}{365} \le 0.06$, which is small. So

$$P(\text{no match}) = \prod_{k=0}^{n-1}\Big(1 - \frac{k}{365}\Big) \approx \prod_{k=0}^{n-1} e^{-k/365}.$$

(c) A product of exponentials is one exponential. Since $e^a e^b = e^{a+b}$, the exponents add:

$$\prod_{k=0}^{n-1} e^{-k/365} = \exp\Big(-\frac{0 + 1 + \cdots + (n-1)}{365}\Big) = \exp\Big(-\frac{n(n-1)}{2 \cdot 365}\Big).$$

Therefore

$$P(\text{at least one match}) \approx 1 - e^{-n(n-1)/730}.$$

(d) Meaning of the exponent. $\frac{n(n-1)}{2 \cdot 365} = \binom{n}{2} \cdot \frac{1}{365}$ is (number of pairs) $\times$ (chance a given pair matches) = the *expected number of matching pairs*, λ. So $P(\text{no match}) \approx e^{-\lambda}$, the Poisson chance of seeing zero events when λ are expected. For $n = 23$: $\lambda = \frac{253}{365} \approx 0.693$ and $e^{-0.693} \approx 0.500$, so $P(\text{match}) \approx 0.500$ (exact: 0.5073).

(e) Solving for the 50% point. Setting $e^{-n(n-1)/730} = \frac{1}{2}$ gives $n(n-1) = 730 \ln 2 \approx 506$, so $n \approx 23$. With d equally likely days, $n^2 \approx 2d \ln 2$, i.e.

$$n_{50\%} \approx \sqrt{2 \ln 2 \; d} \approx 1.18\sqrt{d} \qquad (d = 365 : \; 1.18 \times 19.1 \approx 22.5).$$

(f) Which way the error goes. For every x, $1 - x \le e^{-x}$, so each true factor is slightly *smaller* than its exponential stand-in. The true no-match probability is therefore slightly smaller, and the approximation slightly *underestimates* the match probability (50.0% vs. 50.7% at $n = 23$; 88.2% vs. 89.1% at $n = 40$). The gap grows with n because the ignored $x^2/2$ terms accumulate.

n	10	20	23	30	40	57
exact $P(\text{match})$	11.7%	41.1%	50.7%	70.6%	89.1%	99.0%
$1 - e^{-n(n-1)/730}$	11.6%	40.6%	50.0%	69.6%	88.2%	98.7%

(g) Common trap. "Will someone share *my* birthday?" involves only your $n - 1$ pairs: $1 - (364/365)^{n-1} \ge \frac{1}{2}$ needs $n \approx 254$ people, not 23.

$P(\textbf{some shared birthday}) = 1 - \prod_{k=0}^{n-1} \frac{365-k}{365} \approx 1 - e^{-n(n-1)/730}$**. With** $n = 23$ **people it is already** $\approx 50.7\%$ **— because 23 people form 253 pairs.**

Concept Summary and Applications

Collisions among n items in d equally likely boxes.

$$P(\text{no repeat}) = \prod_{k=0}^{n-1}\left(1-\frac{k}{d}\right) \;\approx\; e^{-n(n-1)/(2d)} = e^{-\lambda}, \qquad \lambda = \binom{n}{2}\cdot\frac{1}{d}$$

(λ = expected number of matching pairs). A repeat becomes likely once $n \approx \sqrt{2\ln 2\, d} \approx 1.18\sqrt{d}$ (the *square-root law*).

Expected number of draws until the first repeat:

$$E[\text{draws}] = \sum_{k=0}^{d}\prod_{j=0}^{k-1}\left(1-\frac{j}{d}\right) \;\approx\; \sqrt{\frac{\pi d}{2}} + \frac{2}{3}.$$

- **Count pairs, not people:** n people form $\binom{n}{2}$ pairs, which grows like n^2; that is why 23 people suffice for 365 days.
- **Certainty vs. probability:** the pigeonhole principle says $d+1$ items *guarantee* a repeat (366 people, 3 socks); the birthday problem says a repeat becomes *likely* after only $\sim\sqrt{d}$.
- **Trap:** matching a *specific* person needs about $d\ln 2 \approx 253$ others, not 23.
- **Applications:** hash collisions and "birthday attacks" in cryptography, duplicate detection, random ID generation (how long IDs must be to avoid clashes), and coincidence estimates.

Additional Problem 1 — Drawing Socks Until a Matching Pair (Q152)

Problem: A drawer holds an unlimited supply of socks in two colours, each colour equally likely on every draw. What is the expected number of socks you must draw to get a matching pair?

Link to the birthday problem. This is a collision problem with only $d = 2$ "boxes" (colours): we draw until two socks share a colour.

Let the two colors be A and B, each drawn with probability $\frac{1}{2}$, independently from draw to draw.

We seek the expected number of draws until the first time a matching pair is obtained.

Step 1: Observe the earliest possible match

- After the first draw, no pair is possible.
- After the second draw, a pair is formed if the second sock matches the first.

Since there are only two colors, if the second sock does **not** match the first, then the two socks drawn are one of each color. In that case, the **third** sock must match one of them, so a matching pair is guaranteed by the third draw.

Thus the number of draws N until a match is found can only be:

- $N = 2$, with probability $\frac{1}{2}$,
- $N = 3$, with probability $\frac{1}{2}$.

Step 2: Compute the expectation

Therefore,

$$E[N] = 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 1 + \frac{3}{2} = \frac{5}{2}.$$

Same answer from the general formula

Let N be the number of draws until the first repeat. Then $E[N] = \sum_{k \geq 0} P(N > k)$, and $N > k$ means the first k draws are all different, which has probability $\prod_{j<k}(1 - \frac{j}{d})$. With $d = 2$:

$$P(N > 0) = 1, \quad P(N > 1) = 1, \quad P(N > 2) = 1 - \tfrac{1}{2} = \tfrac{1}{2}, \quad P(N > 3) = 0 \quad \Longrightarrow \quad E[N] = 1 + 1 + \tfrac{1}{2} = \tfrac{5}{2}.$$

The same formula gives $\frac{26}{9} \approx 2.89$ draws for 3 colours, ≈ 3.77 rolls of a die until a repeated face, ≈ 5.04 people until a shared birth month, and ≈ 24.6 people until a shared birthday.

$E[\text{draws}] = 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = \frac{5}{2}$. A match is certain by the third sock (pigeonhole), and it comes on the second sock half the time.

Additional Problem 2 — Rolling a Die Until a Face Repeats

Problem: You roll a fair six-sided die repeatedly. What is the expected number of rolls until some face appears for the second time (i.e. until the first repeat)?

Link to the birthday problem. This is a collision problem with $d = 6$ "boxes" (faces): each roll is a "person" and each face a "birthday". By pigeonhole a repeat is certain by the 7th roll.

Step 1 — the tail probabilities. Let N be the roll on which the first repeat occurs. $N > k$ means the first k rolls are all different. The first roll can be anything, the second must avoid 1 face, the third 2 faces, and so on:

$$P(N > k) = \frac{6}{6} \cdot \frac{5}{6} \cdot \frac{4}{6} \cdots \frac{6 - k + 1}{6} = \prod_{j=0}^{k-1} \Big(1 - \frac{j}{6}\Big).$$

k	0	1	2	3	4	5	6
$P(N > k)$	1	1	$\frac{5}{6}$	$\frac{5}{9}$	$\frac{5}{18}$	$\frac{5}{54}$	$\frac{5}{324}$

(For example, $P(N > 3) = 1 \cdot \frac{5}{6} \cdot \frac{4}{6} = \frac{5}{9}$: the first three rolls are all different.) And $P(N > 7) = 0$.

Step 2 — tail-sum formula. For a count N, $E[N] = \sum_{k \geq 0} P(N > k)$ (write N as a row of 1's, one for each "not yet"):

$$E[N] = 1 + 1 + \frac{5}{6} + \frac{5}{9} + \frac{5}{18} + \frac{5}{54} + \frac{5}{324} = \frac{324 + 324 + 270 + 180 + 90 + 30 + 5}{324} = \frac{1223}{324} \approx 3.77.$$

Step 3 — the distribution (as a check). $P(N = k) = P(N > k - 1) - P(N > k)$:

first repeat on roll k	2	3	4	5	6	7
$P(N = k)$	$\frac{1}{6}$	$\frac{5}{18}$	$\frac{5}{18}$	$\frac{5}{27}$	$\frac{25}{324}$	$\frac{5}{324}$

These add to 1, and $\sum k\,P(N=k) = \frac{1223}{324}$, as before. The most likely rolls for the first repeat are the 3rd and 4th, and the median is 4.

E[**rolls until a repeated face**] $= \frac{1223}{324} \approx 3.77$**. A repeat is certain by the 7th roll, but on average it comes before the 4th.**

Square-root check. The approximation $\sqrt{\pi d/2} + \frac{2}{3}$ gives $\sqrt{3\pi} + 0.67 \approx 3.07 + 0.67 = 3.74$ — close to 3.77 even for $d = 6$. Compare: about 24.6 people until a shared birthday ($d = 365$), and 2.5 socks ($d = 2$, Additional Problem 1).

6 Simpson's Paradox

Problem: Is it possible for one option to be better than another in *every* subgroup, yet worse when the subgroups are pooled? Give an example.

Yes — Two Kidney-Stone Treatments

Yes. A comparison that holds *within every subgroup* can reverse once the groups are combined, because a lurking variable is spread unevenly. Recovery rates:

	Small stones	**Large stones**	**Overall**
Treatment A	$\frac{81}{87} = \mathbf{93}\%$	$\frac{192}{263} = \mathbf{73}\%$	$\frac{273}{350} = 78\%$
Treatment B	$\frac{234}{270} = 87\%$	$\frac{55}{80} = 69\%$	$\frac{289}{350} = \mathbf{83}\%$

Treatment A wins in *both* subgroups (93% vs. 87% and 73% vs. 69%), yet pooled, Treatment B looks better (83% vs. 78%).

Lurking Variables and Confounders

The reversal is driven by a third variable hiding behind the table, so it is worth defining the terms precisely before explaining the mechanism.

A **lurking variable** is a variable that influences the outcome but is *not shown* in the analysis — it "lurks" outside the two columns you are comparing. Here the lurking variable is **stone size**. The headline comparison (Treatment A vs. B, overall) never mentions stone size, yet stone size strongly affects whether a patient recovers: small stones are easy to cure ($\sim 90\%$), large stones are hard ($\sim 70\%$). A variable can lurk simply because the analyst forgot it, never measured it, or chose to average over it.

A **confounder** (or confounding variable) is the important special case of a lurking variable that is associated with *both* the "cause" being studied *and* the outcome. It thereby tangles up ("confounds") the apparent relationship between them. For stone size to be a confounder here, it must be linked to both:

- **linked to the treatment:** the doctors did not assign treatments at random. Surgery (Treatment A) was chosen mostly for the *severe*, large-stone cases (263 of A's 350 patients), while B was reserved mostly for the *easy*, small-stone cases (270 of 350).
- **linked to the outcome:** large stones are harder to cure regardless of treatment.

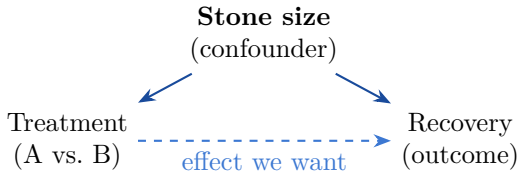
Because stone size is tied to *both* the treatment received and the chance of recovery, it confounds the raw A-vs-B comparison: Treatment A looks worse overall not because it is worse, but because it was handed the harder patients.

Lurking variable: any unshown variable that affects the outcome.

Confounder: a lurking variable linked to *both* the treatment/group *and* the outcome, so it distorts their measured association.

Every confounder is a lurking variable; not every lurking variable is a confounder (it must touch *both* sides).

A quick picture of the causal wiring, with stone size sitting upstream of both:



The dashed arrow is the treatment effect we actually care about; the two solid arrows are the confounding path that contaminates the naive overall comparison. The fix is to "control for" the confounder — compare A vs. B *within* each stone-size group, which is exactly what the subgroup rows do.

Lurking vs. confounding, in one line: *a lurking variable is merely absent from the analysis; it becomes a confounder only when it is correlated with both the grouping and the outcome. Randomised assignment (a coin flip deciding A or B) breaks the confounder's link to the treatment, which is precisely why randomised trials defeat Simpson's paradox.*

Why It Happens — Weighted Averages

The overall rate is a **weighted average** of the subgroup rates, weighted by group sizes. The lurking variable is **stone size**, confounded with treatment choice: large stones are harder to cure, and Treatment A was used mostly on the *hard* cases (263 of its 350 patients), while B was used mostly on the *easy* ones (270 of 350):

$$\text{A} \approx \frac{87(93\%) + 263(73\%)}{350} \text{ (mostly the 73\%)}, \qquad \text{B} \approx \frac{270(87\%) + 80(69\%)}{350} \text{ (mostly the 87\%)}.$$

Better per-group rates lose to worse group weights.

Which number to trust? *When the groups are a* pre-existing confounder *(as here), the subgroup comparison is honest: A is genuinely better, since severity — not treatment — drives the aggregate gap. If instead the grouping variable were a* consequence *of the choice, the pooled figure could be correct. It depends on the causal structure, not the numbers alone.*

Concept Summary and Applications

- **Mechanism:** an aggregate is a size-weighted average of parts; reversal occurs when the weights are correlated with the rates in the opposite direction to the within-group ordering.
- **Red flag:** always ask whether a confounder is distributed unevenly across the things being pooled before trusting a headline average.

- **Applications:** A/B tests with unequal traffic mixes, drug-trial analysis, admissions/hiring fairness (the 1973 U.C. Berkeley case), and any KPI aggregated over heterogeneous segments.

Additional Problem — Batting Averages

Problem: Player A beats Player B in batting average in *both* the first and second halves of a season. Can B finish with the higher average overall?

Yes — the season average is $\frac{H_1+H_2}{N_1+N_2} = \frac{N_1a_1+N_2a_2}{N_1+N_2}$ (hits over at-bats), a weight-by-at-bats average. A real example (Justice vs. Jeter):

	Year 1	Year 2	Combined
Player A	$\frac{104}{411} = .253$	$\frac{45}{140} = .321$	$\frac{149}{551} = .270$
Player B	$\frac{12}{48} = .250$	$\frac{183}{582} = .314$	$\frac{195}{630} = \mathbf{.310}$

A hits better both years, but B — who took most of his at-bats in the higher-average year — wins the combined average.

Yes: overall averages are at-bat-weighted, so unequal weights let the season-long order reverse the within-season order. (This is Simpson's paradox.)

7 The Airplane (Train) Seating Problem

Representative Problem — Will the Last Passenger Get Their Seat? (Q384)

Problem: A train (or plane) has 100 seats and is fully booked; passenger i is assigned seat i. Passengers board in order $1, 2, \ldots, 100$. The first passenger has lost their ticket and sits in a *random* seat. Every later passenger sits in their own seat if it is free; otherwise they pick one of the remaining empty seats at random. What is the probability that the last passenger ends up in their own seat?

Warm-up with small cases.

- $n = 2$: passenger 1 takes seat 1 or seat 2 with probability $\frac{1}{2}$ each, so the last passenger gets their seat with probability $\frac{1}{2}$.
- $n = 3$: passenger 1 picks seat 1 (prob. $\frac{1}{3}$): everyone is fine. Seat 3 (prob. $\frac{1}{3}$): passenger 3 loses. Seat 2 (prob. $\frac{1}{3}$): passenger 2 must choose between seats 1 and 3 at random, so passenger 3 is fine half of that time. Total: $\frac{1}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

Both give $\frac{1}{2}$. The number of seats seems not to matter — here is why.

Step 1 — only two seats matter: seat 1 and seat 100. Whenever someone has to choose at random (the first passenger, or anyone who finds their seat taken), the choice falls into one of three kinds:

- **Seat** 1 (the first passenger's own seat): this "closes the loop." Every later passenger now finds their own seat free, so passenger 100 gets seat 100.
- **Seat** 100: passenger 100 has lost their seat for good.
- **Some other seat** k (between 2 and 99): nothing is decided yet. Passengers before k sit normally; passenger k then finds their seat taken and becomes the next random chooser.

So the process keeps passing the "random chooser" role down the line until someone picks seat 1 or seat 100. (It must happen by the end: every seat $2, \ldots, 99$ is taken by its owner or earlier, so when passenger 100 boards, the one empty seat is either 1 or 100.)

Step 2 — symmetry. At every random choice, seats 1 and 100 are both still empty and neither belongs to the chooser, so they are equally likely to be picked. The first time the process lands on one of them, it is seat 1 or seat 100 with probability $\frac{1}{2}$ each. Therefore

$$P(\text{last passenger gets their own seat}) = \frac{1}{2}.$$

Step 3 — a check by recursion (for any n). Let p_n be the answer with n seats. Passenger 1 picks seat 1 (prob. $\frac{1}{n}$, last passenger succeeds), seat n (prob. $\frac{1}{n}$, fails), or seat k with $2 \le k \le n-1$ (prob. $\frac{1}{n}$ each). In the last case passengers $2, \ldots, k-1$ sit normally and passenger k faces the same problem with the $n - k + 1$ seats $\{1, k+1, \ldots, n\}$ (seat 1 now plays the role of k's own seat). So

$$p_n = \frac{1}{n} \cdot 1 + \frac{1}{n} \cdot 0 + \sum_{k=2}^{n-1} \frac{1}{n} p_{n-k+1}.$$

With $p_2 = \frac{1}{2}$ and assuming $p_m = \frac{1}{2}$ for all smaller m,

$$p_n = \frac{1}{n} + \frac{n-2}{n} \cdot \frac{1}{2} = \frac{2+n-2}{2n} = \frac{1}{2}.$$

The last passenger sits in their own seat with probability $\boxed{\frac{1}{2}}$ — for any number of seats $n \geq 2$ (the "100" is a red herring).

A Useful Extension

The same reasoning gives the chance for *every* passenger. Fix passenger k ($2 \leq k \leq n$) and look at the $n-k+2$ seats $S = \{1, k, k+1, \ldots, n\}$. Consider the first time any random chooser picks a seat in S — by symmetry, each seat of S is equally likely to be that first pick.

- If it is seat 1, the loop closes and passenger k gets seat k.
- If it is some seat $j > k$, the displaced person is seated and nobody is displaced again until passenger j boards — after passenger k — so k gets seat k.
- Only if it is seat k itself is passenger k displaced.

So passenger k loses their seat with probability $\frac{1}{n-k+2}$, i.e.

$$P(\text{passenger } k \text{ gets their own seat}) = \frac{n-k+1}{n-k+2}.$$

For $k = n$ this is $\frac{1}{2}$; for $k = 2$ it is $\frac{n-1}{n}$ (passenger 2 loses only if passenger 1 took seat 2). With $n = 10$, a simulation gives $0.899,\ 0.857,\ 0.667,\ 0.501$ for passengers $2, 5, 9, 10$, matching $\frac{9}{10}, \frac{6}{7}, \frac{2}{3}, \frac{1}{2}$.

Concept Summary and Applications

- **Reduce to what matters:** most of the process is irrelevant; the outcome is decided the first time a random chooser picks one of two special seats.
- **Symmetry beats computation:** those two seats are always treated identically, so each wins with probability $\frac{1}{2}$.
- **Check small cases and recurse:** $n = 2, 3$ suggest the answer; the recursion $p_n = \frac{1}{n} + \frac{n-2}{n} p$ confirms it for all n.
- **Applications:** random-displacement and "chain of evictions" problems, hashing collisions, and a classic interview test of spotting symmetry instead of brute-force enumeration.

8 Urn-Splitting Games: Urns Are Weighted Equally, Not Balls

Someone splits a fixed collection of "winning" and "losing" balls between two urns; a player who cannot see the split picks an urn at random and draws one ball. The surprise is that the split matters a lot, even though the total number of each colour is fixed — because the player chooses an *urn*, not a ball.

Representative Problem — Split the Marbles to Help the Player (Q53)

Problem: You have 100 green and 100 red marbles and divide them however you like between two bags, with at least one marble in each bag. A player who does not know your split picks one bag at random and draws one marble from it. A green marble wins \$100; a red marble wins nothing. If you split the marbles to give the player the best chance, what is the fair price of the game?

Step 1 — write the winning probability. Let bag i hold g_i green and r_i red marbles ($g_1+g_2 = 100$, $r_1+r_2=100$). The player picks each bag with probability $\frac{1}{2}$, *whatever its size*, then draws a marble from it:

$$P(\text{green}) = \tfrac{1}{2}\cdot\frac{g_1}{g_1+r_1} + \tfrac{1}{2}\cdot\frac{g_2}{g_2+r_2}.$$

The fair price is $100\cdot P(\text{green})$, so we maximise $P(\text{green})$.

Step 2 — the key observation. If all the marbles were in one big pile, the chance of green would be $\frac{100}{200}=\frac{1}{2}$. With two bags, each bag gets weight $\frac{1}{2}$ in the formula, *regardless of how many marbles it contains*. So a tiny bag counts as much as a huge one. We can exploit this by making one bag tiny but perfect.

Step 3 — the lone-green trick. Put **one green marble alone** in bag 1 and the other 99 green and all 100 red in bag 2:

$$P(\text{green}) = \tfrac{1}{2}\cdot 1 + \tfrac{1}{2}\cdot\frac{99}{199} = \frac{1}{2}+\frac{99}{398} = \frac{298}{398} = \frac{149}{199} \approx 74.9\%.$$

Step 4 — why nothing does better. To push bag 1's green fraction to 1, bag 1 must contain only green marbles, say g of them. Then bag 2 holds $100-g$ green and 100 red, with green fraction $\frac{100-g}{200-g}$, which is *largest* when g is smallest, i.e. $g=1$. Any red marble in bag 1 lowers bag 1's fraction below 1 and costs more than it gains. (A check of every possible split confirms that $g_1=1$, $r_1=0$ is the maximum.)

Put a single green marble alone in one bag. The player wins with probability $\frac{149}{199}\approx 74.9\%$, so the fair price is $100\cdot\frac{149}{199}=\frac{14900}{199}\approx\$74.87$ — far above the \$50 you might expect from "half the marbles are green".

Concept Summary and Applications

Urns are weighted equally, not balls. When the player picks an urn at random and then a ball,

$$P(\text{win}) = \tfrac{1}{2}\,p_1 + \tfrac{1}{2}\,p_2, \qquad p_i = \text{winning fraction in urn } i.$$

The overall winning fraction is an average weighted by *urn size*, but the player's chance is an average with *equal* weights, so it can be pushed far from the overall fraction by making one urn tiny.

- **To help the player:** put one *winning* ball alone ($\approx \frac{1}{2} + \frac{1}{4}$).
- **To hurt the player:** put one *losing* ball alone ($\approx \frac{1}{4}$).

- **Watch the wording:** who arranges the balls (friend or opponent), and whether an urn may be empty, change the answer.
- **Same idea elsewhere:** an average of averages is not the overall average when group sizes differ — the root of Simpson's paradox and of many "average of rates" mistakes in data analysis.
- **Applications:** survey sampling (picking a group, then a member, gives small groups too much weight), designing games and promotions, and spotting when an equal-weighted average can be manipulated.

Additional Problem — An Opponent Splits the Balls (Q433)

Problem: There are 30 blue and 30 red balls and two urns. Your opponent arranges the balls in the two urns any way he likes (each urn non-empty) without telling you. You choose one urn and draw a ball at random from it; a blue ball wins \$10, a red ball wins nothing. How much should you be willing to pay to play?

How this differs from Q53. Now the person arranging the balls wants you to *lose*, and you choose the urn. It is the same equal-weighting effect, used against you: the opponent puts a single *losing* ball alone.

1. What kind of game this is

The opponent chooses how to split the balls, you choose an urn, and the opponent wants you to lose. It is a zero-sum game, so the fair price is the *value*: the win probability you can guarantee against his best arrangement.

The key point: you choose an **urn**, not a ball. Each urn gets weight $\frac{1}{2}$ no matter how many balls it holds, so "half the balls are blue" does *not* mean a $\frac{1}{2}$ chance of winning.

2. Your best strategy: pick an urn at random

You cannot tell the urns apart, and the opponent could label them at random. Any preference for one urn can therefore be exploited, so your best play is to choose each urn with probability $\frac{1}{2}$. If urn i holds b_i blue and r_i red balls ($b_1 + b_2 = r_1 + r_2 = 30$), your win probability is

$$P = \tfrac{1}{2} \cdot \frac{b_1}{b_1 + r_1} + \tfrac{1}{2} \cdot \frac{b_2}{b_2 + r_2}.$$

3. The opponent's best arrangement

He wants to waste one of your two equally likely choices. So he puts **a single red ball alone** in one urn and the other 59 balls (30 blue, 29 red) in the other:

$$P = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2} \cdot \frac{30}{59} = \frac{15}{59} \approx 25.4\%.$$

Checking every possible split (both urns non-empty) confirms that no arrangement pushes your chance below $\frac{15}{59}$.

4. Why this is the value of the game

- By choosing 50/50, you guarantee at least $\frac{15}{59}$ whatever he does.
- By using the "lone red ball" arrangement (with random labels), he holds you to at most $\frac{15}{59}$ whatever you do.

Both sides can guarantee the same number, so the value is $\frac{15}{59}$ and the fair price is

$$10 \cdot \frac{15}{59} = \frac{150}{59} \approx \$2.54.$$

5. Why "\$5" is wrong

It is true that the two urns cannot *both* be better than $\frac{1}{2}$ (the overall blue fraction is $\frac{1}{2}$, and it is an average of the two urns weighted by their sizes). But your pick weights the urns *equally*, not by size. A tiny urn with no blue balls counts just as much as the big one, which is exactly what the opponent exploits.

6. Mirror image and assumptions

- **Q53** is the friendly version: if you arrange 100 green and 100 red marbles to *help* the drawer, put one green alone, giving $\frac{1}{2} + \frac{1}{2} \cdot \frac{99}{199} = \frac{149}{199} \approx 74.9\%$.
- If an urn may be left *empty* (and choosing it means you lose), the opponent can put all 60 balls in one urn, giving $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, i.e. \$2.50. State which rule you assume.

See Concept BO.

Pay at most $10 \cdot \frac{15}{59} = \frac{150}{59} \approx \$2.54$ (not \$5). The opponent puts a single red ball alone, and choosing each urn with probability $\frac{1}{2}$ guarantees you $\frac{15}{59}$; neither side can do better.

9 The Pigeonhole Principle

Problem: What is the minimum number of people in a room to guarantee that at least 5 of them share the same birth month?

Fill the Holes as Evenly as Possible

The 12 months are the "holes." To *avoid* 5 in any single month, the most you can place is 4 per month:

$$12 \times 4 = 48 \text{ people with no month reaching 5.}$$

The very next person forces some month to 5. Hence

$12 \times 4 + 1 =$ **49 people are needed to guarantee** 5 **share a birth month.**

The Principle

If n items are placed into k boxes, some box holds at least $\lceil n/k \rceil$ items.
To **force** at least m items in some box, you need $k(m-1)+1$ items.

Here $k = 12$, $m = 5$, giving $12(5-1)+1 = 49$. (The same formula gives 73 for "7 share a month.")

Additional Problem — The Handshake Pigeonhole

Problem: At a party of $n \geq 2$ people, show that at least two people have shaken the same number of hands.

Each person shakes somewhere between 0 and $n-1$ hands — that looks like n possible values for n people, not enough for pigeonhole on its own. The trick: 0 and $n-1$ **cannot both occur**. If someone shook all $n-1$ others, then nobody shook 0 hands; if someone shook 0, nobody shook everyone. So the handshake counts actually lie in a set of only $n-1$ possible values, shared among n people.

n **people into** $n-1$ **possible handshake counts** $\Rightarrow$ **by pigeonhole, at least two people have the same handshake number.**

Concept Summary and Applications

- **Existence, not construction:** pigeonhole proves *that* a coincidence must exist without finding it. The art is choosing the right "holes," and sometimes shrinking their count (as with $0/n-1$ above).

- **Applications:** guaranteed collisions (hashing, birthdays), approximation results (two of $n+1$ reals within $1/n$), the existence of monochromatic structures (Ramsey-type arguments), and "some substring/subsum must repeat" proofs.

Part II — Expected Value and Key Distributions

10 Binomial Distribution

$B(n,p)$ models the number of successes in n independent Bernoulli(p) trials.

PMF and Key Properties

PMF ($k = 0, 1, \ldots, n$):

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np, \qquad \mathrm{Var}(X) = np(1-p), \qquad M_X(t) = \left((1-p) + pe^t\right)^n$$

Property	Formula / Value
Notation	$B(n,p)$
Support	$k \in \{0, 1, \ldots, n\}$
PMF	$\binom{n}{k} p^k (1-p)^{n-k}$
Mean	np
Variance	$np(1-p)$
Skewness	$(1-2p)/\sqrt{np(1-p)}$
Ex. kurtosis	$(1-6p(1-p))/(np(1-p))$

Understanding the PMF

$$\binom{n}{k} p^k (1-p)^{n-k} = \underbrace{\binom{n}{k}}_{\text{ways to choose } k \text{ successes}} \times \underbrace{p^k}_{\text{prob. of } k \text{ successes}} \times \underbrace{(1-p)^{n-k}}_{\text{prob. of } (n-k) \text{ failures}}$$

Normal Approximation (CLT)

For large n: $B(n,p) \approx \mathcal{N}(np,\, np(1-p))$. Reliable when $np \geq 5$ and $n(1-p) \geq 5$.

Example: Multiple-Choice Quiz

Quiz scenario: *You take a multiple-choice quiz with 5 questions. Each question has 4 choices and you guess randomly, so the probability of getting any one question correct is* $p = \frac{1}{4}$. *Let* X = *number of correct answers. Then* $X \sim B(5, 1/4)$.

Key statistics:

$$E[X] = np = 5 \cdot \tfrac{1}{4} = 1.25, \qquad \mathrm{Var}(X) = np(1-p) = 5 \cdot \tfrac{1}{4} \cdot \tfrac{3}{4} = \tfrac{15}{16} \approx 0.94.$$

Probability of getting exactly 3 correct:

$$P(X = 3) = \binom{5}{3}\left(\tfrac{1}{4}\right)^3\left(\tfrac{3}{4}\right)^2 = 10 \cdot \frac{1}{64} \cdot \frac{9}{16} = \frac{90}{1024} \approx 8.8\%.$$

Probability of passing (3 or more correct):

$$P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) \approx 8.8\% + 1.5\% + 0.1\% \approx 10.4\%.$$

Additional Problem 1 — Counting Heads in a Few Coin Flips (Q83, Q337, Q280, Q345)

Problems: All coins are fair.

(a) (Q83) What is the probability of exactly 4 heads in 9 tosses?
(b) (Q337) What is the probability of at least 4 heads in 7 flips?
(c) (Q280) What is the probability of at most 2 tails when flipping 5 coins?
(d) (Q345) How many tosses are needed so that the probability of getting at least 4 tails is at least 0.7?

The common tool

For a fair coin, every one of the 2^n head/tail sequences of length n has probability $(\frac{1}{2})^n$. A sequence with exactly k heads is fixed by choosing *which* k of the n positions are heads, which can be done in $\binom{n}{k}$ ways. So

$$P(X = k) = \binom{n}{k}\left(\frac{1}{2}\right)^n = \frac{\binom{n}{k}}{2^n}.$$

The numbers $\binom{n}{k}$ form row n of Pascal's triangle (each entry is the sum of the two above it):

$n = 5$	1, 5, 10, 10, 5, 1 (sum 32)
$n = 6$	1, 6, 15, 20, 15, 6, 1 (sum 64)
$n = 7$	1, 7, 21, 35, 35, 21, 7, 1 (sum 128)
$n = 8$	1, 8, 28, 56, 70, 56, 28, 8, 1 (sum 256)
$n = 9$	1, 9, 36, 84, 126, 126, 84, 36, 9, 1 (sum 512)

For "at least" or "at most" questions, add the relevant entries of the row.

(a) Exactly 4 heads in 9 tosses

$$\binom{9}{4} = \frac{9!}{4!\,5!} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} = 126, \qquad P(X = 4) = \frac{126}{512} = \frac{63}{256} \approx 0.246.$$

(b) At least 4 heads in 7 flips — use symmetry

Direct sum: $\dfrac{\binom{7}{4}+\binom{7}{5}+\binom{7}{6}+\binom{7}{7}}{2^7} = \dfrac{35+21+7+1}{128} = \dfrac{64}{128} = \dfrac{1}{2}.$

Symmetry shortcut: with 7 flips there can be no tie between heads and tails. Either heads are the majority (≥ 4 heads) or tails are (≥ 4 tails), and for a fair coin these two events are mirror images, so each has probability $\frac{1}{2}$.

(c) At most 2 tails in 5 flips

$$P(T \leq 2) = \frac{\binom{5}{0}+\binom{5}{1}+\binom{5}{2}}{32} = \frac{1+5+10}{32} = \frac{1}{2}.$$

Same symmetry: "at most 2 tails" is the same as "at least 3 heads", i.e. heads are the majority of 5.

(d) Smallest number of tosses with $P(\geq 4$ tails$) \geq 0.7$

Use the complement $P(T \geq 4) = 1 - P(T \leq 3)$ and try increasing n:

n	$P(T \leq 3) = \dfrac{\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\binom{n}{3}}{2^n}$	$P(T \geq 4)$
6	$\frac{1+6+15+20}{64} = \frac{42}{64}$	0.344
7	$\frac{1+7+21+35}{128} = \frac{64}{128}$	0.500
8	$\frac{1+8+28+56}{256} = \frac{93}{256}$	0.637
9	$\frac{1+9+36+84}{512} = \frac{130}{512}$	**0.746**

$n = 8$ falls short and $n = 9$ first exceeds 0.7. (Quick sanity check: we need about 4 tails, and n tosses give $n/2$ tails on average with a spread of $\sqrt{n}/2 \approx 1.5$; $n = 9$ puts the mean 0.5 above the target — plausible for a 75% chance.)

(a) $\frac{63}{256} \approx 0.246$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{2}$ **(d) 9 tosses (0.746; 8 tosses give only 0.637).**

Additional Problem 2 — One Six in 4 Throws vs. Double Six in 24 Throws (Q498)

Problem: I throw one die 4 times and win if at least one 6 appears. You throw two dice 24 times and win if at least one double six appears. Who is more likely to win? (This is the classic Chevalier de Méré problem.)

Step 1: "At least one" — use the complement

"At least one success" includes $1, 2, 3, \ldots$ successes — messy to add up. Its complement, "no success at all", is a single case. If each try fails with probability $1 - p$ independently, n tries all fail with probability $(1-p)^n$, so

$$P(\text{at least one success in } n \text{ tries}) = 1 - (1-p)^n.$$

Step 2: Compute both

- **One die, 4 throws:** $p = \frac{1}{6}$, so $P(A) = 1 - (\frac{5}{6})^4 = 1 - \frac{625}{1296} = \frac{671}{1296} \approx 0.5177$.
- **Two dice, 24 throws:** a double six has $p = \frac{1}{36}$ (one of 36 equally likely pairs), so $P(B) = 1 - (\frac{35}{36})^{24} \approx 1 - 0.5086 = 0.4914$.

So $P(A) > P(B)$: **the one-die player ("I") is more likely to win.**

Step 3: Why the intuition "they are equal" is wrong

The *expected number* of successes is indeed the same: $4 \cdot \frac{1}{6} = \frac{2}{3}$ and $24 \cdot \frac{1}{36} = \frac{2}{3}$. But "probability of at least one" is not the expected number. With many tries you sometimes get two or more successes; those extra successes add to the expected count but not to the chance of winning (one is enough). With more, rarer tries, more of the expected count is "wasted" on repeats.

A smaller example: one throw of one die ($P = \frac{1}{6} \approx 0.167$) vs. six throws of two dice for a double six ($P = 1 - (\frac{35}{36})^6 \approx 0.156$). Same expected count $\frac{1}{6}$, but the single throw cannot waste anything. In general, with np fixed, $1-(1-p)^n$ decreases as n grows, toward $1-e^{-np}$ (here $1-e^{-2/3} \approx 0.487$).

Step 4: How many double-six throws would make it a good bet?

$1 - (\frac{35}{36})^{24} \approx 0.491 < \frac{1}{2}$ but $1 - (\frac{35}{36})^{25} \approx 0.506 > \frac{1}{2}$: the two-dice player needs 25 throws.

$1 - (\frac{5}{6})^4 \approx 51.8\% \;>\; 1 - (\frac{35}{36})^{24} \approx 49.1\%$**: one die four times is better, even though both expect $\frac{2}{3}$ successes. The double-six bet becomes favourable only at 25 throws.**

Additional Problem 3 — Exactly Half Heads: the Central Term and Stirling's Approximation (Q283)

Problem: Toss n fair coins. What is the probability that exactly half are heads? Answer for $n = 2$, 3 and 20,000.

Step 1: Exact formula

"Exactly half" means $X = n/2$, which is only possible for even n. Then

$$P\Big(X = \frac{n}{2}\Big) = \binom{n}{n/2} 2^{-n} \quad (n \text{ even}), \qquad 0 \quad (n \text{ odd}).$$

- $n = 2$: $\binom{2}{1}/4 = \frac{1}{2}$ (HT or TH out of HH, HT, TH, TT).
- $n = 3$: odd, so 0.
- $n = 20{,}000$: $\binom{20000}{10000}/2^{20000}$ — exact but impossible to evaluate by hand. We need an approximation.

Step 2: Stirling's formula for $n!$

For large n,

$$n! \approx \sqrt{2\pi n}\,\Big(\frac{n}{e}\Big)^n.$$

Where it comes from. (i) The main factor: $\ln n! = \sum_{k=1}^{n} \ln k \approx \int_1^n \ln x\,dx = n\ln n - n + 1$, so $n! \approx (n/e)^n$ up to a smaller factor. (ii) The prefactor: write $n! = \int_0^\infty x^n e^{-x} dx = \int_0^\infty e^{f(x)} dx$ with

$f(x) = n\ln x - x$. The exponent peaks at $x = n$ (where $f'(x) = n/x - 1 = 0$), and near the peak $f(x) \approx f(n) - \frac{(x-n)^2}{2n}$ (since $f''(n) = -1/n$). The integrand is then a Gaussian bump of width $\sqrt{n}$, and $\int e^{-(x-n)^2/(2n)}dx = \sqrt{2\pi n}$, giving $n! \approx (n/e)^n\sqrt{2\pi n}$.

Step 3: Apply it to the central term

Let $n = 2m$. Then

$$\binom{2m}{m} = \frac{(2m)!}{(m!)^2} \approx \frac{\sqrt{4\pi m}\,(2m/e)^{2m}}{\left[\sqrt{2\pi m}\,(m/e)^m\right]^2} = \frac{\sqrt{4\pi m}}{2\pi m}\cdot\frac{(2m)^{2m}}{m^{2m}} = \frac{4^m}{\sqrt{\pi m}}.$$

The huge powers cancel against $2^n = 4^m$:

$$\boxed{P\Big(X = \frac{n}{2}\Big) \approx \frac{1}{\sqrt{\pi m}} = \sqrt{\frac{2}{\pi n}}}$$

Step 4: Check and apply

n	10	20	100	20,000
exact $\binom{n}{n/2}2^{-n}$	0.2461	0.1762	0.0796	—
$\sqrt{2/(\pi n)}$	0.2523	0.1784	0.0798	0.00564

The approximation is already within 3% at $n = 10$. For $n = 20{,}000$: $\sqrt{2/(20000\pi)} = 1/\sqrt{10000\pi} \approx 0.00564$.

Interpretation. "Exactly half" is the single most likely count, yet its probability shrinks like $1/\sqrt{n}$: the number of heads spreads over a window of width about $\sqrt{n}$ (standard deviation $\sqrt{n}/2$), so each value in that window gets only $\sim 1/\sqrt{n}$ of the probability. The normal curve gives the same answer: its peak height is $\frac{1}{\sigma\sqrt{2\pi}}$ with $\sigma = \frac{\sqrt{n}}{2}$, i.e. $\sqrt{2/(\pi n)}$.

P(exactly half) $= \binom{n}{n/2}2^{-n}$ for even n (0 for odd n) $\approx \sqrt{2/(\pi n)}$. $n = 2$: $\frac{1}{2}$; $n = 3$: 0; $n = 20{,}000$: ≈ 0.0056.

Additional Problem 4 — Normal Approximation for Large n: Heads × Tails (Q346)

Problem: Flip 100 fair coins and multiply the number of heads by the number of tails. What is the expected value of this product? Give an approximate 95% range for it.

Step 1: Rewrite in terms of heads only

Let $H \sim B(100, \frac{1}{2})$ be the number of heads; then tails $T = 100 - H$ and the product is $HT = H(100 - H) = 100H - H^2$.

Step 2: The trap and the fix

It is tempting to say $E[H]E[T] = 50 \cdot 50 = 2500$, but H and T are *not* independent (they always add to 100), so $E[HT] \neq E[H]E[T]$. Instead use linearity on $100H - H^2$; we need $E[H^2]$, which comes from the variance:

$$E[H] = np = 50, \qquad \mathrm{Var}(H) = np(1-p) = 25, \qquad E[H^2] = \mathrm{Var}(H) + (E[H])^2 = 25 + 2500 = 2525.$$

$$E[HT] = 100 \cdot 50 - 2525 = \boxed{2475}.$$

General rule: for n fair coins, $E[HT] = \frac{n^2}{4} - \frac{n}{4} = \frac{n(n-1)}{4}$ — below $(\frac{n}{2})^2$ by exactly $\mathrm{Var}(H)$. *Counting check:* HT counts ordered pairs of different coins (first heads, second tails); there are $n(n-1)$ such pairs, each with probability $\frac{1}{4}$.

Step 3: Normal approximation for the range

For large n, the binomial is close to a normal distribution with the same mean and variance: $H \approx N(50,\, 5^2)$. About 95% of a normal lies within 1.96 standard deviations of the mean:

$$H \in 50 \pm 1.96 \cdot 5 \approx [40,\, 60].$$

Now map this to the product. Since $HT = 2500 - (H - 50)^2$, the product is largest (2500) at $H = 50$ and smallest at the ends of the range, $40 \cdot 60 = 2400$. So

$$HT \in [2400,\, 2500] \quad \text{with probability} \approx 95\%$$

(the exact binomial probability of $40 \le H \le 60$ is 96.5%). The interval is lopsided around the mean 2475 because 2500 is a hard ceiling.

$E[HT] = \dfrac{n(n-1)}{4} = 2475$ **for** $n = 100$**; about** 95% **of the time** $2400 \le HT \le 2500$ **(from** $H \approx N(50, 25)$**,** $H \in [40, 60]$**). Related: Q9, Q569, Q207 (Section 14).**

Additional Problem 5 — How the Spread Shrinks with n: 10 or 100 Free Throws? (Q548)

Problem: Each of your free throws goes in independently with probability p (your true shooting rate). You receive \$100 if you make *more* than n% of your throws and pay \$100 if you make *less* than n% (exactly n% pays nothing). Would you rather take 10 throws or 100 throws?

Step 1: The made percentage is a scaled binomial

With m throws, the number made is $X \sim B(m, p)$ and your percentage is X/m. Dividing by m:

$$E\Big[\frac{X}{m}\Big] = p, \qquad \mathrm{SD}\left(\frac{X}{m}\right) = \frac{\sqrt{mp(1-p)}}{m} = \sqrt{\frac{p(1-p)}{m}}.$$

The centre does not move with m, but the spread shrinks like $1/\sqrt{m}$. For $p = 0.6$: about $\pm 15\%$ with 10 throws and $\pm 5\%$ with 100.

Step 2: Who wants more spread?

- If $p > n\%$ (you are the **favourite**), your percentage is centred *above* the target. Less spread means less chance of an unlucky dip below it $\Rightarrow$ choose **100 throws**.
- If $p < n\%$ (you are the **underdog**), you are centred below the target and need luck. More spread gives more chance of a lucky run above it $\Rightarrow$ choose **10 throws**.

This is the law of large numbers: the favourite wants a long match, the underdog a short one.

Step 3: Numbers for a 50% target

Expected payoff = \$100 × [P(above) − P(below)], computed exactly from the binomial:

true rate p	10 throws	100 throws	better choice
0.4	−\$47	−\$96	**10**
0.5	\$0	\$0	either
0.6	+\$47	+\$96	**100**
0.7	+\$80	≈ +\$100	**100**

Example ($p = 0.6$, 10 throws): you win with 6 or more makes, $P = 0.633$; you lose with 4 or fewer, $P = 0.166$; the rest is a tie at exactly 5. Payoff $100(0.633 - 0.166) \approx +\$47$.

Step 4: Caveats

- **Ties pay \$0.** Hitting exactly $n\%$ is much more likely with few throws (≈ 25% for 5/10 at $p = 0.5$, ≈ 8% for 50/100).
- **When $p = n\%$**, the game is exactly fair for $n = 50$ by symmetry; for other targets the binomial's skew tilts it slightly (e.g. $p = n = 70\%$: +\$3.2 with 10 throws, +\$1.1 with 100). These effects are small.

Your percentage has mean p and spread $\sqrt{p(1-p)/m}$, shrinking with more throws. If $p > n\%$, take 100 throws; if $p < n\%$, take 10 throws; if $p = n\%$, there is essentially no edge either way.

11 Geometric Distribution

Models the number of independent Bernoulli trials needed for the *first success*, each succeeding with probability $p \in (0, 1]$.

PMF and Key Properties

PMF (trials until first success, $k = 1, 2, 3, \ldots$):

$$P(X = k) = (1-p)^{k-1}\, p$$

$$E[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}, \qquad M_X(t) = \frac{p\, e^t}{1-(1-p)e^t} \quad (t < -\ln(1-p))$$

Property	Formula / Value
Notation	$\mathrm{Geom}(p)$
Support	$k \in \{1, 2, 3, \ldots\}$
PMF	$(1-p)^{k-1}\, p$
Mean	$1/p$
Variance	$(1-p)/p^2$
Mode	1
MGF	$pe^t/[1-(1-p)e^t], \quad t < -\ln(1-p)$

Alternative: Failures Before First Success

Support $\{0, 1, 2, \ldots\}$:

$$P(X = k) = (1-p)^k\, p, \qquad E[X] = \frac{1-p}{p}.$$

Memoryless Property

The geometric distribution is the **only** memoryless discrete distribution:

$$P(X > m + n \mid X > m) = P(X > n).$$

Recursive Derivation of the Mean

$$E = p \cdot 1 + (1-p)(1+E) \implies pE = 1 \implies \boxed{E[X] = 1/p.}$$

Example: Rolling a Six-Sided Die

Gaming/Gambling: *How many times do you need to roll a standard six-sided die before you roll a "2"?*
Each roll succeeds with probability $p = \frac{1}{6}$*, so* $X \sim \text{Geom}(1/6)$.

$$P(X = k) = \left(\frac{5}{6}\right)^{k-1} \cdot \frac{1}{6}, \qquad k = 1, 2, 3, \ldots$$

Expected number of rolls:

$$E[X] = \frac{1}{p} = \frac{1}{1/6} = 6.$$

On average, you need to roll the die ***6 times*** *to get your first "2". For example, the probability it takes exactly 3 rolls is* $P(X = 3) = (5/6)^2 \cdot (1/6) \approx 11.6\%$.

12 The Coupon Collector Problem

You buy cereal boxes, each containing one of n equally likely coupons. How many boxes until you have every coupon? This *coupon collector* problem — and its relatives (rolling a die until every face appears, drawing cards until every suit appears) — is solved by breaking the waiting time into stages, each a simple "wait for a success" with a known success probability. Variants change what you collect (two copies of each), whether draws are replaced, or ask the reverse question (how many types do m draws reach).

Representative Problem — Every Face Once, Then Every Face Twice (Q665)

Problem: (a) What is the expected number of rolls of a fair six-sided die until every face has appeared at least once? (b) Until every face has appeared at least twice?

Part (a), Step 1: Split the wait into stages

Stage k ($k = 1, \ldots, 6$) is the wait for the k-th *new* face. With $k-1$ faces already seen, a roll shows a new face with probability $\frac{7-k}{6}$.

Part (a), Step 2: Each stage is a geometric wait

If each roll succeeds independently with probability q, the number of rolls until the first success has mean $\frac{1}{q}$ (e.g. waiting for a six takes 6 rolls on average). So stage k takes $\frac{6}{7-k}$ rolls on average:

stage	1	2	3	4	5	6	total
success prob.	$\frac{6}{6}$	$\frac{5}{6}$	$\frac{4}{6}$	$\frac{3}{6}$	$\frac{2}{6}$	$\frac{1}{6}$	
expected rolls	1	1.2	1.5	2	3	6	**14.7**

Part (a), Step 3: Add the stages

By linearity of expectation,

$$E[T_1] = \frac{6}{6} + \frac{6}{5} + \frac{6}{4} + \frac{6}{3} + \frac{6}{2} + \frac{6}{1} = 6\Big(1 + \frac{1}{2} + \cdots + \frac{1}{6}\Big) = 6H_6 = \frac{147}{10} = 14.7.$$

The last face alone takes 6 rolls on average — 40% of the total. The stages are independent, so their variances add: a geometric wait with success probability q has variance $\frac{1-q}{q^2}$, giving $\mathrm{Var}(T_1) = 36\sum_{k=1}^{6}\frac{1}{k^2} - 14.7 \approx 39$, a standard deviation of about 6.2 rolls.

Part (b), Step 1: Why "twice" is not 2×14.7

Running the once-collector twice would take 29.4 throws on average, but that ignores overlap: while you wait for the last new faces, the early faces are already showing up a second time. So the answer is well below 29.4. We need to track both kinds of progress.

Part (b), Step 2: A Markov chain on two counts

Let the state be (a, b): a faces not yet seen, b faces seen exactly once (the rest seen at least twice). Start $(6, 0)$, finish $(0, 0)$. One throw moves

- to $(a-1, b+1)$ with probability $\frac{a}{6}$ (a new face),

- to $(a, b-1)$ with probability $\frac{b}{6}$ (a face's second appearance),
- nowhere with probability $\frac{6-a-b}{6}$ (a face already seen twice).

With $E(a,b)$ the expected remaining throws, $E(a,b) = 1 + \frac{a}{6}E(a-1,b+1) + \frac{b}{6}E(a,b-1) + \frac{6-a-b}{6}E(a,b)$, i.e.

$$E(a,b) = \frac{6 + a\,E(a-1,b+1) + b\,E(a,b-1)}{a+b}, \qquad E(0,0) = 0.$$

Part (b), Step 3: Fill in the table

Row $a = 0$ is the once-collector on b faces: $E(0,b) = 6H_b$. Each later row uses the row above and the entry to its left:

$E(a,b)$	$b=0$	1	2	3	4	5	6
$a=0$	0	6	9	11	12.5	13.7	14.7
$a=1$	12	13.5	14.67	15.63	16.44	17.15	
$a=2$	16.5	17.28	17.95	18.55	19.08		
$a=3$	19.28	19.78	20.24	20.66			
$a=4$	21.28	21.65	21.99				
$a=5$	22.85	23.13					
$a=6$	**24.13**						

For example $E(1,0) = \frac{6+E(0,1)}{1} = 12$ and $E(1,1) = \frac{6+E(0,2)+E(1,0)}{2} = \frac{6+9+12}{2} = 13.5$. The exact value is $E(6,0) = \frac{390968681}{16200000} \approx 24.13$.

Part (b), Step 4: Independent check (Poissonization)

Pretend throws arrive at rate 1 in continuous time; then each face's count by time t is an independent Poisson$(t/6)$ variable. The chance a face has appeared at least twice is $1 - e^{-t/6}(1 + \frac{t}{6})$, so

$$E[T] = \int_0^\infty \left[1 - \left(1 - e^{-t/6}(1 + \tfrac{t}{6})\right)^6\right] dt \approx 24.13,$$

matching the chain (and simulation).

(a) Every face once: $6H_6 = \boxed{14.7}$ rolls (standard deviation ≈ 6.2).
(b) Every face twice: $\frac{390968681}{16200000} \approx \boxed{24.13}$ rolls — about 9.4 more than (a), and well below the naive $2 \times 14.7 = 29.4$. (Part (b) alone is Q90.)

Concept Summary and Applications

Coupon collector (n equally likely types, with replacement):

$$E[T] = \sum_{k=0}^{n-1} \frac{n}{n-k} = nH_n \approx n\ln n + 0.577n, \qquad \mathrm{Var}(T) = \sum_{k=1}^{n} \frac{n(n-k)}{k^2} \quad (\approx 39 \text{ for } n=6).$$

Each type twice: Markov chain on (unseen, seen once), or $E[T] = \int_0^\infty [1-(1-e^{-t/n}(1+\frac{t}{n}))^n]dt$.
Reverse question (m draws, how many types?): $E[\#\text{distinct}] = n(1-(1-\frac{1}{n})^m)$ by indicators.
Without replacement: use max–min inclusion–exclusion with $E[\text{first of } m \text{ special among } n] = \frac{n+1}{m+1}$.

- **Stages:** the last few types dominate the wait (the final face alone takes 6 throws on average).
- **Linearity of expectation** handles "how many" questions without any independence.
- $\frac{1}{e}$ **again:** $m = n$ draws see about $1 - \frac{1}{e} \approx 63\%$ of the types.
- **Applications:** testing coverage, hash-table loading, sampling until every category appears, collecting cards or stickers.

Additional Problem 1 — Minimum Value Before the First 5 (Q125)

Problem: You roll a die until a 5 appears. What is the expected value of the smallest number rolled during the process (including the final 5)?

Step 1: Tail-sum formula

For a variable M taking values $1, 2, \ldots$: $E[M] = \sum_{k\geq 1} P(M \geq k)$ (each value m is counted once for every $k \leq m$). Here $M \in \{1, \ldots, 5\}$.

Step 2: Race between faces

$M \geq k$ means none of $1, \ldots, k-1$ appears before the first 5. Rolls showing any other face are irrelevant to this race; among the faces $\{1, \ldots, k-1, 5\}$, each is equally likely to be the first to appear. So

$$P(M \geq k) = \frac{1}{k} \qquad (k = 1, \ldots, 5).$$

(E.g. $P(M \geq 2) = \frac{1}{2}$: 1 and 5 are equally likely to appear first; 6s and other faces just delay.)

Step 3: Sum

$$E[M] = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = H_5 = \frac{137}{60} \approx 2.28.$$

$P(M \geq k) = \frac{1}{k}$**, so** $E[M] = H_5 = \dfrac{137}{60} \approx 2.28$**.**

Additional Problem 2 — 100 Doors, 100 Rolls of a d100 (Q426)

Problem: 100 doors each hide \$1. You roll a 100-sided die 100 times and may open every door whose number was rolled. What is your expected winnings?

Step 1: Indicators

Your winnings X equal the number of *distinct* numbers rolled. Let $I_i = 1$ if door i is rolled at least once. Then $X = \sum_i I_i$ and, by linearity of expectation (no independence needed), $E[X] = \sum_i P(I_i = 1)$.

Step 2: One door

Door i is missed on one roll with probability $\frac{99}{100}$, on all 100 rolls with probability $(0.99)^{100} \approx e^{-1} \approx 0.366$. So $P(I_i = 1) \approx 0.634$.

Step 3: Total

$$E[X] = 100(1 - 0.99^{100}) \approx 63.4.$$

$E[X] = 100(1 - (\frac{99}{100})^{100}) \approx \$63.4$ — **about $1 - \frac{1}{e}$ of the doors.**

Additional Problem 3 — Cards Until All Four Suits (Q545)

Problem: You draw cards from a shuffled standard deck (no replacement) until you have seen all four suits. What is the expected number of cards drawn?

Step 1: The finish time is a maximum

Let F_s be the position of the first card of suit s. You are done when the slowest suit appears: $T = \max(F_\spadesuit, F_\heartsuit, F_\diamondsuit, F_\clubsuit)$.

Step 2: Max–min inclusion–exclusion

For two numbers, $\max(a, b) = a + b - \min(a, b)$. For four, with alternating signs over all subsets:

$$\max = \sum_i x_i - \sum_{i<j} \min(x_i, x_j) + \sum_{i<j<k} \min(x_i, x_j, x_k) - \min(x_1, \ldots, x_4).$$

The minimum over j suits is the position of the first card among $13j$ "special" cards.

Step 3: Where does the first special card land?

If m special cards are shuffled into n cards, they split the other $n - m$ cards into $m + 1$ gaps of equal average size $\frac{n-m}{m+1}$; the first special card follows the first gap:

$$E[\text{first position}] = \frac{n-m}{m+1} + 1 = \frac{n+1}{m+1}.$$

Step 4: Put it together ($n = 52$)

$$E[T] = 4 \cdot \frac{53}{14} - 6 \cdot \frac{53}{27} + 4 \cdot \frac{53}{40} - 1 = 53\Big(\frac{180 - 140 + 63}{630}\Big) - 1 = \frac{5459}{630} - 1 = \frac{4829}{630} \approx 7.665.$$

Step 5: With vs. without replacement

With replacement (a 4-coupon collector) the answer would be $4H_4 = \frac{25}{3} \approx 8.33$. Without replacement every extra card of a seen suit is removed, so the unseen suits become *more* likely and you finish sooner. (The shortcut $1 + \frac{51}{39} + \frac{50}{26} + \frac{49}{13} = 8$ wrongly freezes those probabilities.)

$E[T] = \dfrac{4829}{630} \approx 7.67$ **cards (vs.** $\frac{25}{3} \approx 8.33$ **with replacement).**

13 Poisson Process and Arrival Rates

Problem: The probability of seeing at least one car in an hour is $\frac{544}{625}$. What is the probability of seeing *no* cars in a 15-minute interval?

Background: the Poisson Process

A **Poisson process** models events (arrivals) occurring in continuous time at a constant average rate λ per unit time, with three defining properties:

- **Independent increments:** counts in disjoint time intervals are independent.
- **Stationary increments:** the count in an interval depends only on its *length*, not its location.
- **Orderliness:** arrivals occur one at a time (no simultaneous events).

Then the number of arrivals in an interval of length t is Poisson(λt), the waiting time between arrivals is Exponential(λ), and in particular

$$P(\text{no arrivals in length } t) = e^{-\lambda t}.$$

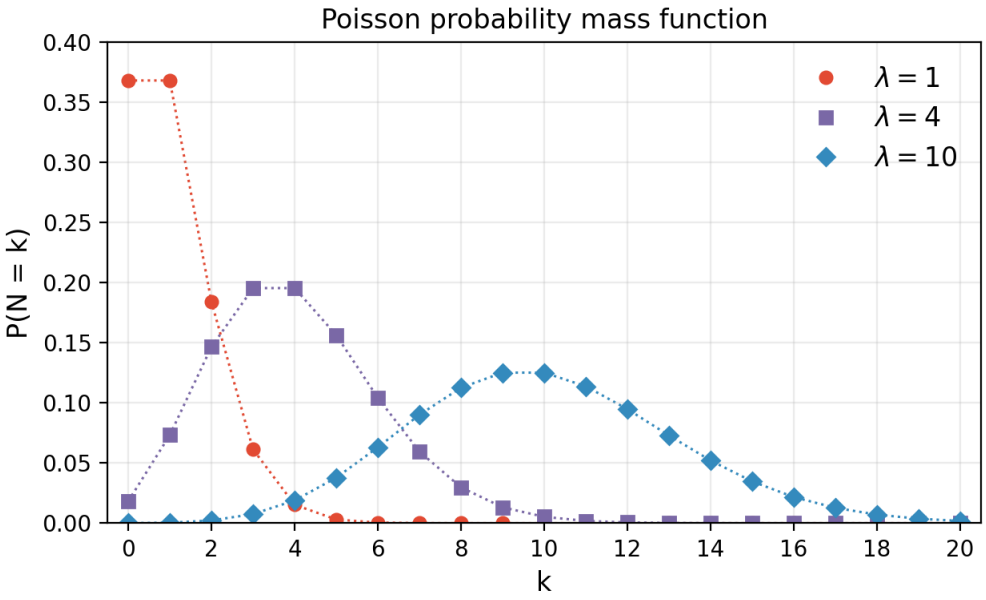
Background: the Poisson Distribution

The Poisson distribution describes a **count** — how many times something happens — when the events are rare per opportunity but there are many opportunities. A single number $\mu > 0$, the expected count, controls it completely:

$N \sim \text{Poisson}(\mu)$: $\quad P(N = k) = e^{-\mu}\,\dfrac{\mu^k}{k!}, \quad k = 0, 1, 2, \ldots$

Mean = Variance = μ.

Reading the formula: $\mu^k/k!$ grows then shrinks as k increases, and the $e^{-\mu}$ prefactor is exactly the constant that makes all the probabilities sum to 1 (it is the Taylor series $\sum_k \mu^k/k! = e^{\mu}$). The distribution peaks near $k = \mu$ and, as μ grows, spreads out and becomes bell-shaped:



For $\mu = 1$ the mass sits on the smallest counts (most likely 0 or 1); for $\mu = 4$ it centres near 4; for $\mu = 10$ it is a wide, near-symmetric hump around 10. In every case the mean and the variance both equal μ, so the spread (standard deviation $\sqrt{\mu}$) grows only like the square root of the mean.

The Law of Rare Events

Where does this formula come from? Imagine chopping a time interval into n tiny slots, each of which independently either contains one arrival (probability p) or none. The total count is then Binomial(n, p), with mean $\mu = np$. Now let the slots become finer and finer: $n \to \infty$ while each slot gets rarer, $p \to 0$, holding the average $np = \mu$ fixed. In that limit the binomial probabilities converge to the Poisson formula:

$$\binom{n}{k} p^k (1-p)^{n-k} \xrightarrow[\substack{n\to\infty,\ p\to 0\\ np\to\mu}]{} e^{-\mu}\,\frac{\mu^k}{k!}.$$

This is the **law of rare events**: a Poisson count is what you get from a huge number of independent trials, each individually very unlikely to fire, when the *expected* number of firings is moderate. The three binomial ingredients map cleanly onto the limit: $\binom{n}{k} p^k \to \mu^k/k!$ counts the ways to place k successes, and $(1-p)^{n-k} \to e^{-\mu}$ is the chance the other slots stay empty (using $(1 - \frac{\mu}{n})^n \to e^{-\mu}$).

Numeric example. Take $\mu = 2$ and watch Binomial$(n, \frac{2}{n})$ approach Poisson(2) for the probability of exactly $k = 1$ event:

	$n = 5,\ p = 0.4$	$n = 20,\ p = 0.1$	$n = 100,\ p = 0.02$	Poisson(2)
$P(N = 1)$	0.2592	0.2702	0.2707	$2e^{-2} = 0.2707$

Even by $n = 100$ the binomial value is indistinguishable from the Poisson limit to four decimals. A classic real-world instance: the yearly number of Prussian cavalrymen killed by horse kicks per

army corps (a rare event with many soldier-years of exposure, averaging about 0.7 deaths) follows a Poisson distribution almost exactly.

Rule of thumb. The approximation Binomial$(n, p) \approx$ Poisson(np) is already excellent when $n \geq 20$ and $p \leq 0.05$, and it turns an awkward binomial (large n, tiny p, huge factorials) into a one-line Poisson evaluation. In our arrivals problem, $\mu = \lambda t$ is the expected number of arrivals in an interval of length t.

Solving the Car Problem

Let $p = P(\text{no cars in 15 min})$. An hour is four disjoint 15-minute intervals, and by **independent increments** the "no car" events multiply:

$$P(\text{no cars in an hour}) = p^4.$$

We are given $P(\geq 1 \text{ car in an hour}) = \frac{544}{625}$, so

$$p^4 = P(\text{no cars in an hour}) = 1 - \tfrac{544}{625} = \tfrac{81}{625} \implies p = \left(\tfrac{81}{625}\right)^{1/4} = \tfrac{3}{5}.$$

$$P(\textbf{no cars in 15 min}) = \frac{3}{5}.$$

Equivalently in Poisson terms: $e^{-\lambda} = \frac{81}{625}$, so $e^{-\lambda/4} = (e^{-\lambda})^{1/4} = \frac{3}{5}$ — the "no arrival" probability over a quarter interval is the fourth root of the hourly one.

Concept Summary and Applications

- **Key handle:** $P(0 \text{ in } t) = e^{-\lambda t}$, and disjoint intervals are independent — so "no event" probabilities multiply and rates add. Splitting/merging Poisson streams keeps them Poisson.
- **Rate scaling:** halving the interval takes the "no arrival" probability to its square root; the count's mean scales linearly with time.
- **Applications:** order-flow and trade-arrival modelling, queueing and call centres, radioactive decay, and defect/failure counts.

14 Expected Product of Heads and Tails

A short calculation that tests two basic tools at once: rewriting a product of two dependent counts in terms of one variable, and using $E[X^2] = \mathrm{Var}(X) + (E[X])^2$.

Representative Problem — Ten Coins (Q9, Q569)

Problem: If you flip a fair coin 10 times, what is the expected value of the product of the number of heads and the number of tails?

Step 1 — write tails in terms of heads. Let H be the number of heads and T the number of tails. Every toss is one or the other, so

$$T = 10 - H, \qquad HT = H(10 - H) = 10H - H^2.$$

By linearity of expectation,

$$E[HT] = 10\,E[H] - E[H^2].$$

Step 2 — the mean and variance of H. $H \sim \mathrm{Binomial}(10, \frac{1}{2})$, so

$$E[H] = np = 10 \cdot \tfrac{1}{2} = 5, \qquad \mathrm{Var}(H) = np(1-p) = 10 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{5}{2}.$$

Step 3 — get $E[H^2]$ from the variance.

$$E[H^2] = \mathrm{Var}(H) + (E[H])^2 = \tfrac{5}{2} + 25 = \tfrac{55}{2}.$$

Step 4 — combine.

$$E[HT] = 10 \cdot 5 - \tfrac{55}{2} = 50 - 27.5 = \tfrac{45}{2} = 22.5.$$

$E[H \cdot T] = \frac{45}{2} = 22.5$.

Why it is not $5 \times 5 = 25$. Heads and tails cannot both be large: when H is above 5, T is below 5 by the same amount. Writing $H = 5 + D$ gives $HT = (5 + D)(5 - D) = 25 - D^2$, so

$$E[HT] = 25 - E[D^2] = 25 - \mathrm{Var}(H) = 25 - 2.5 = 22.5.$$

The shortfall from 25 is exactly the variance.

General Formula

For n fair coins, with H heads and $T = n - H$ tails:

$$\boldsymbol{E[H \cdot T] = n \cdot \frac{n}{2} - \left(\frac{n}{4} + \frac{n^2}{4}\right) = \frac{n^2}{4} - \mathrm{Var}(H) = \frac{n(n-1)}{4}.}$$

For a coin with $P(\text{heads}) = p$: $E[H \cdot T] = n(n-1)\,p(1-p)$.

Counting view (no algebra). $H \cdot T$ counts the ordered pairs of *different* coins (i, j) with coin i heads and coin j tails. There are $n(n-1)$ ordered pairs of different coins, and each pair is "heads, tails" with probability $p(1-p) = \frac{1}{4}$. By linearity of expectation, $E[H \cdot T] = n(n-1) \cdot \frac{1}{4}$.

coins n	10	12	100	general
$E[H \cdot T]$	22.5	33	2475	$\frac{n(n-1)}{4}$
compare $(n/2)^2$	25	36	2500	$\frac{n^2}{4}$
deck questions	Q9, Q569	Q207	Q346	

Additional Problem — 100 Coins, with an Interval (Q346)

Problem: If you flip 100 coins and multiply the number of heads by the number of tails, what is the expected value of that product? Can you give an interval for it?

Expected value. With $H \sim \text{Binomial}(100, \frac{1}{2})$: $E[H] = 50$, $\text{Var}(H) = 100 \cdot \frac{1}{2} \cdot \frac{1}{2} = 25$, so $E[H^2] = 25 + 50^2 = 2525$ and

$$E[HT] = 100 \cdot 50 - 2525 = 2475 = \frac{100 \cdot 99}{4}.$$

An interval. Write $H = 50 + D$, so

$$HT = (50 + D)(50 - D) = 2500 - D^2.$$

The product is at most 2500 (when $H = 50$) and falls off quadratically as H moves away from 50. Using the normal approximation $H \approx N(50, 25)$ (standard deviation 5), about 95% of the time H is within $50 \pm 1.96 \cdot 5 \approx 50 \pm 10$, i.e. $|D| \leq 10$. Then $D^2 \leq 100$, so

$$HT \in [2500 - 100,\ 2500] = [2400,\ 2500]$$

with probability about 95% (exactly $P(40 \leq H \leq 60) \approx 96.5\%$).

$E[H \cdot T] = 2475$**; an approximate** 95% **interval for the product is** [2400, 2500]**.**

Remarks.

- This is a *prediction* interval for one experiment's $H \cdot T$, not a confidence interval for an unknown parameter.
- The interval is lopsided around the mean 2475 because $HT = 2500 - D^2$ is capped at 2500: the distribution piles up just below the cap and has a long tail downward.

Concept Summary and Applications

- **Reduce to one variable:** $T = n - H$, so $E[HT] = nE[H] - E[H^2]$.
- **Second moment from the variance:** $E[H^2] = \text{Var}(H) + (E[H])^2$.

- **Product of negatively linked counts:** $E[HT] = (n/2)^2 - \mathrm{Var}(H)$ — always below $(n/2)^2$, because when one count is high the other is low.
- **Counting view:** $H \cdot T$ counts (heads, tails) pairs, giving $n(n-1)p(1-p)$ by linearity.
- **Applications:** expected number of mixed pairs or cross-group interactions; the same pair counting gives $E[H(H-1)] = n(n-1)p^2$ and hence the binomial variance; quick sanity checks when making markets on functions of coin flips.

15 Correlation Bounds: Valid Correlation Matrices and the Angle Picture

Correlation measures how strongly two variables move together, on a scale from -1 (perfectly opposite) to $+1$ (perfectly together). A natural question is whether correlation is "transitive": if X moves with Y and Y moves with Z, must X move with Z? The answer is *partly*: the two known correlations do not fix the third, but they confine it to an interval. This section derives that interval in two ways — with matrices and with angles.

Representative Problem — Range of the Third Correlation (Q518)

Problem: X and Y have correlation 0.9, and Y and Z have correlation 0.8. What are the smallest and largest possible values of the correlation between X and Z?

Step 1: Reduce to standardised variables

Correlation does not change if we shift or rescale a variable, so assume each of X, Y, Z has mean 0 and variance 1. Then the correlation is simply $\rho_{XY} = E[XY]$, and the *correlation matrix*

$$R = \begin{pmatrix} 1 & \rho_{XY} & \rho_{XZ} \\ \rho_{XY} & 1 & \rho_{YZ} \\ \rho_{XZ} & \rho_{YZ} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0.9 & \rho \\ 0.9 & 1 & 0.8 \\ \rho & 0.8 & 1 \end{pmatrix}$$

is the covariance matrix of (X, Y, Z), where $\rho = \rho_{XZ}$ is unknown.

Step 2: Matrix view — variances cannot be negative

For any numbers w_1, w_2, w_3, the combination $W = w_1X + w_2Y + w_3Z$ is a random variable, so its variance is ≥ 0. Expanding,

$$\mathrm{Var}(W) = \sum_{i,j} w_i w_j \,\mathrm{Cov}(\cdot,\cdot) = w^{\mathsf{T}} R w \;\geq\; 0 \quad \text{for every } w.$$

A matrix with this property is called **positive semidefinite** (PSD). So ρ is possible exactly when R is PSD. For a symmetric 3×3 matrix, PSD means all its principal minors (determinants of the square sub-matrices on the diagonal) are ≥ 0:

- 1×1: the diagonal 1's — fine.
- 2×2: $1 - 0.9^2$, $1 - 0.8^2$, $1 - \rho^2$ — all ≥ 0 automatically when $|\rho| \leq 1$.
- 3×3: $\det R \geq 0$ — the only real condition.

Step 3: Compute the determinant

For a correlation matrix with off-diagonal entries a, b, ρ (here $a = \rho_{XY}$, $b = \rho_{YZ}$), expanding along the first row,

$$\det R = 1 \cdot (1 - b^2) - a(a - b\rho) + \rho(ab - \rho) = 1 + 2ab\rho - a^2 - b^2 - \rho^2.$$

With $a = 0.9$, $b = 0.8$:

$$\det R = 1 + 1.44\rho - 0.81 - 0.64 - \rho^2 = -\rho^2 + 1.44\rho - 0.45 \;\geq\; 0 \iff \rho^2 - 1.44\rho + 0.45 \leq 0.$$

Step 4: Solve the quadratic

The roots are

$$\rho = \frac{1.44 \pm \sqrt{1.44^2 - 4(0.45)}}{2} = \frac{1.44 \pm \sqrt{0.2736}}{2} = \frac{1.44 \pm 0.5231}{2},$$

and the quadratic is ≤ 0 between them:

$$0.4585 \;\leq\; \rho_{XZ} \;\leq\; 0.9815.$$

General formula. Solving $1 + 2ab\rho - a^2 - b^2 - \rho^2 \geq 0$ for ρ in general:

$$\boxed{\rho_{XZ} \in \left[ab - \sqrt{(1-a^2)(1-b^2)},\quad ab + \sqrt{(1-a^2)(1-b^2)} \right]}$$

Here $ab = 0.72$ and $\sqrt{0.19 \cdot 0.36} = \sqrt{0.0684} = 0.2615$, giving 0.72 ± 0.2615.

Step 5: Angle view — correlations are cosines

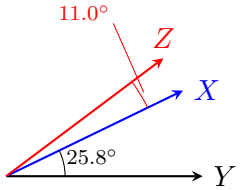
Think of each standardised variable as an arrow of length 1 (formally a vector in the space of random variables, with dot product $E[XY]$). The correlation of two variables is then the dot product of two unit arrows, which is the cosine of the angle between them: $\rho = \cos\theta$.

- $\rho_{XY} = 0.9 \;\Rightarrow\; \theta_{XY} = \arccos 0.9 \approx 25.8°$;
- $\rho_{YZ} = 0.8 \;\Rightarrow\; \theta_{YZ} = \arccos 0.8 \approx 36.9°$.

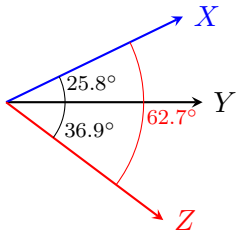
Angles between arrows obey a triangle inequality: the angle from X to Z is at least the difference and at most the sum,

$$|25.8° - 36.9°| = 11.0° \;\leq\; \theta_{XZ} \;\leq\; 25.8° + 36.9° = 62.7°.$$

The extremes occur when all three arrows lie in one plane, with X and Z on the same side of Y or on opposite sides:



same side:
$\cos 11.0° \approx 0.9815$ (max)



opposite sides:
$\cos 62.7° \approx 0.4585$ (min)

Writing $a = \cos\alpha$, $b = \cos\beta$, the cosine addition formula gives

$$\cos(\alpha \mp \beta) = \cos\alpha\cos\beta \pm \sin\alpha\sin\beta = ab \pm \sqrt{1-a^2}\sqrt{1-b^2},$$

exactly the bounds from the matrix view.

Step 6: Both bounds are attainable — an explicit construction

Let Y and U be independent standard variables. Set

$$X = 0.9\,Y + \sqrt{0.19}\,U, \qquad Z = 0.8\,Y \pm \sqrt{0.36}\,U.$$

Each has variance 1 (e.g. $0.81 + 0.19$), $\mathrm{Corr}(X,Y) = 0.9$, $\mathrm{Corr}(Z,Y) = 0.8$, and $\mathrm{Corr}(X,Z) = 0.72 \pm \sqrt{0.19}\sqrt{0.36} = 0.72 \pm 0.2615$. The "+" sign puts X, Z on the same side of Y (max); the "−" sign on opposite sides (min). Any value in between is reached by mixing in a third independent variable (tilting Z out of the plane).

$\rho_{XZ} \in [\,0.72-0.2615,\ 0.72+0.2615\,] = [\,0.4585,\ 0.9815\,]$. **Matrix view:** $\det R \geq 0$. **Angle view:** θ_{XZ} **between** $36.9° - 25.8°$ **and** $36.9° + 25.8°$. **Two strong correlations force** X, Z **to be positively correlated, but only at least** 0.46.

Concept Summary and Applications

Given $\rho_{XY} = a$, $\rho_{YZ} = b$:

$$\rho_{XZ} \in \left[\, ab - \sqrt{(1-a^2)(1-b^2)},\ ab + \sqrt{(1-a^2)(1-b^2)} \,\right] = \left[\cos(\alpha+\beta),\ \cos(\alpha-\beta)\right], \quad a = \cos\alpha,\ b = \cos\beta.$$

Matrix view: a correlation matrix must be positive semidefinite (every combination has variance ≥ 0); for 3×3: $1 + 2ab\rho - a^2 - b^2 - \rho^2 \geq 0$.
Angle view: standardised variables are unit vectors; $\rho = \cos\theta$; angles obey the triangle inequality.

- **Forced sign:** ρ_{XZ} must be positive iff the lower bound is positive, i.e. $ab > \sqrt{(1-a^2)(1-b^2)}$, which simplifies to $a^2 + b^2 > 1$ (for $a, b > 0$). In angles: $\alpha + \beta < 90°$.
- **Special cases:** $a = 1$ forces $\rho_{XZ} = b$ exactly; $a = b = 0$ allows anything in $[-1, 1]$.
- **Equal correlations:** n variables all with pairwise correlation ρ need $\rho \geq -\frac{1}{n-1}$ (Additional Problem 2).
- **Applications:** checking that a correlation matrix used in risk models or simulations is valid (PSD), stress-testing correlations, hedging with a proxy (how well can Y stand in for X?), and factor models.

Additional Problem 1 — When Is the Sign Forced?

Problem: (a) If $\rho_{XY} = \rho_{YZ} = 0.6$, can X and Z be negatively correlated? (b) For which values $a = \rho_{XY} > 0$, $b = \rho_{YZ} > 0$ must ρ_{XZ} be positive?

(a) Plug into the formula

$ab = 0.36$ and $\sqrt{(1-0.36)(1-0.36)} = 0.64$, so

$$\rho_{XZ} \in [\,0.36 - 0.64,\ 0.36 + 0.64\,] = [\,-0.28,\ 1\,].$$

Yes: X and Z can be negatively correlated (down to -0.28), even though each is fairly strongly correlated with Y. In angles: $\arccos 0.6 \approx 53.1°$, and on opposite sides of Y the angle is $106.3° > 90°$, whose cosine is negative.

(b) The general condition

ρ_{XZ} is forced positive iff the lower bound is positive:

$$ab > \sqrt{(1-a^2)(1-b^2)} \iff a^2b^2 > 1 - a^2 - b^2 + a^2b^2 \iff a^2 + b^2 > 1.$$

In angles: $\alpha + \beta < 90°$, so even the "opposite sides" configuration stays within a right angle. Q518 has $0.81 + 0.64 = 1.45 > 1$ (forced positive); here $0.36 + 0.36 = 0.72 < 1$ (not forced).

(a) Yes: $\rho_{XZ} \in [-0.28, 1]$. (b) ρ_{XZ} is forced positive exactly when $a^2 + b^2 > 1$ (equivalently $\alpha + \beta < 90°$).

Additional Problem 2 — How Negative Can Equal Correlations Be?

Problem: Three variables X, Y, Z all have the same pairwise correlation ρ. What is the smallest possible ρ? What about n variables?

Step 1: Use the variance of the sum

Standardise all variables (variance 1). The sum $S = X_1 + \cdots + X_n$ has

$$\mathrm{Var}(S) = \sum_i \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j) = n + n(n-1)\rho \;\geq\; 0 \quad \Longrightarrow \quad \rho \geq -\frac{1}{n-1}.$$

For $n = 3$: $\rho \geq -\frac{1}{2}$.

Step 2: It is attainable

Take three unit arrows in a plane at $120°$ to each other (like a Mercedes logo): each pair has $\cos 120° = -\frac{1}{2}$, and the arrows add to zero. In random-variable terms: with independent standard U_1, U_2, U_3 and mean $\bar{U}$, the variables $X_i = (U_i - \bar{U})/\sqrt{2/3}$ have pairwise correlation exactly $-\frac{1}{2}$ and sum to 0. (Check with the matrix view: with $a = b = \rho = -\frac{1}{2}$, $\det R = 1 + 2(-\frac{1}{8}) - 3 \cdot \frac{1}{4} = 0$, the borderline.)

Intuition

Two variables can be perfectly opposite ($\rho = -1$), but three cannot all be opposite to each other: if X is opposite to Y and Y opposite to Z, then X and Z move together. The more variables, the less room there is for all of them to be negatively correlated — the limit is $-\frac{1}{n-1} \to 0$.

Three variables: $\rho \geq -\frac{1}{2}$ (attained by three arrows at $120°$). In general n equally correlated variables need $\rho \geq -\dfrac{1}{n-1}$, from $\mathrm{Var}(X_1 + \cdots + X_n) = n + n(n-1)\rho \geq 0$.

16 Renewal Theory and the Inspection Paradox

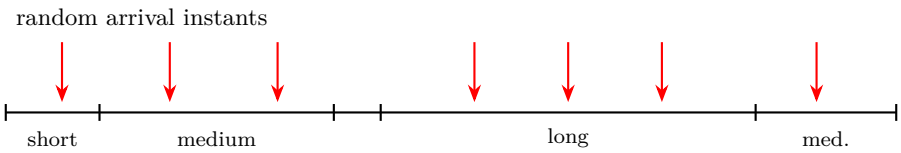
What Renewal Theory Is (in Plain Words)

Renewal theory studies things that happen over and over, separated by *random* waits. Buses arriving at a stop, light bulbs burning out and being replaced, machines failing, customers walking into a shop, earthquakes on a fault line — each is a stream of events with a random gap between one and the next. Call those gaps $L_1, L_2, L_3, \ldots$ and assume they are independent and share the same distribution, with average length $\mathbb{E}[L]$. The word "renewal" captures the key idea: each time an event happens, the process *starts over* statistically — it forgets the past, and the waiting clock resets. Renewal theory answers questions like "how many events by time t?" and, the one we care about, "if I show up at a random moment, how long until the next event?"

The Inspection Paradox (in Plain Words)

Here is the surprise. Suppose buses come *on average* every 10 minutes, and you arrive at the stop at a random time. How long do you wait? The tempting answer is "half the average gap," i.e. 5 minutes. **But you typically wait longer than that.** This is the **inspection paradox**.

Why? Because a random arrival instant is *not* equally likely to fall in every gap — it is more likely to fall inside a *long* gap than a short one, simply because a long gap covers more of the timeline. Imagine throwing a dart at the time axis, which is chopped into intervals of different widths: the dart is more likely to land in a wide interval than a narrow one. So the gap you happen to land in is *not* a typical gap; it is biased toward being long. This is called **length-biasing** (or size-biased sampling).



Seven random arrivals dropped on a timeline of unequal gaps: the long gap catches three of them, the tiny gap none. The gap you experience tends to be longer than a typical gap — that is the paradox.

Everyday versions of the same effect. Ask students "how big is your class?" and the average answer exceeds the true average class size, because more students sit in the big classes. Your friends have, on average, more friends than you do (the "friendship paradox"), because popular people appear on many friend lists. Each is length-biasing in disguise: sampling *by a random member* over-weights the large groups.

Turning the Intuition into a Formula

Your waiting time is the *residual life*: the time from your random arrival to the next event. Two effects combine.

(i) You land in long gaps more often. The gap you fall into has a **length-biased** distribution — a gap of length ℓ is ℓ times more likely to be the one you hit — with density $\tilde{f}(\ell) = \ell\, f(\ell)/\mathbb{E}[L]$ and mean

$$\mathbb{E}[\tilde{L}] = \frac{\mathbb{E}[L^2]}{\mathbb{E}[L]} \;\geq\; \mathbb{E}[L].$$

(ii) Within that gap you sit at a uniform spot. Given the gap you are in has length ℓ, your arrival is uniformly placed inside it, so your wait averages $\ell/2$. Combining,

$$\mathbb{E}[W] = \tfrac{1}{2}\,\mathbb{E}[\tilde{L}] = \frac{\mathbb{E}[L^2]}{2\,\mathbb{E}[L]}.$$

Splitting $\mathbb{E}[L^2] = \mathrm{Var}(L) + \mathbb{E}[L]^2$ exposes exactly where the paradox lives:

$$\mathbb{E}[W] = \underbrace{\frac{\mathbb{E}[L]}{2}}_{\text{naive guess}} + \underbrace{\frac{\mathrm{Var}(L)}{2\,\mathbb{E}[L]}}_{\text{paradox penalty}} = \frac{\mathbb{E}[L]}{2}(1 + \mathrm{CV}^2), \qquad \mathrm{CV}^2 = \frac{\mathrm{Var}(L)}{\mathbb{E}[L]^2}.$$

Here CV is the **coefficient of variation** — the standard deviation divided by the mean, $\mathrm{CV} = \sqrt{\mathrm{Var}(L)}/\mathbb{E}[L]$ — a unitless measure of how spread out the gaps are relative to their average. The extra wait beyond the naive $\mathbb{E}[L]/2$ comes *entirely* from the variability of the gaps. Two extremes: **constant** gaps ($\mathrm{CV} = 0$) give $\mathbb{E}[W] = \mathbb{E}[L]/2$ — no paradox; **exponential** gaps (a Poisson process, $\mathrm{CV} = 1$) give $\mathbb{E}[W] = \mathbb{E}[L]$ — by memorylessness you wait a *whole* mean gap, no matter how long you have already stood there.

Representative Problem — The Bus and the Ice-Cream Stop

Problem: A bus runs a loop that takes exactly 10 minutes, restarting as soon as it finishes. (a) If you arrive at a uniformly random time, what is your expected wait? (b) Now suppose there is an ice-cream shop on the route; each time the bus passes it, the driver flips a fair coin and, on heads, stops for 10 minutes to eat ice cream before continuing. What is your expected wait now? (c) How does the answer change if the coin is biased?

Part (a) — a clockwork 10-minute loop. The bus visits your stop once every 10 minutes exactly. Arriving uniformly at random, the time *since* the last bus is uniform on $[0, 10]$, so the time *until* the next is also uniform on $[0, 10]$:

$$\mathbb{E}[W] = \frac{10}{2} = 5 \text{ minutes}.$$

Here every gap is the same length, so there is no length-biasing — the naive answer is correct.

Part (b) — the ice-cream stop makes gaps random. Between two successive visits to your stop, the bus passes the shop exactly once, so there is one coin flip per cycle. With stop-probability p (fair coin: $p = \frac{1}{2}$), the gap between visits is

$$L = \begin{cases} 10, & \text{probability } 1-p, \\ 20, & \text{probability } p, \end{cases} \qquad \mathbb{E}[L] = 10(1-p) + 20p = 10(1+p).$$

The tempting answer $\mathbb{E}[L]/2$ is **wrong** — this is exactly the inspection paradox. Use the residual-life formula. First the second moment:

$$\mathbb{E}[L^2] = 10^2(1-p) + 20^2 p = 100 + 300p,$$

so

$$\mathbb{E}[W] = \frac{\mathbb{E}[L^2]}{2\,\mathbb{E}[L]} = \frac{100+300p}{2\cdot 10(1+p)} = 5\,\frac{1+3p}{1+p} \text{ minutes.}$$

For the fair coin $p = \frac{1}{2}$:

$$\mathbb{E}[W] = 5\cdot\frac{1+3/2}{1+1/2} = 5\cdot\frac{5/2}{3/2} = \frac{25}{3} \approx 8.33 \text{ minutes.}$$

Sanity check via the variance form. Here $\mathbb{E}[L] = 15$ and $\mathrm{Var}(L) = \mathbb{E}[L^2] - \mathbb{E}[L]^2 = 250 - 225 = 25$, so $\mathrm{CV}^2 = \frac{25}{225} = \frac{1}{9}$ and $\mathbb{E}[W] = \frac{15}{2}(1+\frac{1}{9}) = \frac{25}{3}$. The 0.83 minutes above the naive 7.5 is precisely the paradox penalty.

Part (c) — biased coin. The general answer $\mathbb{E}[W] = 5\,\frac{1+3p}{1+p}$ rises smoothly from 5 to 10 minutes as p goes from 0 to 1:

- $p = 0$ (never stops): every gap is 10, so $\mathbb{E}[W] = 5$ min. ✓
- $p = 1$ (always stops): every gap is 20, so $\mathbb{E}[W] = 10$ min. ✓

Both endpoints have *constant* gaps, so the paradox penalty vanishes there; it is largest for intermediate p, where the gap length is most uncertain.

Plain loop: $\mathbb{E}[W] = 5$ **min.** **Fair-coin stop:** $\mathbb{E}[W] = \frac{25}{3} \approx 8.33$ **min.** **Biased coin (stop prob.** p**):** $\mathbb{E}[W] = 5\,\frac{1+3p}{1+p}$ **min.**

Concept Summary and Applications

- **Key formula:** a random observer's wait to the next event is $\mathbb{E}[W] = \frac{\mathbb{E}[L^2]}{2\,\mathbb{E}[L]} = \frac{\mathbb{E}[L]}{2}(1+\mathrm{CV}^2)$ — always at least the naive $\mathbb{E}[L]/2$, with the excess set by the gaps' variability.
- **Why:** sampling by a random time-point (or a random member) over-weights long gaps (large groups); this length-biasing raises the mean of the sampled gap to $\mathbb{E}[L^2]/\mathbb{E}[L]$.
- **Two anchors:** deterministic gaps ⇒ no paradox; a Poisson process ⇒ you wait a full mean gap ($\mathbb{E}[W] = \mathbb{E}[L]$).
- **Applications:** bus/subway waiting times, time-to-failure and maintenance planning, queueing and server load, network sampling (friendship paradox), and any "sampled-by-size" bias in data.

17 Nonlinear Random Maps: Recursion on the State and Tracking the Big Paths

In many expectation problems you can update the *average* directly: if each step adds a random amount, the expected total just adds up (linearity of expectation). That shortcut breaks as soon as a step applies a *nonlinear* function to your current value — for example $x \mapsto 1/x$ — because in general $E[g(X)] \neq g(E[X])$. This section shows the two tools that work instead:

1. **Recursion on the state:** write the expected final value as a function of the *current value and number of steps left*, and recurse on that function.
2. **Tracking the big paths:** when one outcome is huge and the rest are small, find exactly which paths keep the huge value; they dominate the mean.

Representative Problem — Add One or Take the Reciprocal (Q459)

Problem: You start with \$100 and flip a fair coin 7 times. On heads you gain \$1 ($x \mapsto x+1$). On tails your amount is replaced by its reciprocal ($x \mapsto 1/x$; e.g. \$100 becomes \$0.01, and \$0.01 becomes \$100 again). What is the expected amount of money after 7 flips?

Step 1: Why the usual "average step" trick fails

For a purely additive game (say $+1$ on heads, -1 on tails), $E[X_n] = E[X_{n-1}] + E[\text{step}]$, so you never need anything but the running mean. Here a tails applies $x \mapsto 1/x$, and

$$E\left[\frac{1}{X}\right] \neq \frac{1}{E[X]}.$$

Example after one flip: $X_1 = 101$ or 0.01, each with probability $\frac{1}{2}$, so $E[X_1] = 50.505$. On the second flip a tails gives $1/X_1$, which is $\frac{1}{101} \approx 0.0099$ or 100, average ≈ 50.005 — not $1/50.505 \approx 0.0198$. The mean alone does not tell you what the next step does; you need to know the whole *distribution* of the current value.

Step 2: Recurse on the state, not on the mean

Let $f_n(x)$ be the expected final amount if you currently hold x and have n flips left. With no flips left you keep what you have; otherwise the next flip is heads or tails with probability $\frac{1}{2}$ each:

$$\boxed{f_0(x) = x, \qquad f_n(x) = \tfrac{1}{2}\, f_{n-1}(x+1) + \tfrac{1}{2}\, f_{n-1}\!\left(\tfrac{1}{x}\right),}$$

and the answer is $f_7(100)$. This is exact, but each level doubles the number of distinct states ($2^7 = 128$ leaves), so by hand it is only practical for small n:

$$f_1(x) = \tfrac{1}{2}(x+1) + \tfrac{1}{2x}, \qquad f_1(100) = 50.5 + 0.005 = 50.505,$$

$$f_2(100) = \tfrac{1}{2} f_1(101) + \tfrac{1}{2} f_1(0.01) = \tfrac{1}{2}(51.005) + \tfrac{1}{2}(50.505) \approx 50.755.$$

Step 3: Warm-up — three flips by listing all paths

Listing all $2^3 = 8$ equally likely sequences shows what really drives the mean:

flips	path of amounts	final	flips	path of amounts	final
HHH	$100 \to 101 \to 102 \to 103$	**103**	THH	$100 \to 0.01 \to 1.01 \to 2.01$	2.01
HHT	$100 \to 101 \to 102 \to \frac{1}{102}$	0.0098	THT	$100 \to 0.01 \to 1.01 \to \frac{100}{101}$	0.99
HTH	$100 \to 101 \to \frac{1}{101} \to \frac{102}{101}$	1.01	TTH	$100 \to 0.01 \to 100 \to 101$	**101**
HTT	$100 \to 101 \to \frac{1}{101} \to 101$	**101**	TTT	$100 \to 0.01 \to 100 \to 0.01$	0.01

$$E[X_3] = \frac{103 + 101 + 101 + (0.0098 + 1.01 + 2.01 + 0.99 + 0.01)}{8} = \frac{305 + 4.03}{8} \approx 38.63.$$

Three of the eight paths (HHH, HTT, TTH) keep the $\approx$ \$100 and contribute $\frac{305}{8} = 38.13$ of the 38.63. The other five end near \$1 or below. *That is the whole story of this game.*

Step 4: When does the \$100 survive?

Classify the amount as **large** (≈ 100) or **small** (≈ 0.01):

- **Large**: heads adds 1 (still large); tails turns it into $\approx \frac{1}{100}$ (small).
- **Small** ($= 1/L$ for a large L): tails turns it back into *exactly* L, because $1/(1/L) = L$. But heads turns it into $1 + \frac{1}{L} \approx 1.01$, and from there the amount can never get back near 100: the remaining flips add at most a few dollars or take reciprocals of numbers near 1 (with 7 flips the lost paths all finish between about \$0.01 and \$6).

So the \$100 survives **exactly when every tails is immediately followed by another tails**: the flip sequence must be built from single **H**'s and **TT** pairs. On such a path each TT pair returns the amount to where it was, and each H adds \$1, so the final amount is $100 + \#\text{heads}$.

Step 5: Count the surviving paths for 7 flips

A surviving sequence with j TT pairs has $h = 7 - 2j$ single heads, so it consists of $h + j$ blocks of which j are TT. Choosing the positions of the TT blocks gives $\binom{h+j}{j}$ sequences:

TT pairs j	heads h	sequences $\binom{h+j}{j}$	final amount	total
0	7	1	107	107
1	5	$\binom{6}{1} = 6$	105	630
2	3	$\binom{5}{2} = 10$	103	1030
3	1	$\binom{4}{3} = 4$	101	404
		21 of 128		**2171**

These 21 paths contribute $\frac{2171}{128} \approx \mathbf{16.96}$ to the mean.

Why Fibonacci? Let a_n be the number of length-n sequences built from H and TT. The first block is either H (then a_{n-1} ways to finish) or TT (then a_{n-2} ways): $a_n = a_{n-1} + a_{n-2}$, with $a_1 = 1$, $a_2 = 2$. So $a_n = 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$ — 3 for $n = 3$ (Step 3), 21 for $n = 7$, 89 for $n = 10$.

Step 6: Add the small paths and finish

The other $128 - 21 = 107$ paths end small. Their values are things like $\frac{1}{101}$, $\frac{102}{101}$, 2.01, 3.01 — reciprocals of large numbers (≈ 0), numbers near 1, or a few dollars added after a near-1 value. A reasonable guess for their average is about \$1.5 (the exact average is 1.555), so they add about

$$\frac{107 \times 1.55}{128} \approx \mathbf{1.30}.$$

Total:

$$E[X_7] \approx 16.96 + 1.30 = 18.26.$$

Evaluating the exact recursion of Step 2 over all 128 paths gives $f_7(100) = 18.2608\ldots$, confirming the estimate.

Step 7: Mean versus typical outcome

Only $21/128 \approx 16\%$ of players keep the \$100; the median outcome is only about \$1.59. The mean of \$18.26 is created almost entirely by the rare surviving paths — a strongly right-skewed distribution.

$E[X_7] = f_7(100) \approx \boxed{\$18.26}$**. The \$100 survives only on the** 21 **(Fibonacci) sequences made of H's and TT pairs, contributing** $\frac{2171}{128} \approx 16.96$**; the remaining** 107 **paths end small and add about** 1.30**.**

Concept Summary and Applications

Nonlinear random map: $X_k = g(X_{k-1}, \text{coin}_k)$ with some g nonlinear (e.g. $x \mapsto 1/x$, $x \mapsto x^2$).

- **Exact method — recurse on the state:**

$$f_0(x) = x, \qquad f_n(x) = \sum_{\text{outcomes } c} P(c)\, f_{n-1}(g(x,c)), \qquad E[X_n] = f_n(x_0).$$

- **Estimation method — track the big paths:** find which outcome sequences keep the large value, count them, and add a rough average for everything else:

$$E[X_n] \approx \frac{\sum_{\text{big paths}} \text{final value}}{2^n} + \frac{\#\text{small paths}}{2^n} \times (\text{typical small value}).$$

- **This game:** survivors are tilings by H and TT — a Fibonacci count $(3, 21, 89$ for $n = 3, 7, 10)$; $E[X_n] = 38.63,\ 18.26,\ 10.81$.

- **Warning sign:** whenever a step multiplies, divides, inverts or squares the current value, $E[g(X)] \neq g(E[X])$ (Jensen's inequality), so you cannot iterate the mean.
- **Mean $\neq$ typical:** the expected value can be driven by a small fraction of paths; the median tells a very different story. Always ask "which paths carry the mass?"
- **Applications:** pricing path-dependent payoffs, leveraged or compounding returns (where volatility drag makes the typical path lose while the mean grows), multiplicative random growth, and any "market-make this game" question with a heavy right tail.

Additional Problem 1 — Ten Flips: What Would You Charge? (Q153, Q499)

Problem: Same game with 10 flips: start with a score of 100; heads adds 1, tails takes the reciprocal. Players are paid their final score in pounds. What is the expected payout, and how much would you charge people to play?

Step 1: Fair price = expected payout

If you run the game many times, your average profit per player is (price) $-$ E[payout]. So the break-even price is $E[X_{10}] = f_{10}(100)$, and you charge a little more for a margin.

Step 2: The big paths

As in the representative problem, the £100 survives exactly on sequences built from single H's and TT pairs. With j TT pairs there are $h = 10 - 2j$ heads, $\binom{h+j}{j}$ sequences, each ending at $100 + h$:

j	h	sequences	final	total
0	10	1	110	110
1	8	$\binom{9}{1} = 9$	108	972
2	6	$\binom{8}{2} = 28$	106	2968
3	4	$\binom{7}{3} = 35$	104	3640
4	2	$\binom{6}{4} = 15$	102	1530
5	0	$\binom{5}{5} = 1$	100	100
		89 of 1024		**9320**

They contribute $\frac{9320}{1024} \approx$ £**9.10**. (Again 89 is the Fibonacci number a_{10}.)

Step 3: The small paths

The other $1024 - 89 = 935$ paths end small. With more flips left after the £100 is lost, a few more dollars can be added, so their average is a little higher than before — about £1.87 (exact). They add

$$\frac{935 \times 1.87}{1024} \approx \text{£}\mathbf{1.71}.$$

Step 4: Total and exact check

$$E[X_{10}] \approx 9.10 + 1.71 = \text{£}10.81.$$

The exact recursion gives $f_{10}(100) = 10.8127\ldots$. For reference, the expected value by number of flips:

n	0	1	2	3	4	5	6	7	8	9	10
$E[X_n]$	100	50.5	50.8	38.6	32.6	26.6	22.0	18.3	15.2	12.8	10.8

Step 5: A second quick estimate — the erosion rate

Why does the value shrink by roughly a fifth per flip? Let L_k be the expected final value starting *large* with k flips left, and S_k starting *small*. Ignoring the \$1–\$2 that small paths are worth:

- from large: heads stays large, tails goes small $\Rightarrow L_k \approx \frac{1}{2}L_{k-1} + \frac{1}{2}S_{k-1}$;

- from small: tails goes back to large, heads loses it $\Rightarrow S_k \approx \frac{1}{2}L_{k-1}$.

Substituting, $L_k \approx \frac{1}{2}L_{k-1} + \frac{1}{4}L_{k-2}$. Trying $L_k \propto r^k$ gives $r^2 = \frac{1}{2}r + \frac{1}{4}$, so

$$r = \frac{1+\sqrt{5}}{4} = \frac{\varphi}{2} \approx 0.81,$$

about a 19% loss per flip (the golden ratio φ appears because of the same Fibonacci structure). The exact ratios $E[X_n]/E[X_{n-1}]$ settle around 0.83–0.84; the small paths add back a little value.

Step 6: What to charge — and the risk

Fair value is about £10.81, so charge roughly £11–£12. Two practical points:

- **Most players lose to the price:** the median payout is only about £1.63, and about 91% of players finish below £10. The mean is carried by the $89/1024 \approx 9\%$ who keep the £100.
- **You need volume and capital:** a single all-heads player is paid £110, so your edge appears only over many players, and you must be able to absorb several large payouts in a row.

$E[\textbf{payout}] = f_{10}(100) \approx £10.81$ **(89 surviving paths give £9.10, the other 935 about £1.71). Charge about £11–£12, knowing the profit comes from many players while a few win about £100 each.**

18 Sicherman Dice and Generating Functions

How Generating Functions Encode Dice

A **generating function** is a way to pack a whole list of counts into a single polynomial, so that ordinary algebra on the polynomial mirrors probability on the dice. The trick is to turn "face values" into *exponents* and "how many faces" into *coefficients*.

A single die. Encode a die by a polynomial in a placeholder variable x: each face showing value a contributes a term x^a. A standard six-sided die, with faces $1, 2, 3, 4, 5, 6$, becomes

$$D(x) = x^1 + x^2 + x^3 + x^4 + x^5 + x^6.$$

More generally, if a die has c_a faces showing value a, its generating function is $\sum_a c_a x^a$: **the exponent records a face value, and the coefficient records how many faces carry that value.** Two handy readings: evaluating at $x = 1$ counts the faces, $D(1) = 6$ (divide by this to turn counts into probabilities); and having only *positive* labels means there is no x^0 term.

The sum of several dice — multiply the polynomials. This is the whole point. Roll two dice with generating functions $A(x) = \sum_a p_a x^a$ and $B(x) = \sum_b q_b x^b$ and multiply:

$$A(x)\,B(x) = \sum_a \sum_b p_a q_b\, x^{a+b} \implies [x^s]\,A(x)B(x) = \sum_{a+b=s} p_a q_b.$$

The coefficient of x^s in the product is exactly the number of ordered face pairs (a, b) that sum to s. In other words, **multiplying generating functions performs the "convolution" that produces the distribution of a sum** — adding independent dice corresponds to multiplying their polynomials. For two standard dice,

$$D(x)^2 = x^2 + 2x^3 + 3x^4 + 4x^5 + 5x^6 + 6x^7 + 5x^8 + 4x^9 + 3x^{10} + 2x^{11} + x^{12},$$

and the coefficients $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ are precisely the familiar frequencies of the sums 2 through 12 (divide by $D(1)^2 = 36$ for probabilities). More dice simply multiply in more factors. Crucially, *the sum distribution depends only on this product, not on how its factors are divided between the two dice* — which is exactly the door the Sicherman puzzle walks through.

Representative Problem — Sicherman Dice (Q220)

Problem: You have two ordinary six-sided dice, whose sums 2 through 12 occur with the usual frequencies. Erase all the faces and rewrite them with a new set of *positive integers*. Can you choose numbers so that the distribution of the sum of the two dice is unchanged? If so, how?

Answer. Yes — and there is exactly one nonstandard way, the **Sicherman dice**

$$\{1, 2, 2, 3, 3, 4\} \qquad \text{and} \qquad \{1, 3, 4, 5, 6, 8\}.$$

The generating-function machinery both finds them and proves they are unique.

Step 1 — reduce the goal to a polynomial equation. Relabelling the two dice means choosing new die polynomials $Q_1(x), Q_2(x)$. Since the sum distribution is the coefficient list of the product, "same sum distribution as two standard dice" means precisely

$$Q_1(x)\,Q_2(x) = D(x)^2.$$

A polynomial Q_i is a *legal die* exactly when it has non-negative integer coefficients, no constant term (positive labels), and $Q_i(1) = 6$ (six faces). So the whole problem is: split the factors of $D(x)^2$ into two legal dice.

Step 2 — factor the standard die. Using $1 + x + \cdots + x^5 = \dfrac{x^6 - 1}{x - 1}$ and $x^6 - 1 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$,

$$D(x) = x\,(1 + x + x^2 + x^3 + x^4 + x^5) = x\,(1 + x)\,(1 + x + x^2)\,(1 - x + x^2).$$

Squaring, two standard dice carry *two copies each* of the four irreducible factors:

$$D(x)^2 = \big[\, x\,(1 + x)\,(1 + x + x^2)\,(1 - x + x^2)\,\big]^2.$$

Because integer polynomials factor *uniquely*, relabelling can only *redistribute* these eight factors between Q_1 and Q_2 — which makes the search finite.

Step 3 — use the constraints to pin the split. Evaluate each irreducible at $x = 1$ to get its "face-weight":

$$x \mapsto 1, \quad (1 + x) \mapsto 2, \quad (1 + x + x^2) \mapsto 3, \quad (1 - x + x^2) \mapsto 1.$$

- **Six faces each** needs $Q_i(1) = 6 = 2 \cdot 3$, so each die must get exactly one $(1 + x)$ and one $(1 + x + x^2)$. (The $(1 - x + x^2)$ factors have weight 1 and do not change the face count, so they may go anywhere.)
- **Positive labels** need a zero constant term, so each die must get one of the two x factors.

The *only* remaining freedom is where the two $(1 - x + x^2)$ factors land:

- **one to each die** $\Rightarrow$ both dice equal $x(1 + x)(1 + x + x^2)(1 - x + x^2) = D(x)$ — the ordinary dice;
- **both to one die** $\Rightarrow$ the Sicherman dice.

> ***The positive-label rule is what makes the answer unique.*** *If a face value of* 0 *(or negative labels) were allowed, the "each die needs an x" constraint disappears, and uniqueness collapses. For instance, moving one x factor from one die to the other gives* $\{0, 1, 2, 3, 4, 5\}$ *and* $\{2, 3, 4, 5, 6, 7\}$ *— shifting one die down by* 1 *and the other up by* 1 *leaves every sum unchanged. More generally you may shift the two dice by $\pm k$ for any k, and shift the Sicherman pair too (e.g.* $\{0, 1, 1, 2, 2, 3\}$ *and* $\{2, 4, 5, 6, 7, 9\}$*), so infinitely many relabellings work once* 0 *is permitted. Insisting on* positive *labels forces each die to be divisible by x, which pins the split and singles out the Sicherman dice.*

Step 4 — read off the Sicherman pair. The die that gets *neither* extra factor is

$$Q_1(x) = x\,(1 + x)\,(1 + x + x^2) = x + 2x^2 + 2x^3 + x^4 \implies \{1, 2, 2, 3, 3, 4\},$$

(coefficients $1, 2, 2, 1$ on x^1, x^2, x^3, x^4: one 1, two 2s, two 3s, one 4), and the die that gets *both* is

$$Q_2(x) = x\,(1 + x)\,(1 + x + x^2)\,(1 - x + x^2)^2 = x + x^3 + x^4 + x^5 + x^6 + x^8 \implies \{1, 3, 4, 5, 6, 8\}.$$

By construction $Q_1(x)Q_2(x) = D(x)^2$, so this pair yields the identical sum distribution — sums $2, \ldots, 12$ with counts $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$.

Step 5 — uniqueness. The constraints forced one x, one $(1 + x)$, and one $(1 + x + x^2)$ onto each die, leaving only the placement of the two $(1 - x + x^2)$ factors; "one each" just rebuilds the standard dice, so "both together" is the *only* nonstandard option. Hence the Sicherman pair is the unique positive-integer relabelling of two d6 preserving the sum distribution.

Yes: the unique nonstandard relabelling is $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$ (the Sicherman dice), found by splitting the factors of $D(x)^2 = [x(1 + x)(1 + x + x^2)(1 - x + x^2)]^2$ into two legal dice.

Concept Summary and Applications

- **Encoding:** a die is a polynomial $\sum_a c_a x^a$ — exponent = face value, coefficient = number of such faces; $D(1)$ = number of faces.
- **Sums multiply:** the generating function of a sum of independent dice is the *product* of their polynomials; the coefficient of x^s counts the ways to reach total s.
- **Relabelling = refactoring:** because the sum only sees the product, redistributing the (uniquely determined) irreducible factors gives every equivalent labelling — the Sicherman dice are one such split.
- **Applications:** counting problems via "coefficient extraction," convolutions of distributions, coin/partition problems, and any question of the form "in how many ways can these pieces sum to a target?"

Part III — Coin Flips, Sequences and Markov Chains

19 Expected Flips for Consecutive Heads — Markov Chain

Problem: How many fair-coin flips are needed (on average) to see 2 consecutive heads?

Recursive Formula for HH

Let E = expected flips. Condition on the first few outcomes:

Outcome	Prob.	Contribution
T	$\frac{1}{2}$	Wasted 1 flip, restart: $\frac{1}{2}(1+E)$
H then T	$\frac{1}{4}$	Wasted 2 flips, restart: $\frac{1}{4}(2+E)$
H then H	$\frac{1}{4}$	Done in 2 flips: $\frac{1}{4}(2)$

$$E = \tfrac{1}{2}(1+E) + \tfrac{1}{4}(2+E) + \tfrac{1}{4}(2) = \tfrac{3}{2} + \tfrac{3E}{4}$$

$$\Rightarrow \quad \tfrac{E}{4} = \tfrac{3}{2}$$

$E[\textbf{flips for HH}] = 6$

Extending to HHH

$$E = \tfrac{1}{2}(1+E) + \tfrac{1}{4}(2+E) + \tfrac{1}{8}(3+E) + \tfrac{1}{8}(3)$$
$$= \tfrac{14}{8} + \tfrac{7E}{8} \;\Rightarrow\; \tfrac{E}{8} = \tfrac{14}{8}$$

$E[\textbf{flips for HHH}] = 14$

General Pattern

Expected flips to get k consecutive heads on a fair coin:

$$E_k = 2^{k+1} - 2$$

Check: $E_1 = 2,\ E_2 = 6,\ E_3 = 14,\ E_4 = 30,\ \ldots$

For a **biased coin** with $P(\mathrm{H}) = p$:

$$E_k = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k} = \sum_{j=1}^{k} \frac{1}{p^j}.$$

Verify at $p = \frac{1}{2}$: $E_2 = 2 + 4 = 6$. ✓

Why This Method Matters

- Works for *any* target pattern (HT, HTH, HTHH, ...).
- Handles biased coins by replacing $\frac{1}{2}$ with p or $1 - p$.
- The key insight: on any "broken" sequence, identify which state you reset to and write an equation for that state.

Additional Problem 1 — Random Walk on the Corners of a Cube (Q527)

Problem: You stand on a corner of a cube. Each second you move to one of the 3 neighbouring corners, each with probability $\frac{1}{3}$. What is the expected number of seconds before you first reach the corner opposite your starting point?

Same method, different states. As with consecutive heads, we do not track the whole history — only a small *state* that determines the future, and write one first-step equation per state.

Step 1 — choose the states. Give each corner a 3-digit binary code $000, \ldots, 111$ (its x, y, z coordinates). Two corners are neighbours exactly when their codes differ in one digit, so every move flips one digit. Start at 000; the target is 111. All that matters is how many digits still differ from the target — the *distance*:

distance	corners	how many	neighbours of each such corner
3 (start)	000	1	all 3 at distance 2
2	100, 010, 001	3	2 at distance 1, 1 back at distance 3
1	110, 101, 011	3	1 is the target, 2 back at distance 2
0 (target)	111	1	—

From distance k, flipping one of the k "wrong" digits moves closer; flipping one of the $3 - k$ "right" digits moves away.

Step 2 — first-step equations. Let E_k be the expected time to reach the target from distance k, with $E_0 = 0$. Each move costs one second:

$$E_3 = 1 + E_2, \qquad E_2 = 1 + \tfrac{2}{3}E_1 + \tfrac{1}{3}E_3, \qquad E_1 = 1 + \tfrac{1}{3}\cdot 0 + \tfrac{2}{3}E_2.$$

Step 3 — solve. Substituting $E_3 = 1 + E_2$ into the middle equation: $\frac{2}{3}E_2 = \frac{4}{3} + \frac{2}{3}E_1$, so $E_2 = 2 + E_1$. Then $E_1 = 1 + \frac{2}{3}(2 + E_1)$ gives $\frac{1}{3}E_1 = \frac{7}{3}$, so $E_1 = 7$, hence $E_2 = 9$ and $E_3 = 10$.

$E = 10$ seconds. States = distance to the target $(3, 2, 1, 0)$; one first-step equation per state, exactly as for consecutive heads.

Why so long when the shortest route is 3 moves? Near the target the walk is pushed away: from distance 1 it steps back with probability $\frac{2}{3}$, from distance 2 with probability $\frac{1}{3}$. The parallel with HHH is direct: there a tail sends you back to the start; here a wrong flip sends you one step back. *Sanity check:* for a random walk on a graph, the expected time to return to the start is $\frac{\text{sum of degrees}}{\text{degree of start}} = \frac{8 \cdot 3}{3} = 8$.

Additional Problem 2 — Two Consecutive Sixes (Q104)

Problem: If you throw a fair die with 6 faces repeatedly, what is the expected number of throws until you get two consecutive sixes?

Same method as HH, with $p = \frac{1}{6}$.

Let E denote the expected number of throws needed to obtain two consecutive 6's.

A convenient way to solve this is to use states.

- **State** A: no previous throw was a 6.
- **State** B: the previous throw was a single 6.
- **Goal**: reach two consecutive 6's.

Let

- E_A = expected additional throws starting from state A,
- E_B = expected additional throws starting from state B.

We want E_A, since initially no throw has occurred.

Step 1: Equation for E_A

From state A:

- with probability $\frac{1}{6}$, the next throw is 6, so we move to state B;
- with probability $\frac{5}{6}$, the next throw is not 6, so we remain in state A.

Thus

$$E_A = 1 + \frac{1}{6}E_B + \frac{5}{6}E_A.$$

Step 2: Equation for E_B

From state B:

- with probability $\frac{1}{6}$, the next throw is again 6, and we are done;
- with probability $\frac{5}{6}$, the next throw is not 6, so we return to state A.

Hence

$$E_B = 1 + \frac{1}{6} \cdot 0 + \frac{5}{6}E_A = 1 + \frac{5}{6}E_A.$$

Step 3: Solve the system

From

$$E_A = 1 + \frac{1}{6}E_B + \frac{5}{6}E_A,$$

subtract $\frac{5}{6}E_A$ from both sides:

$$\frac{1}{6}E_A = 1 + \frac{1}{6}E_B.$$

Multiply by 6:

$$E_A = 6 + E_B.$$

Now substitute $E_B = 1 + \frac{5}{6}E_A$:

$$E_A = 6 + 1 + \frac{5}{6}E_A.$$

So

$$E_A = 7 + \frac{5}{6}E_A.$$

Therefore

$$E_A - \frac{5}{6}E_A = 7,$$

$$\frac{1}{6}E_A = 7,$$

and thus

$$E_A = 42.$$

$E = 42$ throws. This is the biased-coin formula of this section with $p = \frac{1}{6}$: $E_2 = \frac{1}{p} + \frac{1}{p^2} = 6 + 36 = 42$.

Intuition. Each "attempt" is: wait for a six (about 6 throws), then throw once more. That next throw is a six only 1 time in 6, so you need about 6 attempts of about 7 throws each: $6 \times 7 = 42$. Compare $E = 6$ for two heads in a row with a fair coin ($p = \frac{1}{2}$: $2 + 4$).

Additional Problem 3 — Rolls Until the Running Total Exceeds n (Q566)

Problem: What is the expected number of rolls of an n-sided die (faces $1, \ldots, n$) required so that the cumulative total first exceeds n?

Let T be the number of rolls needed for the running total to first exceed n, where each roll is uniformly distributed on $\{1, 2, \ldots, n\}$.

The natural quantity to compute is the expected value $E[T]$.

Step 1: Express $E[T]$ using tail probabilities

For any positive-integer-valued random variable,

$$E[T] = \sum_{k \geq 0} P(T > k).$$

Here, $T > k$ means that after k rolls, the total is still at most n.

Let $X_1, X_2, \ldots$ be independent rolls, each uniform on $\{1, \ldots, n\}$, and let

$$S_k = X_1 + \cdots + X_k.$$

Then

$$P(T > k) = P(S_k \leq n).$$

Hence

$$E[T] = \sum_{k \geq 0} P(S_k \leq n).$$

What $P(T > k)$ means. $P(\cdot)$ is "the probability of". Since T is the number of rolls you end up needing, $P(T > k)$ is the chance that k *rolls are not enough* — after k rolls the total is still at most n. For a d6 ($n = 6$), using the count from Step 2:

k	meaning of "$T > k$"	$P(T > k) = \binom{6}{k}/6^k$
0	you have not rolled yet — certainly need more	1
1	first roll ≤ 6 (always)	$\frac{6}{6} = 1$
2	first two rolls total ≤ 6	$\frac{15}{36} \approx 0.417$
3	first three rolls total ≤ 6	$\frac{20}{216} \approx 0.093$
4	first four rolls total ≤ 6	$\frac{15}{1296} \approx 0.012$
5	first five rolls total ≤ 6	$\frac{6}{7776} \approx 0.0008$
6	all six rolls are 1	$\frac{1}{46656}$

Why adding these gives the average. Write T as a row of 1's, one for each "not yet":

$$T = \mathbf{1}(T > 0) + \mathbf{1}(T > 1) + \mathbf{1}(T > 2) + \cdots,$$

where $\mathbf{1}(\cdot)$ is 1 if the statement is true and 0 otherwise. For example, if $T = 3$ then "$T > 0$", "$T > 1$", "$T > 2$" are true and the rest false: $1 + 1 + 1 + 0 + \cdots = 3$. Taking averages, the average of each $\mathbf{1}(\cdot)$ is just its probability, so $E[T] = \sum_{k \geq 0} P(T > k)$. For the d6: $1 + 1 + 0.417 + 0.093 + 0.012 + 0.0008 + 0.00002 \approx 2.52$, matching $(7/6)^6$ below.

Step 2: Count the sequences with sum at most n

For fixed $k \geq 1$, we must count the number of k-tuples $(x_1, \ldots, x_k)$ with each $x_i \in \{1, \ldots, n\}$ and

$$x_1 + \cdots + x_k \leq n.$$

Since the sum is at most n, the upper bound $x_i \leq n$ is automatic. So this is just the number of compositions into k positive parts with total at most n.

The number of positive integer solutions to

$$x_1 + \cdots + x_k = m$$

is

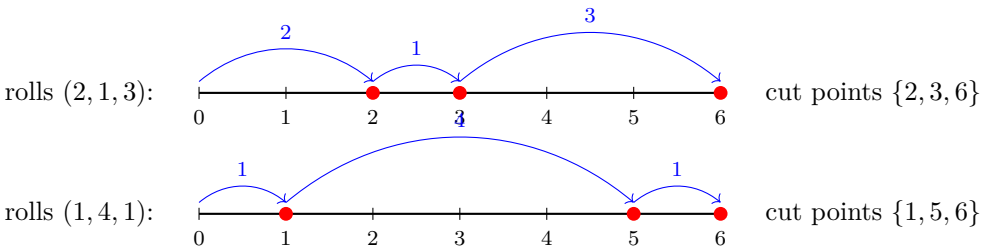
$$\binom{m-1}{k-1}.$$

Summing over $m = k, k+1, \ldots, n$,

$$\#\{(x_1, \ldots, x_k) : x_i \geq 1,\ x_1 + \cdots + x_k \leq n\} = \sum_{m=k}^{n} \binom{m-1}{k-1} = \binom{n}{k},$$

by the hockey-stick identity.

Picture — rolls as cut points. There is an even quicker way to see $\binom{n}{k}$. Record the *running totals* $s_1 = x_1,\ s_2 = x_1 + x_2,\ \ldots,\ s_k = x_1 + \cdots + x_k$. Because every roll is at least 1, these are k *different* numbers with $1 \leq s_1 < s_2 < \cdots < s_k \leq n$ — k cut points chosen from $\{1, \ldots, n\}$. Conversely, any k cut points give back the rolls as the gaps between them. For example, with $n = 6$, $k = 3$:



The blue jumps are the rolls and the red dots are the running totals. Each choice of k red dots among the n positions $1, \ldots, n$ is exactly one sequence of k rolls whose total stays $\leq n$, so there are $\binom{n}{k}$ of them.

Since there are n^k total possible outcomes of k rolls,

$$P(S_k \leq n) = \frac{\binom{n}{k}}{n^k}.$$

Also,

$$P(T > 0) = 1.$$

Therefore

$$E[T] = 1 + \sum_{k=1}^{n} \frac{\binom{n}{k}}{n^k}.$$

Step 3: Final formula

Thus the expected number of rolls needed to exceed a total of n is

$$\boxed{E[T] = \sum_{k=0}^{n} \frac{\binom{n}{k}}{n^k}}.$$

Equivalently,

$$\boxed{E[T] = 1 + \frac{\binom{n}{1}}{n} + \frac{\binom{n}{2}}{n^2} + \cdots + \frac{\binom{n}{n}}{n^n}}.$$

Step 3′: Closed form via the binomial theorem

The sum is exactly a binomial expansion with $x = \frac{1}{n}$:

$$E[T] = \sum_{k=0}^{n} \binom{n}{k} \left(\frac{1}{n}\right)^k \cdot 1^{n-k} = \boxed{\left(1 + \frac{1}{n}\right)^n}.$$

n	1	2	3	6	10	∞
$E[T] = (1+\frac{1}{n})^n$	2	$\frac{9}{4} = 2.25$	$\frac{64}{27} \approx 2.37$	$\frac{7^6}{6^6} \approx 2.52$	≈ 2.594	$e \approx 2.718$

So a standard die needs about 2.52 rolls to pass 6. (Values confirmed by exact backward recursion over the running total.)

Step 4: Asymptotic behavior

As $n \to \infty$,

$$\frac{\binom{n}{k}}{n^k} \to \frac{1}{k!}$$

for each fixed k, so

$$E[T] \to \sum_{k=0}^{\infty} \frac{1}{k!} = e.$$

So for large n, the expected number of rolls is approximately

$$\boxed{e \approx 2.71828},$$

which is also the familiar limit $(1+\frac{1}{n})^n \to e$.

Continuous version. Adding Uniform$(0,1)$ numbers until the total exceeds 1 takes exactly e draws on average: $P(U_1 + \cdots + U_k \le 1) = \frac{1}{k!}$ (the volume of a simplex), so $E = \sum_{k\ge 0} \frac{1}{k!} = e$. The die problem is its discrete version, which is where the e comes from.

$$E[T] = \sum_{k=0}^{n} \frac{\binom{n}{k}}{n^k} = \left(1+\frac{1}{n}\right)^n \textbf{, about } 2.52 \textbf{ for a d6, rising to } e \approx 2.718 \textbf{ as } n \to \infty.$$

Link to this section: like the consecutive-heads problems, the answer comes from tracking "how far you have got" — here through $E[T] = \sum_k P(T > k)$ rather than first-step equations.

20 Pattern Stopping Times (Penney's Game)

Problem: You repeatedly flip a fair coin until either **HHT** or **HTT** appears. If you may bet on which pattern shows up first, which should you choose, and with what probability does it win?

Set Up a Markov Chain on "Progress" States

A **Markov chain** is a random process that hops between a finite set of *states*, where the probability of the next move depends *only on the current state* — not on the full history of how you arrived there. This is the "memoryless" property. The art is to define the states so that this actually holds: here we track only the useful *suffix* of recent flips (the tail that could still grow into a target) and discard everything before it, because only that suffix affects what happens next.

Both targets begin with H, so the relevant states are S — the **start state**, meaning no useful suffix yet — together with H, HH, and HT. Let $p_\bullet$ be the probability that **HHT** appears before **HTT** starting from that state. (This head-to-head race is the classic Penney's game.)

State	One-step equation
S	$p_S = \frac{1}{2}p_H + \frac{1}{2}p_S \;\Rightarrow\; p_S = p_H$
H	$p_H = \frac{1}{2}p_{HH} + \frac{1}{2}p_{HT}$
HH	$p_{HH} = \frac{1}{2}(1) + \frac{1}{2}p_{HH} \;\Rightarrow\; p_{HH} = 1$ (a T completes HHT)
HT	$p_{HT} = \frac{1}{2}(0) + \frac{1}{2}p_H = \frac{1}{2}p_H$ (a T completes HTT, a loss)

Substitute $p_{HH} = 1$ and $p_{HT} = \frac{1}{2}p_H$ into the H equation:

$$p_H = \tfrac{1}{2}(1) + \tfrac{1}{2}\left(\tfrac{1}{2}p_H\right) = \tfrac{1}{2} + \tfrac{1}{4}p_H \implies \tfrac{3}{4}p_H = \tfrac{1}{2} \implies p_H = \tfrac{2}{3}.$$

Since $p_S = p_H$:

$P(\textbf{HHT before HTT}) = \dfrac{2}{3}$, **so bet on HHT (twice as likely to appear first as HTT).**

The Paradox: Equal Waiting Times, Unequal Race

A natural guess is that the pattern with the shorter *expected wait* wins the race. That intuition is wrong. Both HHT and HTT have the *same* expected waiting time of 8 flips, yet HHT wins the head-to-head 2/3 of the time.

$E[\textbf{flips to HHT}]$. Let a, b, c be expected flips from states $\varnothing, H, HH$:

$$c = 1 + \tfrac{1}{2}c \Rightarrow c = 2, \qquad b = 1 + \tfrac{1}{2}c + \tfrac{1}{2}a = 2 + \tfrac{1}{2}a, \qquad a = 1 + \tfrac{1}{2}b + \tfrac{1}{2}a \Rightarrow a = 2 + b.$$

So $a = 2 + (2 + \frac{1}{2}a) = 4 + \frac{1}{2}a \Rightarrow a = 8$.

E[**flips to HTT**]. With states $\varnothing, H, HT$:

$$c = 1 + \tfrac{1}{2}b \text{ (a H returns to } H), \quad b = 1 + \tfrac{1}{2}b + \tfrac{1}{2}c \Rightarrow b = 2 + c, \quad a = 1 + \tfrac{1}{2}b + \tfrac{1}{2}a \Rightarrow a = 2 + b,$$

giving $b = 2 + (1 + \frac{1}{2}b) \Rightarrow b = 6$, then $a = 2 + 6 = 8$.

> *Both wait 8 flips on average, but HHT beats HTT 2/3 vs. 1/3 head-to-head. "Which comes first on average" and "which comes first in a direct race" are different questions — overlaps between the patterns, not the mean wait, decide the race.*

Concept Summary and Applications

- **Method:** model the race as a finite Markov chain whose states are the longest current suffix that is a prefix of a target. Absorbing states are the two completed patterns. Solve the linear system for the win probability.
- **Penney's game:** these races are **non-transitive** — for any length-3 pattern the first player picks, the second player can pick one that beats it. There is no "best" pattern.
- **Expected waiting time** of a pattern is governed by its self-overlaps (Conway's leading-number / correlation method): for a fair coin $E = \sum 2^k$ over the overlap lengths k where a proper prefix equals a suffix, plus the full length. E.g. HHT and HTT have no self-overlap, so $E = 2^3 = 8$; HTH overlaps in 1, so $E = 2^3 + 2^1 = 10$; HHH overlaps in $1, 2$, so $E = 2 + 4 + 8 = 14$.
- **This generalises the "Expected Flips for Consecutive Heads" section.** A run of k heads is the special case where *every* proper prefix of $\mathrm{H}\cdots\mathrm{H}$ is also a suffix, so *all* overlaps $k, k-1, \ldots, 1$ are present and $E = 2^k + 2^{k-1} + \cdots + 2^1 = 2^{k+1} - 2$ (recovering $\mathrm{HH} \to 6$, $\mathrm{HHH} \to 14$). A general pattern simply keeps only the overlaps it actually has — the earlier recursion is this same idea specialised to a single, maximally self-overlapping target.
- **Applications:** genetics/DNA motif waiting times, quality-control run detection, and any "which signal fires first" comparison.

Additional Problem 1 — HHT vs. THH

Problem: Race HHT against THH. Which is favored?

Same state method (states $\varnothing, H, T, HH, TH$; $p = P(\text{HHT first})$):

$$p_{HH} = 1, \quad p_{TH} = \tfrac{1}{2}(0) + \tfrac{1}{2}p_T, \quad p_T = \tfrac{1}{2}p_{TH} + \tfrac{1}{2}p_T \Rightarrow p_T = p_{TH}.$$

Then $p_T = \frac{1}{2}p_T \Rightarrow p_T = 0$, so $p_H = \frac{1}{2}p_{HH} + \frac{1}{2}p_T = \frac{1}{2}$ and $p_\varnothing = \frac{1}{2}p_H + \frac{1}{2}p_T = \frac{1}{4}$.

> $P(\textbf{HHT before THH}) = \dfrac{1}{4}$ **— i.e. THH beats HHT 3 to 1.**

Why so lopsided: once any HH appears it was almost surely preceded by a T (from an earlier tail), completing THH before the trailing T of HHT can land. This is the textbook Penney's-game upset.

Additional Problem 2 — HHT vs. HTH

Problem: Race HHT against HTH (fair coin, then a biased coin with $P(H) = p$).

States $\varnothing, H, HH, HT$; with $P(H) = p$, $q_{HH} = 1$, $q_{HT} = (1-p)q_{\varnothing}$, and $q_{\varnothing} = q_H = p \cdot 1 + (1-p)q_{HT}$. Substituting,

$$q_{\varnothing} = p + (1-p)^2 q_{\varnothing} \implies q_{\varnothing}(1-(1-p)^2) = p, \quad 1-(1-p)^2 = p(2-p),$$

so $q_{\varnothing} = \dfrac{1}{2-p}$.

$$P(\textbf{HHT before HTH}) = \frac{1}{2-p}; \quad \textbf{for a fair coin } \left(p = \tfrac{1}{2}\right) \textbf{ this is } \frac{2}{3}.$$

21 Alternating Coin Games (Geometric Series)

Problem: We play a game and *you go first*. You flip a coin: heads, I pay you \$30. Tails, you hand me the coin and I flip: heads, you pay me \$30; tails, the coin comes back to you. We continue until someone flips heads. What is the most you would pay to go first?

Reduce to "Who Gets the First Heads?"

Nothing is paid until a head appears, so the whole game collapses to a single question: on whose turn does the first head land? You win +\$30 if it is yours, lose \$30 if it is mine.

Let $p = P(\text{you get the first heads})$. On your turn you either flip heads now $\left(\frac{1}{2}\right)$, or both of us flip tails $\left(\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{4}\right)$ and the game resets to the start:

$$p = \tfrac{1}{2} + \tfrac{1}{4}\,p \implies \tfrac{3}{4}p = \tfrac{1}{2} \implies p = \tfrac{2}{3}.$$

Equivalently, as a geometric series over your turns (flips $1, 3, 5, \dots$):

$$p = \tfrac{1}{2} + \tfrac{1}{4}\cdot\tfrac{1}{2} + \left(\tfrac{1}{4}\right)^2\cdot\tfrac{1}{2} + \cdots = \frac{1/2}{1-1/4} = \frac{2}{3}.$$

Expected Value

$$E[X] = 30\cdot\tfrac{2}{3} + (-30)\cdot\tfrac{1}{3} = 20 - 10 = \$10.$$

Going first is worth $E[X] = \$10$, so pay up to \$10. The first mover gets the first heads with probability $2/3$.

Concept Summary and Applications

- **Memoryless reset:** when a game returns to an identical state after a fixed-probability "both miss" event, write a self-referential equation $V = (\text{immediate term}) + (\text{reset prob})\cdot V$ and solve for V. This is the one-line substitute for summing an infinite series.
- **Geometric-series view:** you win the instant *your* flip is the first head, and your turns are rounds $1, 2, 3, \dots$. To win on your k-th turn, two things must happen: (i) the game survives the previous $k-1$ rounds — both players flip tails each time, probability r^{k-1} with $r = P(\text{both miss}) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$; then (ii) you flip heads now, probability $\frac{1}{2}$. Summing over all your turns,

$$p = \sum_{k\ge 1} \underbrace{r^{k-1}}_{\text{reach turn } k} \cdot \underbrace{\tfrac{1}{2}}_{\text{win now}} = \frac{\frac{1}{2}}{1-r} = \frac{\frac{1}{2}}{3/4} = \frac{2}{3}.$$

  The self-referential equation $p = \frac{1}{2} + r\,p$ above is just this geometric sum in closed form.

- **First-mover advantage** is real whenever a round can end on the first player's action: going first is worth a strictly positive premium (here \$10 on a ±\$30 game).
- **Applications:** auctions and races decided by "first success," turn-based duels, and any renewal game with a geometric number of rounds.

Additional Problem — Robot Branching and Extinction (Q525)

Problem: A robot wakes up every morning and, with equal probability $\frac{1}{4}$ each, does one of the following: (1) self-destructs; (2) does nothing; (3) clones itself (giving 2 robots); (4) clones itself twice (giving 3 robots). Every robot alive acts the same way, independently. Starting with one robot, what is the probability that eventually no robots remain?

Same idea as above, with a twist. As in the coin game, condition on the first step and notice that afterwards you face the original problem again. The twist: one step can leave you facing *several independent copies* of the problem at once.

Step 1 — why the terms are q^0, q^1, q^2, q^3. Let q be the probability that the family of a single robot eventually dies out. Look at the first morning:

- **Self-destructs** (0 robots): already extinct, probability $1 = q^0$.
- **Does nothing** (1 robot): back to the start, probability $q = q^1$.
- **Clones once** (2 robots): each robot starts its own family, which behaves exactly like the original process. Everything dies out only if *both* families die out; they are independent, so the probability is $q \cdot q = q^2$.
- **Clones twice** (3 robots): three independent families must all die out: q^3.

Each case has probability $\frac{1}{4}$, so

$$q = \tfrac{1}{4}(1 + q + q^2 + q^3) \iff q^3 + q^2 - 3q + 1 = 0 \iff (q-1)(q^2 + 2q - 1) = 0.$$

The roots are $q = 1$ and $q = -1 \pm \sqrt{2}$; in $[0, 1]$ the candidates are 1 and $\sqrt{2} - 1 \approx 0.414$.

Step 2 — which root? Always the smallest one in $[0, 1]$. $q = 1$ solves the equation for *any* branching problem ("everything dies" is self-consistent), so the equation alone does not decide. Build the answer day by day instead. Let $q_n = P(\text{extinct by day } n)$ and $f(s) = \frac{1}{4}(1 + s + s^2 + s^3)$. Then $q_0 = 0$ and $q_{n+1} = f(q_n)$:

$$q_1 = 0.25, \quad q_2 \approx 0.332, \quad q_3 \approx 0.370, \quad q_4 \approx 0.389, \quad \ldots, \quad q_8 \approx 0.411 \;\to\; 0.4142.$$

The q_n increase toward q, but can never pass the smaller root $r = \sqrt{2} - 1$: f is increasing, so $q_n \le r$ implies $q_{n+1} = f(q_n) \le f(r) = r$. Hence the limit is r.

Quick check with the mean. One robot leaves on average $\frac{1}{4}(0 + 1 + 2 + 3) = 1.5 > 1$ robots, so the population tends to grow and extinction is *not* certain — consistent with $q < 1$. (If the mean were ≤ 1, the only root in $[0, 1]$ would be 1: certain extinction, apart from the trivial case where every robot always leaves exactly one.)

$q = \sqrt{2} - 1 \approx 0.414$. One step can create several independent copies of the problem, so their probabilities multiply (q^k); take the *smallest* root of $q = f(q)$ in $[0, 1]$.

22 Elegant Symmetry: $n{+}1$ vs. n Coins and Beyond

Problem: Player A tosses $n+1$ fair coins and player B tosses n fair coins. B wins if he gets *at least as many* heads as A. Find $P(B \text{ wins})$. (Equivalently, A gets *strictly more* heads with the complementary probability.)

Solution: Condition on A's Extra Coin

Peel off A's one extra coin. Write

$$A' = \#\text{heads among } A\text{'s first } n \text{ coins}, \qquad X = A\text{'s extra coin (1 if head, 0 if tail)},$$

so A's total is $A = A' + X$. The key facts: A' and B are **identically distributed** (each is the head count of n fair coins), and X is an independent fair bit.

B wins when $B \geq A = A' + X$. Condition on the extra coin:

- $X = 0$ (prob $\frac{1}{2}$): B wins iff $B \geq A'$;
- $X = 1$ (prob $\frac{1}{2}$): B wins iff $B \geq A' + 1$, i.e. $B > A'$.

Therefore

$$P(B \text{ wins}) = \tfrac{1}{2}\,P(B \geq A') + \tfrac{1}{2}\,P(B > A').$$

Now use that A' and B are i.i.d., so relabelling $A' \leftrightarrow B$ gives $P(A' > B) = P(B > A')$. Hence

$$P(B \geq A') = 1 - P(A' > B) = 1 - P(B > A'),$$

i.e. the two terms $P(B \geq A')$ and $P(B > A')$ are complementary and sum to 1. Thus

$$P(B \text{ wins}) = \tfrac{1}{2}\big[P(B \geq A') + P(B > A')\big] = \tfrac{1}{2} \cdot 1 = \tfrac{1}{2}.$$

> **$P(B$ has at least as many heads$) = \frac{1}{2}$, and hence $P(A$ strictly more$) = \frac{1}{2}$ — *independent of* n. The lone extra coin is exactly what removes the tie and splits the two outcomes evenly.**

The Heads–Tails Symmetry Shortcut

The same $\frac{1}{2}$ drops out in one line, with no conditioning. Let H_A, H_B be the head counts and T_A, T_B the tail counts. Because A has exactly **one more coin** than B, exactly one of the following always holds:

$$H_A > H_B \qquad \text{or} \qquad T_A > T_B,$$

mutually exclusive and exhaustive because

$$T_A > T_B \iff (n{+}1) - H_A > n - H_B \iff H_A \leq H_B,$$

which is precisely the negation of $H_A > H_B$. Now **flip every coin**, swapping heads and tails: this is a probability-preserving symmetry, so

$$P(H_A > H_B) = P(T_A > T_B).$$

Two equal probabilities that sum to 1 are each $\frac{1}{2}$, so $P(A \text{ strictly more heads}) = \frac{1}{2}$. No binomial sums and no casework — the same answer as the conditioning argument, but the single extra coin and one flip symmetry do all the work.

Illustrative Verification (5 vs. 4 coins)

Take $n = 4$, so $X = H_A \sim \text{Bin}(5, \frac{1}{2})$ and $Y = H_B \sim \text{Bin}(4, \frac{1}{2})$, independent. We check $P(Y \geq X) = \frac{1}{2}$ by brute force. The tail probabilities of Y (out of 16) are

$$P(Y \geq 0, 1, 2, 3, 4) = \tfrac{16}{16}, \tfrac{15}{16}, \tfrac{11}{16}, \tfrac{5}{16}, \tfrac{1}{16}, \qquad P(Y \geq 5) = 0,$$

and $P(X = k) = \frac{1,5,10,10,5,1}{32}$ for $k = 0, \ldots, 5$. Then

$$P(Y \geq X) = \sum_{k=0}^{5} P(X = k)\, P(Y \geq k) = \frac{16 + 75 + 110 + 50 + 5 + 0}{512} = \frac{256}{512} = \frac{1}{2}. \checkmark$$

The direct computation confirms the symmetry answer — but the one-line symmetry argument is what you give in an interview.

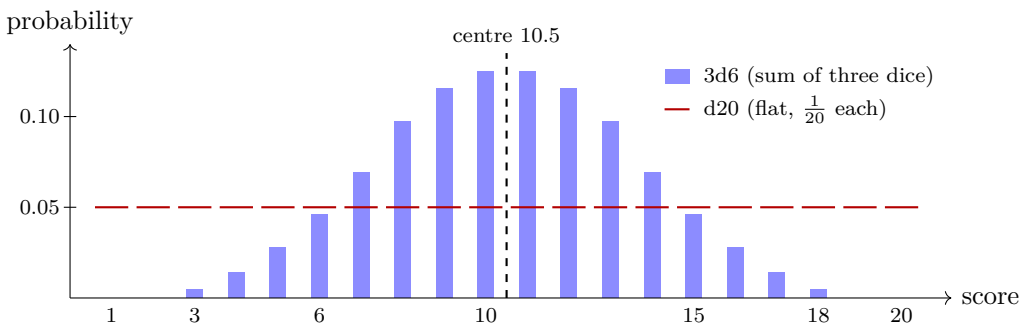
Additional Problem — 3d6 versus 1d20 (Q423)

Problem: One player rolls three six-sided dice and adds them (3d6); the other rolls one twenty-sided die (1d20). Which player is more likely to get the strictly higher score?

First instinct (and why it misleads). The d20 can reach 19 and 20, which 3d6 never can, so it is tempting to say the d20 wins more often. But the d20 can also roll 1 and 2, which 3d6 never can. The question is not "who can go higher" but "who is more often higher," and that is settled by symmetry.

Step 1 — both scores have the same centre. Let X = 3d6 total (3 to 18) and Y = d20 roll (1 to 20). Both distributions are symmetric about the same point, 10.5:

- d20: every face $1, \ldots, 20$ has probability $\frac{1}{20}$, and $1 \leftrightarrow 20$, $2 \leftrightarrow 19$, … pair up around 10.5.
- 3d6: replacing each die face f by $7 - f$ turns a total x into $21 - x$ and is a one-to-one map on outcomes, so $P(X = x) = P(X = 21 - x)$ — again symmetric about 10.5.



Step 2 — symmetry of the difference. Reflect both scores about 10.5: $X \mapsto 21 - X$ and $Y \mapsto 21 - Y$. This leaves each distribution unchanged (Step 1), and since X and Y are independent it leaves their joint distribution unchanged too. But it turns the event $\{X > Y\}$ into $\{21 - X > 21 - Y\} = \{Y > X\}$. Hence

$$P(X > Y) = P(Y > X).$$

This is the same move as the heads↔tails flip in the main problem: find a symmetry that swaps the two "win" events.

Step 3 — the tie. The two probabilities are equal but do not sum to 1, because ties are possible. A tie needs $Y = X$; whatever the 3d6 total x ($3 \le x \le 18$), the d20 matches it with probability $\frac{1}{20}$, so

$$P(X = Y) = \tfrac{1}{20}, \qquad P(X > Y) = P(Y > X) = \frac{1 - \frac{1}{20}}{2} = \frac{19}{40}.$$

Check by direct counting. The 3d6 counts for totals $3, \ldots, 18$ are

$$1,\ 3,\ 6,\ 10,\ 15,\ 21,\ 25,\ 27,\ 27,\ 25,\ 21,\ 15,\ 10,\ 6,\ 3,\ 1 \qquad \text{(out of 216)}.$$

Given total x, the d20 is lower in $x - 1$ of its 20 faces, so

$$P(X > Y) = \frac{\sum_x \#(x)\,(x-1)}{216 \cdot 20} = \frac{2052}{4320} = \frac{19}{40}. \ \checkmark$$

Neither: each player wins with probability $\frac{19}{40} = 47.5\%$, and they tie with probability $\frac{1}{20} = 5\%$.

Takeaway. Spread does not decide "who is more often higher" — the d20 is far more spread out (SD ≈ 5.8 vs. 3.0), but it wins big or loses big, and the two effects cancel. When two independent scores are symmetric about the *same* centre, each is equally likely to be the larger one. (If the centres differ, the one with the higher centre need not win more often in general, and you must compute.)

Concept Summary and Applications

- **Pattern:** whenever two symmetric processes differ by exactly one extra "trial," look for a parity/complement dichotomy that the extra trial resolves, then invoke a symmetry (here head↔tail) to split the remaining probability evenly.
- **Key identity:** with one extra coin, "A leads in heads" and "A leads in tails" partition all outcomes — there are no ties to worry about, which is what makes the split exactly $\frac{1}{2}$.
- **Same centre ⇒ equal odds (Q423):** if two independent scores are symmetric about a common point, reflecting both swaps "$X > Y$" and "$Y > X$", so they are equally likely (3d6 vs. 1d20: $\frac{19}{40}$ each, tie $\frac{1}{20}$).
- **Applications:** best-of races with a first-mover extra draw, "leader after one more sample" problems, and any comparison where an odd total rules out ties.

23 Parity of Heads

Problem: You flip n fair coins. What is the probability that the number of heads is **even** (i.e. $0, 2, 4, \ldots$; note 0 counts as even)?

Background

Each flip is heads or tails, so n flips produce one of 2^n equally likely outcome sequences (e.g. HTTH). The number of sequences with exactly k heads is the **binomial coefficient** $\binom{n}{k}$ (the number of ways to choose which k of the n positions are heads). "Even number of heads" just means $k \in \{0, 2, 4, \ldots\}$. We want the total probability of all such outcomes.

Fair Coin — Method 1: Count with the Binomial Theorem

Since every one of the 2^n sequences is equally likely,

$$P(\text{even}) = \frac{\#\{\text{sequences with an even number of heads}\}}{2^n} = \frac{1}{2^n} \sum_{k \text{ even}} \binom{n}{k}.$$

To evaluate $\sum_{k \text{ even}} \binom{n}{k}$, apply the **binomial theorem** $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$ at two values of x:

$$x = 1: \quad (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} = 2^n \quad \text{(all sequences)},$$

$$x = -1: \quad (1-1)^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k = \sum_{k \text{ even}} \binom{n}{k} - \sum_{k \text{ odd}} \binom{n}{k} = 0.$$

The second line says the even-k and odd-k binomial coefficients sum to the *same* amount. Adding the two equations,

$$2 \sum_{k \text{ even}} \binom{n}{k} = 2^n \implies \sum_{k \text{ even}} \binom{n}{k} = 2^{n-1},$$

so

$$P(\text{even}) = \frac{2^{n-1}}{2^n} = \frac{1}{2}.$$

Check, $n = 4$: even-head counts $0, 2, 4$ give $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 1 + 6 + 1 = 8 = 2^3$, while odd counts $1, 3$ give $\binom{4}{1} + \binom{4}{3} = 4 + 4 = 8$; indeed $8/16 = \frac{1}{2}$.

Fair Coin — Method 2: The One-Flip Bijection

Here is the same answer in one line. Pair each outcome with the outcome you get by *toggling the first coin* (H $\leftrightarrow$ T). That single toggle changes the head count by ± 1, so it flips the parity — turning every even-head outcome into an odd-head one, and vice versa. This is a perfect one-to-one pairing of even outcomes with odd outcomes, so exactly half of all outcomes are even.

$P(\textbf{even number of heads}) = \dfrac{1}{2}$, **for every** $n \geq 1$.

Additional 1 — All Coins Biased with $P(H) = p$

Now every coin lands heads with the *same* probability p (tails $q = 1 - p$), independently. The counting shortcut above breaks because sequences are no longer equally likely: a sequence with k heads has probability $p^k q^{n-k}$, so

$$P(k \text{ heads}) = \binom{n}{k} p^k q^{n-k}.$$

We again isolate the even terms, but now with a **signed (alternating) sum**. Consider

$$S = \sum_{k=0}^{n} (-1)^k \, P(k \text{ heads}) = \sum_{k=0}^{n} \binom{n}{k} (-p)^k q^{n-k}.$$

By the binomial theorem $\sum_k \binom{n}{k} a^k b^{n-k} = (a+b)^n$ with $a = -p$, $b = q$,

$$S = (q - p)^n = (1 - 2p)^n.$$

On the other hand, grouping the sum S by parity, the even terms carry $(-1)^k = +1$ and the odd terms carry -1:

$$S = \underbrace{\sum_{k \text{ even}} P(k)}_{P(\text{even})} - \underbrace{\sum_{k \text{ odd}} P(k)}_{P(\text{odd})} = P(\text{even}) - P(\text{odd}).$$

Together with the total-probability equation $P(\text{even}) + P(\text{odd}) = 1$, we have a simple 2×2 system:

$$P(\text{even}) - P(\text{odd}) = (1 - 2p)^n, \qquad P(\text{even}) + P(\text{odd}) = 1.$$

Adding and dividing by 2:

$$P(\textbf{even}) = \frac{1 + (1 - 2p)^n}{2}, \qquad P(\textbf{odd}) = \frac{1 - (1 - 2p)^n}{2}.$$

Sanity checks. A fair coin has $p = \frac{1}{2}$, so $1 - 2p = 0$ and $P(\text{even}) = \frac{1+0}{2} = \frac{1}{2}$, recovering the earlier result. A single coin ($n = 1$) gives $\frac{1+(1-2p)}{2} = 1 - p = P(0 \text{ heads})$, as it must. (Equivalently, this is the generating function $f(x) = (q + px)^n$ evaluated at the two parity-filtering points: $P(\text{even}) = \frac{1}{2}[f(1) + f(-1)]$, since $f(1) = 1$ and $f(-1) = (q - p)^n$. With *different* biases p_i the same argument gives $P(\text{even}) = \frac{1}{2}[1 + \prod_i (1 - 2p_i)]$.)

Additional 2 — The Most General Case: A Different p_i for Every Coin (Q424)

Now coin i lands heads with its *own* probability p_i ($i = 1, \dots, n$), all coins independent. The binomial formula no longer applies (the coins are not identical), but the signed-sum idea from Additional 1 still works, and in fact becomes simpler.

Step 1 — turn parity into a product. Give each coin a sign

$$s_i = (-1)^{H_i} = \begin{cases} +1, & \text{coin } i \text{ is tails}, \\ -1, & \text{coin } i \text{ is heads}, \end{cases}$$

where $H_i \in \{0, 1\}$ counts that coin's heads. Multiplying the signs gives $(-1)^H$ with $H = H_1 + \cdots + H_n$, which is $+1$ when the total is even and -1 when it is odd:

$$s_1 s_2 \cdots s_n = (-1)^H.$$

Step 2 — take expectations. On one side,

$$\mathbb{E}\big[(-1)^H\big] = (+1)\,P(\text{even}) + (-1)\,P(\text{odd}) = P(\text{even}) - P(\text{odd}).$$

On the other side, the coins are independent, so the expectation of the product is the product of the expectations, and each factor is

$$\mathbb{E}[s_i] = (+1)(1 - p_i) + (-1)\,p_i = 1 - 2p_i.$$

Therefore

$$P(\text{even}) - P(\text{odd}) = \prod_{i=1}^{n} (1 - 2p_i).$$

Step 3 — solve the 2×2 system. Combining with $P(\text{even}) + P(\text{odd}) = 1$ and adding:

$$P(\text{even}) = \frac{1 + \prod_{i=1}^{n}(1 - 2p_i)}{2}, \qquad P(\text{odd}) = \frac{1 - \prod_{i=1}^{n}(1 - 2p_i)}{2}.$$

(In generating-function language: $f(x) = \prod_i \big((1 - p_i) + p_i x\big)$ and $P(\text{even}) = \frac{1}{2}[f(1) + f(-1)]$ with $f(1) = 1$, $f(-1) = \prod_i (1 - 2p_i)$.)

Worked example. Three coins with $p = 0.1,\ 0.2,\ 0.3$:

$$\prod (1 - 2p_i) = 0.8 \times 0.6 \times 0.4 = 0.192, \qquad P(\text{even}) = \frac{1 + 0.192}{2} = 0.596.$$

Check by listing the even outcomes (heads count 0 or 2): $0.9 \cdot 0.8 \cdot 0.7 + 0.9 \cdot 0.2 \cdot 0.3 + 0.1 \cdot 0.8 \cdot 0.3 + 0.1 \cdot 0.2 \cdot 0.7 = 0.504 + 0.054 + 0.024 + 0.014 = 0.596$. ✓

Every earlier case is a special case.

Case	$\prod(1 - 2p_i)$	$P(\textbf{even})$
all fair ($p_i = \frac{1}{2}$)	0	$\frac{1}{2}$
all equal bias p (Additional 1)	$(1 - 2p)^n$	$\frac{1 + (1-2p)^n}{2}$
at least one fair coin (Additionals 3 and 4)	0 (one factor is 0)	$\frac{1}{2}$
all coins certain heads ($p_i = 1$)	$(-1)^n$	1 if n even, 0 if n odd

So the answer is exactly $\frac{1}{2}$ *if and only if* some $p_i = \frac{1}{2}$ (the product is zero only when a factor is zero). In particular, to guarantee a 50% chance of an even number of heads, **one fair coin is enough** — and, if the other coins are biased, one fair coin is also *necessary*.

P(even number of heads) $= \dfrac{1+\prod_{i=1}^{n}(1-2p_i)}{2}$ for independent coins with head probabilities $p_1, \ldots, p_n$.

Additional 3 — One Biased Coin Among Fair Ones

Suppose one coin is biased with $P(H) = p$ and the other $n-1$ coins are fair ($n \geq 2$). Split the total head count into the biased coin plus the fair group, and condition on the biased coin's outcome:

$$P(\text{even total}) = \underbrace{P(\text{biased} = T)}_{1-p} \cdot P(\text{fair group even}) + \underbrace{P(\text{biased} = H)}_{p} \cdot P(\text{fair group odd}).$$

(If the biased coin is a tail it adds 0 heads, so the total is even exactly when the fair group is even; if it is a head it adds 1, so the total is even exactly when the fair group is *odd*.) But the fair group has $n - 1 \geq 1$ fair coins, and by the fair-coin result its head count is even or odd each with probability $\frac{1}{2}$. Hence

$$P(\text{even total}) = (1-p) \cdot \tfrac{1}{2} + p \cdot \tfrac{1}{2} = \tfrac{1}{2}.$$

The bias p cancels completely: whatever the biased coin does, the fair group has already randomised the parity.

Additional 4 — Only One Fair Coin

Now flip the situation: exactly *one* coin is fair and the other $n-1$ are biased with *arbitrary* (possibly different, possibly unknown) probabilities. Let the biased group have head-count parity "even" with some probability e and "odd" with probability $1-e$ — we will not need e. Condition on the single fair coin:

$$P(\text{even total}) = \underbrace{P(\text{fair} = T)}_{1/2} P(\text{biased group even}) + \underbrace{P(\text{fair} = H)}_{1/2} P(\text{biased group odd}) = \tfrac{1}{2}e + \tfrac{1}{2}(1-e) = \tfrac{1}{2}.$$

The lone fair coin is a perfect parity randomiser: mixing any fixed parity distribution with an independent fair ± 1 flip yields $\frac{1}{2}$, no matter what the biased coins do.

One fair coin is enough. *As long as at least one coin is fair, the total parity is exactly* 50/50 *— that single fair coin washes out everything else. Bias only matters when* every *coin is biased, giving* $P(\text{even}) = \frac{1+\prod_i (1-2p_i)}{2}$*; its deviation from* $\frac{1}{2}$ *is the product* $\prod_i (1-2p_i)$*, which collapses to* 0 *the instant any one* $p_i = \frac{1}{2}$.

Concept Summary and Applications

- **Two tools for the fair case:** count directly with $\sum_{k \text{ even}} \binom{n}{k} = 2^{n-1}$ (from the binomial theorem at $x = \pm 1$), or use the one-flip *bijection*. Both give $\frac{1}{2}$.
- **Parity filter for the biased case:** the signed sum $\sum_k (-1)^k P(k)$ equals $P(\text{even}) - P(\text{odd})$ *and*, by the binomial theorem, equals $(1-2p)^n$; combine with $P(\text{even}) + P(\text{odd}) = 1$ to solve. In generating-function language this is $\frac{1}{2}[f(1) \pm f(-1)]$ with $f(x) = (q+px)^n$ — the "roots-of-unity filter," which extends to any modulus.
- **Key intuition:** a single fair coin forces parity to $\frac{1}{2}$; bias shows up only through the product $\prod(1 - 2p_i)$.
- **Applications:** error-detecting parity bits, the XOR of random bits, and any "is the count even or odd?" question over independent trials.

24 Coin Simulation and Randomness Extraction

Problem: You have a *biased* coin (heads with unknown probability p). How can you simulate a perfectly **fair** coin?

Von Neumann's Trick — Pair the Flips

Flip the biased coin **twice** and look at the ordered pair:

- $HT \Rightarrow$ output **H**;
- $TH \Rightarrow$ output **T**;
- HH or $TT \Rightarrow$ discard both and repeat.

Why it is exactly fair: the two kept patterns have *equal* probability,

$$P(HT) = p(1-p) = P(TH),$$

so conditioning on "HT or TH" gives $P(\text{output } H) = P(\text{output } T) = \frac{1}{2}$ — regardless of the unknown bias p.

Output H on HT, T on TH, discard HH/TT and repeat. Each output bit is fair for *any* bias $p \in (0,1)$.

Concept Summary and Applications

Two complementary techniques cover almost every "simulate one source from another" problem:

- **Extraction by symmetry (von Neumann):** turn a biased or unknown source into fair bits using $P(HT) = P(TH)$ — no knowledge of p required. The same "compare two independent draws" idea makes a fair coin from *any* continuous source.
- **Rejection sampling:** build a target uniform distribution from an easy, *larger* equiprobable source by discarding out-of-range outcomes and retrying. Every kept outcome stays equally likely, so the result is exactly uniform.
- **Applications:** de-biasing hardware RNGs, fair decisions from unfair sources, simulating dice/-cards from coins (and vice versa), and Monte-Carlo seeding.

Additional Problem 1 — Fair d6 from a Fair Coin (Rejection Sampling)

Problem: Simulate a fair six-sided die using only fair coin flips.

Flip 3 coins to get one of $2^3 = 8$ equally likely outcomes; read them as a binary number 0–7. Keep 1–6 as the die face; **reject** 0 (000) and 7 (111) and reflip. Each kept value has probability $\frac{1}{8}$, and conditioning on acceptance gives $\frac{1/8}{6/8} = \frac{1}{6}$ per face — a fair die.

$$P(\text{accept per round}) = \tfrac{6}{8} = \tfrac{3}{4}, \quad E[\text{rounds}] = \tfrac{8}{6} = \tfrac{4}{3}, \quad E[\text{flips}] = 3 \cdot \tfrac{4}{3} = 4.$$

Three coin flips → a number 1–6 (reject 0 and 7, reflip). Uses 4 flips on average.

Because 6 is not a power of 2, no fixed-length scheme can be exactly fair — rejection (unbounded flips) is unavoidable. The simple 3-flip method uses 4 flips because it throws away the two rejected outcomes' entropy; recycling them (the Knuth–Yao construction) averages $\frac{11}{3} \approx 3.67$, and the information floor is $\log_2 6 \approx 2.585$ flips.

Additional Problem 2 — Fair Coin from a Continuous Source

Problem: Given i.i.d. draws $U_1, U_2, \ldots$ from some continuous distribution, simulate a fair coin.

The distribution does not need to be uniform. This is the key point, so let us be explicit about it. The draws could come from *any* continuous distribution at all — Uniform$[0,1]$, a bell-shaped Normal, a lop-sided Exponential, a heavily skewed heavy-tailed law — and the method below is still *exactly* fair. You do not need to know which distribution it is, and you do not need it to be symmetric or centred anywhere in particular.

Draw two values and compare: output **H** if $U_1 < U_2$, **T** if $U_1 > U_2$ (the tie $U_1 = U_2$ has probability 0 for a continuous law; redraw in the measure-zero event it occurs).

Why it is fair for any continuous law. The only fact we use is that U_1 and U_2 are *independent and identically distributed*. Because they share the same distribution, relabelling "draw 1" and "draw 2" cannot change any probability — the pair (U_1, U_2) and the swapped pair (U_2, U_1) are statistically indistinguishable. This symmetry (called *exchangeability*) forces

$$P(U_1 < U_2) = P(U_2 < U_1),$$

and since a continuous law makes ties impossible, these two probabilities are the complementary halves of certainty:

$$P(U_1 < U_2) + P(U_1 > U_2) = 1 \implies P(U_1 < U_2) = P(U_1 > U_2) = \tfrac{1}{2}.$$

Notice the argument never mentions the *shape* of the distribution — only that both draws follow the same one. That is exactly why uniformity is irrelevant. It is the same von Neumann pairwise comparison as before, now on real numbers instead of coin pairs.

Compare two independent draws from *any* continuous distribution (need not be uniform, or even symmetric): output H if the first is smaller, T if larger. Fair for every continuous source, with no knowledge of it required.

25 Spreading Processes: Doubling, Parity and Stalling

Paint on hands, a rumour, or an infection: something spreads when an "infected" person meets an "uninfected" one. If contacts happen in rounds of pairwise meetings, three ideas answer almost every question: **doubling** bounds how fast it can spread, **parity** (odd/even counting) restricts how the count can change, and **stalling** — infected people meeting each other — shows how slow it can be.

Representative Problem — Painted Hands in a Hall (Q554)

Problem: There are 1000 people in a hall. One person's hand is painted. Every minute, everyone shakes hands with someone else (so the 1000 people form 500 pairs). Paint passes from a painted hand to an unpainted hand when they shake. How long until everyone's hand is painted, in the best case and in the worst case?

Step 1: Best case — the count at most doubles

Let P_t be the number of painted people after t minutes. Each painted person shakes exactly one hand per minute, so can paint at most one new person. Hence

$$P_{t+1} \le 2P_t, \qquad P_0 = 1 \quad \Longrightarrow \quad P_t \le 2^t.$$

To paint all 1000 we need $2^t \ge 1000$. Since $2^9 = 512 < 1000 \le 1024 = 2^{10}$, at least **10** minutes are needed.

It is achievable: for 9 minutes every painted person shakes an unpainted hand: $1, 2, 4, \ldots, 512$. In minute 10, 488 of the 512 painted people shake the 488 unpainted people, and the remaining 24 painted people pair among themselves. All 1000 are painted after 10 minutes. In general, n people need exactly $\lceil \log_2 n \rceil$ minutes in the best case.

Step 2: A parity rule

Suppose k people are painted and, in some minute, c handshakes are *cross* pairs (painted with unpainted). The other $k - c$ painted people must shake hands with each other, in pairs, so

$$k - c \text{ is even.}$$

- **Minute 1:** $k = 1$, so c must be odd, i.e. $c = 1$: the painted person *must* paint someone. Now $k = 2$.
- **After that:** k is even, so c is even, and k grows by an even number each minute: it stays even, and each minute paints either 0 or at least 2 new people. "Exactly one new person per minute" is impossible.

Step 3: Worst case — three versions of the rules

(i) Repeated handshakes allowed. The two painted people can shake each other every minute while the 998 others pair among themselves. Then $c = 0$ forever: **the paint never spreads**.

(ii) At least one new hand must be painted each minute. By parity, progress after minute 1 is at least 2 per minute, and the slowest path is

$$1 \to 2 \to 4 \to 6 \to \cdots \to 1000,$$

taking $1 + \frac{1000-2}{2} = \mathbf{500}$ minutes. (The tempting answer "999 minutes, one person at a time" violates the parity rule.)

(iii) No pair may shake hands twice. Now stalling cannot last:

- *Upper bound* 999*:* the first painted person meets a different person every minute, and everyone they meet becomes painted; after 999 minutes they have met all 999 others.
- *At least* 500*:* a computer schedule (play the round of a round-robin tournament that paints the fewest new people) takes 500 minutes.
- *Small exact cases* (computer search) give the slowest times $2, 3, 5, 6, 8, 9, 11$ for $n = 4, 6, \ldots, 16$ people, fitting $\lfloor 3n/4 \rfloor - 1$: painted people stall by meeting each other until about half are painted; after that the unpainted can no longer avoid painted hands. For 1000 people this suggests about 750 minutes (a well-supported pattern, not a proof).

Step 4: Typical case

With completely random pairings each minute, it takes about 13 minutes (simulation: average 13.1, range 12–15); Additional Problem 2 derives this by hand.

Best case: 10 minutes ($2^{10} \geq 1000$; the count at most doubles). Worst case: never, if repeats are allowed (two painted people keep shaking each other); 500 minutes if paint must spread every minute (parity: after the first minute, growth comes in steps of 2); between 500 and 999 (likely ≈ 750) if no handshake may repeat. Random pairings: about 13 minutes.

Concept Summary and Applications

Pairwise-contact spreading among n people (rounds of perfect pairings):

- **Fastest:** the infected count at most doubles $\Rightarrow \lceil \log_2 n \rceil$ rounds, achievable.
- **Parity:** with k infected and c cross pairs, $k - c$ is even; after round 1, k stays even.
- **Slowest:** stalls forever if infected people may keep meeting; $\frac{n}{2}$ rounds if progress is forced each round; at most $n - 1$ rounds if no pair repeats.
- **Random pairings:** $k \mapsto k + \frac{k(n-k)}{n-1}$ on average — about $\log_2 n + \log_2 \log_2 n$ rounds.

- **Upper bounds from capacity:** each infected person can pass it on to at most one person per round.
- **Invariants:** parity (or any quantity that cannot change in certain ways) rules out tempting answers like "one new person per round".
- **Worst case depends on the rules:** always state whether repeats are allowed and whether progress is forced.
- **Applications:** gossip/broadcast algorithms in networks, epidemic models, information flow in markets, and "how many rounds" interview puzzles.

Additional Problem 1 — The Same Questions for $n = 100$ People

Problem: Repeat the representative problem with 100 people: (a) the best case, (b) the worst case when at least one new hand must be painted every minute, (c) an upper bound when no handshake may repeat.

(a) $2^6 = 64 < 100 \le 128 = 2^7$, so at least 7 minutes. Achievable: double for 6 minutes $(1, \dots, 64)$; in minute 7, 36 painted people shake the 36 unpainted, and the other 28 painted pair up. 7 **minutes.**

(b) Minute 1 forces $1 \to 2$. Afterwards parity forces growth in even steps, slowest $2 \to 4 \to \cdots \to 100$: 49 more minutes. $1 + 49 = 50 = n/2$ **minutes.**

(c) The first painted person meets someone new every minute, so after 99 minutes everyone has been met. **At most** 99 **minutes** (and by the small-case pattern, the true worst case is likely about $\lfloor 3 \cdot 100/4 \rfloor - 1 = 74$).

(a) 7 minutes (b) 50 minutes (c) at most 99 minutes (likely about 74).

Additional Problem 2 — Random Handshakes: Why About 13 Minutes?

Problem: In the representative problem, suppose the 500 pairs are chosen completely at random each minute. Estimate, without a computer, how many minutes it takes to paint all 1000 hands.

Step 1: Expected growth in one minute

With k painted people, a given painted person's partner is one of the other $n - 1 = 999$ people, chosen at random; it is unpainted with probability $\frac{n-k}{n-1}$. So the expected number of new painted people is

$$E[\text{new}] = k \cdot \frac{n-k}{n-1} \quad \Longrightarrow \quad k_{t+1} \approx k_t + \frac{k_t(n-k_t)}{n-1}.$$

Step 2: Early phase — almost doubling

While k is small, $\frac{n-k}{n-1} \approx 1$, so k nearly doubles: $1, 2, 4, 8, 16, 32, 62, 121, 227, 402$ after 9 minutes — about $\log_2(n/2) \approx 9$ minutes to reach half the room.

Step 3: Late phase — the unpainted fraction squares

Track the unpainted fraction $u = 1 - k/n$. An unpainted person stays unpainted only if their partner is also unpainted, which has probability $\approx u$. So

$$u_{t+1} \approx u_t^2.$$

From $u \approx \frac{1}{2}$: $\frac{1}{4}$, $\frac{1}{16}$, $\frac{1}{256}$, $\frac{1}{65536}$. With 1000 people the last unpainted person disappears once $u < \frac{1}{1000}$, i.e. about 4 more minutes ($\frac{1}{256} \cdot 1000 \approx 4$ people left after 3, under 1 after 4). This squaring takes about $\log_2 \log_2 n \approx \log_2 10 \approx 3.3$ steps.

Step 4: Total

$$\underbrace{\approx 9}_{\text{doubling to } n/2} \quad + \quad \underbrace{\approx 4}_{\text{squaring away the rest}} \quad \approx 13 \text{ minutes.}$$

Iterating the expected-growth formula exactly gives $k_{13} \approx 999.8$, and a simulation of 500 random games averages 13.1 minutes (range 12–15).

About 13 minutes: roughly $\log_2 n \approx 9$–10 minutes of near-doubling, then the unpainted fraction squares each minute ($u \to u^2$), clearing the rest in about $\log_2 \log_2 n \approx 3$–$4$ minutes. Only 3 minutes slower than the best case.

Part IV — Optimal Stopping and Dynamic Programming

26 Optimal Stopping and Threshold Rules (Backward Induction)

Problem: You roll a fair six-sided die.

(a) With no choices, what is the expected value of a single roll?
(b) Now you may, after seeing the first roll, either *keep* it or *reroll once* and take that second roll. Playing optimally, what is your expected value?
(c) Now you may roll up to *three* times total: after each of the first two rolls you may keep the current value or roll again. What is your expected value under the best strategy?
(d) What if you may roll *infinitely many* times (keep or roll again after every roll, at no cost)?

This is an **optimal stopping** problem: you see a random outcome and must decide whether to *stop* and take it, or *continue* in the hope of something better. The winning idea is to compare the value in your hand against the **continuation value** — the expected payoff of playing on optimally — and to work **backward** from the last decision, where there is no more choice to make.

Part (a) — a single roll

With no decision to make, the value is just the average face of a fair die:

$$\mathbb{E}[X] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Part (b) — one optional reroll

Suppose your first roll shows n. You have two options:

- **keep** it: you walk away with n;
- **reroll**: you give up n and take a fresh roll, worth on average 3.5 (the part-(a) value).

A rational player keeps whichever is larger, so you keep when n beats the continuation value 3.5 and reroll otherwise:

$$\text{keep if } n \geq 4, \qquad \text{reroll if } n \leq 3.$$

This is a **threshold rule**: accept anything at or above the cutoff 4, throw back anything below it. The resulting payoff is 3.5 for each of the three low first-rolls (you reroll) and $4, 5, 6$ for the three high ones (you keep), so

$$\mathbb{E} = \frac{1}{6}\Big(\underbrace{3.5+3.5+3.5}_{n=1,2,3:\ \text{reroll}} + \underbrace{4+5+6}_{n=4,5,6:\ \text{keep}}\Big) = \frac{25.5}{6} = 4.25.$$

> ***This is* not *the same as "take the larger of two rolls."*** *In Section 45 (Discrete Order Statistics) you roll twice and keep whichever is bigger,* $\max(roll_1, roll_2)$*, worth* $\frac{161}{36} \approx 4.47$*. Here you must decide to reroll* before *seeing the second roll, and once you reroll you are* ***stuck with*** *that second value even if it is worse than the first. Giving up the safety net of hindsight costs you:* $4.25 < 4.47$*. The max-of-two only applies when you may look at both outcomes and then choose.*

Part (c) — up to three rolls (backward induction)

With more rolls allowed we cannot just eyeball the answer; we compute it **backward**, one stage at a time. Let V_k denote the expected value when you are about to roll and are allowed *up to k rolls in total* from here on, playing optimally. Each stage obeys the same rule — keep the current roll if it beats the value of continuing:

$$V_k = \mathbb{E}\big[\max(X,\ V_{k-1})\big].$$

Base case $V_1 = 3.5$: with only one roll left you must accept it, so the value is the plain average. **One step back**, V_2: keep the roll if it beats $V_1 = 3.5$ (that is, $n \geq 4$), otherwise reroll into a position worth 3.5. This is exactly part (b), so $V_2 = 4.25$. **One more step back**, V_3: keep the roll if it beats $V_2 = 4.25$ (that is, $n \geq 5$), otherwise reroll into a position worth 4.25:

$$V_3 = \frac{1}{6}\Big(\underbrace{4.25 + 4.25 + 4.25 + 4.25}_{n=1,2,3,4:\ \text{reroll}} + \underbrace{5 + 6}_{n=5,6:\ \text{keep}}\Big) = \frac{28}{6} = \frac{14}{3} \approx 4.667.$$

Notice the threshold *rose* from 4 to 5: with an extra roll in reserve you can afford to be pickier.

Part (d) — unlimited rolls

Now suppose there is **no limit** on the number of rolls (and rolling is free): after every roll you may keep the value or roll again.

Step 1 — keep extending the backward induction. The recursion $V_k = \mathbb{E}[\max(X, V_{k-1})]$ still applies for any finite limit k; the threshold is always "keep iff the roll is at least V_{k-1}":

Rolls allowed k	Continuation value V_{k-1}	Keep if roll $\geq$	Value V_k
1	—	(must keep)	3.5
2	3.5	4	4.25
3	4.25	5	4.667
4	4.667	5	$\frac{89}{18} \approx 4.944$
5	4.944	5	$\frac{277}{54} \approx 5.130$
6	5.130	6	5.275
10	5.580	6	5.650

Once $V_{k-1} > 5$ (from $k = 6$ on) only a 6 is worth keeping, so

$$V_k = \tfrac{5}{6}\,V_{k-1} + \tfrac{1}{6}\cdot 6 \quad\Longrightarrow\quad 6 - V_k = \tfrac{5}{6}\,(6 - V_{k-1}) \quad\Longrightarrow\quad V_k = 6 - \tfrac{47}{54}\left(\tfrac{5}{6}\right)^{k-5} \ \ (k \geq 5).$$

The gap to 6 shrinks by a factor $\frac{5}{6}$ with every extra roll, so $V_k \to 6$ as $k \to \infty$.

Step 2 — the unlimited game directly. With no limit, the continuation value no longer depends on how many rolls are left: it is the same number V every time. So V must satisfy the *fixed-point* (stationary) equation

$$V = \mathbb{E}[\max(X, V)].$$

Subtract V from both sides: $\mathbb{E}[\max(X,V)] - V = \mathbb{E}[(X-V)^+]$, the expected amount by which the roll beats V. If $V < 6$, a 6 beats it with probability $\frac{1}{6}$, so this is strictly positive and the equation fails. The only solution is $V = 6$ (check: $\mathbb{E}[\max(X,6)] = 6$).

Step 3 — a strategy that achieves it. "*Roll until you get a* 6*, then stop.*" Each roll is a 6 with probability $\frac{1}{6}$, so the number of rolls is geometric with mean 6 and is finite with probability 1; you always end with a 6.

> ***Why this is (almost) trivial — and how interviews make it interesting.*** *Waiting costs nothing here, so you can hold out for the best face. Real optimal-stopping problems add a price for waiting: a fee per reroll, discounting, or a deadline. With a* cost *c per reroll the stationary equation becomes $V = \mathbb{E}[\max(X, V-c)]$ and the threshold drops (e.g. $c = \$1$: keep any roll ≥ 3, $V = 4$); see Section 27 (Optimal Stopping with a Reroll Cost).*

No reroll: 3.5. One optional reroll: 4.25. Up to three rolls: $\frac{14}{3} \approx 4.67$. Unlimited free rolls: 6 (roll until a 6).

The Backward-Induction Procedure

The same recipe solves any finite-horizon optimal-stopping problem:

1. **Set up the stages and horizon.** Identify the states and the *last* moment a decision is made.
2. **Solve the final stage first.** At the end there is no choice left, so its value is just the expected payoff of whatever you are forced to take.
3. **Step backward.** At each earlier stage the value is $V_{\text{now}} = \mathbb{E}[\max(\text{take it now},\ V_{\text{next}})]$, where V_{next} is the already-computed value of continuing.
4. **Read off the threshold.** The optimal action is a cutoff: *stop* exactly when the current outcome is at least the continuation value V_{next}, otherwise *continue*.
5. **Repeat to the first stage.** That first-stage value is the answer.

$V_k = \mathbb{E}[\max(X, V_{k-1})]$, with the base case $V_1 = \mathbb{E}[X]$; stop iff the current draw $\geq V_{k-1}$.

A useful sanity check: **more attempts can only help**, so V_k increases with k, and the acceptance threshold rises with it (more tries in hand $\Rightarrow$ hold out for more).

Concept Summary and Applications

- **Method:** work *backward* from the final decision; at each stage compare "take it now" against the continuation value V_{next}.

- **Threshold rule:** the optimum is always a cutoff — accept iff the current outcome is at least the continuation value.
- **Monotonicity:** more rounds remaining raise both the value and the cutoff; fewer rounds make you less choosy.
- **Applications:** hiring / secretary problems, sequential search with the option to keep a past offer, "keep or redraw" games, and early exercise of American options.

Additional Problem 1 — Four Coins with a Full Retoss

Problem: You toss four fair coins and are paid \$1 for each head (so \$0 to \$4). (a) With no options, what is a fair price to play? (b) Now, after seeing the first toss, you may either keep that result or *retoss all four coins* once and take the new result. What is a fair price now?

Part (a). The payoff equals the number of heads H, and the number of heads in four fair tosses averages

$$\mathbb{E}[H] = 4 \cdot \tfrac{1}{2} = 2 \quad \Longrightarrow \quad \text{fair price } \$2.$$

Part (b) — one backward-induction step. Let H_1 be the heads on the first toss. Keeping pays H_1; retossing throws it away for a fresh toss worth $\mathbb{E}[H_2] = 2$ on average. So the continuation value is 2, and the threshold rule says

$$\text{keep if } H_1 \geq 2, \qquad \text{retoss if } H_1 \leq 1.$$

The number of heads in four tosses follows the counts $\binom{4}{k} = 1, 4, 6, 4, 1$ out of 16:

$$\mathbb{P}(H_1 = 0, 1, 2, 3, 4) = \tfrac{1}{16}, \tfrac{4}{16}, \tfrac{6}{16}, \tfrac{4}{16}, \tfrac{1}{16}.$$

When $H_1 \in \{0, 1\}$ (probability $\frac{5}{16}$) you retoss and collect the continuation value 2; otherwise you keep H_1. Hence

$$\mathbb{E}[\text{payoff}] = \underbrace{2 \cdot \tfrac{5}{16}}_{H_1 \leq 1\text{: retoss}} + \underbrace{2 \cdot \tfrac{6}{16} + 3 \cdot \tfrac{4}{16} + 4 \cdot \tfrac{1}{16}}_{H_1 \geq 2\text{: keep}} = \frac{10 + 12 + 12 + 4}{16} = \frac{38}{16} = \frac{19}{8}.$$

Without the option: \$2. With one full retoss (keep if ≥ 2 heads): $\frac{19}{8} = \$2.375$.

Additional Problem 2 — Four Coins, Unlimited Reflips at $1 Each (Q418)

Problem: You flip four fair coins and get $1 for each head. You may reflip *all four* coins **as many times as you want**, but each reflip costs $1. What is the optimal strategy, and what is the game worth?

Step 1 — the distribution. As in Additional Problem 1, the number of heads H takes the values $0, 1, 2, 3, 4$ with probabilities $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$, and $\mathbb{E}[H] = 2$.

Step 2 — the common mistake. It is tempting to reason: "a reflip costs $1 and a fresh flip is worth $\mathbb{E}[H] = 2$ on average, so reflipping is worth $2 - 1 = 1$; keep anything ≥ 1 and reflip only on 0." This is **wrong**, because after a reflip you are *not* stuck with the new result — you may reflip again. The value of reflipping is not $\mathbb{E}[H] - 1$ but the value of *the whole game* minus $1.

Step 3 — the stationary (fixed-point) equation. Let V be the expected payoff of the game (net of all future fees) just before a flip, under optimal play. Because reflips are unlimited, the situation after any reflip is identical to the start, so the value is the *same* V every time — exactly the idea of part (d) of the main problem. After seeing H:

- **keep**: receive H;
- **reflip**: pay $1 and face the same game again, worth $V - 1$.

Hence

$$V = \mathbb{E}\big[\max(H,\ V - 1)\big], \qquad \text{keep iff } H \geq V - 1.$$

Step 4 — test each threshold for consistency. Since H is an integer, the rule is "keep iff $H \geq t$" for some t. For each t, solve for V, then check that the implied cutoff $V - 1$ gives back the same t.

Rule	Equation for V	V and $V - 1$	Consistent?
keep ≥ 1 (reflip 0)	$V = \frac{32}{16} + \frac{1}{16}(V - 1)$	$V = \frac{31}{15} \approx 2.07$, $V - 1 \approx 1.07$	no: $1 < 1.07$
keep ≥ 2 (reflip $0, 1$)	$V = \frac{28}{16} + \frac{5}{16}(V - 1)$	$V = \frac{23}{11} \approx 2.09$, $V - 1 \approx 1.09$	**yes**
keep ≥ 3 (reflip $0, 1, 2$)	$V = \frac{16}{16} + \frac{11}{16}(V - 1)$	$V = 1$, $V - 1 = 0$	no: keep all

For the consistent rule, the numerator $28 = 2{\cdot}6 + 3{\cdot}4 + 4{\cdot}1$ collects the kept outcomes and $\frac{5}{16} = \frac{1+4}{16}$ is the chance of reflipping. Multiplying by 16: $16V = 28 + 5V - 5$, so $11V = 23$ and $V = \frac{23}{11}$. The cutoff $V - 1 = \frac{12}{11}$ lies between 1 and 2, so indeed a 1 is reflipped and a 2 is kept.

Step 5 — sanity check by counting reflips. Each flip is reflipped with probability $\frac{5}{16}$, so the number of reflips is geometric with mean $\frac{5/16}{11/16} = \frac{5}{11}$. When you stop, $H \geq 2$, with $\mathbb{E}[H \mid H \geq 2] = \frac{28/16}{11/16} = \frac{28}{11}$. Net value $= \frac{28}{11} - \frac{5}{11} \cdot \$1 = \frac{23}{11}$. ✓

Reflip on 0 or 1 head; keep 2, 3 or 4. Value $V = \dfrac{23}{11} \approx \$2.09$ (the unlimited-reflip option is worth $\frac{1}{11} \approx \$0.09$ over playing once).

Takeaway. With repeated options, compare the current outcome with the *continuation value of*

the whole game ($V - \text{cost}$), not with the one-shot average ($\mathbb{E}[H] - \text{cost}$). Here the continuation value $\frac{12}{11}$ is slightly above 1, which tips $H = 1$ into "reflip." The same fixed-point method handles any per-attempt cost; see also Section 27 (Optimal Stopping with a Reroll Cost).

Additional Problem 3 — Four Coins and a Magic Wand (Q472, Q539)

Flip 4 fair coins and get paid \$1 for each head. Without any wand the expected earnings are $4 \cdot \frac{1}{2} = \$2$ (linearity of expectation). Two different wands:

(a) The wand must re-flip every head–tail pair (Q472). Group the coins into two pairs; whenever a pair shows one head and one tail, both coins are re-flipped once. What are the expected earnings?

Key fact. A head–tail pair holds exactly 1 head, and after re-flipping both coins it holds $2 \cdot \frac{1}{2} = 1$ head on average — the same. For one pair:

Pair shows	Probability	Wand?	Expected heads after
HH	$\frac{1}{4}$	no	2
TT	$\frac{1}{4}$	no	0
HT or TH	$\frac{1}{2}$	re-flip both	1

$E[\text{heads per pair}] = \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 0 + \frac{1}{2} \cdot 1 = 1$, so with two pairs the expected earnings are still \$2.

Repeated tapping. Even if you keep tapping until no head–tail pair is left, the answer stays \$2: each tap swaps one head and one tail for an expected one head, so the expected number of heads never changes. The process ends at all heads (\$4) or all tails (\$0), each with probability $\frac{1}{2}$ by symmetry, and $4 \cdot \frac{1}{2} = 2$.

(a) \$2 — the forced wand re-flips one head for every tail, so it cannot change the expected value.

(b) The wand may re-flip any 2 coins, or all 4, once, after seeing the result (Q539). What are the expected earnings with optimal use?

Key fact. Re-flipping a tail gains $\frac{1}{2}$ in expectation ($0 \to \frac{1}{2}$); re-flipping a head loses $\frac{1}{2}$ ($1 \to \frac{1}{2}$). So re-flip an even-sized set only if it contains more tails than heads — ideally exactly the tails.

Heads k	$P(k)$	Best use of the wand	Expected heads after
0	$\frac{1}{16}$	re-flip all 4	2
1	$\frac{4}{16}$	re-flip 2 tails	$1 + 1 = 2$
2	$\frac{6}{16}$	re-flip the 2 tails	$2 + 1 = 3$
3	$\frac{4}{16}$	no gain possible	3
4	$\frac{1}{16}$	do nothing	4

$$E = \frac{1 \cdot 2 + 4 \cdot 2 + 6 \cdot 3 + 4 \cdot 3 + 1 \cdot 4}{16} = \frac{44}{16} = \frac{11}{4}.$$

(b) $\frac{11}{4} = \$2.75$ — a \$0.75 improvement, because you can aim the wand at tails.

Takeaway. Re-flipping helps only when you can *choose* to re-flip tails rather than heads. A wand that re-flips coins without regard to their faces (like the head–tail pairs in (a)) leaves the expected value unchanged; a wand that targets tails (b) raises it — the same idea as Additional Problems 1 and 2, where you keep good results and re-toss bad ones.

Additional Problem 4 — d12 vs. d20, Two Rolls Each (Q447)

Problem: Player A has a 12-sided die and player B a 20-sided die. Each gets up to two rolls and may stop after the first roll, keeping that number (otherwise they must keep the second). Neither sees the other's rolls. The higher final number wins; **ties go to** A (the d12). What is the probability that B wins?

Key idea — value a roll by its chance of winning. Unlike the main problem, you are not paid the number you keep; you are paid for *winning*. So the right comparison after the first roll is

$$\text{keep: } P(\text{win with this roll}) \quad \text{vs.} \quad \text{reroll: average of } P(\text{win}) \text{ over a fresh roll.}$$

Each player still uses a threshold (keep large rolls, reroll small ones), but the threshold depends on the *opponent's* final distribution. We find the pair of thresholds by **best-response iteration**: guess, compute each player's best reply, and repeat until nothing changes.

Tool — final distribution from a threshold. If a player with an n-sided die keeps the first roll iff it is at least k:

$$P(\text{final} = x) = \begin{cases} \dfrac{k-1}{n^2}, & x < k \quad \text{(reroll, then roll } x\text{)}, \\ \dfrac{n+k-1}{n^2}, & x \geq k \quad \text{(keep } x\text{, or reroll and roll } x\text{)}. \end{cases}$$

Step 1 — start from the "average" guess: A keeps ≥ 7. Then $P(A = x) = \frac{6}{144}$ for $x \leq 6$ and $\frac{18}{144}$ for $x \geq 7$. Keeping y wins for B with probability $F(y-1) = P(A \leq y-1)$; averaging over a fresh d20 roll gives B's reroll value $\frac{1}{20}\sum_{y=1}^{20} F(y-1) = \frac{12}{20} = 0.600$. Since $F(9) = \frac{5}{8} = 0.625 \geq 0.600$ but $F(8) = \frac{1}{2} < 0.600$, B **keeps** ≥ 10.

Step 2 — A's best reply to B keeping ≥ 10. Now $P(B = y) = \frac{9}{400}$ for $y \leq 9$ and $\frac{29}{400}$ for $y \geq 10$. Because A wins ties, keeping x wins with probability $G(x) = P(B \leq x)$: $\frac{9x}{400}$ for $x \leq 9$, then $\frac{110}{400}, \frac{139}{400}, \frac{168}{400}$. A's reroll value is

$$\frac{1}{12}\sum_{x=1}^{12} G(x) = \frac{9 \cdot 45 + 110 + 139 + 168}{12 \cdot 400} = \frac{822}{4800} \approx 0.171.$$

Since $G(8) = 0.180 \geq 0.171$ but $G(7) = 0.1575 < 0.171$, A **keeps** ≥ 8 — one higher than the guess.

Step 3 — recheck B against A keeping ≥ 8. Now $F(t) = \frac{7t}{144}$ for $t \leq 7$ and $F(8), \ldots, F(11) = \frac{68}{144}, \frac{87}{144}, \frac{106}{144}, \frac{125}{144}$. B's reroll value is $\frac{1}{20}\left(\frac{196+386}{144} + 8\right) \approx 0.602$, and $F(9) = \frac{87}{144} \approx 0.604 \geq 0.602$ while $F(8) \approx 0.472 < 0.602$: B *still* keeps ≥ 10. Nothing changes, so the thresholds are settled:

Player	Keeps	Rerolls
A (d12)	8–12	1–7
B (d20)	10–20	1–9

Step 4 — probability that B wins. With these thresholds, $P(A = x) = \frac{7}{144}$ $(x \le 7)$, $\frac{19}{144}$ $(x \ge 8)$, and B wins when $A \le y - 1$:

$$P(B \text{ wins}) = \sum_{y=1}^{20} P(B = y)\, F(y-1).$$

- $y = 1, \dots, 9$ (weight $\frac{9}{400}$): $F(0) + \cdots + F(8) = \frac{196+68}{144} = \frac{11}{6}$, contributing $\frac{9}{400} \cdot \frac{11}{6} = \frac{396}{9600}$.
- $y = 10, \dots, 20$ (weight $\frac{29}{400}$): 11 terms, $F(9)+F(10)+F(11) = \frac{318}{144}$ plus eight 1's $(F(12), \dots, F(19))$, total $\frac{245}{24}$, contributing $\frac{29}{400} \cdot \frac{245}{24} = \frac{7105}{9600}$.

> **A keeps 8–12, B keeps 10–20, and $P(B \text{ wins}) = \dfrac{396 + 7105}{9600} = \dfrac{7501}{9600} \approx 78.1\%$.**

Takeaway. When the prize is *winning*, a roll's worth is its win probability against the opponent, so the thresholds are not the dice averages: the d20 settles for 10 (it already beats most of the d12's range), while the d12 must gamble up to 8. Best-response iteration from the naive guess converges in one round, and the optimum is flat — the naive thresholds (7 and 11) give $\frac{25}{32} = 0.78125$, almost the same. (Checked by solving the full 13×21 threshold game by computer: the pair $(8, 10)$ is a saddle point.)

Additional Problem 5 — The Bucket Game: Fixed Throws and When to Stop (Q452, Q571)

Set-up. There are n buckets, each able to hold one ball, all starting empty. Each throw lands in one of the n buckets uniformly at random, independently of earlier throws. If the bucket is empty, the ball stays and you gain +\$1; if it is already filled, the ball bounces out and you gain −\$1.

(a) (Q452) With $n = 3$ buckets and exactly 2 throws, what is your expected total gain?

(b) (Q571) With $n = 5$ buckets and unlimited balls, you may stop whenever you like. What is the optimal exit strategy, and what is the expected return?

Key formula — the value of one more throw. With k of the n buckets filled, the next ball lands in an empty bucket with probability $\frac{n-k}{n}$ $(+1)$ and in a filled one with probability $\frac{k}{n}$ (-1):

$$E[\text{next throw} \mid k \text{ filled}] = \frac{n-k}{n} - \frac{k}{n} = \frac{n-2k}{n}.$$

A throw is worth making only while this is positive, i.e. while fewer than half the buckets are filled.

(a) Three buckets, two throws (Q452).

- *First throw:* all buckets are empty $(k = 0)$, so it is certain to land in an empty bucket: gain $+1$ $(= \frac{3-0}{3})$. Now 1 bucket is filled.
- *Second throw:* $k = 1$, so with probability $\frac{2}{3}$ it lands in an empty bucket $(+1)$ and with probability $\frac{1}{3}$ in the filled one (-1): expected gain $\frac{2}{3} - \frac{1}{3} = \frac{1}{3}$ $(= \frac{3-2}{3})$.

By linearity of expectation,

$$E[\text{total}] = 1 + \tfrac{1}{3} = \tfrac{4}{3}.$$

(a) Expected total gain $= \frac{4}{3}$.

(b) Five buckets, unlimited balls, choose when to stop (Q571).

Step 1 — one-step values. With $n = 5$, the next throw is worth $\frac{5-2k}{5}$:

filled k	0	1	2	3	4	5
value of next throw	1	$\frac{3}{5}$	$\frac{1}{5}$	$-\frac{1}{5}$	$-\frac{3}{5}$	-1

Step 2 — why the one-step rule is optimal. The number of filled buckets k never goes down, so the value of a throw only gets worse over time. Once a throw has negative value, every later throw does too, so there is no reason to continue. (This is a *monotone* stopping problem: looking one step ahead is enough.) Hence: **keep throwing while $k \le 2$, stop as soon as 3 buckets are filled.**

Step 3 — the expected return, stage by stage. Going from k to $k+1$ filled buckets, each throw succeeds with probability $\frac{5-k}{5}$, so the number of throws N_k is geometric with mean $\frac{5}{5-k}$. Exactly one of those throws gains $+1$ and the other $N_k - 1$ lose 1 each, so the stage is worth

$$E[1 - (N_k - 1)] = 2 - \frac{5}{5-k}.$$

Summing the stages $k = 0, 1, 2$:

$$\left(2 - \tfrac{5}{5}\right) + \left(2 - \tfrac{5}{4}\right) + \left(2 - \tfrac{5}{3}\right) = 1 + \tfrac{3}{4} + \tfrac{1}{3} = \tfrac{12+9+4}{12} = \tfrac{25}{12}.$$

(b) Keep throwing until exactly 3 buckets are filled, then stop. Expected return $= \frac{25}{12} \approx \$2.08$.

General rule. For n buckets: throw while $k < \frac{n}{2}$, stop once $k \ge \frac{n}{2}$ (indifferent at $k = \frac{n}{2}$ when n is even), and

$$E = \sum_{0 \le k < n/2} \left(2 - \frac{n}{n-k}\right).$$

The stages are the waiting times of the coupon collector, here with a payoff attached and the option to quit. **Takeaway:** when the state only moves in one direction and the one-step value only gets worse, the optimal rule is the simple myopic one — stop the first time the next step has negative expected value.

27 Optimal Stopping with a Reroll Cost

Problem: You roll a 100-sided die. You may accept the face value in dollars, or pay \$1 to re-roll — as many times as you like. What is the optimal strategy, and what is the fair value of the game?

Before any formulas, get a feel for the game. The faces run 1 to 100, so a blind single roll averages 50.5. But you are not forced to keep a bad roll: for just \$1 you can throw it back and try again. Because a reroll is so cheap compared with the spread of possible values (up to \$100), you should be *picky* — keep only high rolls and pay the small fee to escape low ones. The whole problem is figuring out exactly *how* picky: what is the cutoff above which you keep, and below which you pay to reroll?

Step 1 — The Value Equation (a fixed point)

Let V be the value of the game: your expected earnings when you play optimally starting from a fresh roll. The trick is that V refers to itself. Suppose you roll and see a face x. You have two choices:

- **keep it** — you walk away with x dollars; or
- **reroll** — you pay \$1 and are back to a fresh roll, which by definition is worth V. So rerolling is worth $V - 1$ (the continuation value minus the \$1 fee).

A rational player takes whichever is larger, so the outcome of seeing face x is worth $\max(x,\, V-1)$. Since each face $1, \ldots, 100$ is equally likely, average over all of them:

$$V = \frac{1}{100} \sum_{x=1}^{100} \max(x,\ V-1).$$

This is a **fixed-point equation**: V appears on both sides. That is not a problem — it is exactly what pins down the value.

Step 2 — Why the Rule Is a Threshold

Look at $\max(x, V-1)$. Write $C = V - 1$ for the continuation value. If x is large it beats C and you keep it; if x is small, C wins and you reroll. So the optimal policy is automatically a **cutoff**: there is an integer k with

$$\text{reroll on } 1, 2, \ldots, k, \qquad \text{keep on } k+1, \ldots, 100,$$

and the switch happens where x crosses C, i.e.

$$k < C \le k+1, \qquad C = V - 1.$$

Now rewrite the value equation using this structure. The k rerolled faces each contribute the continuation value $C = V - 1$ (*not* V — rerolling costs a dollar), and the kept faces contribute

their own value:

$$V = \frac{1}{100}\left(\underbrace{k(V-1)}_{k \text{ rerolled faces}} + \underbrace{\sum_{x=k+1}^{100} x}_{\text{kept faces}} \right).$$

Step 3 — The Value for a Given Cutoff

Clear the fraction and the sum. Multiply through by 100:

$$100V = k(V-1) + \sum_{x=k+1}^{100} x \implies (100-k)V = \sum_{x=k+1}^{100} x - k.$$

The remaining sum is the total $1 + \cdots + 100 = 5050$ minus the first k terms:

$$\sum_{x=k+1}^{100} x = 5050 - \frac{k(k+1)}{2}.$$

Substituting and combining the two k-terms ($-\frac{k(k+1)}{2} - k = -\frac{k(k+3)}{2}$) gives the key equation

$$(100-k)\,V = 5050 - \frac{k(k+3)}{2}. \qquad (\star)$$

This is exact once we know the correct integer k.

Step 4 — Solve for the Cutoff with a Quadratic

We could test values of k one by one, but there is a cleaner route. Recall the consistency condition $k < V - 1 \le k + 1$. Subtract 1 throughout:

$$k - 1 < V - 2 \le k,$$

so the integer k and the continuous quantity $V - 2$ differ by less than 1. At the threshold they essentially coincide, so **relax k to its continuous value** and set $k = V - 2$. Substitute into $(\star)$, using $100 - k = 102 - V$ and $\frac{k(k+3)}{2} = \frac{(V-2)(V+1)}{2}$:

$$V(102 - V) = 5050 - \frac{(V-2)(V+1)}{2}.$$

Multiply by 2 and expand $(V-2)(V+1) = V^2 - V - 2$:

$$204V - 2V^2 = 10100 - (V^2 - V - 2) = 10102 - V^2 + V,$$

which collapses to a single quadratic:

$$V^2 - 203V + 10102 = 0.$$

By the quadratic formula,

$$V = \frac{203 \pm \sqrt{203^2 - 4 \cdot 10102}}{2} = \frac{203 \pm \sqrt{801}}{2} = \frac{203 \pm 28.30}{2}.$$

The upper root ≈ 115.7 is impossible (a die value can be at most 100), so the relevant root is $V = \frac{203-28.30}{2} \approx 87.35$.

Step 5 — Pin the Integer Cutoff and Get the Exact Value

The relaxed answer $V \approx 87.35$ tells us where the cutoff sits: keep $x \geq C = V - 1 \approx 86.35$, i.e. keep $x \geq 87$ and reroll $x \leq 86$, so the true integer count of rerolled faces is $k = 86$. Put $k = 86$ back into the *exact* equation $(\star)$ to remove the continuous approximation:

$$(100 - 86)V = 5050 - \frac{86 \cdot 89}{2} \implies 14V = 5050 - 3827 = 1223 \implies V = \frac{1223}{14} \approx 87.357.$$

The quadratic's 87.35 and the exact $1223/14 = 87.357$ agree to two decimals; the tiny gap is only the continuous relaxation of the integer k.

Keep any roll of 87 or higher; otherwise pay \$1 and reroll. Fair value $V = \frac{1223}{14} \approx$ \$87.36.

Intuition. You stop only on the top 14 faces (about the top 14%). A reroll is cheap — \$1 against a spread up to \$100 — so it pays to hold out for a value far above the naive mean of 50.5; the \$1 fee per reroll is exactly what stops you from waiting forever for a 100.

Concept Summary and Applications

- **Method:** the value solves the fixed point $V = \mathbb{E}[\max(x, V - c)]$, where c is the reroll cost. The optimal policy is always a threshold; the cutoff is where the continuation value $C = V - c$ falls between two consecutive faces.
- **The quadratic shortcut:** write the value equation for a general cutoff, then *relax the integer cutoff to its continuous value* (set it equal to the continuation value). This turns a search into one quadratic; solve it, discard the root that exceeds the maximum face, and round to pin the exact integer cutoff.
- **Intuition:** cheap rerolls $\Rightarrow$ a high, picky threshold (hold out for a top value); expensive rerolls $\Rightarrow$ settle sooner.
- **Applications:** job-offer / house-hunting search with a search cost, "keep or redraw" games, and reservation-price setting.

Additional Problem 1 — 64-Sided Die, \$1 Reroll

Problem: Same game, but with a 64-sided die and a \$1 reroll cost.

The setup is identical. With cutoff m (keep $x \geq m$, reroll $x \leq m-1$), the $m-1$ rerolled faces each contribute $V-1$, so

$$64V = \sum_{x=m}^{64} x + (m-1)(V-1) \implies (65-m)V = \sum_{x=m}^{64} x - (m-1).$$

Writing the sum as $\sum_{x=m}^{64} x = 2080 - \frac{m(m-1)}{2}$ (the total $1+\cdots+64 = 2080$ minus the first $m-1$ terms) and combining k-terms gives the key equation

$$(65-m)\,V = 2080 - \frac{(m-1)(m+2)}{2}. \qquad (\star)$$

The cutoff is self-consistent when $m-1 < V-1 \leq m$, so relax m to its continuous value and set $m = V-1$. Then $65-m = 66-V$ and $(m-1)(m+2) = (V-2)(V+1)$, and $(\star)$ becomes $V(66-V) = 2080 - \frac{(V-2)(V+1)}{2}$. Multiplying by 2 and simplifying yields

$$V^2 - 131V + 4162 = 0.$$

The discriminant is $131^2 - 4\cdot 4162 = 513$, so $V = \frac{131 \pm \sqrt{513}}{2} = \frac{131 \pm 22.65}{2}$. The upper root ≈ 76.8 exceeds 64 and is spurious; the relevant root is $V \approx 54.18$. This places the cutoff at $m = 54$ (keep $x \geq 54$, since $V-1 \approx 53.18$). Putting $m = 54$ back into $(\star)$ gives the exact value:

$$(65-54)V = 2080 - \frac{53\cdot 56}{2} = 596 \implies 11V = 596 \implies V = \frac{596}{11} \approx 54.18.$$

Keep any roll of 54 or higher; value $V = \dfrac{596}{11} \approx \$54.18$.

Additional Problem 2 — n-Sided Die, Reroll Costs $1/n$

Problem: An n-sided die, but now each reroll costs only $\frac{1}{n}$ dollars. What is the value of the game?

Everything is the same, with the reroll cost 1 replaced by $c = \frac{1}{n}$. After seeing k you take $\max\!\left(k,\, V - \frac{1}{n}\right)$, so

$$V = \frac{1}{n}\sum_{k=1}^{n} \max\!\left(k,\; V - \tfrac{1}{n}\right).$$

With cutoff m (keep $x \geq m$), the $m-1$ rerolled faces each contribute $V - \frac{1}{n}$, and the self-consistency condition is $m-1 < V - \frac{1}{n} \leq m$. Writing out the value equation and the sum $\sum_{k=m}^{n} k = \frac{n(n+1)}{2} - \frac{(m-1)m}{2}$ gives, after moving the $(m-1)V$ term left,

$$(n-m+1)\,V = \frac{n(n+1)}{2} - \frac{(m-1)m}{2} - \frac{m-1}{n}. \qquad (\star)$$

(The last term $-\frac{m-1}{n}$ is the total reroll cost: $m-1$ rerolled faces, each paying $\frac{1}{n}$.)

The quadratic shortcut. Relax the cutoff to its continuous value and set $m = V - \frac{1}{n} =: u$. It is cleanest to substitute *before* dividing: from the undivided value equation $nV = (m-1)(V - \frac{1}{n}) + \frac{n(n+1)}{2} - \frac{(m-1)m}{2}$, both mixed terms become $(u-1)u$ (since $m = u$ and $V - \frac{1}{n} = u$), giving

$$nV = (u-1)u + \frac{n(n+1)}{2} - \frac{(u-1)u}{2} = \frac{(u-1)u}{2} + \frac{n(n+1)}{2}.$$

Multiply by 2 and use $2nV = 2n\big(u + \frac{1}{n}\big) = 2nu + 2$:

$$2nu + 2 = (u-1)u + n(n+1) = u^2 - u + n^2 + n,$$

which collapses to a remarkably clean quadratic in u:

$$u^2 - (2n+1)\,u + (n+2)(n-1) = 0.$$

Its discriminant is a perfect square, $(2n+1)^2 - 4(n+2)(n-1) = 9$, so

$$u = \frac{(2n+1) \pm 3}{2} \in \{\, n-1,\ n+2 \,\}.$$

The root $u = n+2$ exceeds n and is spurious; the valid one is $u = n-1$. Hence

$$V = u + \frac{1}{n} = n - 1 + \frac{1}{n}.$$

Here the relaxation is not even an approximation: $u = V - \frac{1}{n} = n-1$ is already an integer, so the continuous cutoff coincides exactly with the true integer cutoff $m = n-1$ (keep the top two faces $n-1, n$; equivalently keep only n, since at $n-1$ you are exactly indifferent).

With reroll cost $1/n$: value $V = n - 1 + \dfrac{1}{n} \approx n - 1$ (hold out for the top). For example $n = 100$ gives $V = 99.01$; $n = 6$ gives $V = 5\frac{1}{6} \approx 5.17$.

28 Optimal Stopping with Forbidden Sums

In these games you roll a die repeatedly, adding to a running total, and you may stop at any time to bank the total. The catch is a set of *forbidden* totals (or faces): land on one and you lose everything. Each roll adds value on average but risks the whole bank, so the question is *when to stop*. The universal tool is **backward induction**: define the value of each total, compare "stop" with "roll once more", and work from the states where the answer is obvious back to the start.

Representative Problem — Keep Rolling Without Exceeding 6 (Q170)

Problem: You roll a fair die repeatedly, adding each roll to a running total. You may stop after any roll and keep your total. If the total ever reaches 7 or more, you bust and get 0. What strategy maximises your expected payoff, and what is the game worth?

Step 1: Define the value of a state

Let $V(s)$ be the best expected payoff when your current total is s ($0 \le s \le 6$). You have two options:

- **stop:** receive s;
- **roll:** each face $x = 1, \dots, 6$ has probability $\frac{1}{6}$; you move to $s + x$ and continue optimally, or receive 0 if $s + x \ge 7$.

So

$$V(s) = \max\Big(s,\ \underbrace{\tfrac{1}{6}\textstyle\sum_{x=1}^{6} W(s+x)}_{\text{roll value}}\Big), \qquad W(t) = \begin{cases} V(t), & t \le 6, \\ 0, & t \ge 7 \text{ (bust)}. \end{cases}$$

This is the *Bellman equation*: the value of a state is the better of its options, assuming you also play optimally afterwards.

Step 2: Start where the answer is obvious

At $s = 6$ any roll busts, so $V(6) = 6$ (stop). Rolls only increase the total, so each $V(s)$ depends only on larger totals — we can solve *backwards*, $s = 5, 4, \dots, 0$.

Step 3: Work backwards

s	safe rolls lead to	roll value $\frac{1}{6}\sum W$	stop	$V(s)$
5	6	$\frac{1}{6}(6) = 1$	5	5 (stop)
4	5, 6	$\frac{1}{6}(5+6) = 1.83$	4	4 (stop)
3	4, 5, 6	$\frac{1}{6}(4+5+6) = 2.5$	3	3 (stop)
2	3, 4, 5, 6	$\frac{1}{6}(3+4+5+6) = 3$	2	**3** (roll)
1	2, …, 6	$\frac{1}{6}(3+3+4+5+6) = 3.5$	1	**3.5** (roll)
0	1, …, 6	$\frac{1}{6}(3.5+3+3+4+5+6) = \frac{49}{12}$	—	$\mathbf{\frac{49}{12} \approx 4.08}$

Note that at $s = 2$ the roll value uses $V(3) = 3$, $V(4) = 4$, … — the values we just found, already including the decision to stop there.

Step 4: Read off the rule and check it

Roll at totals $0, 1, 2$; stop at $3, 4, 5, 6$. Intuition: from 2, four of the six rolls are safe and land on 3–6 (average 4.5), so rolling is worth $\frac{4}{6} \cdot 4.5 = 3 > 2$. From 3, only three rolls are safe (average 5): $\frac{3}{6} \cdot 5 = 2.5 < 3$, so the bust risk now outweighs the gain.

Roll while your total is 0, 1 or 2; stop as soon as it reaches 3 or more. The game is worth $V(0) = \frac{49}{12} \approx 4.08$.

Concept Summary and Applications

Bellman equation with a forbidden set B:

$$V(s) = \max\Big(s,\ \tfrac{1}{6}\sum_{x=1}^{6} W(s+x)\Big), \qquad W(t) = \begin{cases} 0, & t \in B \text{ (bust)}, \\ V(t), & \text{otherwise.} \end{cases}$$

Solve **backwards** from states where stopping is obviously right (or from a far horizon), then read off: roll iff the roll value exceeds s.
One-step test: rolling once more is worth $\frac{1}{6}\sum_{\text{safe } x}(s+x)$ if you would stop afterwards. Compare with s. This myopic test is exact when stopping regions are "absorbing" (once stopping is right, it stays right at every later total), as in Q39 and Q170.

- **Types of forbidden set:** a ceiling (bust above c: Q170, Q573, Q645), a forbidden face (Q39), periodic sums (multiples of m: Q401), irregular sums (perfect squares: Q222).
- **Value $\ll$ threshold:** the threshold is where one more roll breaks even, but the journey there often busts; the start value is much lower.
- **Stop regions need not be a single threshold:** with periodic or irregular hazards they can come in pockets (Q401, Q222).
- **One reroll vs. unlimited rolls:** the first decision is often the same; unlimited rolls add value later in the game (Q573).
- **Applications:** when to take profits on a position that can be stopped out, blackjack-style decisions, and any accumulate-or-bank choice.

Additional Problem 1 — A Six Wipes You Out (Q39)

Problem: You may roll a die as many times as you like, adding the rolls to your score, and stop whenever you choose. But if you ever roll a 6, the game ends and your score becomes 0. What is the optimal strategy, and what is the game worth?

Step 1: One-step comparison

At total s, one more roll gives $s+1, \ldots, s+5$ (each $\frac{1}{6}$) or 0 (a six):

$$E[\text{roll once, then stop}] = \frac{1}{6}[(s+1) + \cdots + (s+5)] + \frac{1}{6} \cdot 0 = \frac{5s+15}{6}.$$

This beats stopping iff $\frac{5s+15}{6} > s \iff s < 15$. Since the total only grows, once $s \geq 15$ it stays ≥ 15, so the one-step rule is the optimal rule: **roll while $s < 15$, stop once $s \geq 15$** (indifferent at exactly 15).

Step 2: Value of the game, backwards

$V(s) = s$ for $s \geq 15$; for $s < 15$, $V(s) = \frac{1}{6}\sum_{k=1}^{5} V(s+k)$ (each value is the average of the five above it, divided by 6 rather than 5 because of the bust). For example $V(14) = \frac{1}{6}(15+16+17+18+19) = \frac{85}{6} \approx 14.17$. Continuing down:

s	14	12	10	8	6	4	2	0
$V(s)$	14.17	12.59	11.16	9.91	8.80	7.81	6.93	**6.15**

Step 3: Why only 6.15 when you aim for 15

Tracking where you end up: you bust before reaching 15 about 62% of the time; otherwise you stop at 15–19 (average about 16.3). So $E \approx 0.38 \times 16.3 \approx 6.15$. The threshold is where one roll breaks even; the climb there is what costs you.

Roll while your total is below 15; stop once it is 15 or more. Expected score $V(0) \approx 6.15$ (you bust about 62% of the time).

Additional Problem 2 — Bust Above 9, One Optional Reroll (Q573)

Problem: Roll a die once. You may stop (keep the result) or roll once more and add the two rolls. If the sum exceeds 9 you get nothing. For each first roll, should you stop or roll again? What is the game worth? What changes if you may keep rolling as often as you like?

Step 1: Value of rolling again after first roll x

$$E_x = \frac{1}{6}\sum_{y=1}^{6}(x+y)\,\mathbf{1}\{x+y \leq 9\}.$$

x	safe second rolls	E_x	decision
1	all	$\frac{2+\cdots+7}{6} = 4.5$	roll
2	all	$\frac{3+\cdots+8}{6} = 5.5$	roll
3	all	$\frac{4+\cdots+9}{6} = 6.5$	roll
4	1–5	$\frac{5+\cdots+9}{6} = 5.83$	roll
5	1–4	$\frac{6+7+8+9}{6} = 5$	either (tie)
6	1–3	$\frac{7+8+9}{6} = 4$	stop

Step 2: Value

$$\frac{1}{6}(4.5 + 5.5 + 6.5 + \tfrac{35}{6} + 5 + 6) = \frac{50}{9} \approx 5.56.$$

Step 3: Unlimited rolls

Solving $V(s) = \max\{s, \frac{1}{6}\sum_k W(s+k)\}$ backwards ($W = 0$ above 9): stop at 5 or more (at 5 rolling is worth exactly 5, a tie; at 6 it is worth 4), roll at 1–4. The first-roll decisions are unchanged, but from small totals you can now roll more than once ($V(1) \approx 6.18$, $V(2) \approx 6.44$, $V(3) \approx 6.81$, $V(4) \approx 5.83$), so the game is worth $\frac{46991}{7776} \approx 6.04$.

Roll again on 1–4, either on 5, stop on 6; value $\frac{50}{9} \approx 5.56$. With unlimited rolls the first decisions are the same and the value rises to ≈ 6.04.

Additional Problem 3 — Ten-Sided Die, Bust Above 13 (Q645)

Problem: Roll a ten-sided die (faces 1–10). You may keep the result or roll once more and add. A total of 13 or less is paid in pounds; above 13 you get nothing. What is the optimal strategy and its value?

Step 1: Rolling again from x

The second roll y is safe iff $y \le 13 - x$.

- $x \le 3$: every y is safe, so $E_x = x + 5.5$: 6.5, 7.5, 8.5.
- $x \ge 4$: $E_x = \frac{1}{10}\sum_{y=1}^{13-x}(x+y) = \frac{1}{10}[(x+1)+\cdots+13]$: $x = 4$: 8.1; 5: 7.6; 6: 7.0; 7: 6.3; 8: 5.5; 9: 4.6; 10: 3.6.

Step 2: Compare with stopping

Roll when $E_x > x$: true for $x \le 6$ ($7.0 > 6$), false from $x = 7$ ($6.3 < 7$). Value:

$$\frac{1}{10}(6.5 + 7.5 + 8.5 + 8.1 + 7.6 + 7.0 + 7 + 8 + 9 + 10) = \frac{79.2}{10} = £7.92.$$

Roll again if the first roll is 1–6; keep 7–10. Expected payout £7.92.

Additional Problem 4 — Multiples of 10 Are Forbidden (Q401)

Problem: Roll a die repeatedly, summing the rolls; stop any time and keep the sum. If the sum is ever a multiple of 10, your score becomes 0 and the game ends. What is the optimal strategy? What if the forbidden multiple is m instead of 10?

Step 1: The recursion

$$E(n) = \begin{cases} 0, & n \equiv 0 \pmod{10}, \\ \max\Big(n,\ \frac{1}{6}\sum_{x=1}^{6} E(n+x)\Big), & \text{otherwise.} \end{cases}$$

For large n a $\frac{1}{6}$ chance of losing everything outweighs any gain, so $E(n) = n$ far out; solve backwards from there.

Step 2: The surprising answer — two stopping pockets

current sum n	action
1–23	roll
24, 25	**stop**
26–33	roll
34 and above	**stop**

Expected score $E(0) \approx 13.22$.

Step 3: Why stop at 24–25 but roll at 26–29?

- From 24 or 25, one face busts at 30 and the others land on 25–31; several of those are themselves close to 30, so the upside is small.
- From 26–29, one face still busts at 30, but the others can jump past it to 31–35, which are *safe for the next roll* (far from 40) and worth banking. That upside makes rolling worthwhile.
- From 34 on, you hold so much that a $\frac{1}{6}$ chance of losing it all is never worth it.

Under this policy the total can never exceed 39 (from 33 a roll reaches at most 39).

Step 4: A hand estimate

The die advances 3.5 per roll on average, so the running total lands on any given far-away number with probability $\approx \frac{1}{3.5} = \frac{2}{7}$ (one hit per 3.5 integers) and jumps over it with probability $\approx \frac{5}{7}$. To bank about 36 you must survive crossing 10, 20 and 30:

$$E \approx 36 \times \left(\tfrac{5}{7}\right)^3 \approx 36 \times 0.36 \approx 13.$$

Step 5: Other forbidden multiples m

The same recursion with 10 replaced by m gives:

m	stop when	$E(0)$
6	$n \geq 19$	7.22
7	$n \geq 15$	8.59
8	$n = 18$ or $n \geq 26$	10.13
9	$n = 21, 22$ or $n \geq 30$	11.68
10	$n = 24, 25$ or $n \geq 34$	13.22
11	$n = 27, 28$ or $n \geq 38$	14.73
12	$n = 30, 31$ or $n \geq 42$	16.28

For small m every total is near a forbidden one and the rule is a single threshold; for larger m an early pocket appears.

Stop at 24 or 25, or at 34 and above; otherwise roll. Expected score ≈ 13.22 (hand estimate $36 \times (5/7)^3 \approx 13$). For general m solve the same recursion; see the table.

Additional Problem 5 — Perfect Squares Are Forbidden (Q222)

Problem: Roll a die repeatedly and add to your score; stop any time. If your total is ever a perfect square $(1, 4, 9, 16, 25, \dots)$, you immediately lose everything. What is the optimal strategy?

Step 1: Same Bellman equation, irregular hazards

$$V(s) = \max\Big(s,\ \tfrac{1}{6}\sum_{k=1}^{6} W(s+k)\Big), \qquad W(t) = 0 \text{ if } t \text{ is a perfect square, else } V(t).$$

The gaps between squares grow ($2k+1$ between k^2 and $(k+1)^2$), so there is no periodic shortcut; solve backwards from a far horizon. Far out, each square you try to cross is hit with probability $\approx \frac{2}{7}$, wiping out a large bank for only 3.5 points per roll, so eventually you always stop.

Step 2: Only the danger zone matters

Let d be the distance to the next square Q. If $d \geq 7$, no single roll can hit Q, so rolling is free: always roll. Decisions arise only for $d \in \{1, \dots, 6\}$:

next square Q	in the zone $Q-6 \leq s \leq Q-1$
$4, 9, 16, 25$	always roll (score too small to protect)
36	stop at $30, 31$ ($d = 6, 5$); roll at $d \leq 4$
49	stop at 43–45 ($d = 6, 5, 4$); roll at $d \leq 3$
64	stop at 58–62; roll only at 63 ($d = 1$)
81 and beyond	always stop once within 6

Why roll when very close? From $d = 1$ (e.g. $s = 35$), the square is hit only by a roll of 1; the other five rolls jump over it into a long safe gap. From $d = 5$ or 6, most rolls land just short of the square, still in danger.

Step 3: Value

The first roll already busts with probability $\frac{1}{3}$ (on 1 or 4), and 9, 16, 25 claim many more runs while the score is small. Backward induction gives

$$V(0) \approx 7.18.$$

Overall you bust about 81% of the time; survivors bank about 38 on average: $0.19 \times 38 \approx 7.2$.

Always roll when the next square is more than 6 away. Near a square: always roll for $4, 9, 16, 25$; stop at 30–31, 43–45, 58–62; from 75 on, stop as soon as the next square is within 6. Expected score ≈ 7.18.

29 Finite-Deck Optimal Stopping (Backward Induction)

Problem: A bag holds 3 black and 2 white balls. You draw one at a time *without replacement* and may **stop whenever you like**. Your payoff is $B - W$: (black balls drawn) − (white balls drawn). When should you stop, and what is the game worth?

Background: Why This Needs Backward Induction

Each black ball drawn is worth +1 and each white ball −1, and you may quit at any point keeping what you have. The order is random, so at every step you face a choice — stop now, or draw again hoping for more black than white from what remains. The catch is that the *right* choice now depends on what you would do later, which depends on still-later choices. The clean way to untangle this is to solve **backward**: figure out the value of the near-empty bag first, then use it to value fuller bags, and so on back to the start. That is **dynamic programming** (a.k.a. backward induction).

State and the Bellman Equation

Describe the situation by what is *left* in the bag: b black and w white. Let $V(b, w)$ be the best expected *additional* payoff achievable from that point. You compare two options:

- **Stop:** take 0 more.
- **Draw:** the next ball is black w.p. $\frac{b}{b+w}$ (+1, then state $(b-1, w)$) or white w.p. $\frac{w}{b+w}$ (−1, then state $(b, w-1)$).

Taking the better of the two gives the **Bellman equation**:

$$V(b,w) = \max\Big\{\,0,\ \tfrac{b}{b+w}\big(1+V(b-1,w)\big) + \tfrac{w}{b+w}\big(-1+V(b,w-1)\big)\Big\},$$

$$V(b,0) = b \text{ (all black left — take them all)}, \qquad V(0,w) = 0 \text{ (all white left — stop)}.$$

Solve by Working Upward

Start from the boundary and build up (each "continue" value is the bracket above; the state's V is that value if positive, else 0):

$$V(1,0) = 1, \quad V(2,0) = 2, \quad V(3,0) = 3, \qquad V(0,1) = V(0,2) = 0.$$

$$V(1,1):\ \tfrac{1}{2}(1+\underbrace{V(0,1)}_{0}) + \tfrac{1}{2}(-1+\underbrace{V(1,0)}_{1}) = \tfrac{1}{2}+0 = \tfrac{1}{2} > 0 \ \Rightarrow\ V(1,1) = \tfrac{1}{2}.$$

$$V(2,1):\ \tfrac{2}{3}(1+\tfrac{1}{2}) + \tfrac{1}{3}(-1+2) = 1+\tfrac{1}{3} = \tfrac{4}{3} \ \Rightarrow\ V(2,1) = \tfrac{4}{3}.$$

$$V(1,2):\ \tfrac{1}{3}(1+0) + \tfrac{2}{3}(-1+\tfrac{1}{2}) = \tfrac{1}{3}-\tfrac{1}{3} = 0 \ \Rightarrow\ V(1,2) = 0 \text{ (indifferent — stop)}.$$

$$V(2,2):\ \tfrac{1}{2}(1+\underbrace{V(1,2)}_{0}) + \tfrac{1}{2}(-1+\underbrace{V(2,1)}_{4/3}) = \tfrac{1}{2}+\tfrac{1}{6} = \tfrac{2}{3} \ \Rightarrow\ V(2,2) = \tfrac{2}{3}.$$

$$V(3,1):\ \tfrac{3}{4}(1+\tfrac{4}{3})+\tfrac{1}{4}(-1+3)=\tfrac{7}{4}+\tfrac{1}{2}=\tfrac{9}{4}\ \Rightarrow\ V(3,1)=\tfrac{9}{4}.$$
$$V(3,2):\ \tfrac{3}{5}(1+\underbrace{V(2,2)}_{2/3})+\tfrac{2}{5}(-1+\underbrace{V(3,1)}_{9/4})=\tfrac{3}{5}\cdot\tfrac{5}{3}+\tfrac{2}{5}\cdot\tfrac{5}{4}=1+\tfrac{1}{2}=\tfrac{3}{2}.$$

Optimal Stopping Rule and Answer

Continue exactly when the continuation value is positive. Scanning the states, the only reachable one where it is not is $(1,2)$ (value 0), plus any $b=0$. So the rule from the starting bag $(3,2)$ is:

> **Keep drawing; stop only if the remaining bag becomes $(1$ black$, 2$ white$)$ or runs out of black. The game is worth $V(3,2)=\dfrac{3}{2}$.**

A simpler rule and an independent check. *Because stopping and continuing tie at $(1,2)$, an equally optimal rule is:* ***keep drawing until all 3 black balls are out, then stop.*** *If you had to draw all five balls you would always end with $3-2=+1$. Under this rule you still collect all 3 blacks but* skip every white ball that comes after the last black *(the "trailing whites"), so*

$$\textit{payoff}=3-\big(2-\#\textit{trailing whites}\big)=1+\#\textit{trailing whites}.$$

The last ball is white with probability $\frac{2}{5}$; the last two are both white with probability $\frac{2}{5}\cdot\frac{1}{4}=\frac{1}{10}$. Hence

$$\mathbb{E}[\#\textit{trailing whites}]=\tfrac{2}{5}+\tfrac{1}{10}=\tfrac{1}{2},\qquad \mathbb{E}[\textit{payoff}]=1+\tfrac{1}{2}=\tfrac{3}{2},$$

matching $V(3,2)$. The option to stop is worth exactly the expected number of white balls you avoid, $\frac{1}{2}$. (This "stop after the last good ball" rule works here because the bag is small and tilted toward black; for balanced or larger decks, such as Additional Problem 1, the optimal rule can stop earlier on a temporary lead. Same game with blue/red balls: Q419, Q285, Q367.)

Generalization

Nothing was special about $(3,2)$: for *any* starting counts (b,w) the same recursion

$$V(b,w)=\max\Big\{0,\ \tfrac{b}{b+w}\big(1+V(b-1,w)\big)+\tfrac{w}{b+w}\big(-1+V(b,w-1)\big)\Big\},\quad V(b,0)=b,\ V(0,w)=0,$$

computed backward from the empty bag, determines both the value and the optimal rule (continue iff the bracket is positive). The hard part is purely the bookkeeping of filling in the $V(b,w)$ table.

Additional Problem 1 — Full Deck (26 Black, 26 Red)

Problem: A standard deck (26 black, 26 red) is shuffled face down; black pays +\$1, red −\$1; stop any time and keep your running profit. Value?

This is the *same* dynamic program at full scale — too big for hand computation, but everything above still applies.

Reframe (a bridge pinned to zero). Your running profit is (blacks turned) − (reds turned). Turning the *whole* deck always ends at $26 - 26 = 0$, so the profit is a symmetric random walk **pinned to** 0 **at both ends**. A "never stop" strategy is therefore worth exactly \$0 — **all the value is in the right to quit early** on a temporary lead. (Handy identity: current profit = reds remaining − blacks remaining, so "ahead" means "more red than black still buried.")

The trap. The myopic rule "the next card is worth $\frac{b-r}{b+r}$, so stop as soon as $b \le r$" is **wrong**: at the start $b = r = 26$ it says to stop for a value of \$0. That ignores **option value** — even when black and red remaining are *equal*, so the next flip has zero immediate expectation, the *right to stop after a good flip* makes continuing strictly worthwhile. Indeed $V(26, 26) > 0$, and the optimum keeps drawing at every balanced state (26–26, 10–10, . . .); the true boundary is whatever the DP gives, and it is more patient than "stop the instant red leads."

Same backward-induction DP. With b black and r red cards left,

$$V(b,r) = \max\Big\{0,\ \tfrac{b}{b+r}\big(1 + V(b-1,r)\big) + \tfrac{r}{b+r}\big(-1 + V(b,r-1)\big)\Big\}, \quad V(0,r) = 0,\ V(b,0) = b,$$

solved up to the full deck, gives

$$V(26,26) \approx \$2.62 \quad (2.6245\ldots).$$

The value grows slowly: $V(n,n) = \frac{1}{2}, \frac{2}{3}, 0.85, 1.12, 1.84, 2.62$ for $n = 1, 2, 3, 5, 13, 26$ (and $V(1,1) = \frac{1}{2}$ is the one-pair game you can do in your head).

Estimating \$2.62 by hand.

- **Order of magnitude:** a sum of $\sim k$ ± 1 steps has typical size $\sqrt{k}$, largest mid-deck: SD $\approx \sqrt{26^2/51} \approx \sqrt{13} \approx 3.6$ — so the answer is a *few dollars*.
- **Scaling law, calibrated at $n = 1$:** you cannot stop exactly at the peak, so you capture a fraction of that swing, giving $V(n,n) \approx \frac{1}{2}\sqrt{n}$ (exact at $n = 1$). A full deck: $\frac{1}{2}\sqrt{26} \approx \$2.5$.
- **Sharper:** the asymptotic $V(n,n) \approx \sqrt{\pi n/12} = \sqrt{6.8} \approx \$2.6$, essentially exact.

Play by backward induction (continue while $V(b,r) > 0$; roughly, stop once red sufficiently outnumbers black). Expected winnings ≈ \$2.62.

Additional Problem 2 — Stop and Match the Next Card's Colour (Q405)

Problem: A shuffled deck has 26 red and 26 black cards. You repeatedly look at the top card and either *discard* it or *end the game*. When you end the game, you win if the *next* card (the one directly below the visible top card) has the **same colour** as the top card; otherwise you lose. What is the optimal strategy, and what is the win probability?

(You must end the game by the time only 2 cards remain — discarding then would leave no "next card.")

Same idea, different payoff. As in the main problem, the cards are drawn *without replacement*, so the state is the remaining composition and we solve the game backward. The difference is the payoff: not a running profit, but a *win probability* for a colour match.

Step 1 — State and the two actions. Let $V(r,b)$ be the best achievable win probability when r red and b black cards remain (top card not yet revealed), and let $n = r+b$. Suppose the top card turns out **red** (probability r/n). Below it there are $r-1$ red and b black cards.

- **Stop:** win iff the next card is red: $\quad S_R = \dfrac{r-1}{n-1}$.
- **Discard:** the game continues from $(r-1,b)$: $\quad C_R = V(r-1,b)$.

A black top card is the same with colours swapped: $S_B = \frac{b-1}{n-1}$, $C_B = V(r,b-1)$.

Step 2 — Bellman recursion. For $n \geq 3$,

$$V(r,b) = \frac{r}{n}\max\left\{\frac{r-1}{n-1},\, V(r-1,b)\right\} + \frac{b}{n}\max\left\{\frac{b-1}{n-1},\, V(r,b-1)\right\},$$

with the forced-stop boundary at $n=2$: $V(2,0) = V(0,2) = 1$, $V(1,1) = 0$.

Step 3 — Claim.

$$V(r,b) = 1 - \frac{rb}{n(n-1)} \qquad (n = r+b \geq 3),$$

and for $n \geq 4$ discarding is always at least as good as stopping.

Step 4 — Base case $n=3$ (by hand).

- $(3,0)$: every card matches, $V = 1 = 1 - 0$. ✓
- $(2,1)$, *red on top*: stop wins $\frac{1}{2}$; discarding leaves one red and one black, which cannot match $(0) \Rightarrow$ **stop**.
  Black on top: stop wins 0; discarding leaves two reds $(1) \Rightarrow$ **discard**.
  $V(2,1) = \frac{2}{3}\cdot\frac{1}{2} + \frac{1}{3}\cdot 1 = \frac{2}{3} = 1 - \frac{2}{6}$. ✓

This is the one place where stopping can be *strictly* better.

Step 5 — Inductive step ($n \geq 4$): continue vs. stop. Assume the formula for decks of size $n-1$ (≥ 3). With red on top,

$$C_R = 1 - \frac{(r-1)b}{(n-1)(n-2)}, \qquad S_R = \frac{r-1}{n-1}.$$

Multiply the difference by the common denominator $(n-1)(n-2)$:

$$\begin{aligned}(n-1)(n-2)(C_R - S_R) &= (n-1)(n-2) - (r-1)b - (r-1)(n-2)\\ &= (n-2)\big[(n-1)-(r-1)\big] - (r-1)b\\ &= (n-2)\,b - (r-1)\,b \qquad (n-r=b)\\ &= b\big[(n-2)-(r-1)\big] = b(b-1) \qquad (n-1-r = b-1).\end{aligned}$$

$$C_R - S_R = \frac{b(b-1)}{(n-1)(n-2)} \geq 0, \qquad b = \text{number of opposite-colour cards below the top card.}$$

opposite cards b	$C_R - S_R$	conclusion
0	0	tie — every remaining card matches
1	0	tie — both win with probability $\frac{n-2}{n-1}$
≥ 2	> 0	waiting is **strictly** better

Why $b = 0$ or $b = 1$ is a tie. Let $m = n - 1$ be the number of cards below the red top card.

- $b = 0$: all m cards below are red. Stopping wins surely; continuing also wins surely (whatever ends on top, the next card is the same colour).
- $b = 1$: the single black card is equally likely to sit in any of the m positions below. *Stop now*: lose only if it is the very next card, probability $\frac{1}{m}$. *Keep waiting*: you lose only if the last three cards are R B R (with 3 left you see a red top card, stop, and the next card is black) — i.e. the black card is second-to-last, again probability $\frac{1}{m}$. Anywhere else you win: it surfaces earlier (discard it; all remaining cards are red) or the ending is R R B (stop, next is R) or B R R (discard B). Same win probability $\frac{n-2}{n-1}$.
- $b \geq 2$: stopping loses whenever *any* of the b black cards is next; waiting keeps extracting information about the composition.

Examples: 5R, 1B, red on top: stop $\frac{4}{5}$, wait $V(4,1) = 1 - \frac{4}{20} = \frac{4}{5}$ (tie). 4R, 2B, red on top: stop $\frac{3}{5}$, wait $V(3,2) = 1 - \frac{6}{20} = \frac{7}{10}$ (wait better).

Step 6 — The formula satisfies the recursion. Since each max equals its continuation value when $n \geq 4$,

$$\begin{aligned} V(r,b) &= \frac{r}{n}\Big[1 - \frac{(r-1)b}{(n-1)(n-2)}\Big] + \frac{b}{n}\Big[1 - \frac{r(b-1)}{(n-1)(n-2)}\Big] \\ &= 1 - \frac{rb\big[(r-1)+(b-1)\big]}{n(n-1)(n-2)} = 1 - \frac{rb(n-2)}{n(n-1)(n-2)} = 1 - \frac{rb}{n(n-1)}. \end{aligned}$$

By induction the formula holds for every $n \geq 3$, and the strategy attains it, so it is optimal.

Step 7 — When do you lose? Under the waiting strategy the only decision is at the last three cards: all one colour $\to$ win; $XXY \to$ stop on X, win; $YXX \to$ discard Y, win; $XYX \to$ stop on X, **lose**. So you lose only if the deck ends in an alternating triple (RBR or BRB):

$$P(\text{lose}) = \frac{r\,b\,(r-1) + b\,r\,(b-1)}{n(n-1)(n-2)} = \frac{rb}{n(n-1)}.$$

Discard until 3 cards remain (stopping earlier only ties, and only when at most one opposite-colour card is left; e.g. "stop once ≤ 1 opposite card remains" is equally optimal). With 3 left, stop iff the top card's colour appears twice among the three. Win probability

$$V(26,26) = 1 - \frac{26 \cdot 26}{52 \cdot 51} = 1 - \frac{13}{51} = \mathbf{\frac{38}{51}} \approx \mathbf{74.5\%}.$$

Common wrong answers: stop on the first card $\to \frac{25}{51}$; never stop (last two cards decide) $\to \frac{25}{51}$; "stop as soon as the top card's colour is the majority of the remaining deck" $\to$ only $\frac{26}{51}$, because it stops with $b \geq 2$, which Step 5 shows is strictly worse. **Lesson (same as the main problem):**

in finite-deck stopping problems, the option to wait has value; check the continue-minus-stop gap rather than acting on the immediate odds.

Additional Problem 3 — Call a Colour for the Next Two Cards (Q526)

Problem: A standard 52-card deck (26 red, 26 black) is dealt face up one card at a time. At any point you may stop and name a colour. You win if the next *two* cards are both that colour. What is the optimal strategy?

Step 1 — for a fixed colour, the timing of the call does not matter. With r red and b black cards left ($n = r + b$), calling red now wins with probability

$$P(\text{next two red}) = \frac{r}{n} \cdot \frac{r-1}{n-1} = \frac{r(r-1)}{n(n-1)}.$$

The unseen cards are in random order, so the *next* two cards are no different from the *last* two: $P(\text{next two red}) = P(\text{last two red})$ at every moment. So if you were going to call red regardless, calling now, later, or at the very end all give the same chance, which from the full deck is $\frac{26}{52} \cdot \frac{25}{51} = \frac{25}{102} \approx 24.5\%$. (This quantity is a martingale.)

Step 2 — what waiting buys: choosing the colour later. Waiting does not change the odds of a fixed colour, but it lets you pick the colour *after* seeing more cards. Wait until 2 cards remain: by counting what has been shown, you know exactly what they are. Name their colour if they match; if they are one of each, no call can win. You win exactly when the last two cards match:

$$P(\text{last two same colour}) = \frac{25}{102} + \frac{25}{102} = \frac{25}{51} \approx 49.0\%.$$

Step 3 — nothing does better. Whenever you stop you name the better colour, so you win with probability

$$\max\left\{\tfrac{r(r-1)}{n(n-1)}, \tfrac{b(b-1)}{n(n-1)}\right\} \le \frac{r(r-1)+b(b-1)}{n(n-1)} = P(\text{last two match} \mid \text{now}).$$

The right side is a sum of two martingales, so its expected value at any stopping time stays $\frac{25}{51}$. No strategy can beat $\frac{25}{51}$, and waiting to the end attains it. Stopping early never helps; it only ties when one colour has at most one card left.

> **Never call early: wait until 2 cards remain and name their colour if they match. Win probability $\frac{25}{51} \approx 49\%$, double the $\frac{25}{102}$ from calling at the start.**

Common wrong answer: "$P(\text{next two red})$ is a martingale, so waiting never helps $\Rightarrow \frac{25}{102}$." That is true for *one fixed* colour, but you choose the colour when you stop, and the better of two martingales grows on average (a submartingale). **Lesson (as in Additional Problem 2):** the option to wait has value whenever waiting reveals information you can still act on.

Contrast with the *With*-Replacement Walk

If the cards were **replaced** after each draw (Section 33), the total would be a *free* recurrent random walk with EV = 0: no stopping rule helps. What makes *these* games have positive value is exactly

that the bag or deck is *finite and not replaced*: the walk is pinned to a fixed endpoint (0 for the balanced deck), and knowing the leftover composition (b vs. w or r) gives you a real, computable edge. **Removing replacement turns a worthless game into a solvable one** — and makes the backward-induction DP well-posed, because the horizon is finite (the bag empties).

Concept Summary and Applications

- **Method:** finite horizon + sampling *without* replacement $\Rightarrow$ let the state be the remaining composition and solve $V = \max(\text{stop},\ \text{expected continue})$ backward from the terminal (empty) states. Small cases (like 3B/2W) are hand-computable; large ones (a full deck) need a computer but obey the identical rule.
- **Watch for option value:** a myopic "stop when the next draw is non-positive" rule undervalues the game, because the right to stop later is itself worth something.
- **Applications:** card-counting / advantage play, secretary-style hiring from a known pool, inventory drawdown, and any "draw without replacement, stop when ahead" problem.

30 Backward Induction as Dynamic Programming: Multi-Path

Section 26 solved stopping problems by working *backward* along a simple line of stages. This section pushes the same tool one step further, to problems where the states branch and rejoin — the natural home of **dynamic programming**.

Backward Induction and Dynamic Programming Are One Idea

Backward induction and dynamic programming (DP) are two names for the same method. Both rest on four pieces:

1. a **state** that captures everything you need to know to act (here, the running total of points);
2. a **value function** $f(\text{state})$ — the best achievable value from that state onward;
3. a **recursion** (the *Bellman equation*) expressing f at a state in terms of the states you can move to, taking a max over your available actions;
4. **boundary values** at the terminal states, from which everything else is built.

You solve the terminal states first and work backward, so every subproblem is ready before the states that depend on it.

The only difference between the two names is the *shape* of the state space. In Section 26 the states formed a single chain (rolls remaining: $3 \to 2 \to 1$), one path from start to end — "backward induction" pictures marching straight back along that line. Here the states form a **branching graph**: from a total s you can move to $s+2$ or $s+3$, and the *same* state can be reached by several different paths (a total of 5 arises both as 2+3 and as 3+2). "Dynamic programming" stresses computing each state's value *once* and reusing it for every path that passes through, instead of recomputing — this reuse ("memoization") is what makes branching problems tractable.

Representative Problem — The Dartboard

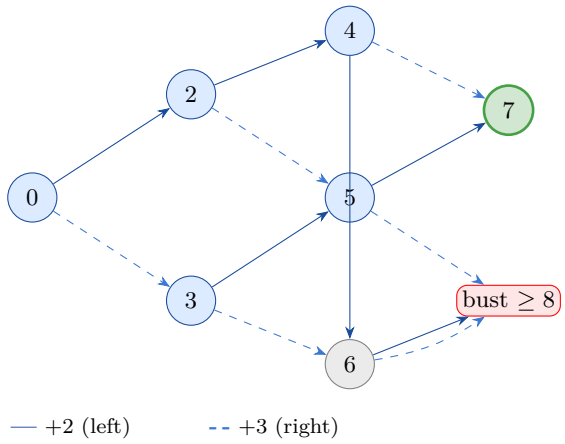
Problem: A dartboard is split in half: the left half is worth 2 points, the right half 3 points. On any throw you hit the board with probability 80%; when you hit, you land on the side you aimed for with probability 70% (and the other side with probability 30%). With unlimited throws, and choosing which side to aim for before each throw, what is the largest probability that your running total is *exactly* 7?

Three facts to see first.

- **7 has only one recipe.** The only way to build 7 from 2s and 3s is $2+2+3$ (since $2x+3y=7$ forces $x=2, y=1$): two left hits and one right hit, in some order.
- **The 80% hit rate is a red herring.** A miss scores 0; with unlimited throws you simply throw again, so misses only delay you, never change whether 7 is reachable. Conditioning on an eventual hit, each hit lands on the aimed side with probability 0.7 and the other side with 0.3.
- **Overshooting is the real enemy.** The total only ever rises in steps of +2 or +3, so if you ever jump *past* 7 you have lost it forever (e.g. $5 \xrightarrow{+3} 8$, or landing on 6, from which any hit gives 8 or 9).

Why the answer is not 1. It is tempting to say "aim left until a 2 comes, again until a second 2, then aim right for the 3 — each eventually happens." This is wrong: while you wait for a 2, every throw *also* has a 0.3 chance of a stray 3, which can shove your total past 7. You cannot wait for a specific value without risking an overshoot, so the probability is strictly below 1 and depends on aiming cleverly.

The state graph (why it is "multi-path"). Track only the running total. Each total leads to two others (+2 or +3), and several totals are reachable by more than one route — e.g. 5 from both 2 and 3:



Because total 5 (and 6, and the goal 7) is reached by two different paths, this is a graph, not a chain — DP computes each $f(s)$ once and reuses it.

The Bellman recursion. Let $f(s)$ be the *maximum* probability of finishing on exactly 7 starting from total s. On each hit you choose a side: aiming *left* gives +2 with probability 0.7 and +3 with 0.3; aiming *right* gives +3 with 0.7 and +2 with 0.3. Taking the better aim,

$$f(s) = \max\Big\{\underbrace{0.7\,f(s{+}2) + 0.3\,f(s{+}3)}_{\text{aim left (for 2)}},\ \underbrace{0.3\,f(s{+}2) + 0.7\,f(s{+}3)}_{\text{aim right (for 3)}}\Big\},$$

with boundary $f(7) = 1$ and $f(s) = 0$ for $s \geq 8$ (overshot). Now work *downward* from the boundary.

$f(6)$: both 6+2=8 and 6+3=9 are busts, so $f(6) = 0$.

$f(5)$: 5+2=7 (f=1), 5+3=8 (f=0). Aim left $= 0.7(1) + 0.3(0) = 0.7$; aim right $= 0.3(1) = 0.3$. So $f(5) = 0.7$ (aim for 2).

$f(4)$: 4+2=6 (0), 4+3=7 (1). Aim left $= 0.3(1) = 0.3$; aim right $= 0.7(1) = 0.7$. So $f(4) = 0.7$ (aim for 3).

$f(3)$: 3+2=5 (0.7), 3+3=6 (0). Aim left $= 0.7(0.7) = 0.49$; aim right $= 0.3(0.7) = 0.21$. So $f(3) = 0.49$ (aim for 2).

$f(2)$: 2+2=4 (0.7), 2+3=5 (0.7). Aim left $= 0.7(0.7) + 0.3(0.7) = 0.70$; aim right $= 0.70$ too. So $f(2) = 0.70$ (either).

$f(0)$: 0+2=2 (0.70), 0+3=3 (0.49).

aim left $= 0.7(0.70){+}0.3(0.49) = 0.490{+}0.147 = 0.637$, aim right $= 0.3(0.70){+}0.7(0.49) = 0.553$.

So $f(0) = \max(0.637,\, 0.553) = 0.637$ (aim for 2). (The total $s = 1$ never occurs, since every step from 0 is +2 or +3.)

Optimal policy and answer.

s	0	2	3	4	5	6
$f(s)$	**0.637**	0.70	0.49	0.70	0.70	0
aim for	2	either	2	3	2	—

$$P(\textbf{score exactly } 7) = \frac{637}{1000} = 0.637\textbf{, aiming for } 2 \textbf{ at totals } 0, 3, 5 \textbf{ and for } 3 \textbf{ at total } 4.$$

Strategy matters. Adapting the aim to the current total is the whole game. Naively *always* aiming for 2s scores exactly 7 with probability only $\binom{3}{2}(0.7)^2(0.3) = 0.441$, and always aiming for 3s gives about 0.189; steering toward 7 lifts this to 0.637 (a direct simulation confirms ≈ 0.637).

Concept Summary and Applications

- **Method:** define a state, a value function, and a Bellman recursion with a max over actions; fix the terminal values and evaluate backward. When states branch and rejoin, store each value once and reuse it (DP / memoization).
- **Policy:** the optimal action at each state falls out of which term achieves the max — here, which side to aim for at each total.
- **Watch the boundary:** terminal "win" states get value 1 and overshoot / "bust" states get value 0; getting these right is half the problem.
- **Applications:** shortest-path and reachability on graphs, optimal play in dice/board games, inventory and resource control, sequence alignment, and any "best expected outcome over a sequence of choices" problem.

31 Rank-Based Stopping: The Secretary Problem

In a *rank-based* stopping problem, options arrive one at a time in random order, and you can only compare each new option with the ones you have already seen — you never learn its absolute quality. Each option must be accepted or rejected on the spot, with no going back. The classic example is hiring a secretary; the same structure appears in choosing an apartment, selling a house to sequential bidders, or picking a partner. The famous answer is the $1/e$ **rule**: look at about 37% of the options without committing, then take the first one that beats everything seen so far.

Representative Problem — Hiring the Best of 100 Candidates

Problem: 100 candidates are interviewed one at a time in a uniformly random order. After each interview you know only how the candidate ranks among those interviewed so far. You must hire or reject immediately; rejected candidates cannot be recalled, and if you reach the last candidate you must hire them. You succeed only if you hire the single best candidate. What strategy maximises the probability of success, and what is that probability?

Step 1: Only "record" candidates are worth hiring

Call a candidate a *record* if they are better than everyone interviewed before them. A non-record cannot be the overall best (someone earlier already beats them), so hiring one guarantees failure. So the only question is *which* record to stop at.

Step 2: A record becomes more valuable the later it appears

If candidate i is a record, they are the best of the first i. The overall best is equally likely to be anywhere, so the chance that the best of all n lies among the first i is $\frac{i}{n}$. Hence

$$P(\text{record at position } i \text{ is the overall best}) = \frac{i}{n},$$

which grows with i. Meanwhile, the value of continuing shrinks as fewer candidates remain. So once stopping at a record is worth it, it stays worth it: the optimal rule is a **threshold rule** — reject the first k candidates, then hire the first record.

Step 3: Success probability of "reject k, then take the first record"

Suppose the best candidate is at position j (probability $\frac{1}{n}$ for each j).

- If $j \le k$, the best is rejected during the look phase: failure.
- If $j > k$, we hire the best exactly when no record appears at positions $k+1, \dots, j-1$, i.e. when the best of the first $j-1$ candidates lies in the look phase (positions $1, \dots, k$). That has probability $\frac{k}{j-1}$.

So

$$P_n(k) = \sum_{j=k+1}^{n} \frac{1}{n} \cdot \frac{k}{j-1} = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j}.$$

Step 4: Optimise — where $1/e$ comes from

For large n write $x = k/n$. The sum $\sum_{j=k}^{n-1} \frac{1}{j} \approx \ln \frac{n}{k} = -\ln x$, so

$$P(x) \approx -x \ln x.$$

Setting the derivative to zero: $-\ln x - 1 = 0 \Rightarrow x = \frac{1}{e} \approx 0.368$, and then $P = -\frac{1}{e} \ln \frac{1}{e} = \frac{1}{e} \approx 36.8\%$.

Step 5: Exact numbers for $n = 100$

Evaluating $P_{100}(k)$ exactly, the best choice is $k = 37$ (reject the first 37, i.e. about $n/e = 36.8$), giving

$$P_{100}(37) \approx 0.3710.$$

Neighbouring choices are almost as good ($k = 36$ or 38 lose less than 0.001): the curve $-x \ln x$ is flat near its peak.

Step 6: Why this is remarkable

Picking at random succeeds with probability $\frac{1}{100}$. The $1/e$ rule succeeds with probability about 37% *no matter how large n is* — a million candidates still give $\approx 36.8\%$.

Reject the first 37 candidates ($\approx n/e$), then hire the first candidate better than all previous ones (or the last if none appears). Success probability ≈ 0.371; as $n \to \infty$ it tends to $1/e \approx 36.8\%$.

Concept Summary and Applications

Classic secretary problem (n candidates, random order, relative ranks only, no recall, succeed only with the best):

reject the first $k \approx n/e$, then take the first record; $\quad P_n(k) = \frac{k}{n} \sum_{j=k}^{n-1} \frac{1}{j} \longrightarrow \frac{1}{e} \approx 0.368.$

Key facts: a record at position i is the overall best with probability $\frac{i}{n}$; the best at position $j > k$ is chosen with probability $\frac{k}{j-1}$.
Small n: $n = 3$: reject 1, $P = \frac{1}{2}$; $n = 4$: reject 1, $P = \frac{11}{24}$; $n = 10$: reject 3, $P \approx 0.399$.

- **Look, then leap:** the look phase calibrates the bar; the leap phase takes the first option that clears it.
- **Different goals, different thresholds:** if second-best is also acceptable, start accepting earlier (Additional Problem 2); with recall allowed, or with full information about values (not just ranks), you can do better than $1/e$.
- **Rank-based vs. value-based stopping:** when you know the distribution of values (e.g. dice or uniform draws), use backward induction on the values instead (see the earlier optimal-stopping sections).
- **Applications:** hiring, apartment or house hunting, selling to sequential bidders, and any one-shot accept/reject decision.

Additional Problem 1 — Small Cases by Hand: 3 and 4 Candidates

Problem: With $n = 3$ candidates, and then $n = 4$, find the best "reject k, then take the first record" rule and its success probability.

$n = 3$: list all $3! = 6$ orders

Label candidates by quality, 3 = best. With $k = 1$ (reject the first, then take the first record):

order	hired	best?
1 2 3	2 (first record after position 1)	no
1 3 2	3	yes
2 1 3	3	yes
2 3 1	3	yes
3 1 2	2 (no record; forced to take last)	no
3 2 1	1 (forced to take last)	no

Success in 3 of 6 orders: $P_3(1) = \frac{1}{2}$. Compare $k = 0$ (take the first): $\frac{1}{3}$; $k = 2$ (take the last): $\frac{1}{3}$. Formula check: $P_3(1) = \frac{1}{3}(1 + \frac{1}{2}) = \frac{1}{2}$. ✓

$n = 4$: use the formula

$$P_4(0) = \tfrac{1}{4}, \qquad P_4(1) = \tfrac{1}{4}(1+\tfrac{1}{2}+\tfrac{1}{3}) = \tfrac{11}{24} \approx 0.458, \qquad P_4(2) = \tfrac{2}{4}(\tfrac{1}{2}+\tfrac{1}{3}) = \tfrac{5}{12} \approx 0.417, \qquad P_4(3) = \tfrac{1}{4}.$$

Best: reject 1, success $\frac{11}{24}$. (Already, $\frac{k}{n} = \frac{1}{4}$ and $P \approx 0.46$ are heading towards $\frac{1}{e}$ and 0.368 as n grows.)

$n = 3$: reject 1, then take the first record: $P = \frac{1}{2}$. $n = 4$: reject 1: $P = \frac{11}{24}$.

Additional Problem 2 — Best or Second-Best Is Good Enough

Problem: Same setting with 100 candidates, but you now succeed if you hire either the best or the second-best candidate. What is the optimal strategy?

Step 1: Which candidates are worth considering?

Only a candidate who is currently 1st or 2nd among those seen can be overall top-2. At position i (out of n):

$$P(\text{overall top-2} \mid \text{currently 1st}) = \frac{i}{n} + \frac{i(n-i)}{n(n-1)}, \qquad P(\text{overall top-2} \mid \text{currently 2nd}) = \frac{i(i-1)}{n(n-1)}.$$

(The first term: the current leader is overall best with probability $\frac{i}{n}$. The second: the leader is overall second when exactly one better candidate is still to come. For a current runner-up, both the overall best and the leader must already be among the first i.)

Step 2: Backward induction gives two thresholds

Compare each stopping value with the value of continuing optimally, working backwards from the last candidate. For $n = 100$:

- reject candidates 1–34;
- from candidate 35 on, hire anyone who is **currently best**;
- from candidate 67 on, also hire anyone who is **currently second-best**.

Success probability ≈ 0.580 (as $n \to \infty$: thresholds $\approx 0.35n$ and $\approx 0.67n$, probability ≈ 0.574).

Intuition

A relaxed goal means you can start leaping a little earlier (35 vs. 38), and late in the process a current runner-up is likely to be overall second (at $i = 67$, $\frac{67 \cdot 66}{100 \cdot 99} \approx 0.45$), which beats waiting for a better candidate who may never come.

Reject the first 34; then hire the first candidate who is best so far; from candidate 67 on, also accept one who is second-best so far. P**(hire a top-2 candidate)** ≈ 0.58 **— much better than** 0.37 **for "best only".**

32 Worst-Case Search with Limited Failures (The Egg-Drop Problem)

Some searches use tests that can *fail destructively*: a dropped egg that breaks cannot be used again. With unlimited tests you would use binary search; with only one test item you must scan one by one. With a small number of items (e.g. two eggs) the best strategy lies in between, and the goal is to minimise the number of tests in the *worst case*.

Representative Problem — Two Eggs and a 100-Storey Building (Q428)

Problem: You have two identical eggs and a 100-storey building. There is some floor F (possibly 0) such that an egg dropped from floor F or below does not break, and dropped from any floor above F it breaks. A broken egg cannot be reused; an unbroken egg can. What is the smallest number of drops that *guarantees* you find F, and how should you drop the eggs?

Step 1 — what happens once one egg breaks. With only one egg left you cannot risk skipping floors: if it broke on a skipped floor you would have no egg left and would not know F. So the last egg must test floors *one by one*, upward from the highest floor known to be safe.

Step 2 — the first egg should jump in shrinking steps. Suppose the worst case is allowed to be n drops. Drop the first egg from floor n:

- if it breaks, test floors $1, 2, \ldots, n-1$ with the second egg: at most $n-1$ more drops, so n in total;
- if it survives, one drop is used up, so the next jump can only be $n-1$ floors (to floor $n+(n-1)$), so that if the egg breaks there the linear scan of the $n-2$ floors in between still fits within n drops.

Continuing, the jumps are $n,\ n-1,\ n-2,\ \ldots,\ 1$, and they cover

$$n + (n-1) + \cdots + 1 = \frac{n(n+1)}{2} \text{ floors.}$$

Step 3 — find the smallest n. We need $\frac{n(n+1)}{2} \geq 100$:

$$n = 13: \ \frac{13 \cdot 14}{2} = 91 < 100, \qquad n = 14: \ \frac{14 \cdot 15}{2} = 105 \geq 100.$$

So $n = 14$ (equivalently, $n^2 + n - 200 \geq 0$ gives $n \geq \frac{-1+\sqrt{801}}{2} \approx 13.66$).

Step 4 — the actual drop floors. Jumps of $14, 13, 12, \ldots$ give the first-egg floors

$$14,\ 27,\ 39,\ 50,\ 60,\ 69,\ 77,\ 84,\ 90,\ 95,\ 99,\ 100.$$

Check two worst cases:

- it breaks at 14: 1 drop, then floors 1–13 one by one: $1 + 13 = 14$ drops;
- it breaks at 27: 2 drops, then floors 15–26 (12 floors): $2 + 12 = 14$ drops.

Every branch costs at most 14 drops; later branches cost less.

Step 5 — why fewer than 14 is impossible. With n drops, the first drop can be no higher than floor n (otherwise a break there leaves too many floors for a one-by-one scan). After it survives,

the problem repeats with $n-1$ drops. So n drops can handle at most $n+(n-1)+\cdots+1=\frac{n(n+1)}{2}$ floors, and 13 drops handle only $91<100$.

14 drops in the worst case. Drop the first egg at floors $14, 27, 39, 50, 60, 69, 77, 84, 90, 95, 99, 100$; when it breaks, test the floors above the last safe floor one by one with the second egg.

Concept Summary and Applications

Let $f(n,k)$ be the largest number of floors that k eggs and n drops can always resolve. Drop from some floor: if the egg breaks, the floors *below* must be resolved with $k-1$ eggs and $n-1$ drops; if it survives, the floors *above* with k eggs and $n-1$ drops. Hence

$$f(n,k)=f(n-1,k-1)+1+f(n-1,k),\qquad f(n,1)=n,\quad f(0,k)=0,$$

$$\text{so}\quad f(n,k)=\binom{n}{1}+\binom{n}{2}+\cdots+\binom{n}{k},\qquad f(n,2)=\frac{n(n+1)}{2}.$$

The answer for N floors is the smallest n with $f(n,k)\geq N$, and the optimal first drop is from floor $f(n-1,k-1)+1$.

From the recursion to the closed form.

Method 1 — Pascal's rule and induction. The only fact needed is Pascal's rule, $\binom{n}{i}=\binom{n-1}{i-1}+\binom{n-1}{i}$ (choosing i items from n either includes the last item or not). Base cases: $f(n,1)=\binom{n}{1}=n$ and $f(0,k)=0$. Suppose the closed form holds for $n-1$ drops and substitute into the recursion:

$$f(n-1,k-1)+1+f(n-1,k)=\left[\tbinom{n-1}{1}+\cdots+\tbinom{n-1}{k-1}\right]+\mathbf{1}+\left[\tbinom{n-1}{1}+\cdots+\tbinom{n-1}{k}\right].$$

Write the lone 1 as $\binom{n-1}{0}$, so the first bracket becomes $\sum_{i=1}^{k}\binom{n-1}{i-1}$. Pairing the two sums term by term,

$$\sum_{i=1}^{k}\left[\tbinom{n-1}{i-1}+\tbinom{n-1}{i}\right]=\sum_{i=1}^{k}\tbinom{n}{i},$$

which is the closed form for n drops.

Method 2 — counting outcomes (why binomials appear). Each drop either breaks (B) or survives (S), and with a fixed strategy the full B/S sequence identifies the floor F. At most k breaks can occur, so n drops produce at most $\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{k}$ different sequences (those with $0,1,\ldots,k$ breaks; a strategy that stops early can be padded with imaginary survivals). For f floors there are $f+1$ possible answers $F=0,1,\ldots,f$, each needing its own sequence:

$$f+1\leq\sum_{i=0}^{k}\tbinom{n}{i}\quad\Longrightarrow\quad f\leq\sum_{i=1}^{k}\tbinom{n}{i},$$

and the recursion's strategy (first drop at floor $f(n-1,k-1)+1$) attains the bound.

Check, two eggs: $f(n,2)=\binom{n}{1}+\binom{n}{2}=n+\frac{n(n-1)}{2}=\frac{n(n+1)}{2}$; e.g. $f(14,2)=14+91=105$.

- **Balance the worst case:** each drop used up shrinks the next jump, so every branch ends at the same worst-case count.
- **Extremes:** one egg means a linear scan (N drops); unlimited eggs means binary search ($\lceil \log_2(N+1) \rceil$ drops, 7 for 100 floors). Two eggs give about $\sqrt{2N}$.
- **Applications:** testing with destructive or costly failures (stress-testing hardware, finding the breaking load), bisecting a bug when each failed build is expensive, and any search where only a few "failures" are affordable. Related worst-case search: balance-scale weighings.

Additional Problem 1 — Two Eggs, 1,000 Floors

Problem: With two eggs, how many drops guarantee finding F in a 1,000-storey building? Where is the first drop?

Solution. We need the smallest n with $\frac{n(n+1)}{2} \geq 1000$:

$$n = 44: \ \frac{44 \cdot 45}{2} = 990 < 1000, \qquad n = 45: \ \frac{45 \cdot 46}{2} = 1035 \geq 1000.$$

So 45 drops. The first drop is from floor 45, the next from $45 + 44 = 89$, then $89 + 43 = 132$, and so on, each jump one floor shorter.

45 drops, first drop at floor 45. (Roughly $\sqrt{2N} = \sqrt{2000} \approx 44.7$.)

Additional Problem 2 — Three Eggs, 100 Floors

Problem: With three eggs and a 100-storey building, what is the minimum worst-case number of drops, and where should the first egg be dropped?

Step 1 — how many floors n drops can handle. Using the formula,

$$f(n,3) = \binom{n}{1} + \binom{n}{2} + \binom{n}{3}: \qquad f(8,3) = 8{+}28{+}56 = 92 < 100, \qquad f(9,3) = 9{+}36{+}84 = 129 \geq 100.$$

So 9 drops are needed.

Step 2 — where to drop first. If the first egg breaks, the floors below must be solved with 2 eggs and 8 drops, which handle $f(8,2) = \frac{8 \cdot 9}{2} = 36$ floors. So the first drop is from floor $36 + 1 = 37$.

Step 3 — the next drops. If it survives, the next jump is $f(7,2) + 1 = 28 + 1 = 29$ floors (to 66), then $f(6,2) + 1 = 22$ floors (to 88), then $f(5,2) + 1 = 16$ floors (beyond 100, so the top floors are reached). Each time the first egg breaks, solve the gap below with the two-egg strategy.

9 drops. Drop the first egg at floors 37, 66, 88, ...; once it breaks, solve the floors below with the two-egg method.

Comparison. 1 egg: 100 drops; 2 eggs: 14; 3 eggs: 9; unlimited eggs: 7 (binary search). Each extra egg helps a lot at first, then less and less.

Part V — Betting, Martingales and Market Making

33 Martingales, Random Walks, and Gambler's Ruin

Problem: A fair ± 1 random walk starts at 0. You *win* if it reaches $+20$ and *lose* if it reaches -10, whichever happens first. What is the probability that you lose?

Background: What Is a Martingale?

A **martingale** is the mathematical model of a *fair game*: a sequence of random values $X_0, X_1, X_2, \ldots$ — picture your running bankroll — with the property that, no matter what has happened so far, your expected value after the next step equals your value right now:

$$\mathbb{E}[X_{n+1} \mid X_0, X_1, \ldots, X_n] = X_n.$$

In words: there is **no drift**. On average you neither gain nor lose, whatever the history. A running total of fair ± 1 coin flips is the archetype; a *biased* game drifts and is not a martingale.

The reason martingales are so useful is the **Optional Stopping Theorem (OST)**: if you stop a fair game at a random time T chosen by a reasonable rule (e.g. "stop when I first hit a boundary"), then the expected value *at that stopping time* still equals the start,

$$\mathbb{E}[X_T] = \mathbb{E}[X_0].$$

This is powerful because it lets us compute where a walk ends up *without* solving any recursion — we just balance the possible endings against the mean.

Solution via Optional Stopping (Step by Step)

Let X_n be the position after n steps, starting at $X_0 = 0$, each step $+1$ or -1 with probability $\frac{1}{2}$. We want $p = P(\text{hit } -10 \text{ before } +20)$.

Step 1 — X_n is a martingale. Each step adds ± 1 with equal probability, independent of the past, so

$$\mathbb{E}[X_{n+1} \mid \text{past}] = X_n + \underbrace{\left(\tfrac{1}{2}(+1) + \tfrac{1}{2}(-1)\right)}_{=0} = X_n.$$

The zero-mean step is exactly where the *fair* coin is used.

Step 2 — the mean is preserved. Taking expectations, $\mathbb{E}[X_{n+1}] = \mathbb{E}[X_n]$, so $\mathbb{E}[X_n] = \mathbb{E}[X_0] = 0$ for all n.

Step 3 — apply OST at the stopping time. Let T be the first time the walk hits -10 or $+20$. The walk is trapped in the bounded band $[-10, 20]$ and T is finite with probability 1, so OST applies and the mean survives the random stopping:

$$\mathbb{E}[X_T] = \mathbb{E}[X_0] = 0.$$

Step 4 — evaluate the two endings. At time T the walk sits on exactly one barrier: $X_T = -10$ (you lose, probability p) or $X_T = +20$ (you win, probability $1-p$), so

$$\mathbb{E}[X_T] = (-10)\,p + (20)(1-p).$$

Step 5 — solve. Set this equal to 0:

$$(-10)p + 20(1-p) = 0 \implies 20 - 30p = 0 \implies p = \tfrac{20}{30} = \tfrac{2}{3}.$$

$$P(\textbf{lose, i.e. hit } -10 \textbf{ first}) = \frac{2}{3}, \qquad P(\textbf{win, hit } +20 \textbf{ first}) = \frac{1}{3}.$$

(Equivalently, shift to $Y_n = X_n + 10$: the barriers become 0 and 30 and the start is 10, the textbook symmetric *gambler's ruin* on $\{0, 1, \ldots, 30\}$.)

Gambler's Ruin — General Formula

For a symmetric walk with an *absorbing* barrier a distance a below and b above the start, each barrier's hit probability is the distance to the **opposite** barrier over the total width:

$$P(\text{hit lower first}) = \frac{b}{a+b}, \qquad P(\text{hit upper first}) = \frac{a}{a+b}.$$

Here $a = 10$, $b = 20$, so $P(\text{hit } -10) = \frac{20}{30} = \frac{2}{3}$. Intuitively $E[X_T] = 0$ forces the outcomes to balance around the start: the far barrier $(+20)$ has more leverage, so the near barrier (-10) must carry proportionally more probability.

Biased Walk: the $(q/p)^{X_n}$ Martingale

If steps are $+1$ with probability $p \neq \frac{1}{2}$ and -1 with probability $q = 1-p$, the walk **drifts**: the mean step is $p(+1) + q(-1) = p - q \neq 0$, so X_n itself is *not* a martingale and the balancing trick fails on it directly. The remedy is to find a *function* of the walk that *is* a martingale. The right choice is

$$M_n = (q/p)^{X_n}.$$

Verify the martingale property from a generic position $X_n = x$ (a step takes $x \to x+1$ w.p. p or $x \to x-1$ w.p. q):

$$\mathbb{E}[M_{n+1} \mid X_n = x] = p\,(q/p)^{x+1} + q\,(q/p)^{x-1} = (q/p)^x \left[p \cdot \tfrac{q}{p} + q \cdot \tfrac{p}{q}\right] = (q/p)^x\,[\,q + p\,] = (q/p)^x = M_n.$$

The bracket collapses to $p + q = 1$ — *that* is why the base q/p is the magic value: it makes the up- and down-contributions cancel, so M_n has no drift even though X_n does.

Numerical illustration. Take $p = \frac{2}{3}$, $q = \frac{1}{3}$, so $q/p = \frac{1}{2}$ and $M_n = \left(\frac{1}{2}\right)^{X_n}$. From any level x,

$$\mathbb{E}[M_{n+1} \mid X_n = x] = \tfrac{2}{3}\left(\tfrac{1}{2}\right)^{x+1} + \tfrac{1}{3}\left(\tfrac{1}{2}\right)^{x-1} = \left(\tfrac{1}{2}\right)^x \left[\tfrac{2}{3} \cdot \tfrac{1}{2} + \tfrac{1}{3} \cdot 2\right] = \left(\tfrac{1}{2}\right)^x \left[\tfrac{1}{3} + \tfrac{2}{3}\right] = \left(\tfrac{1}{2}\right)^x = M_n.$$

Even though the walk drifts *upward* (heads twice as likely as tails), the quantity $\left(\frac{1}{2}\right)^{X_n}$ keeps a constant expectation — a martingale, confirmed numerically.

Gambler's ruin from it. Applying OST to M_n between barriers 0 and N gives $\mathbb{E}[M_T] = M_0 = (q/p)^k$ when starting at k; splitting into the two endings ($M = 1$ at level 0, $M = (q/p)^N$ at level N) and solving,

$$P(\text{ruin from } k) = P(\text{hit 0 before } N) = \frac{(q/p)^k - (q/p)^N}{1 - (q/p)^N}.$$

For instance with $p = \frac{2}{3}$ ($q/p = \frac{1}{2}$), start $k = 1$, barriers 0 and $N = 3$:

$$P(\text{ruin}) = \frac{(\frac{1}{2})^1 - (\frac{1}{2})^3}{1 - (\frac{1}{2})^3} = \frac{\frac{1}{2} - \frac{1}{8}}{1 - \frac{1}{8}} = \frac{3/8}{7/8} = \frac{3}{7} \approx 0.43,$$

comfortably below $\frac{1}{2}$ — the upward drift makes reaching +3 before 0 the more likely outcome, exactly as it should.

Additional Problem — 52-Card Walk *With* Replacement

Problem: Draw from a shuffled 52-card deck (26 black, 26 red): black = +1 point, red = −1 point. After each draw you **replace and reshuffle**, so you may draw indefinitely. What strategy maximises your total?

The words "with replacement" are the whole problem. Because every draw is from the full deck, the draws are i.i.d. ± 1 with probability $\frac{1}{2}$ each, and the running score $S_n = X_1 + \cdots + X_n$ is a **simple symmetric random walk** — a martingale. Two facts now pull in opposite directions.

Fact 1 (fairness). For any stopping rule τ with finite expected duration $\mathbb{E}[\tau] < \infty$, optional stopping gives $\mathbb{E}[S_\tau] = 0$: no time-bounded strategy earns a positive *expected* score.

Fact 2 (recurrence). A symmetric walk on the integers visits *every* integer with probability 1. So *any* target level is reached almost surely — not only +1, but +2, +100, $+100^9$, and even the astronomically distant negative level -100^{10}: each of these is hit eventually with probability 1 (the walk wanders arbitrarily far in *both* directions).

Optimal strategy. Pick a positive target and simply wait for it. The cleanest rule is

$$\boxed{\text{quit the first moment you are ahead } (S_n = +1),}$$

which finishes with a guaranteed +1. Since $+2, +100, +100^9, \ldots$ are all reached a.s. as well, **there is no finite maximum**: the supremum of what you can guarantee is $+\infty$, and no single strategy attains it.

The catch. The free lunch is paid for in *time*. The expected number of draws to reach any positive level is infinite, $\mathbb{E}[\tau_{+1}] = \infty$. That is precisely why guaranteeing $+T$ almost surely does *not* contradict Fact 1: the winning rule's stopping time has infinite mean, so the theorem "$\mathbb{E}[S_\tau] = 0$" (which *requires* $\mathbb{E}[\tau] < \infty$) does not apply to it.

With replacement: stop the first time you are ahead — you reach $+1$ (or any target you name) with probability 1, so you can guarantee an arbitrarily large gain, but only at the cost of an unbounded expected number of draws. Every strategy that must finish in finite *expected* time has value $0.

With vs. Without Replacement — Why It Matters

- **With replacement** (this problem): the total is a *free* random walk that can wander arbitrarily far in either direction. A martingale with no endpoint constraint has EV $= 0$ — stopping cannot manufacture value.
- **Without replacement** (finite deck, Section 29): the walk is *pinned* to end at 0 (26 black − 26 red), a **martingale bridge**. That pinning — knowing the leftover composition — is exactly what creates a real, finite "option value" ($\approx$ \$2.62 for a full deck). Removing replacement turns a worthless game into a solvable one.

Concept Summary and Applications

- **Recipe:** spot a martingale (fair walk X_n; or $(q/p)^{X_n}$ / $X_n^2 - n$ for biased or variance problems), then apply optional stopping $E[X_T] = X_0$ and read off the barrier probabilities.
- **Applications:** risk of ruin and bankroll survival, first-passage and hitting probabilities, sequential-testing boundaries, and pricing by no-arbitrage (a fairly priced game is a martingale).

34 Market Making and Optimal Bidding

What "Making a Market" Means

To **make a market** on an uncertain quantity X is to quote two prices at once:

- a **bid** — the price at which *you will buy* X;
- an **ask** (or offer) — the price at which *you will sell* X;

together with a **size** — how many units you will trade at those prices. It is written "bid @ ask, size," e.g. "6.5 at 7.5, 10 up" means "I buy up to 10 at 6.5 and sell up to 10 at 7.5." The difference ask − bid is the **spread**. The counterparty chooses whether to buy from you or sell to you (or neither) *after* hearing your quote.

Representative Problem — Market on the Sum or Product of Two Dice (Q149)

Problem: Two fair six-sided dice will be rolled. Make a market — bid, ask, and size — on (a) the **sum** of the two dice, and (b) the **product** of the two dice. Justify your prices and quantities.

The recipe. (1) Find the *fair value* $\mathbb{E}[X]$ — the centre of the market. (2) Measure the *risk* with the variance / standard deviation — this drives the spread and the size. (3) Quote a spread around fair value and a size that reflects that risk.

Part (a) — the sum $S = D_1 + D_2$.

Fair value. One die averages $\mathbb{E}[D] = \frac{1+2+\cdots+6}{6} = 3.5$, and expectation adds, so

$$\mathbb{E}[S] = \mathbb{E}[D_1] + \mathbb{E}[D_2] = 3.5 + 3.5 = 7.$$

Distribution. The 36 equally likely outcomes give

S	2	3	4	5	6	7	8	9	10	11	12
#ways	1	2	3	4	5	6	5	4	3	2	1

— symmetric around 7 and fairly concentrated.

Risk. For one die, $\mathbb{E}[D^2] = \frac{1^2+2^2+\cdots+6^2}{6} = \frac{91}{6}$, so

$$\mathrm{Var}(D) = \frac{91}{6} - 3.5^2 = \frac{35}{12}.$$

Variances of independent variables add, so

$$\mathrm{Var}(S) = 2 \cdot \frac{35}{12} = \frac{35}{6} \approx 5.83, \qquad \mathrm{SD}(S) = \sqrt{35/6} \approx 2.42.$$

Market. Centre on 7 with a modest spread and decent size:

$$\boxed{\text{6.5 bid, 7.5 ask, size 10.}}$$

Buying at 6.5 or selling at 7.5 earns 0.5 of *edge* per unit on average (the distance from fair value). A tighter 6.8–7.2 would be more competitive; a wider quote is justified if the counterparty might be informed or you already hold inventory.

Part (b) — the product $P = D_1 D_2$.

Fair value. For independent variables, the expectation of a product is the product of expectations:

$$\mathbb{E}[P] = \mathbb{E}[D_1]\,\mathbb{E}[D_2] = 3.5 \times 3.5 = 12.25.$$

Distribution. The product takes 18 different values:

P	1	2	3	4	5	6	8	9	10
#ways	1	2	2	3	2	4	2	1	2

P	12	15	16	18	20	24	25	30	36
#ways	4	2	1	2	2	2	1	2	1

It is **right-skewed**: most outcomes are small, but a long tail reaches 36. In fact $P \le 11$ about 53% of the time, so the *median* sits below the mean 12.25.

Risk. Independence also gives $\mathbb{E}[P^2] = \mathbb{E}[D_1^2]\,\mathbb{E}[D_2^2] = \left(\frac{91}{6}\right)^2 = \frac{8281}{36} \approx 230.03$, so

$$\mathrm{Var}(P) = 230.03 - 12.25^2 = 230.03 - 150.06 \approx 79.97, \qquad \mathrm{SD}(P) \approx 8.94.$$

That is almost *four times* the standard deviation of the sum.

Market. Centre on 12.25 but go wider and smaller:

$$\boxed{11 \text{ bid},\ 13.5 \text{ ask},\ \text{size } 5.}$$

(A rounder 10.5–14 for size 10 is also defensible.)

Sum: fair value 7, SD ≈ 2.4 — quote about 6.5 at 7.5, size 10. Product: fair value 12.25, SD ≈ 8.9 and right-skewed — quote wider and smaller, about 11 at 13.5, size 5.

Concept Summary — Expected Value, Variance, Spread, and Size

	Fair value	SD	Quote	Size
Sum of two dice	7	2.42	6.5 – 7.5	10
Product of two dice	12.25	8.94	11 – 13.5	5

- **Expected value sets the centre.** If you bought and sold many times at a price p, your average profit per unit would be $\mathbb{E}[X] - p$ (when buying) or $p - \mathbb{E}[X]$ (when selling). Centring the market on $\mathbb{E}[X]$ makes both sides profitable by half the spread.
- **Variance sets the spread.** The spread is your compensation for risk and for being wrong. The more X can move (SD), the more a single trade can hurt, and the more room the other side has to know something you don't — so quote wider. A useful habit is to think of the spread as a fraction of the SD.
- **Variance also sets the size.** Risk grows with size × SD. To keep the risk of a fill roughly constant, size should go *down* as SD goes up — big size on the tame sum, small size on the volatile product.
- **Adverse selection and skew.** The counterparty picks the side *after* seeing your quote, so they only trade when your price is wrong in their favour. Quote wider if they may be informed; with a skewed payoff (like the product), remember the median differs from the mean.

- **Inventory.** If you are already long, shade both prices down (to encourage buyers and discourage sellers); if short, shade them up.

Additional Problem — The Optional Bet (Q140)

Problem: Two players, A and B, each roll a fair die privately. Whoever has the higher number wins a unit bet (a tie is a push), but nobody is forced to bet. A looks at their roll and chooses whether to offer the bet; if A declines, the round ends and nobody pays anything. B knows whether A offered, looks at their own roll, and then accepts or declines; if B declines, again nothing is paid. Which player has the higher expected payout?

Step 1 — the offer is information. A only offers when A's roll is good. So when B sees an offer, B should update: "A's number is probably high." B then accepts only if, *given that A offered*, B is at least as likely to win as to lose.

Step 2 — try "A offers with 5 or 6." If an offer means $a \in \{5, 6\}$ (each equally likely):

- B with $b = 6$ beats 5 and ties 6 — positive for B, so B accepts;
- B with $b = 5$ ties 5 and loses to 6 — negative, so B declines;
- smaller b are even worse, so they decline.

So B only ever accepts holding a 6. But then A holding a 5 *loses* every accepted bet — A would rather not offer with a 5. The strategy falls apart.

Step 3 — the unravelling argument (any strategy). Suppose A offers with some set of values, and let $a_{\min}$ be the smallest of them, with $a_{\min} < 6$.

- B never accepts with $b < a_{\min}$: such a b loses to *every* value A offers with.
- B always accepts with $b = 6$: a 6 never loses and beats $a_{\min}$.

So whenever the bet is accepted, B holds $b \geq a_{\min}$: A's $a_{\min}$ never wins and sometimes loses (to the 6). A would do better by *not* offering with $a_{\min}$. Remove it and repeat — the offering set shrinks until only $a = 6$ is left. But if A offers only with 6, every B declines (B can at best tie). So no bet is ever accepted.

Neither player has an advantage: in equilibrium no bet is ever accepted, so both expected payouts are 0. (B's information makes B impossible to exploit, but it does not give B a positive payout — there is nothing to win when bets are declined.)

The no-trade / adverse-selection theorem. This is a textbook case of the **no-trade theorem** (Milgrom–Stokey): when two rational players share the same beliefs about the world and a bet is purely speculative and zero-sum, *no* bet can be profitable for the person proposing it once the other side is allowed to read the proposal as information. The engine is **adverse selection**: the counterparty accepts *only* in the cases that are good for them — which are exactly the cases bad for you. Every bet B takes is one A expects to lose, and every bet A would win, B declines. Since each player can always guarantee 0 by declining, the result is 0 for both. In market-making language: *if your quote gets hit, ask why* — the willingness to trade is itself a signal.

Caveat — it depends on the folding rule. *The* 0–0 *result relies on the rule that* declining is free*: a declined bet simply ends the round with no money moving. Change that and the answer changes. In a* ***poker-style*** *game — say both players first put an ante into a pot, and a player who* folds *(declines) gives up that pot — declining now* costs *something. Then A can profit by* bluffing *(offering with a weak roll to make B fold and hand over the pot), B must sometimes call weak-looking offers to prevent this, and the equilibrium involves mixed betting/calling ranges with a non-zero edge for one side. So "nobody wins" is a property of* free, voidable *bets, not of betting games in general.*

Applications

- **Trading interviews:** quote a two-sided market on any random quantity — centre on the mean, widen and shrink size with the variance and skew.
- **Real markets:** bid–ask spreads on stocks and options widen with volatility and with the risk of informed traders (adverse selection).
- **Negotiation and betting:** a counterparty eager to take your bet or hit your quote is information — update before you trade.

35 The Kelly Criterion: How Much to Bet

Knowing that a bet is favourable tells you *whether* to bet; it does not tell you *how much*. Bet too little and you waste your edge; bet too much and a few losses can wipe you out. The **Kelly criterion** answers "how much" for a bettor who plays many times and reinvests the winnings.

Representative Problem — Betting on a Die (Q234)

Problem: What is the optimal way to bet on the outcomes of a six-sided die roll?

The question is deliberately open: "optimal" depends on what the bets *pay* and on what you are trying to maximise. A good answer sets up the payoff, finds when a bet is worth taking at all, and then sizes it.

Step 1 — the model. Each face $1, \dots, 6$ has probability $\frac{1}{6}$. You can bet on an event A (for example "the roll is a 4" or "the roll is even") with probability $p = P(A)$. Staking 1 unit, you receive r units of *net profit* ("r to 1") if A happens and lose your 1 unit if it does not. The expected profit per unit staked is

$$\mathbb{E}[\text{profit}] = p\,r - (1-p),$$

so the bet is favourable exactly when

$$p\,r - (1-p) > 0 \quad \Longleftrightarrow \quad r > \frac{1-p}{p}.$$

(Equivalently, if $b = r + 1$ is the total return including your stake, the bet is favourable when $p\,b > 1$.)

Step 2 — fair odds. For a single face, $p = \frac{1}{6}$, so the *fair* payout is $r = \frac{1-p}{p} = 5$, i.e. $5:1$. At exactly $5:1$ the expected profit is

$$\tfrac{1}{6} \cdot 5 - \tfrac{5}{6} \cdot 1 = 0.$$

The same holds for any correctly priced bet on a fair die. Hence:

- at **fair odds**, every betting strategy has expected profit 0;
- at **worse than fair odds** (e.g. a casino paying $4:1$ on a face), every strategy loses on average — the optimal bet is *no bet*;
- only if some bet pays **better than fair odds** is there anything to gain.

Step 3 — two different meanings of "optimal."

- **One roll, maximise expected money.** Bet nothing if no bet has positive expected value; otherwise put *everything* on the bet with the highest expected return. (This is reckless over many rounds, as shown below.)
- **Many rolls, maximise long-run growth.** Bet a *fraction* of your current wealth each time — the Kelly fraction. For a bet with win probability p and net odds r,

$$f^* = \frac{p\,r - (1-p)}{r} \qquad \text{(bet nothing if this is } \le 0\text{)}.$$

Step 4 — apply it to one face. With $p = \frac{1}{6}$,

$$f^* = \frac{\frac{r}{6} - \frac{5}{6}}{r} = \frac{r-5}{6r}.$$

So: if $r < 5$, do not bet; if $r = 5$, the edge is zero, so do not bet; if $r > 5$, bet the fraction $\frac{r-5}{6r}$ of your current wealth on that face. For example, if a face paid $7:1$, you would stake $\frac{7-5}{42} = \frac{1}{21} \approx 4.8\%$ of your wealth each roll.

There is no single "optimal" bet until the payouts are given. Bet only when the offered odds beat the fair odds implied by $\frac{1}{6}$ (for one face: net payout above $5:1$); at fair or worse odds, don't bet. When a bet is favourable and you play repeatedly, stake the Kelly fraction $f^* = \frac{pr-(1-p)}{r}$ of your wealth — $\frac{r-5}{6r}$ for a single face.

Kelly Criterion — The Setup

You start with wealth W_0 and face the same favourable bet over and over:

- it wins with probability p and loses with probability $q = 1 - p$;
- it pays net odds $b:1$ — stake 1, win b more, or lose the 1;
- each round you bet a fixed *fraction* f of your *current* wealth.

After a win your wealth is multiplied by $(1 + bf)$; after a loss, by $(1 - f)$. After n rounds with k wins,

$$W_n = W_0\,(1+bf)^k\,(1-f)^{n-k}.$$

Wealth *multiplies*; it does not add. That single fact drives everything below.

Why Maximise the Logarithm of Wealth

Maximising the average wealth is a trap. Take $p = 0.6$ at even money ($b = 1$). Betting *everything* ($f = 1$) every round maximises expected wealth: it grows like $E[W_n] = W_0\,(1.2)^n$. But you survive n rounds only if you win all of them, with probability $0.6^n \to 0$. The average is huge only because of an astronomically rare lucky path; you are almost certainly broke.

What matters is the growth rate of a typical path. Take logarithms — they turn products into sums:

$$\ln \frac{W_n}{W_0} = k \ln(1+bf) + (n-k)\ln(1-f).$$

Divide by n. By the law of large numbers, $k/n \to p$, so

$$\frac{1}{n} \ln \frac{W_n}{W_0} \;\longrightarrow\; G(f) = p\ln(1+bf) + q\ln(1-f) \qquad \text{(with probability 1)}.$$

So $W_n \approx W_0\, e^{nG(f)}$: $G(f)$ is the **long-run growth rate per bet**. Maximising it is the same as maximising the expected *log* of wealth, $\mathbb{E}[\ln W_n] = \ln W_0 + n\,G(f)$. This is why Kelly uses **log utility**:

- it maximises the compound growth rate of a typical path — over enough rounds, the Kelly bettor ends up ahead of any other fixed-fraction strategy with probability approaching 1;
- $\ln W \to -\infty$ as $W \to 0$, so it never risks total ruin (it keeps $f < 1$ whenever a loss is possible);
- it has *diminishing marginal utility*: an extra dollar matters less when you are rich, which matches how most people and firms feel about money;
- it is scale-free: the right bet is a *fraction* of wealth, whatever the wealth.

Deriving the Kelly Fraction

Maximise $G(f) = p\ln(1+bf) + q\ln(1-f)$. Differentiate:

$$G'(f) = \frac{p\,b}{1+bf} - \frac{q}{1-f}.$$

Set $G'(f) = 0$ and cross-multiply:

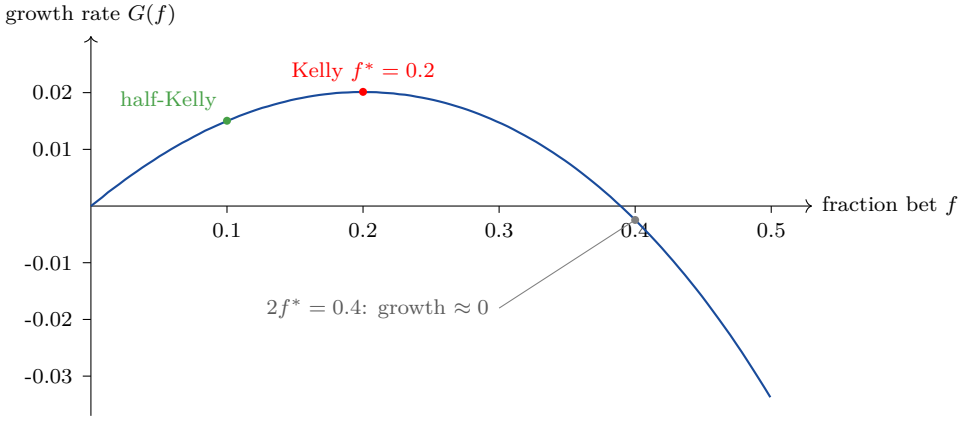
$$p\,b\,(1-f) = q\,(1+bf) \implies pb - pbf = q + qbf \implies pb - q = bf\,(p+q) = bf.$$

$$f^* = \frac{p\,b - q}{b} = p - \frac{q}{b} \qquad \left(= \frac{\text{expected profit per unit}}{\text{profit if you win}}\right).$$

For even money ($b = 1$): $f^* = p - q = 2p - 1$, the size of your edge.

The second derivative $G''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{q}{(1-f)^2} < 0$, so G is concave and f^* is the maximum.

Example. $p = 0.6$, even money: $f^* = 0.6 - 0.4 = 0.20$. Betting 20% each round gives growth $G(0.2) = 0.6\ln 1.2 + 0.4\ln 0.8 \approx 0.0201$ per bet — about 2% compound growth per round.



Growth rate per bet for $p = 0.6$ at even money. It peaks at $f^* = 0.2$; half-Kelly (0.1) keeps 75% of the peak growth; betting $2f^* = 0.4$ gives roughly zero growth, and anything more makes your wealth shrink over time.

Intuition and Properties

- **Bet your edge over your odds.** $f^* = \frac{pb-q}{b}$: the numerator is the expected profit per unit bet (your edge); dividing by b scales it down when the payout is long (long odds $\Rightarrow$ more variance per unit bet).
- **No edge, no bet.** $f^* > 0$ exactly when $pb > q$, i.e. the bet has positive expected value. With no edge, Kelly says bet nothing, however tempting the odds.
- **Never bet it all.** $f^* < 1$ whenever losing is possible. On a fair coin ($p = \frac{1}{2}$), even at huge odds $f^* = \frac{1}{2} - \frac{1}{2b} < \frac{1}{2}$: you never stake more than half, because you still lose half the time.
- **Proportional betting.** After wins you bet more dollars, after losses fewer — the bet size automatically shrinks as you lose, which is what prevents ruin.
- **Overbetting is worse than underbetting.** Near the peak, G is roughly a parabola: betting $c\,f^*$ gives about $(2c - c^2)$ of the maximum growth. So half Kelly keeps 75% of the growth, while double Kelly ($c = 2$) gives about *zero* growth, and more than that gives negative growth even though every bet is favourable.
- **Continuous version.** For an investment with expected excess return μ and volatility σ, the Kelly fraction of wealth to hold is $f^* = \mu/\sigma^2$.

Risks, Limitations, and Why Fractional Kelly

Full Kelly maximises long-run growth, but the ride is rough and the result depends on inputs you rarely know exactly.

- **Large drawdowns.** Even at full Kelly, deep losing streaks are normal. In the continuous-time approximation, the chance that your wealth *ever* falls to half of what you started with is about 50%; to a third, about 33%. Many people and funds cannot tolerate that.
- **You don't know p exactly.** Kelly assumes the true probabilities. If you overestimate your edge, you overbet — and overbetting is costly (at twice the true f^*, growth is about zero). Estimation error makes the *effective* bet too big.
- **Long-run guarantee, short-run noise.** Kelly's optimality holds "in the long run"; over a short horizon, a Kelly bettor can easily do badly.
- **Model assumptions.** Independent, repeatable bets; infinitely divisible stakes; no transaction costs; known, stable odds; no fat tails. Real markets break some of these.
- **Log utility may not be yours.** Someone more risk-averse than log utility should bet less.

Fractional Kelly. Bet $f = c\,f^*$ with $0 < c < 1$ (commonly $c = \frac{1}{2}$).

- Growth: about $(2c - c^2)$ of the maximum — 75% at half Kelly.
- Volatility: scales with c — half Kelly has half the standard deviation (a quarter of the variance).
- Drawdowns: the chance of ever falling to a fraction x of your starting wealth drops from about x to about $x^{\,2/c-1}$ — for half Kelly, the chance of ever being halved falls from $\frac{1}{2}$ to $\frac{1}{8}$.
- Safety margin: it protects you if your estimate of p is too optimistic.

Giving up a quarter of the growth to halve the volatility and sharply cut the risk of big drawdowns is why practitioners usually bet half Kelly or less.

Concept Summary and Applications

- **First decide whether to bet:** only if the odds beat the fair odds ($pb > q$, or $r > \frac{1-p}{p}$).
- **Then decide how much:** maximise the growth rate $G(f) = p\ln(1+bf) + q\ln(1-f)$, giving $f^* = p - \frac{q}{b}$ ($= 2p - 1$ at even money).
- **Log utility** targets the growth of a typical wealth path, not the average (which a rare lucky path dominates) — and never risks ruin.
- **In practice, bet a fraction of Kelly:** half Kelly keeps 75% of the growth with far smaller drawdowns and protects against overestimating your edge.
- **Applications:** position sizing in trading and portfolio management, sports and casino betting with an edge, capital allocation across strategies, and interview questions of the form "you have an edge — how much do you bet?"

36 Martingale and Guaranteed-Outcome Betting

Suppose you can only make simple even-money bets on individual events (single games, single coin flips), but you want a payoff that depends on a *combination* of events (who wins a whole series). You can build that payoff exactly by choosing each bet's size based on the current situation — this is called *dynamic replication*. The tool is a value table filled in by backward induction: the amount you should be holding in each state is the average of the amounts in the two states it can lead to. It is the discrete version of delta hedging an option.

Representative Problem — Betting on a Seven-Game Series (Q15)

Problem: Two teams play a best-of-seven series (first to 4 wins). You may bet any amount at even money on each individual game. Devise a betting strategy so that if team 1 wins the series you end up exactly +\$1000, and if team 1 loses you end up exactly −\$1000, whatever the sequence of game results.

Step 1: Describe the state and the target

Let (a, b) mean team 1 still needs a wins and team 2 still needs b wins; the series starts at $(4, 4)$. Let $f(a, b)$ be the bankroll (cumulative profit) you must hold in state (a, b) so that the strategy can still finish on target. At the end:

$$f(0, b) = +1000 \ \text{(team 1 has clinched)}, \qquad f(a, 0) = -1000 \ \text{(team 2 has clinched)}.$$

Step 2: One even-money bet can reach both next states only if f is their average

In state (a, b) you hold $f(a, b)$ and bet x on team 1. If team 1 wins the game you have $f(a, b) + x$ and need $f(a-1, b)$; if it loses you have $f(a, b) - x$ and need $f(a, b-1)$. Adding the two equations gives

$$f(a, b) = \tfrac{1}{2}\big[f(a-1, b) + f(a, b-1)\big], \qquad x = \text{bet}(a, b) = \tfrac{1}{2}\big[f(a-1, b) - f(a, b-1)\big].$$

So: **each value is the average of its two successors**, and **the bet is half their difference**. A bankroll process in which each value is the average of the next possible values is called a *martingale*.

Step 3: Fill the table backwards

Start from the boundary values and average. For example $f(1,1) = \frac{1}{2}(1000 - 1000) = 0$, $f(1,2) = \frac{1}{2}(f(0,2) + f(1,1)) = \frac{1}{2}(1000 + 0) = 500$, $f(1,3) = \frac{1}{2}(1000 + 500) = 750$, and so on:

$f(a,b)$	$b=1$	$b=2$	$b=3$	$b=4$
$a=1$	0	500	750	875
$a=2$	−500	0	375	625
$a=3$	−750	−375	0	312.5
$a=4$	−875	−625	−312.5	**0**

$f(4, 4) = 0$: **you need no money up front**.

Step 4: The bets

Using $\text{bet}(a, b) = \frac{1}{2}[f(a-1, b) - f(a, b-1)]$ (amount staked on team 1):

bet ($)	$b=4$	$b=3$	$b=2$	$b=1$
$a=4$	**312.5**	312.5	250	125
$a=3$	312.5	375	375	250
$a=2$	250	375	500	500
$a=1$	125	250	500	1000

Game 1: bet \$312.50 on team 1. The stakes grow as the series tightens, up to \$1000 on a deciding game 7 (state $(1,1)$).

Step 5: Check one path

A 4–0 sweep by team 1 passes $(4,4) \to (3,4) \to (2,4) \to (1,4) \to (0,4)$ with bets $312.5+312.5+250+125 = \$1000$. ✓ A path where team 1 loses game 1 and then wins four: -312.5 (at $(4,4)$), then at $(4,3)$: $+312.5$, $(3,3)$: $+375$, $(2,3)$: $+375$, $(1,3)$: $+250$; total $-312.5+312.5+375+375+250 = 1000$. ✓ All 70 possible series paths end exactly at $\pm\$1000$.

Step 6: Meaning of the numbers

With fair games, $f(a,b) = 1000\,(2p(a,b)-1)$, where $p(a,b)$ is team 1's chance of winning the series from (a,b): the bankroll equals the expected final payoff. The bet equals 1000 times the probability that this game decides the series:

$$\text{bet}(a,b) = 1000 \cdot \frac{\binom{a+b-2}{a-1}}{2^{a+b-2}},$$

because the current game matters only if the other $a+b-2$ remaining games would split $a-1$ to team 1 and $b-1$ to team 2. Check: $(4,4)$: $1000\binom{6}{3}/2^6 = 312.5$; $(1,1)$: 1000.

Step 7: The hedge does not depend on who is more likely to win

The identities $f(a,b) \pm \text{bet} = \text{next value}$ are pure algebra; they hold whatever the true chance of each game. The 50/50 assumption only gives the table a probability interpretation. You are replicating, not forecasting.

Hold $f(a,b)$ defined by $f(0,\cdot) = 1000$, $f(\cdot,0) = -1000$, $f(a,b) = \frac{1}{2}[f(a-1,b)+f(a,b-1)]$, and before each game bet $\frac{1}{2}[f(a-1,b) - f(a,b-1)] = 1000\binom{a+b-2}{a-1}/2^{a+b-2}$ on team 1. Start with \$0 and bet \$312.50 on game 1; every path ends at exactly $\pm\$1000$. (Same question: Q111.)

Concept Summary and Applications

Replicating a target payoff with even-money bets:

$$f(\text{state}) = \tfrac{1}{2}\big[f(\text{up}) + f(\text{down})\big], \qquad \text{bet} = \tfrac{1}{2}\big[f(\text{up}) - f(\text{down})\big],$$

filled in backwards from the target payoffs. The starting value $f(\text{start})$ is the fair price of the target payoff (here 0).
Martingale: a fair game's bankroll; its expected future value equals its current value. You cannot create positive expected value from fair bets by choosing stakes (optional stopping) — unless time or losses are unbounded, or someone gives you an extra asymmetric right (a token, an option).

- **Bet = sensitivity:** the stake equals how much the target payoff changes with this game's result — exactly the "delta" of an option.
- **Path independence:** the final result depends only on who wins the series, not the order of games.
- **Related paradox (Q629):** rebuying after every bust guarantees a profit eventually, but the expected number of buy-ins is infinite — the fair-game rule fails only because nothing is bounded.
- **Applications:** delta hedging and option replication, binomial option pricing, matched betting, and hedging a parlay with single-event bets.

Additional Problem — What Is the Insurance Token Worth? (Q42)

Problem: You may buy a token that lets you propose a bet. If you win the bet, you keep the token for future use. If you lose, you may use the token to recover half of your stake (the token is then used up). Given the chips each side holds, how much is the token worth? Assume each bet is a fair coin flip at even money, you have A chips and your opponent has B, and no bet can exceed what either side can pay.

Step 1: The edge on a single bet

Without the token a fair even-money bet of size x is worth 0. With the token:

$$\tfrac{1}{2}(+x) + \tfrac{1}{2}(-x + \tfrac{x}{2}) = \frac{x}{4}.$$

The token turns each bet into one with expected gain $\frac{x}{4}$ — so you want to bet as much as possible, and as often as possible. Note the token survives only if you win.

Step 2: Single use (the token is lost or the game ends after one bet)

The largest possible bet is $m = \min(A, B)$, so the token is worth

$$\frac{m}{4} = \frac{\min(A, B)}{4},$$

or $\frac{1-p}{2}\,m$ if your win probability is p (the token only pays, $\frac{m}{2}$, when you lose). This also covers $A \geq B$: winning an all-in bet bankrupts the opponent, so there is no second bet.

Step 3: Reusable token when you are the smaller stack ($A < B$)

Bet all-in each time. A win doubles your stack and keeps the token, so the stakes run $A, 2A, 4A, \ldots$ while you keep winning, until the opponent cannot cover a full double. The k-th bet happens only if you won the previous $k-1$ (probability $2^{-(k-1)}$) and contributes $\frac{\text{stake}}{4}$:

$$\text{value} = \frac{A}{4} + \frac{1}{2} \cdot \frac{2A}{4} + \frac{1}{4} \cdot \frac{4A}{4} + \cdots,$$

so **each full doubling round adds** $\frac{A}{4}$, and the last round adds $\frac{1}{4} \times$ (remaining opponent chips) $\times$ (probability of reaching it). Backward induction over all possible bet sizes confirms that betting the maximum each time is optimal. Examples:

chips (A, B)	bets while winning	token value
$(10, 10)$	10	2.5
$(10, 20)$	10, then 10 (opponent has only 10 left)	$2.5 + \frac{1}{2} \cdot 2.5 = 3.75$
$(10, 30)$	10, 20	$2.5 + 2.5 = 5$
$(10, 70)$	10, 20, 40	$2.5 + 2.5 + 2.5 = 7.5$
$(30, 10)$	10	2.5

Roughly, the token is worth $\frac{A}{4} \log_2 \frac{A+B}{A}$ when you are the smaller stack: every doubling of your stack before the opponent runs out adds $\frac{A}{4}$.

Step 4: Limits

- Against an opponent with unlimited chips and no bet limit, the number of doubling rounds is unbounded and the token's value is **infinite** (the doubling paradox again).
- With a fixed maximum bet c and deep pockets on both sides, the value G satisfies $G = \frac{c}{4} + \frac{1}{2}G$, so $G = \frac{c}{2}$.

Each fair bet with the token is worth $\frac{x}{4}$ to you, so bet the maximum. One use: $\frac{1}{4}\min(A, B)$. Reusable, as the smaller stack: $\frac{A}{4}$ for each doubling of your stack the opponent can cover (e.g. $(10, 70) \to 7.5$); as the larger stack: $\frac{B}{4}$. Capped bets of size c with deep pockets: $\frac{c}{2}$; uncapped against unlimited chips: infinite.

37 Single-Variable Optimisation: Probability vs. Payoff Trade-off

Many interview games make you choose one number — a bid, a guess, a stake — where a bigger choice makes the prize *larger* but *less likely* (or more likely but smaller). The expected value is a product

$$E(x) = P(\text{win} \mid x) \times \text{payoff}(x),$$

and the best choice balances the two factors. Neither the most likely outcome nor the biggest prize is optimal on its own.

Representative Problem — Guess the Sum of a d6 and a d10 (Q327)

Problem: You roll a standard 6-sided die (d6) and a 10-sided die (d10). Before seeing the result, you name a number b. If the sum of the two dice equals b, you win $\$b$; otherwise you win nothing. Which b maximises your expected winnings?

Step 1 — write the expected value. The payoff is all-or-nothing, so

$$E(b) = b \cdot P(\text{sum} = b).$$

This is the key: you are *not* looking for the average sum, and not simply for the most likely sum, but for the largest product of payoff and probability.

Step 2 — the distribution of the sum. The d6 shows $a \in \{1, \dots, 6\}$ and the d10 shows $c \in \{1, \dots, 10\}$, independently, so there are $6 \times 10 = 60$ equally likely pairs. For a sum s, count the values a for which $c = s - a$ lies between 1 and 10:

$$\#(s) = \min(6,\, s-1) - \max(1,\, s-10) + 1.$$

sum s	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ways	1	2	3	4	5	6	6	6	6	6	5	4	3	2	1

The distribution is a *trapezoid*: it rises, then stays flat at 6 ways for every sum from 7 to 11 (for each of these sums, all six d6 values can be completed by some d10 value), then falls. So $P(\text{sum} = s) = \frac{6}{60} = \frac{1}{10}$ on the whole plateau $s = 7, \dots, 11$.

Step 3 — maximise $b \cdot P(b)$. On the plateau the probability is constant, so $E(b) = \frac{b}{10}$ simply grows with the payoff: choose the *largest* sum that is still on the plateau.

b	7	9	10	**11**	12	13
$E(b) = b \cdot \frac{\#(b)}{60}$	\$0.70	\$0.90	\$1.00	**\$1.10**	\$1.00	\$0.87

One step beyond the plateau ($b = 12$) the probability drops from $\frac{6}{60}$ to $\frac{5}{60}$, and the extra \$1 of payoff no longer compensates.

Bid 11, for expected winnings $\frac{11}{10} = \$1.10$.

Traps. Guessing the *most likely* sum (7, or anywhere on the plateau) ignores the payoff: 7 gives only \$0.70. Guessing the *mean* sum ($3.5 + 5.5 = 9$) gives \$0.90. With unequal dice the distribution has a flat top, and the optimum is the *right-hand end of the plateau*.

Concept Summary and Applications

Probability vs. payoff trade-off. If choosing x changes both the chance of winning and the size of the win,

$$E(x) = P(\text{win} \mid x) \cdot \text{payoff}(x).$$

Method: (1) write $E(x)$ explicitly; (2) for a smooth problem, treat x as continuous and solve $E'(x) = 0$; (3) check the integers on either side of the root and the endpoints. For a discrete distribution, tabulate $x \cdot P(x)$ and look where it peaks.

- **Both extremes are bad:** stake nothing and you never win; stake everything and the prize is zero. The optimum sits in between.
- **The most likely outcome is rarely optimal** when the payoff grows with your choice; on a flat stretch of probability, go to the high end.
- **Flat optima:** near the best choice $E(x)$ is usually flat, so nearby choices lose very little — useful for quick interview estimates.
- **The other side of the table:** if *you* design the distribution and someone else picks the best x, you want to make $x \cdot P(x)$ as *flat* as possible (Additional Problem 2).
- **Applications:** contest and lottery entries, choosing a bid or quote size (bigger size, worse fill probability), pricing with demand that falls as price rises, and Tullock-style contests.

Additional Problem 1 — How Many Balls to Put in the Pot? (Q425)

Problem: You have 100 white balls and your opponent has 100 black balls. Each of you puts some of your balls into a common pot. One ball is drawn at random from the pot; whoever owns it wins an amount of money equal to the number of balls they have *left over* (not put in the pot). Your opponent puts in 99 balls. How many should you put in?

Step 1 — the trade-off. If you put in x balls, the pot holds $x + 99$ balls, of which x are yours, so

$$P(\text{you win}) = \frac{x}{x+99}, \qquad \text{payoff if you win} = 100 - x.$$

More balls in the pot raise your chance of winning but shrink the prize. Hence

$$E(x) = \frac{x}{x+99}\,(100 - x) = \frac{x(100-x)}{x+99}, \qquad 0 \le x \le 100.$$

Step 2 — differentiate. By the quotient rule,

$$E'(x) = \frac{(100 - 2x)(x+99) - (100x - x^2)}{(x+99)^2}.$$

Expanding the numerator:

$$(100-2x)(x+99)-(100x-x^2)=100x+9900-2x^2-198x-100x+x^2=9900-198x-x^2.$$

Step 3 — solve $E'(x)=0$.

$$x^2+198x-9900=0 \quad\Longrightarrow\quad x=\frac{-198+\sqrt{198^2+4\cdot 9900}}{2}=-99+\sqrt{19701}\approx 41.36.$$

(Only the positive root is relevant.) $E'(x)>0$ to the left of the root and <0 to the right, so this is a maximum.

Step 4 — check the neighbouring integers.

$$E(41)=\frac{41\cdot 59}{140}=\frac{2419}{140}\approx \$17.279, \qquad E(42)=\frac{42\cdot 58}{141}=\frac{2436}{141}\approx \$17.277.$$

Put in 41 balls, for expected winnings ≈ \$17.28.

Interpretation. The optimum is very flat: anything from 38 to 45 is within about 5¢ of the best. And it is far below the opponent's 99 — matching a large stake is not worth it, because every extra ball you add raises your chance only a little while costing you a full \$1 of prize.

Extension (both choose at once). If both players choose simultaneously, with your payoff $\frac{x}{x+y}(100-x)$, the first-order condition at a symmetric point $x=y$ gives $100x-3x^2=0$, i.e. each puts in about $\frac{100}{3}\approx 33$ balls.

Additional Problem 2 — Write the Cards to Minimise the Guesser's Return (Q607)

Problem: You have 100 blank cards and write one positive integer on each. The interviewer sees how many cards carry each number, the deck is shuffled, and he names a number. If the top card shows that number, he wins that many dollars. What should you write to make his expected winnings as small as possible?

The tension in one sentence. You want to write *small* numbers, because a correct guess then pays very little — but you cannot write the same small number on too many cards, because then it becomes too *likely* to be guessed. (Writing "1" on all 100 cards makes the guesser certain to win \$1.) So small values should appear often, larger values more rarely, and the goal is to balance value against frequency so that no single guess is attractive.

Step 1 — the guesser's side. Let m_v be the number of cards showing v. If he names v, he wins v with probability $\frac{m_v}{100}$, so he expects $\frac{v\,m_v}{100}$. He picks the best v, so his return is

$$\frac{1}{100}\max_v\,(v\,m_v).$$

This is exactly the representative problem from the *other* side: he maximises payoff × probability, and you want that maximum to be small.

Step 2 — reformulate. Let $M=\max_v(v\,m_v)$. To keep every $v\,m_v\le M$, value v may appear on at most $\lfloor M/v\rfloor$ cards (and values above M cannot appear at all). So 100 cards fit if and only if

$$\sum_{v=1}^{M}\left\lfloor\frac{M}{v}\right\rfloor \;\ge\; 100.$$

Step 3 — find the smallest M.

- $M = 27$: $27 + 13 + 9 + 6 + 5 + 4 + 3 + 3 + 3 + 2 + 2 + 2 + 2 + \underbrace{1 + \cdots + 1}_{v=14,\ldots,27:\ 14} = 95 < 100$. Not enough.
- $M = 28$: $28 + 14 + 9 + 7 + 5 + 4 + 4 + 3 + 3 + 2 + 2 + 2 + 2 + 2 + \underbrace{1 + \cdots + 1}_{v=15,\ldots,28:\ 14} = 101 \geq 100$. Enough.

So the smallest possible M is 28, and the guesser's best return is $\frac{28}{100}$.

Step 4 — an optimal deck.

value v	1	2	3	4	5	6	7	8	9	10–14	15–27
cards m_v	28	14	9	7	5	4	4	3	3	2 each	1 each

That is $87 + 13 = 100$ cards, and every value has $v\, m_v \leq 28$.

Spread the values so that $v \times$ (number of cards with v) ≤ 28 for every value, e.g. the deck above. The guesser's best expected return is then $\frac{28}{100} = \$0.28$, the minimum possible.

Intuition. A value written on many cards is easy to guess, so it must be small; a large value must be rare. The ideal is to make $v \cdot m_v$ the same for every value — a harmonic spread $m_v \approx \frac{M}{v}$ — so that no guess stands out. You cannot go below \$0.28 because with only 27 to "spend" per value you can fit 95 cards, not 100. **Lesson:** the guesser *maximises* $x \cdot P(x)$ (as in the representative problem); the designer *flattens* it.

38 Auction Theory and the Winner's Curse

In an auction the thing being sold often has a value you cannot see. The trap is to bid as if you will receive an *average* item. But you only get the item when your bid wins — and *when* your bid wins tells you something about the item. That hidden information is the **winner's curse**: winning is itself news, and usually bad news.

Representative Problem — Bidding for a Car (Q28)

Problem: A seller offers you a car whose true value V is uniformly distributed between \$0 and \$1000; you do not know V. You make one bid b. If your bid is at least the car's value ($V \le b$), you buy the car at price b and can resell it for 1.5 times its true value, $1.5V$. Otherwise no trade happens. What bid maximises your expected profit?

The naive (wrong) reasoning. "On average the car is worth \$500, I can resell it for $1.5 \times 500 =$ \$750, so any bid below \$750 is profitable." This ignores *which* cars you actually end up buying.

Step 1 — write down the profit. If you win ($V \le b$) your profit is the resale value minus what you paid, $1.5V - b$. If you lose, the profit is 0.

Step 2 — average over the unknown value. V is uniform on $[0, 1000]$, so its density is $\frac{1}{1000}$. You win exactly when $V \le b$, so for a bid $0 \le b \le 1000$

$$\mathbb{E}[\Pi(b)] = \int_0^b (1.5v - b)\,\frac{1}{1000}\,dv = \frac{1}{1000}\Big[\tfrac{1.5}{2}v^2 - bv\Big]_0^b = \frac{0.75b^2 - b^2}{1000} = -\frac{b^2}{4000}.$$

Step 3 — optimise. $-\frac{b^2}{4000}$ is negative for every $b > 0$ and equals 0 only at $b = 0$. Bids above \$1000 win for sure but earn $1.5\,\mathbb{E}[V] - b = 750 - b \le -250$, even worse. So the best bid is \$0 — do not bid at all.

Step 4 — why: condition on winning. You get the car only when $V \le b$. Given that, V is uniform on $[0, b]$, so

$$\mathbb{E}[V \mid \text{you win}] = \frac{b}{2}.$$

Winning tells you the car is worth only *half your bid* on average. Reselling at 1.5× returns $1.5 \cdot \frac{b}{2} = 0.75b$, which is less than the b you paid — you lose about $0.25b$ on every car you win. The "average car" worth \$500 is irrelevant: you never get to buy the good ones cheaply, because a low bid only wins when the car is worth even less.

Optimal bid = $\boxed{\$0}$. Conditional on winning, the car is worth $b/2$ on average, and $1.5 \cdot \frac{b}{2} = 0.75b < b$, so every positive bid loses money in expectation.

The Profit Threshold — When Does Bidding Pay?

Replace the resale multiple 1.5 by a general k (e.g. a mechanic who can raise the car's value, or a buyer who values it more). The same calculation gives

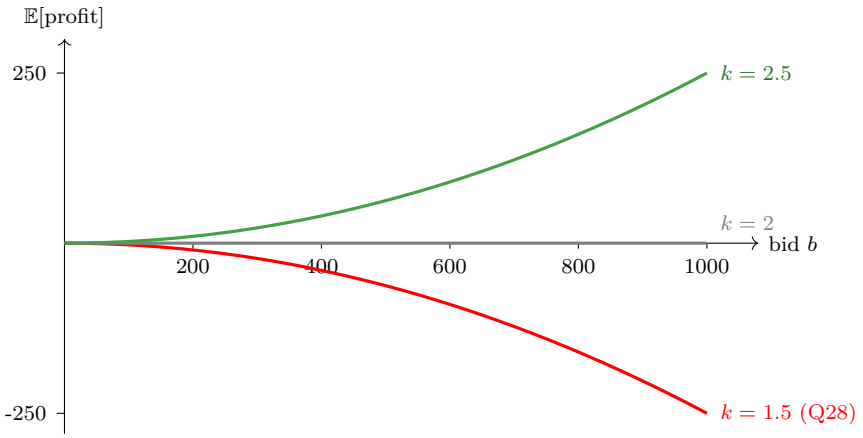
$$\mathbb{E}[\Pi(b)] = \int_0^b (kv - b)\,\frac{1}{1000}\,dv = \frac{1}{1000}\Big(\frac{k}{2} - 1\Big)b^2.$$

Everything hinges on the sign of $\frac{k}{2} - 1$:

$$\begin{aligned} k < 2: &\quad \text{every positive bid loses } \Rightarrow \text{ bid 0;} \\ k = 2: &\quad \text{break-even for any bid;} \\ k > 2: &\quad \text{profit grows with } b \Rightarrow \text{ bid as high as possible.} \end{aligned}$$

Threshold: bidding is profitable *only if* the resale multiple exceeds **2**, because winning with bid b means the item is worth $b/2$ on average, and you need $k \cdot \frac{b}{2} > b$.

When $k > 2$ the profit keeps rising up to $b = 1000$, where you win every time and earn $k \cdot 500 - 1000$ (e.g. $k = 2.5$ gives \$250). The threshold 2 does not depend on the \$1000 ceiling: for V uniform on $[0, M]$ it is still $k > 2$.



Expected profit versus bid. Below the threshold ($k = 1.5$) it only goes down as you bid more, so bid nothing; at $k = 2$ it is zero for every bid; above the threshold ($k = 2.5$) it rises, so bid the maximum.

Concept Summary and Applications

- **The winner's curse:** in a *common-value* setting (the item is worth the same to everyone, but nobody knows exactly how much), winning selects the cases where the item is worth *less* than you bid. Always value the item *conditional on winning*, $\mathbb{E}[V \mid \text{win}]$, not the unconditional average $\mathbb{E}[V]$.
- **Uniform benchmark:** with $V \sim U[0, M]$ and a winning rule $V \le b$, $\mathbb{E}[V \mid \text{win}] = b/2$; expected profit is $\frac{1}{M}(\frac{k}{2} - 1)b^2$.
- **Profit threshold:** bid only if the value multiplier $k > 2$; for $k < 2$ the optimal bid is 0, for $k > 2$ bid as high as the range allows.
- **Adverse selection:** the same logic appears whenever the other side accepts *only* when it is good for them — your quote gets hit when you are wrong. Ask "what does it mean that this trade happened?"
- **Both sides informed (wallet game):** when each bidder holds part of the value, the ascending-auction equilibrium is to quit at the price that breaks even against an opponent *with your own signal* — $2x$ in the problem below.

- **Applications:** market making (getting filled is informative), oil-lease and spectrum auctions, IPO allocations, used-car ("lemons") markets, and takeover bids.

Additional Problem — Bidding for a Shared Pot (Q139)

Problem: You and another player each independently draw a number uniformly from $[0, 1]$; each number is private (you see yours, not theirs). Both numbers go into a pot, and you take turns bidding for the pot. What is your optimal bidding strategy?

Rules of the game. Your number is x, the opponent's is y; the pot is worth $x+y$, but each player sees only their own number. Bids go back and forth with a rising price (an ascending, or "English," auction). A player may drop out at any time; **the last player still bidding wins the pot and pays the price at which the other dropped out**. The winner's payoff is $(x + y) -$ price; the loser's payoff is 0.

Step 1 — the naive answer, $x + \frac{1}{2}$, is wrong. The tempting reasoning: you know x, and y averages $\frac{1}{2}$, so the pot is "worth" $x + \frac{1}{2}$ and you should pay up to that. But this uses the *unconditional* average of y. You only win when the opponent drops out first, and players drop out early precisely when their number is *low*. So conditional on winning, y is below average — exactly the winner's curse from the car problem. Valuing the pot at $x + \frac{1}{2}$ makes you overpay.

Step 2 — guess a symmetric strategy and check it. Suppose each player stays in until the price reaches $2\times$(their own number), and test whether that is a best response. If the opponent quits at price $2y$ and you plan to quit at price T, you win exactly when the opponent quits first,

$$2y < T \iff y < \tfrac{T}{2},$$

and in that case you pay $2y$. Your expected profit is

$$\int_0^{T/2} [(x + y) - 2y]\, dy = \int_0^{T/2} (x - y)\, dy = x \cdot \frac{T}{2} - \frac{1}{2}\Big(\frac{T}{2}\Big)^2.$$

Differentiate in T: $\frac{x}{2} - \frac{T}{4} = 0 \Rightarrow T = 2x$. (Equivalently, the integrand $x - y$ is positive while $y < x$ and negative after, so you want to keep winning exactly up to $y = x$, i.e. $T/2 = x$.) Your best response to "$2y$" is "$2x$", so

$$\boxed{\text{stay in until the price reaches } 2x}$$

is a symmetric equilibrium.

Step 3 — intuition: think about the marginal opponent. At the moment you are about to drop out at price p, ask: "what if my opponent were about to drop out at this same price?" With both players using the same increasing rule, that opponent has the *same* number, $y = x$. Then the pot is worth $x + x = 2x$, so you should be willing to pay up to exactly $2x$ and no more. Conditioning on this "tie" case builds the winner's-curse correction in automatically.

Step 4 — check the profit. With both players using $2\times$, you win exactly when $y < x$, paying $2y$ for a pot worth $x + y$, so you keep $x - y > 0$ on every win. Your expected profit is

$$\int_0^x (x - y)\, dy = \frac{x^2}{2} \geq 0,$$

and, conditional on winning, $\mathbb{E}[y \mid \text{win}] = \frac{x}{2}$ — well below the naive $\frac{1}{2}$.

Stay in until the price reaches twice your own number, $2x$ (not $x + \frac{1}{2}$). Expected profit $x^2/2$.

Sealed-bid variant. *If instead each player submits one sealed bid and the higher bidder wins the pot and pays* their own *bid, the equilibrium bid is $b(x) = x$. Check: if the opponent bids $b(y) = y$ and you bid z, you win when $y < z$ and your expected profit is $\int_0^z (x + y - z)\, dy = xz - \frac{z^2}{2}$, maximised at $z = x$. Conditional on winning, the pot is worth $x + \frac{x}{2} = 1.5x$ on average; you pay x and keep $\frac{x}{2}$. Expected profit is again $\int_0^x (x + y - x)\, dy = \frac{x^2}{2}$ — the same as the ascending auction (*revenue equivalence*: the ascending winner pays $2y$, which averages x given $y < x$). The rule "bid half your value" belongs to a different problem (private values, where you know the item's worth to you exactly) and does not apply here.*

39 The Doubling Cube: When to Raise the Stakes

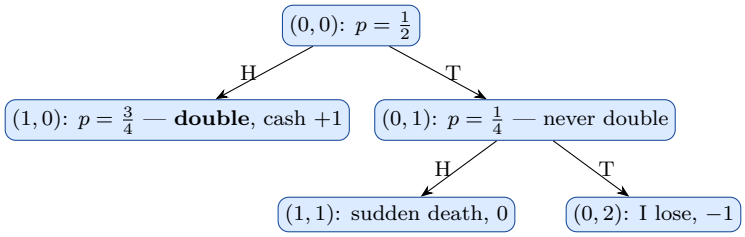
In backgammon (and many trading-style games) a player may offer to **double** the stake mid-game. The opponent then either **takes** (play continues for twice the stake) or **drops** (resigns now and pays the current stake). The right to double is an *option*: it has value because you can wait and use it only when the game has turned in your favour.

Representative Problem — A Coin Race with a Doubling Option (Q395)

Problem: A coin is flipped repeatedly. Heads gives me 1 point, tails gives you 1 point. The first to reach 2 points wins the game, and the loser pays the winner \$1. However, I have a one-time option to raise the stake to \$2. What is the value of this option?

Setup. Track the score as $(a, b) = (\text{my points}, \text{your points})$; each flip adds a point to one side with probability $\frac{1}{2}$. Without the option the game is symmetric and worth \$0 to me, so *the option's value is exactly the extra value it creates.* I may raise the stake from \$1 to \$2 once, at any point.

Warning — the score difference is not enough. It is tempting to track only the difference $b - a \in \{-1, 0, 1\}$. But the tied states $(0, 0)$ and $(1, 1)$ are *not* interchangeable: from $(1, 1)$ the very next flip ends the game (sudden death), so there is no later chance to double from ahead, while from $(0, 0)$ there is. Merging them gives the wrong answer $\frac{1}{3}$, so we keep the full score.



Step 1 — win probabilities and what a double is worth. Let $p(a, b) = \Pr(\text{I win} \mid (a, b))$. One flip at a time,

$$p(1,1) = \tfrac{1}{2}, \qquad p(1,0) = \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot p(1,1) = \tfrac{3}{4}, \qquad p(0,1) = \tfrac{1}{2}\, p(1,1) + \tfrac{1}{2} \cdot 0 = \tfrac{1}{4}, \qquad p(0,0) = \tfrac{1}{2}.$$

If the stake is already \$2, my value in a state is $2(2p - 1)$ (win \$2 with probability p, lose \$2 otherwise):

$$(1,0)\colon\ 2(2 \cdot \tfrac{3}{4} - 1) = 1, \qquad (0,0)\colon\ 0, \qquad (1,1)\colon\ 0, \qquad (0,1)\colon\ 2(2 \cdot \tfrac{1}{4} - 1) = -1.$$

So a double is worth at most \$1, reached only when I lead 1–0 (win chance $\frac{3}{4}$). At $\frac{3}{4}$ this is exactly the *take/drop point*: if you play on at \$2 my value is $2 \cdot \frac{1}{2} = \$1$; if you resign you pay the \$1 stake. Either way I collect \$1, so the answer is the same whether or not you may decline the double.

Step 2 — value with the option, state by state. Let $V(a, b)$ be my value while I still hold the option (stake \$1). In each state I take the better of *doubling now* or *flipping on with the option kept.*

- $(1, 1)$ — **sudden death.** Doubling is worth 0; flipping gives $\frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$. So $V(1, 1) = 0$.
- $(1, 0)$ — **I lead.** Doubling locks in \$1. Flipping on instead gives $\frac{1}{2}(+1) + \frac{1}{2}V(1, 1) = \frac{1}{2}$. Since $1 > \frac{1}{2}$, **double here**: $V(1, 0) = 1$.

- $(0,1)$ — **I trail.** Doubling is worth $-\$1$, so never; flip on:

$$V(0,1) = \tfrac{1}{2}V(1,1) + \tfrac{1}{2}(-1) = \tfrac{1}{2}\cdot 0 - \tfrac{1}{2} = -\tfrac{1}{2}.$$

- $(0,0)$ — **the start.** Doubling is worth 0; flip on, keeping the option:

$$V(0,0) = \tfrac{1}{2}V(1,0) + \tfrac{1}{2}V(0,1) = \tfrac{1}{2}(1) + \tfrac{1}{2}(-\tfrac{1}{2}) = \tfrac{1}{4}.$$

The game without the option is worth \$0, so the option is worth $V(0,0) = \frac{1}{4} = \boxed{\$0.25}$.
Strategy: do nothing at the start; if the first flip puts me ahead 1–0, double immediately (banking \$1); if it puts me behind, never double.

Why not $\frac{1}{3}$? That answer comes from writing $V(\text{behind}) = \frac{1}{2}V(\text{tied}) + \frac{1}{2}(-1)$ with a single "tied" value. But from behind (0–1) a winning flip leads to $(1,1)$ — sudden death, value 0 — not back to the start $(0,0)$, value $\frac{1}{4}$. Using the correct $V(1,1) = 0$ gives $V(0,1) = -\frac{1}{2}$ and hence $\frac{1}{4}$.

The General Principle — Double at the Opponent's Take Point

When should the opponent take? Suppose you double when the opponent's chance of winning is q (yours is $p = 1-q$), with stake 1 and no further doubling. If the opponent *drops*, they lose 1 for sure. If they *take*, their value is $2(q\cdot 1 - (1-q)\cdot 1) = 2(2q-1)$. Taking is at least as good as dropping when

$$2(2q-1) \;\geq\; -1 \quad\Longleftrightarrow\quad q \;\geq\; \tfrac{1}{4}.$$

So the opponent should take while they win at least 25% of the time, and drop below that.

What a double is worth to you. The opponent picks whichever response is worse for you, so a double at win probability p is worth

$$\min\{\underbrace{2(2p-1)}_{\text{they take}},\ \underbrace{1}_{\text{they drop}}\}.$$

This is capped at 1 (the current stake): once the opponent can simply resign, doubling never earns more than cashing the game.

Threshold rule (one double, final stake). The opponent's take point is $q = \frac{1}{4}$, i.e. **offer the double when your win probability reaches $\mathbf{p = \frac{3}{4}}$.** Below $p = \frac{1}{2}$ never double (it multiplies a losing position); between $\frac{1}{2}$ and $\frac{3}{4}$ doubling raises your immediate value but throws away the option; at $p = \frac{3}{4}$ doubling *cashes* the game for the full stake.

Why wait until $\frac{3}{4}$ rather than double as soon as you are ahead? Keep the option and double exactly when p first reaches $\frac{3}{4}$ (worth $+1$), or lose at 0 (worth -1). Because p is a fair (martingale) process, the chance of reaching $\frac{3}{4}$ before 0 is $\frac{p}{3/4} = \frac{4p}{3}$, so waiting is worth

$$\tfrac{4p}{3}(1) + \left(1 - \tfrac{4p}{3}\right)(-1) = \tfrac{8p}{3} - 1,$$

whereas doubling now (they take) is worth $4p-2$. Waiting is better exactly when $\frac{8p}{3}-1 \geq 4p-2$, i.e. $p \leq \frac{3}{4}$ — and the two agree at $p=\frac{3}{4}$. The right moment is the last moment your opponent can still accept.

Discrete games: don't "lose your market." When p moves in jumps (as with coin flips), you may leap from below $\frac{3}{4}$ to well above it in one step. Then the opponent drops and you collect only 1, whereas an earlier double they had accepted would have been worth $2(2p-1) > 1$ in that branch. So in jumpy games you sometimes double slightly *before* $\frac{3}{4}$. In Q395 the lead jumps from $p=\frac{1}{2}$ exactly to $\frac{3}{4}$, so doubling at 1–0 is exactly right.

Concept Summary and Applications

- **The double is an option:** its value comes from waiting and exercising only in good states; value it by backward induction over the *full* state (not just the score difference).
- **Take point $\frac{1}{4}$:** with one final double, the opponent takes iff their win chance is at least 25%; a double is worth $\min\{2(2p-1), 1\}$ to you.
- **Doubling point $\frac{3}{4}$:** offer when your win probability reaches $\frac{3}{4}$ (earlier only if the next move could jump you past it). With re-doubles allowed (a "live" cube, as in backgammon) the take point drops to about 20% because the taker gains the right to redouble.
- **Applications:** backgammon cube decisions, raising bets in poker-style games, early exercise of options, and any "commit more capital once the odds are clearly in your favour" decision.

Part VI — Game Theory

40 Zero-Sum Games and Mixed Strategies

In a *zero-sum game* two players move at the same time and one player's gain is exactly the other's loss (Rock–Paper–Scissors, matching pennies, sealed bids). A *pure strategy* is a single fixed move; a *mixed strategy* chooses among moves at random with chosen probabilities. The central lesson of this section: **when every fixed move can be exploited, the best you can do is randomise — and the right probabilities are the ones that make your opponent indifferent.**

Representative Problem — Rock–Paper–Scissors Without Rock: Why Randomise? (Q455)

Problem: You play Rock–Paper–Scissors against an opponent who is *not allowed to play Rock*. You win \$1 for a win, lose \$1 for a loss, and get \$0 for a tie. What should you play to maximise your expected profit?

Step 1: Write the payoff table

Rules: Rock beats Scissors, Scissors beats Paper, Paper beats Rock. The opponent can play only Paper or Scissors. Your payoff in each case:

You \ Opponent	Paper	Scissors
Rock	-1	$+1$
Paper	0	-1
Scissors	$+1$	0

Step 2: Remove dominated moves

A move is *dominated* if some other move does at least as well against every opponent move (and strictly better against at least one). Compare your Paper and Scissors row by row: against Paper, Scissors gives $+1 > 0$; against Scissors, it gives $0 > -1$. Scissors is better in *both* columns, so you should never play Paper. Only Rock and Scissors remain:

You \ Opponent	Paper	Scissors
Rock	-1	$+1$
Scissors	$+1$	0

Step 3: Why you must randomise

Suppose you commit to a single move and the opponent figures it out:

- always Scissors $\Rightarrow$ the opponent plays Scissors, every game is a tie, you earn 0;
- always Rock $\Rightarrow$ the opponent plays Paper, you lose \$1 every game.

Each pure strategy has a counter. A random mix makes you unpredictable: the opponent cannot know which move to counter.

Step 4: Choose the mix that makes the opponent indifferent

Play Rock with probability p and Scissors with probability $1-p$. Your expected payoff against each opponent move is

$$\text{vs Paper: } p(-1)+(1-p)(+1)=1-2p, \qquad \text{vs Scissors: } p(+1)+(1-p)(0)=p.$$

If these two numbers differ, the opponent simply plays the column that is worse for you. The best you can guarantee is when they are equal:

$$1-2p=p \quad\Longrightarrow\quad p=\tfrac{1}{3}.$$

With Rock $\frac{1}{3}$, Scissors $\frac{2}{3}$ you earn $\frac{1}{3}$ *whatever* the opponent does — even if they know your probabilities.

Why equalising is optimal: your guaranteed payoff is $\min(1-2p,\ p)$. The first term falls as p rises and the second rises, so the minimum is largest where the two lines cross, at $p=\frac{1}{3}$, giving $\frac{1}{3}$.

Step 5: The opponent's side and the value of the game

Now let the opponent play Paper with probability q and Scissors with $1-q$. Your Rock earns $-q+(1-q)=1-2q$ and your Scissors earns q. The opponent wants to minimise your best reply $\max(1-2q,\ q)$, which again happens where they are equal: $q=\frac{1}{3}$. Then you earn at most $\frac{1}{3}$ whatever you do.

So you can *guarantee* $\frac{1}{3}$, and the opponent can *hold you to* $\frac{1}{3}$. This common number is the **value of the game** (von Neumann's minimax theorem says such a value always exists in finite zero-sum games once mixing is allowed).

Step 6: What if you know the opponent's mix?

If you *know* the opponent plays Paper with a fixed probability q, you don't need to mix: just play the better row. Rock earns $1-2q$, Scissors earns q, so play Rock if $q<\frac{1}{3}$ and Scissors if $q>\frac{1}{3}$. Against a 50/50 opponent, Scissors earns $\$\frac{1}{2}$. But this best response can be exploited if the opponent adapts; the $\frac{1}{3}/\frac{2}{3}$ mix is the safe strategy when you don't know.

Never play Paper. Play Rock with probability $\frac{1}{3}$, Scissors with probability $\frac{2}{3}$; this guarantees an expected profit of $\boxed{\$\frac{1}{3}\text{ per game}}$, the value of the game. (The opponent's optimal mix is Paper $\frac{1}{3}$, Scissors $\frac{2}{3}$.)

Concept Summary and Applications

Solving a small zero-sum game.

1. Write the payoff table (your payoff; the opponent's is its negative).
2. **Remove dominated moves** — moves worse than another move in every column.
3. Look for a **safe pure move** (one that cannot be beaten). If there is none, **mix so the opponent is indifferent**: for a 2×2 game with your mix $(p, 1-p)$,

$$p\,a + (1-p)\,c \;=\; p\,b + (1-p)\,d \qquad \text{for payoffs } \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

4. Do the same from the opponent's side; the common payoff is the **value** V:

$$\max_{\text{your mix}} \min_{\text{their move}} E[\text{payoff}] \;=\; \min_{\text{their mix}} \max_{\text{your move}} E[\text{payoff}] \;=\; V.$$

5. If the opponent's mix is **known**, best-respond with a pure move (it is never worse than any mix) — but be aware it is exploitable.

- **Indifference principle:** in equilibrium every move you play with positive probability earns the same expected payoff; otherwise you would drop the worse one.
- **Symmetric games have value** 0 (or $\frac{1}{2}$ for win probability): neither player can have an edge, so the goal is to find a mix that cannot be exploited.
- **Restrictions create an edge:** removing Rock from the opponent is worth $\$\frac{1}{3}$ per game to you (Q269, Q455, Q461; the mirror case Q558, opponent has only Rock/Paper, is solved the same way).
- **Change the rules, change the answer:** if draws are replayed (Additional Problem 1), a pure move can become optimal.
- **Applications:** bluffing frequencies in poker, penalty kicks, random audits, sealed-bid auctions (Additional Problem 3), and any market-making setting where a predictable quote can be picked off.

Additional Problem 1 — No Draws Allowed: A Definitive Choice (Q520)

Problem: You repeatedly play Rock–Paper–Scissors against an opponent who cannot choose Rock. If a round is a draw, you play again; the game ends at the first round that someone wins. What strategy maximises your chance of winning the game?

Step 1: What happens in one round

The opponent plays Paper or Scissors. Your outcomes:

You \ Opponent	Paper	Scissors
Rock	lose	win
Paper	draw (replay)	lose
Scissors	win	draw (replay)

Paper can never win a round, so it is useless. Compare Rock and Scissors:

- **Rock** can *lose* (to Paper).
- **Scissors** can *never lose*: it wins against Paper and draws against Scissors.

Step 2: Only decisive rounds matter

A draw just restarts the same game, so the winner is decided by the first decisive round. If in each round you win with probability w and lose with probability ℓ (and draw otherwise), then

$$P(\text{you win the game}) = \frac{w}{w+\ell}$$

(condition on the first round that is not a draw).

Let your per-round mix be Rock with probability r, Scissors with $1-r$, and the opponent's be Paper with probability q, Scissors with $1-q$. Then

$$w = r(1-q) + (1-r)q, \qquad \ell = rq, \qquad P(\text{win}) = \frac{r(1-q)+(1-r)q}{r(1-q)+(1-r)q+rq}.$$

Step 3: Any Rock is exploitable; pure Scissors is not

- If $r > 0$, the opponent can always play Paper ($q = 1$): then $w = 1-r$, $\ell = r$, and $P(\text{win}) = 1-r < 1$. Every bit of Rock gives the opponent a way to beat you.
- If $r = 0$ (always Scissors), $\ell = 0$: **you can never lose**. You win the first time the opponent plays Paper; if they play Paper with any positive probability q, you win with probability 1 (after $1/q$ rounds on average).

The opponent's only defence against Scissors is to play Scissors forever, which produces an endless draw: they can stall but never win.

Step 4: Why the answer differs from the representative problem

In Q455 every round is scored separately, and a draw is worth \$0; the opponent exploits "always Scissors" by drawing every round, so you had to mix in Rock. Here a draw costs nothing — it is simply replayed — and only the first decisive result counts. Scissors' only weakness (drawing) disappears, so it becomes a pure, definitive choice.

Always play Scissors. It never loses to an opponent without Rock: it beats Paper and draws (replays) against Scissors. Any Rock you add can be punished by Paper. You win with probability 1 unless the opponent plays Scissors forever (an endless draw).

Additional Problem 2 — Best Response to Known Probabilities (Q469)

Problem: Your opponent plays Rock, Paper and Scissors with known probabilities p_R, p_P, p_S ($p_R + p_P + p_S = 1$). With win $= +1$, loss $= -1$, tie $= 0$, how should you play to maximise your expected winnings?

Step 1: Expected payoff of each move

Each of your moves beats one opponent move and loses to another:

Your move	beats	loses to	expected payoff
Rock	Scissors	Paper	$E_R = p_S - p_P$
Paper	Rock	Scissors	$E_P = p_R - p_S$
Scissors	Paper	Rock	$E_S = p_P - p_R$

In words: $E[\text{move}] = P(\text{opponent plays what it beats}) - P(\text{opponent plays what beats it})$.

Step 2: A pure move is enough

If you mix with probabilities (x_R, x_P, x_S), your expected payoff is $x_R E_R + x_P E_P + x_S E_S$, a weighted average of the three numbers. An average is never larger than the largest number, so simply **play the move with the largest** E. Note $E_R + E_P + E_S = 0$, so the best move always earns at least 0.

Step 3: A clean rule using the threshold $\frac{1}{3}$

Rock is best when $E_R \geq E_P$ and $E_R \geq E_S$:

$$p_S - p_P \geq p_R - p_S \iff 2p_S \geq p_R + p_P = 1 - p_S \iff p_S \geq \tfrac{1}{3},$$

$$p_S - p_P \geq p_P - p_R \iff p_S + p_R \geq 2p_P \iff 1 - p_P \geq 2p_P \iff p_P \leq \tfrac{1}{3}.$$

By the same algebra for the other moves:

Play	when
Rock	$p_S \geq \frac{1}{3}$ and $p_P \leq \frac{1}{3}$
Paper	$p_R \geq \frac{1}{3}$ and $p_S \leq \frac{1}{3}$
Scissors	$p_P \geq \frac{1}{3}$ and $p_R \leq \frac{1}{3}$

That is: play the move whose *prey* is played at least a third of the time and whose *predator* at most a third of the time.

Step 4: Examples — and a common trap

- $(p_R, p_P, p_S) = (0.5, 0.3, 0.2)$: $E_R = -0.1$, $E_P = +0.3$, $E_S = -0.2$. Play **Paper**, earning 0.3 per game. Here "beat the most likely move" works.
- $(p_R, p_P, p_S) = (0.40, 0.25, 0.35)$: $E_R = 0.35 - 0.25 = +0.10$, $E_P = 0.40 - 0.35 = +0.05$, $E_S = 0.25 - 0.40 = -0.15$. Play **Rock**, even though the opponent's most likely move is Rock! Paper beats the 40% Rock but loses to the 35% Scissors; Rock beats the 35% Scissors and loses only to the 25% Paper.
- $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: every move earns 0 — the uniform mix cannot be exploited.

Trap: "beat the opponent's most likely move" is a good heuristic but *not* the exact rule, because it ignores how often your move gets beaten. Always compare $P(\text{win}) - P(\text{lose})$.

Compute $E_R = p_S - p_P$, $E_P = p_R - p_S$, $E_S = p_P - p_R$ and play the largest (no need to mix). Equivalently, play Rock if $p_S \geq \frac{1}{3} \geq p_P$, Paper if $p_R \geq \frac{1}{3} \geq p_S$, Scissors if $p_P \geq \frac{1}{3} \geq p_R$. If the opponent can adapt, fall back to the unexploitable $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

Additional Problem 3 — The Bid-Reducing Auction (Q473)

Problem: Two people each secretly bid an integer from 1 to 100. The higher bid is then reduced by 10, and whoever has the larger number after the reduction wins (equal numbers are a tie). What bid strategy maximises your probability of winning?

Step 1: Translate the rule

Let the higher bid be h and the lower ℓ, with gap $d = h - \ell$. After the reduction the comparison is $h - 10$ vs. ℓ:

- $d > 10$ (gap 11 or more): $h - 10 > \ell$, the **higher** bid wins;
- $1 \leq d \leq 9$ (gap less than 10): $h - 10 < \ell$, the **lower** bid wins;
- $d = 10$ or $d = 0$: tie.

So bidding high only pays if you are *far* above your opponent; bidding just below them is the best place to be.

The answer depends on what the opponent does, so we consider two readings.

Reading 1: the opponent bids uniformly at random

Fix your bid x; the opponent's bid y is equally likely to be any of $1, \ldots, 100$. You win when

- $y \leq x - 11$ (you are 11 or more above): $\max(x - 11, 0)$ values;
- $x + 1 \leq y \leq x + 9$ (they are 1–9 above you): $\min(9,\ 100 - x)$ values.

So the number of opponent bids you beat is

$$W(x) = \max(x - 11, 0) + \min(9, 100 - x) = \begin{cases} 9, & 1 \leq x \leq 11, \\ x - 2, & 11 \leq x \leq 91, \\ (x - 11) + (100 - x) = 89, & 91 \leq x \leq 100. \end{cases}$$

W increases up to $x = 91$ and then stays at 89. Any bid from 91 to 100 wins against 89 of the 100 opponent bids, ties 2 ($y = x$ and $y = x - 10$) and loses 9:

$$P(\text{win}) = 89\%, \qquad P(\text{win}) + \tfrac{1}{2}P(\text{tie}) = 90\%.$$

Intuition: a high bid is more than 10 above most random bids; the only danger is an opponent just above you, and at 91–100 there are at most 9 such bids.

Reading 2: a rational opponent (game theory)

If the opponent also plays to win, the game is symmetric and zero-sum, so its value is $\frac{1}{2}$ (counting a tie as half): neither player can guarantee more than an even chance.

No fixed bid is safe. Whatever $x \geq 2$ you choose, the opponent can bid $x-1$: the gap is $1 < 10$, so the *lower* bid — theirs — wins. (Bidding $x+1$ would be a mistake for them: then *you* are the lower bid within 9.) And $x = 1$ is beaten by any bid from 12 to 100, which is more than 10 above it. A predictable bidder always loses.

Optimal mix: bid 90 or 100, each with probability $\frac{1}{2}$. This guarantees at least $\frac{1}{2}$ against every opponent bid:

Opponent bids	Your 90	Your 100	Average
1–79	win (you are 11+ above)	win	1
80	tie (exactly 10 apart)	win	$\frac{3}{4}$
81–89	lose (you are 1–9 above)	win (11–19 above)	$\frac{1}{2}$
90	tie	tie (exactly 10 apart)	$\frac{1}{2}$
91–99	win (they are 1–9 above)	lose (you are 1–9 above)	$\frac{1}{2}$
100	tie (exactly 10 apart)	tie	$\frac{1}{2}$

The two bids cover each other's weak spots: against bids just *above* 90, your 90 is the lower bid within 9 and wins; against bids just *below* 90, your 100 is more than 10 above and wins. Since the guaranteed average is never below $\frac{1}{2}$ and the game's value is $\frac{1}{2}$, this mix is optimal (a full computer solution of the 100×100 game confirms it).

Random opponent: bid anything from 91 to 100 — win 89%, tie 2% (about 90% counting ties as half). Rational opponent: every fixed bid x is beaten by $x-1$; randomise between 90 and 100 (each $\frac{1}{2}$), which guarantees the fair value $\frac{1}{2}$ of this symmetric game.

41 Closest-Guess Games: The Median Rule and Mover Advantage

In a *closest-guess* game each player picks a number, a random value X is drawn (a die, a sum of dice, ...), and whoever is *closer* to X wins. Two picks split the number line at their midpoint: each player "owns" the outcomes on their side. So the contest is about *how much probability (or value) lies on each side*, not about the size of the distances. This leads to the **median rule**, and to questions about whether moving first or second is better.

Representative Problem — Closest to the Sum of Two Dice (Q436)

Problem: A and B each choose a *different* integer from 2 to 12. Two dice are rolled and the sum is computed. The player whose number is closer to the sum wins. Would you rather choose first or second, and which number should you choose?

Let the two chosen numbers be distinct integers. Since the sum of two fair dice can be any integer from 2 to 12, only choices in $\{2, 3, \ldots, 12\}$ are relevant.

The problem has two parts:

1. Is it better to choose first or second?
2. What number should be chosen?

1. Best response to a given first choice

Suppose the first player chooses a number x. The second player should choose a number $y \neq x$ to maximize the probability that the dice sum is closer to y than to x.

The probabilities for the sum S of two dice are:

S	2	3	4	5	6	7	8	9	10	11	12
$P(S)$	1	2	3	4	5	6	5	4	3	2	1

These are proportional to the listed counts out of 36.

If two chosen numbers are $a < b$, then:

- sums less than the midpoint $\frac{a+b}{2}$ favor a,
- sums greater than the midpoint favor b,
- if the midpoint is a half-integer, there is no tie,
- if the midpoint is an integer, that sum is tied.

Thus, the contest between two numbers depends on which side of the midpoint carries more dice probability.

Because the distribution is symmetric and peaked at 7, the strongest locations are near the center.

2. Optimal pair of choices

Consider choosing 7.

If the opponent chooses any other number, compare who wins more often.

Opponent chooses 6

The midpoint of 6 and 7 is 6.5, which is not an integer, so there are **no ties**:

- 6 wins for sums $2, 3, 4, 5, 6$,
- 7 wins for sums $7, 8, 9, 10, 11, 12$.

So

$$P(6 \text{ wins}) = \frac{1+2+3+4+5}{36} = \frac{15}{36}, \qquad P(7 \text{ wins}) = \frac{6+5+4+3+2+1}{36} = \frac{21}{36} = \frac{7}{12}.$$

Opponent chooses 8

By symmetry, 7 again wins with probability $\frac{21}{36}$ and 8 with $\frac{15}{36}$.

Opponent chooses anything farther away

For example, against 5 the midpoint is 6 (a tie on sum 6):

$$P(7 \text{ wins}) = \frac{6+5+4+3+2+1}{36} = \frac{21}{36}, \qquad P(5 \text{ wins}) = \frac{1+2+3+4}{36} = \frac{10}{36}.$$

Moving farther away only makes the opponent worse. So 7 beats *every* other number, and the opponent's best reply (6 or 8) still loses 15 to 21.

3. First or second?

Because the picks must be **different**, the first player can take 7, the median of the sum, and the second player cannot also take it. Whatever the second player does, the first player wins with probability at least $\frac{21}{36} = \frac{7}{12} \approx 58\%$.

If you go second against an optimal first player, the best you can do is 6 or 8, winning only $\frac{15}{36} \approx 42\%$.

Go first and choose 7 (the median of the sum). You win with probability at least $\frac{21}{36} = \frac{7}{12} \approx 58\%$, even against the best reply (6 or 8). If identical picks were allowed, the second player could copy 7 and force a tie on every roll; the first-mover edge comes from the rule that the numbers must differ.

Concept Summary and Applications

Closest-guess games. Two picks $a < b$ split the outcomes at the midpoint $\frac{a+b}{2}$; each player owns the outcomes on their side (an outcome exactly at the midpoint is a tie).

- **Median rule:** pick the *median of what you are paid*.
  - fixed prize (just win or lose): the *probability* median — the point with at least half the probability on each side;
  - paid the value rolled: the *value-weighted* median — the point where the running total of value × probability first reaches half.

  A pick at the median wins at least half wherever the opponent sits, because the opponent can only take one side of it.
- **Mode and mean are traps:** the most likely face, or the average, can be beaten by a number just next to it.
- **Who moves first?** If picks must *differ*, the first mover takes the median and cannot be beaten; if copying is allowed, the second mover copies for a guaranteed tie (value 0).
- **Three or more players:** the median becomes the *worst* first pick — the others squeeze it from both sides.

- **Hotelling / median-voter connection:** shops on a street, or candidates on a policy line, each capture the customers (voters) closer to them; with two competitors both drift to the median.
- **Applications:** positioning a product or price against one competitor, choosing a quote relative to another market maker, and any "closest guess wins" contest.

Additional Problem 1 — A Skewed 12-Sided Die (Q157)

Problem: An unfair 12-sided die rolls 12 with probability 40%; the other eleven faces are equally likely. Two people each choose a number from 1 to 12, and whoever is closer to the roll wins. Which number should you choose?

Let the die outcome be $X \in \{1, 2, \ldots, 12\}$ with

- $P(X = 12) = 0.4$,
- and the remaining probability 0.6 spread equally over $1, 2, \ldots, 11$.

Thus, for each $k = 1, \ldots, 11$,

$$P(X = k) = \frac{0.6}{11}.$$

Two players each choose a number, and the one whose choice is closer to X wins. To determine the best single number to choose, one should select a number minimizing the expected absolute distance to the outcome. Equivalently, one should choose a **median** of the distribution of X.

Step 1: Use the median criterion

A number m is a median if

$$P(X \le m) \ge \frac{1}{2} \qquad \text{and} \qquad P(X \ge m) \ge \frac{1}{2}.$$

Compute the cumulative probabilities.

For $k = 1, \ldots, 11$,

$$P(X \le k) = k \cdot \frac{0.6}{11}.$$

In particular,

$$P(X \le 9) = 9 \cdot \frac{0.6}{11} = \frac{5.4}{11} \approx 0.4909,$$

while

$$P(X \le 10) = 10 \cdot \frac{0.6}{11} = \frac{6}{11} \approx 0.5455.$$

So the cumulative probability first reaches $\frac{1}{2}$ at 10. Hence the median is

$$m = 10.$$

Step 2: Interpretation for the game

If you choose 10, then outcomes 1 through 10 are at or below your choice with total probability exceeding $\frac{1}{2}$, while the heavy mass at 12 pulls the distribution to the high end. This makes 10 the point that best balances the distribution in terms of closeness.

Therefore, the optimal choice is

$$\boxed{10}.$$

If ties are possible and broken fairly, this conclusion remains the same.

Why the median — and why *not* the mean

The heavy 40% mass at 12 is a decoy. It pulls the *mean* up to

$$\mathbb{E}[X] = 0.4 \cdot 12 + \frac{0.6}{11}(1 + 2 + \cdots + 11) = 4.8 + \frac{0.6}{11} \cdot 66 = 4.8 + 3.6 = 8.4,$$

but the mean is *not* the answer. "Closer wins" is a **mass-splitting** contest: only *which side* of the midpoint between the two picks the roll lands on matters — the *magnitude* of the distance is irrelevant, since a roll 1 away and a roll 5 away are each just one lump of probability on your side. The point you can never be outflanked from — where whichever side the opponent takes, your side still holds $\ge \frac{1}{2}$ — is by definition the point with half the mass on each side: the **median**. A single face at 40% cannot hold a side because $40\% < 50\%$, and the median ignores *how big* a spike is, only *where* it sits.

Hotelling's law (why two rational players both sit at the median)

This is the discrete **Hotelling / median-voter** result. Model the probability mass as "voters" on the line $1, \ldots, 12$; each player is a "shop" and captures the voters closer to it. If your opponent sits anywhere off the median, you move to the median side that holds $> \frac{1}{2}$ and win. The only configuration with no profitable deviation is *both* players at the median, splitting 50/50. So 10 is the symmetric Nash equilibrium and the maximin choice: against *any* opponent number it wins with probability $\ge \frac{1}{2}$ (e.g. 10 beats 12 with 0.573, beats 11 with 0.545, beats 9 with 0.509), with equality only against 10 itself.

Median or mean? It depends on what the distance does

The key is whether the rule performs an *ordinal comparison* of distances or *aggregates a distance-weighted expectation*:

- **Minimize $\mathbb{E}\,|c-X|$ (absolute error) $\Rightarrow$ median.** Here is the "balancing forces" reason it lands on the median even though the objective contains distances. Nudge your guess up by a tiny ε. Every outcome *below* c becomes ε *farther* (its $|c-X|$ rises by ε), and every outcome *above* c becomes ε *closer* (its $|c-X|$ falls by ε) — and crucially, that change is ε *regardless of how far the point is*, because $|c-X|$ has slope ± 1 everywhere. So the net change in expected error per unit move is

$$\frac{d}{dc}\mathbb{E}\,|c-X| = \underbrace{P(X<c)}_{\text{points pushed away, } +1 \text{ each}} - \underbrace{P(X>c)}_{\text{points pulled closer, } -1 \text{ each}}.$$

  You keep moving until the two *probabilities* balance, $P(X<c)=P(X>c)$ — which is exactly the median. The distances themselves never enter: each point "votes" with weight 1 for the direction it wants c to go, so it is a pure head-count of mass, and the balance point is the median. (Contrast the square, whose slope $2(c-X)$ grows with distance, so far points vote *harder* and the balance shifts to the mean.)
- **Minimize $\mathbb{E}\,(c-X)^2$ (squared error):** $\frac{d}{dc}\mathbb{E}(c-X)^2 = 2(c-\mathbb{E}[X]) = 0 \Rightarrow c=\mathbb{E}[X]$, which balances *mass* $\times$ *distance* $\Rightarrow$ **mean**. Far outcomes push harder, so the 40% at 12 finally exerts leverage.
- **Closer-wins competition (this problem):** an ordinal "who is closer" test; the magnitude is discarded $\Rightarrow$ **median**.
- **"Smaller *squared* difference wins" competition:** since $t \mapsto t^2$ is increasing for $t \geq 0$, $(c-X)^2 < (b-X)^2 \Leftrightarrow |c-X| < |b-X|$, i.e. still "who is closer." A monotone relabelling never flips the winner $\Rightarrow$ **still the median**. (Squaring gives the mean only when it is *aggregated into an expectation you minimize*, not when it sits inside a beat-your-opponent comparison.)

Summary

Rule	What the distance does	Optimal
Closer wins / smaller $\lvert\cdot\rvert$ / smaller $(\cdot)^2$ wins	ordinal "who is closer" comparison	**10** (median)
Single player, minimize $\mathbb{E}\,\lvert c-X\rvert$	margin balances *mass*	**10** (median)
Single player, minimize $\mathbb{E}\,(c-X)^2$	margin balances *mass* $\times$ *distance*	**8.4** (mean)

Additional Problem 2 — Paid the Value Rolled: A 30-Sided Die (Q25)

Problem: Two players pick *distinct* numbers from 1 to 30, then a fair 30-sided die is rolled. If the roll is closer to your number, the other player pays you *the number rolled* in dollars; if it is closer to theirs, you pay them that amount; if equidistant, nobody pays. Should you go first or second, which number should you pick, and what is your expected value?

How this differs from the other problems. In Q436 and Q157 the prize is fixed — you simply win or lose — so every outcome counts equally and you want the *probability* median. Here you are paid the *value* of the roll: winning a roll of 30 is worth 30 times as much as winning a roll of 1. So the high faces matter more, and the right target is the **value-weighted median** — the point

that splits the *total value* $1+2+\cdots+30=465$ in half — rather than the probability median (15 or 16). That pushes the best pick far up, to 22.

Setup. The two picks must be *distinct*. You are paid the die *value*, so the high faces are the prize, and a pick claims the contiguous block of faces closest to it, worth their sum. The faces total $\sum_1^{30}=465$, half of which is 232.5.

The second player's best reply is to sit adjacent. Given the opponent's number a, your best response is to pick $a{+}1$ (seizing the whole upper tail $\{a{+}1,\ldots,30\}$) or $a{-}1$ (seizing the whole lower tail) — whichever tail carries the larger value. Picking farther away only cedes faces back across the new midpoint, so adjacent is optimal.

The first player picks the pivot 22. Anticipating the adjacent grab, the first player chooses the number that makes *both* tails unattractive to the opponent while keeping the pivotal face on their own side. That number is the value-weighted median 22:

$$\sum_{1}^{21}=231<232.5, \qquad \sum_{1}^{22}=253\geq 232.5.$$

If the first player picks 22:

- opponent replies 21 $\Rightarrow$ opponent takes $\{1,\ldots,21\}=231$, you keep $\{22,\ldots,30\}=234$: your net $=\dfrac{234-231}{30}=+\$0.10$;
- opponent replies 23 $\Rightarrow$ opponent takes $\{23,\ldots,30\}=212$, you keep $\{1,\ldots,22\}=253$: your net $=+\$1.37$.

Either way **you win**, and the opponent minimizes the damage by taking the low side, leaving you $+\$0.10$. A full search confirms 22 is the *unique* first move with a positive value — every other first pick hands the opponent the win (e.g. picking 21 gives *them* $+\$0.10$).

Why the first mover wins here. The nine high faces $\{22,\ldots,30\}$ sum to 234, just edging out the twenty-one low faces $\{1,\ldots,21\}=231$. Claiming that block first secures the (slim) majority of the value. Because the picks must be *distinct*, the second player cannot neutralize this by copying 22 — distinctness is exactly what gives the first mover the edge. (Drop the distinctness rule and the second player copies for a \$0 tie; see Q432/Q438.)

Go first and pick 22 (the value-weighted median: $1+\cdots+21=231<232.5\leq 253=1+\cdots+22$). The second player's best reply is 21, leaving you an expected $\frac{234-231}{30}=+\$0.10$ per roll. Without the value weighting the answer would be the probability median, 15 or 16.

Additional Problem 3 — Three Players and a 21-Sided Die (Q553, Game 2)

Problem: You compete with two other players and a 21-sided die numbered 1 to 21. Each player picks a number, then the die is rolled; whoever's number is closest to the roll wins. What is your optimal strategy? How does it change if all players can communicate?

How this differs. With two players the median is unbeatable. With *three*, a central pick can be squeezed from both sides, so the median becomes the worst first move.

A fair 21-sided die; three players each pick a number, the die is rolled, and the closest number wins. Assume the usual set-up: players pick **in turn** (you first), numbers must be **different**, and ties are split.

Do not pick the centre

Picking 11 looks natural, but it is the *worst* first pick: the next two players take 10 and 12 and squeeze you, so you win only when the die shows exactly 11 — probability $\frac{1}{21} \approx 5\%$. With three players, a central pick is boxed in from both sides (the "squeeze").

Best first pick: 6 (or 16)

Solving the game backwards (player 3 best-responds to the first two picks, player 2 anticipates player 3, and you anticipate both), the best first pick is about a quarter of the way in from an end:

- You pick 6. Player 2's best reply is the mirror spot 16.
- Player 3 then gets $\frac{5}{21}$ whatever they do — squeezing into the gap 7–15 or stepping outside at 5 or 17 — so they are indifferent.
- Your share: $\frac{8}{21} \approx 38\%$ on average over player 3's equally good choices; even if player 3 breaks ties *against* you, at least $\frac{11}{42} \approx 26\%$ — still the best guaranteed share of any first pick.

your first pick	1	3	4	5	**6**	7	8	9	11
your share (avg.)	19%	27%	31%	35%	**38%**	24%	19%	14%	**5%**

(Picks 12–21 mirror 10–1. Values from exact backward induction, ties for player 3 broken at random.)

Why 6 *works:* it leaves the long stretch 7–21 open. Player 2 prefers to claim the far side at 16 rather than squeeze you, and player 3 then has no move that hurts you much.

If all players can communicate

The total chance of winning is always 1 (someone always wins), so cooperation cannot create value — it only decides how the 21 outcomes are shared. The natural agreement is an even split,

$$\boxed{4,\ 11,\ 18},$$

with 4 covering 1–7, 11 covering 8–14 and 18 covering 15–21: exactly 7 faces, i.e. $\frac{1}{3}$, each.

Note the twist: as the first mover you get about 38% without talking, so an equal-split agreement actually *hurts* you. Communication helps the later movers; the first mover should agree only if it protects them from the worst case (26%), or insist on keeping the 6/16 advantage.

Pick first at 6 (or 16), worth about $\frac{8}{21} \approx 38\%$; never the centre ($11$ gets only $\frac{1}{21}$). With communication the natural agreement is an even split, $4, 11, 18$ ($\frac{1}{3}$ each), which is actually worse for the first mover.

42 The Pirate Game (Backward Induction with Voting)

Representative Problem — Five Pirates, 100 Coins (Q249)

Problem: Five pirates of different ages share 100 gold coins. The oldest proposes how to split the coins, and *all* pirates (including the proposer) vote. If 50% **or more** vote for it, the coins are shared that way. Otherwise the proposer is thrown overboard, and the next-oldest pirate proposes, and so on. A pirate who would get the *same* number of coins either way votes *against* (he enjoys seeing the proposer thrown overboard). All pirates are rational, greedy, want to survive, and are perfect at mathematics. What happens?

Label the pirates from oldest to youngest P_1, P_2, P_3, P_4, P_5; P_1 proposes first.

The voting rule in one line. With n pirates alive, a proposal passes with at least $\lceil n/2 \rceil$ votes, and the proposer votes for himself. Each other pirate compares the offer with what he would get *if the proposer died* (the outcome of the game with one fewer pirate). He votes

- **for** if the offer is *more* than that,
- **against** if it is *less* or *the same* (the tie-breaking rule).

So to buy a pirate's vote, the proposer must give him **strictly more** than he would get in the next round. That is why we must first know the next round — we reason **backward**, starting from the smallest game.

Step 1 — 1 pirate (P_5). He keeps everything: $P_5 = 100$.

Step 2 — 2 pirates (P_4, P_5). P_4 needs $\lceil 2/2 \rceil = 1$ vote — his own. So he takes everything:

$$P_4 = 100, \qquad P_5 = 0.$$

Step 3 — 3 pirates (P_3, P_4, P_5). P_3 needs 2 votes: his own plus one more. If P_3 dies, the Step 2 outcome is $(P_4, P_5) = (100, 0)$. So:

- P_4 would need 101 coins — impossible;
- P_5 gets 0 if P_3 dies, so *one* coin buys his vote.

P_3 proposes $(P_3, P_4, P_5) = (99, 0, 1)$: P_3 and P_5 vote for, P_4 against — it passes 2–1.

Step 4 — 4 pirates ($P_2, \ldots, P_5$). P_2 needs $\lceil 4/2 \rceil = 2$ votes: his own plus one. If P_2 dies, the Step 3 outcome is $(P_3, P_4, P_5) = (99, 0, 1)$. The prices of the votes are:

- P_3: would need 100 — too expensive;
- P_4: gets 0 otherwise, so 1 coin suffices;
- P_5: gets 1 otherwise, so he would need 2.

P_2 buys the cheapest vote, P_4's: $(P_2, P_3, P_4, P_5) = (99, 0, 1, 0)$. Votes: P_2 and P_4 for, P_3 and P_5 against — 2 of 4, exactly 50%, so it passes.

Step 5 — all 5 pirates. P_1 needs $\lceil 5/2 \rceil = 3$ votes: his own plus two. If P_1 dies, the Step 4 outcome is $(P_2, P_3, P_4, P_5) = (99, 0, 1, 0)$. The prices:

- P_2: would need 100 — too expensive;
- P_3: gets 0 otherwise, so 1 coin;
- P_4: gets 1 otherwise, so 2 coins;
- P_5: gets 0 otherwise, so 1 coin.

P_1 buys the two cheapest votes, P_3 and P_5:

$$(P_1, P_2, P_3, P_4, P_5) = (98,\ 0,\ 1,\ 0,\ 1).$$

Votes: P_1, P_3, P_5 for; P_2, P_4 against — it passes 3–2.

The oldest pirate survives and proposes $(98, 0, 1, 0, 1)$ from oldest to youngest: he keeps 98 coins and gives 1 coin each to the third and fifth pirates.

The General Principle — Build Up from the Smallest Game

pirates	P_1	P_2	P_3	P_4	P_5	votes needed
1					100	1
2				100	0	1
3			99	0	1	2
4		99	0	1	0	2
5	98	0	1	0	1	3

1. **Start with the fewest players.** With one (or two) pirates the answer is obvious. Each larger game only needs the answer to the game one size smaller, because that is what happens if the current proposer is thrown overboard.
2. **Each voter's price is "next round $+1$".** A pirate votes yes only if the offer beats what he would get in the smaller game (strictly, because of the tie-breaking rule). So his vote costs (his next-round payoff) $+1$.
3. **Buy the cheapest votes.** The proposer needs $\lceil n/2 \rceil$ votes, i.e. $\lceil n/2 \rceil - 1$ besides his own. The cheapest are the pirates who get *nothing* in the next round — each costs just 1 coin. Everyone else gets 0.
4. **Read off the pattern.** The pirates who get 0 in one round are exactly the ones bought for 1 coin in the next, so the 1-coin gifts **alternate**: they go to every other pirate counting from the proposer. The proposer keeps
$$100 - (\lceil n/2 \rceil - 1) \text{ coins.}$$
For $n = 5$: $100 - 2 = 98$. This keeps working until the bribes use up all 100 coins (roughly 200 pirates); beyond that the analysis changes because the proposer can no longer afford enough votes.
5. **Check the rules.** Small changes flip the answer: if a tie (exactly 50%) *fails*, the proposer needs one more vote; if indifferent pirates vote *for*, votes cost 0 instead of 1. Always redo the smallest cases under the stated rules.

Concept Summary and Applications

- **Backward induction:** solve the smallest version first; each bigger version depends only on the next smaller one.
- **Price of a vote = outside option +1:** a rational voter compares your offer with what he gets if your proposal fails.
- **Cheapest coalition:** buy exactly the votes you need from those with the worst outside option; give everyone else nothing.
- **Applications:** bargaining and coalition-building (legislatures, shareholder votes), ultimatum-style negotiations, and any sequential game where a failed offer passes the move to the next player.

Additional Problem — 99 Lions and One Sheep (Q536)

Problem: On an island there are 99 lions and one sheep. When a lion eats the sheep, it turns into a sheep. Every lion's first goal is to survive; its second goal is to eat a sheep. Only one lion can eat at a time, lions never starve, and all lions are perfectly rational. How many lions will be on the island after some time?

Same idea as the pirates. Each lion compares "eat now" with what happens in the *smaller* game that eating creates — exactly like a pirate comparing an offer with the next round. So start with the fewest lions and work upward.

Step 1 — tiny islands.

- **1 lion:** it eats. It becomes a sheep, but no lion is left to eat it, so it is safe.
- **2 lions:** if one eats, it becomes a sheep facing 1 lion, which (by the case above) eats it. Survival comes first, so **nobody eats**.
- **3 lions:** if one eats, the island becomes 2 lions and 1 sheep, where nobody eats — so the new sheep is safe. **A lion eats.**
- **4 lions:** eating would leave 3 lions, where someone eats, so the new sheep would be eaten. **Nobody eats.**

Step 2 — the pattern. A lion eats only if the island it leaves behind ($n-1$ lions) is one where nobody eats. So the answer flips with each extra lion:

lions n	1	2	3	4	$\cdots$	99
a lion eats?	yes	no	yes	no	$\cdots$	yes

Odd n: a lion eats. Even n: nobody eats.

Step 3 — apply to 99. 99 is odd, so one lion eats the sheep and becomes a sheep. That leaves 98 lions — even — so nobody eats again.

98 **lions (and** 1 **sheep) remain. Work up from** 1 **lion: a lion eats exactly when the number of lions is odd.**

Comparison with the pirates. Both puzzles are solved by the same backward induction: the smallest case is obvious, each larger case depends only on the next smaller one, and the answer settles into an alternating pattern (every other pirate is bribed with 1 coin; every other lion count is safe to eat). The assumptions matter: if lions could starve, or several lions could eat at once, the answer could change.

43 Colonel Blotto and Resource-Allocation Games

In a *Colonel Blotto* game two players simultaneously split a fixed budget (soldiers) across several battlefields of known values; each battlefield goes to whoever commits more there. What matters is not your allocation on its own but how it compares with the opponent's — and you must commit before seeing it. This makes it a game of strategy, not a single-player optimisation.

Representative Problem — Win the Round (Q276)

Problem: There are 10 battlefields worth $1, 2, \ldots, 10$ points (total 55). You and your opponent each have 100 soldiers to place on them. On each battlefield the side with *strictly more* soldiers captures its points. You win the round if you capture at least 28 of the 55 points. Develop a strategy.

There are 10 battlefields worth $1, 2, \ldots, 10$ (total $V = \sum i = 55$) and 100 soldiers each. You capture a battlefield by placing **strictly more** soldiers on it, and you **win the round by capturing at least** 28 **of the** 55 **points**. The game is symmetric, so its value is 0 and there is no winning pure strategy — the answer must be a *randomized* strategy.

1. The tempting first answer: proportional spread

A natural heuristic is to allocate soldiers *proportionally to value*, $E[X_i] = 100 \cdot \frac{i}{55}$, giving the baseline

$$(2, 4, 5, 7, 9, 11, 13, 15, 16, 18) \qquad \text{(sums to 100)},$$

and then to "mix" by jittering 1–3 soldiers between nearby fields from game to game. This spreads force across all ten fields with heavier garrisons on the valuable ones.

2. Why proportional spread is weak

Spreading across all ten fields contests everything but wins nothing decisively, and it is crushed by **concentration**. Concretely, an opponent who simply puts 25 **soldiers on each of fields** $7, 8, 9, 10$ (and 0 elsewhere) allocates

$$(0, 0, 0, 0, 0, 0, 25, 25, 25, 25),$$

and beats the proportional garrisons there ($13, 15, 16, 18$, all < 25), capturing fields worth 34 (that is, $7 + 8 + 9 + 10$). The proportional player wins only the abandoned cheap fields 1–6, worth $1 + \ldots + 6 = 21$. Result: 34 **to** 21 — **the "balanced" strategy loses**, and small ± 1–3 jitter does nothing to save it. In simulation, a proportional spread with jitter loses to a concentration strategy essentially 100% of the time. The flaw is fundamental: you only need 28 points, so surrendering the six cheapest fields (worth $21 < 28$) costs nothing, while thin garrisons lose every field that is actually contested.

3. The right idea: random winning coalitions

Two forces matter — **concentration** (to win decisively) and **unpredictability** (to stay unexploitable):

- You do not need all 10 fields, only a subset worth ≥ 28. Pour the whole army onto ~ 4 high-value fields (~ 25 each) and you win a value-majority against anyone who spreads out.

- If you *always* contested the same four fields, the opponent would match you there and take the rest — so **randomize which coalition you contest each round**.

Recipe.

1. Each round, randomly pick a small set of (mostly high-value) fields whose values sum to ≥ 28, e.g. $\{7, 8, 9, 10\} = 34$, $\{10, 9, 8, 1\} = 28$, or $\{10, 8, 7, 4\} = 29$.
2. Put *all* 100 soldiers on just those fields (roughly equal large garrisons, $\sim$20–30 each); put 0 on the rest.
3. Re-randomize the coalition every round, tilting toward *fewer, higher-value* fields (the cheapest way to reach 28).

4. Why it works (simulation)

Against a field of opponents, including ones built to exploit it:

random winning-coalition strategy vs	win rate
even split (10 each)	96%
proportional spread	100%
"cheap-field grabber"	93%
"contest the big fields" counter	99%
another coalition mixer	$\approx 50\%$

It beats every naive or exploitative opponent decisively and plays $\approx 50/50$ against an equally clever concentrator — exactly right for a symmetric zero-sum game of value 0. The $\approx$50% against itself confirms coalition play is near-equilibrium: **you cannot be exploited, and you punish every mistake**, but you cannot force a win against someone equally sharp.

5. Refinements and caveats

- **Do not over-garrison:** winning a field by 1 soldier is as good as by 20; keep coalitions lean so each contested field gets a strong garrison.
- **Occasional feint:** now and then overload one cheap field to steal a point and blur your pattern.
- **Best-respond to known opponents:** the random coalition is the *unexploitable* baseline; against a predictable opponent, concede where they are heavy and outbid them exactly where it reaches 28.
- **Ties:** you need *strictly* more soldiers and ties split value, so budget +1 over the force you expect to face on a contested field.

Do not spread proportionally across all ten fields — that loses to concentration (e.g. to 25 soldiers on each of fields $7, 8, 9, 10$). Instead, randomly choose a set of high-value battlefields worth at least 28, put your whole army on just those, abandon the rest, and change the set every round: concentration to win, randomisation to stay unexploitable.

Concept Summary and Applications

Colonel Blotto. Two players split a budget across fields of known value; each field goes to the larger commitment.

- **No pure strategy is optimal.** The game is simultaneous and symmetric, so any fixed allocation can be beaten by someone who knows it (beat it by one soldier where it matters, concede the rest). The equilibrium is a *mixed* (randomised) strategy.
- **Spreading evenly or proportionally is weak:** it contests everything but wins nothing decisively, and loses to concentration.
- **Concede cheap fields;** spend soldiers on the valuable ones.
- **The objective changes the answer:** "win the round" (reach a target) favours a random winning *coalition*; "maximise score with a penalty for excess" favours *narrow* wins weighted by value.

- **In practice:** choose a strong allocation for the opponents you expect, avoid round numbers (to win ties), and randomise. Say explicitly that the best allocation depends on the opponent field.
- **Applications:** advertising and marketing budgets across markets, election campaigning across districts, allocating security resources, and splitting trading capital across competitive opportunities.

Additional Problem — Maximise Expected Score with a Wastage Penalty (Q430)

Problem: There are 10 castles worth 1 to 10 points. You have 100 soldiers to distribute (all must be deployed), and your opponent does the same independently. On each castle the side with more soldiers wins its points; a tie gives nobody points. In addition, for each castle you win, you lose 0.2 points for every soldier you have *beyond* your opponent's number there. Formulate a strategy to maximise your expected score.

Solution

This is a variant of the Colonel Blotto game with a *wastage penalty*. Let the castles have values

$$v_i = i, \qquad i = 1, 2, \ldots, 10,$$

so the total available value is

$$\sum_{i=1}^{10} i = 55.$$

Each player allocates exactly 100 soldiers. If at castle i you deploy x_i and your opponent deploys y_i, then your payoff from that castle is

- 0, if $x_i \le y_i$,
- $v_i - 0.2(x_i - y_i)$, if $x_i > y_i$.

Thus the total payoff is

$$\sum_{i:x_i>y_i} (v_i - 0.2(x_i - y_i)).$$

The question is what kind of allocation is optimal or strategically reasonable.

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1. Reformulating the payoff

Write the payoff on castle i as

$$u_i(x_i, y_i) = \mathbf{1}_{\{x_i>y_i\}}\, v_i \;-\; 0.2\,\mathbf{1}_{\{x_i>y_i\}}(x_i - y_i).$$

The first term is the usual Blotto reward for winning the castle. The second term penalizes overkill.

A key observation is that if you win a castle by a large margin, the net reward may become negative. Indeed, winning castle i by margin m yields net gain

$$v_i - 0.2m.$$

This is positive only when

$$m < 5v_i.$$

So, for example:

- castle 1 is worth winning only if the margin is at most 4,
- castle 10 is worth winning only if the margin is at most 49.

This already shows that piling huge numbers onto low-value castles is bad.

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2. A universal upper bound

Suppose you win a set S of castles. Your total score is

$$\sum_{i\in S} v_i - 0.2\sum_{i\in S}(x_i - y_i).$$

Since both players use exactly 100 soldiers,

$$\sum_{i=1}^{10} x_i = \sum_{i=1}^{10} y_i = 100.$$

Hence the total excess on castles you win is balanced by the total deficit on castles you lose:

$$\sum_{i:x_i>y_i} (x_i - y_i) = \sum_{i:x_i<y_i} (y_i - x_i).$$

In particular, the excess on won castles cannot be made arbitrarily small unless the two allocations are nearly identical. Therefore an optimal strategy must try to win valuable castles *narrowly* and avoid wasting soldiers on castles that are either lost anyway or won by too much.

The maximum raw point total is 55, but every won castle contributes a wastage deduction. So any realistic optimum must prioritize efficient wins on high-value castles.

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3. Marginal value of an extra soldier

At a given castle of value v, the only reason to add another soldier is to increase the chance of changing the outcome from loss/tie to narrow win. Once a castle is already likely to be won, extra soldiers have negative marginal effect because they increase the penalty without increasing the castle's value.

Thus, compared to the ordinary Blotto game, the penalty changes the objective from

- "win as many points as possible"

to

- "win valuable castles by the smallest possible margins."

This strongly favors a **spread-out allocation**, especially concentrated toward the higher-value castles, but not with large excesses.

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4. Which castles should be prioritized?

Since castle i gives value i, the high castles are much more efficient. If one had to choose which castles to contest seriously, one should favor castles 6 through 10, whose total value is

$$6 + 7 + 8 + 9 + 10 = 40,$$

and castles 7 through 10 already total

$$7 + 8 + 9 + 10 = 34.$$

By contrast, castles 1 through 4 total only

$$1 + 2 + 3 + 4 = 10.$$

Because an excess of only 5 soldiers costs 1 point, low-value castles are easily made unprofitable by even modest overdeployment. So rational play deemphasizes the small castles.

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5. Symmetry and equilibrium intuition

Because the two players are identical, no deterministic pure strategy can be secure: if one fixed allocation were known, the opponent could counter it by slightly overbidding on selected castles and underbidding elsewhere, winning more net value.

So the game's equilibrium must be **mixed**: one should randomize allocations.

The penalty implies that the support of the mixed strategy should avoid extreme concentrations. In equilibrium, the expected allocation to each castle should be increasing with its value, but not too steeply, because overcommitting to a castle is directly punished.

A standard heuristic in Blotto-type games is to allocate expected force proportional to value. Here that heuristic is even more compelling, since proportional allocation reduces the risk of large waste while giving more support to more valuable castles.

Since total value is 55, proportional allocation of 100 soldiers gives expected allocations

$$\frac{100}{55}(1,2,3,4,5,6,7,8,9,10) = \left(\frac{20}{11}, \frac{40}{11}, \frac{60}{11}, \frac{80}{11}, \frac{100}{11}, \frac{120}{11}, \frac{140}{11}, \frac{160}{11}, \frac{180}{11}, \frac{200}{11}\right).$$

Numerically, this is approximately

$$(1.82,\ 3.64,\ 5.45,\ 7.27,\ 9.09,\ 10.91,\ 12.73,\ 14.55,\ 16.36,\ 18.18).$$

A deterministic version close to this is

$$(2, 4, 5, 7, 9, 11, 13, 15, 16, 18),$$

which indeed sums to 100.

This kind of allocation has the right qualitative features:

- more soldiers on higher-value castles,
- no severe overconcentration,
- low-value castles not ignored completely, but not overfunded.

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6. Why this is preferable to extreme allocations

Consider putting, say, 30 soldiers on castle 10. If the opponent puts only 15, you win castle 10, but your net gain there is

$$10 - 0.2(30 - 15) = 10 - 3 = 7.$$

So 15 of those soldiers generated only 7 net points, while the remaining 15 soldiers might have helped contest several other castles. If the opponent puts even fewer, say 5, then the net gain is

$$10 - 0.2(25) = 5,$$

which is worse still.

By contrast, using those soldiers to obtain several narrow wins is more efficient. For example, winning castles 7 and 8 narrowly yields raw value

$$7 + 8 = 15,$$

and if the total excess is only, say, 4 soldiers across both, the penalty is only

$$0.2 \cdot 4 = 0.8,$$

so the net is

$$15 - 0.8 = 14.2,$$

far better than overkilling castle 10.

Thus narrow victories on multiple medium/high castles dominate huge margins on a single castle.

7. Exploitability: why no single allocation is "the" answer

The game is simultaneous and symmetric, so **every fixed allocation can be beaten**. If your opponent knew your allocation, they would beat you by one soldier on the castles that matter and concede the rest. In particular, the value-proportional allocation above is *not* safe: an opponent who concedes castles 1–3 (worth only 6 points) and moves those soldiers onto castles 4–10 wins almost everything that matters.

A simulation makes this concrete. Each allocation below was scored against 4,000 random opponents of each type, and head-to-head against the proportional allocation:

Your allocation (castles $1 \to 10$)	vs uniform random	vs value-prop. random	vs proportional
Proportional $(2, 4, 5, 7, 9, 11, 13, 15, 16, 18)$	32.4	22.6	0 (mirror)
Even $(10, \ldots, 10)$	22.4	10.9	10.4
Skip 1–3: $(0, 0, 0, 8, 10, 12, 15, 17, 18, 20)$	33.8	28.2	**46.8**
Top-heavy: $(0, 0, 0, 0, 0, 16, 18, 20, 22, 24)$	29.7	**31.1**	34.6

(Scores are your expected points after the 0.2-per-soldier penalty. Head-to-head, "skip 1–3" scores 46.8 against the proportional allocation, which scores only 3.8 back.)

Lessons:

- Conceding the cheap castles 1–3 and shifting those soldiers upward is best or near-best against every opponent type tested.
- The even split is the weakest: it overspends on low-value castles.
- The best allocation depends on the opponent field, so in equilibrium you must **randomize**; in practice you choose a strong allocation for the opponents you expect and add some noise.

8. Final strategic prescription

1. **Win narrowly.** Each soldier beyond what is needed to win a castle costs 0.2 points; a margin of 5 costs a full point.
2. **Concede or barely contest castles** 1–3 (only 6 of the 55 points).
3. **Allocate roughly in proportion to value on castles** 4–10, tilted toward 7–10.
4. **Use "odd" counts** (e.g. 17, 21 rather than 15, 20) to win ties against opponents who pick round numbers.
5. **Randomize** around this plan so that you cannot be exploited.

There is no single optimal pure allocation: every fixed plan can be exploited, and the equilibrium is a mixed strategy. Because of the wastage penalty, aim for narrow wins on the valuable castles: concede or lightly contest castles 1–3, weight castles 4–10 roughly by value (tilted toward 7–10), e.g. $(0, 0, 0, 8, 10, 12, 15, 17, 18, 20)$, avoid round numbers, and randomise around this plan.

How to answer in an interview

The interviewer is not looking for one "correct" allocation; they want to see that you recognise the game, reason about the opponent, and commit to a sensible plan. A strong answer, in about a minute:

1. **Name the game.** "This is a Colonel Blotto game with an overkill penalty. It is simultaneous and symmetric, so there is no single optimal allocation; any fixed plan can be exploited, and the equilibrium is a mixed strategy."
2. **Read the payoff.** "Winning castle i by m soldiers nets $i - 0.2m$, so a margin of 5 costs a whole point. I want narrow wins on valuable castles, not big margins."
3. **Reason about the opponent.** "My best allocation depends on what others do. Many people spread proportionally or evenly, so I can beat them by conceding the cheap castles and slightly outbidding them on the valuable ones."
4. **Commit to a concrete plan.** "Castles 1–3 are only 6 of the 55 points, so I concede or barely contest them and weight 4–10 by value, something like $(0, 0, 0, 8, 10, 12, 15, 17, 18, 20)$. I avoid round numbers so I win ties, and I randomize a little so I am not predictable."
5. **Show you would test and adapt.** "I would check the plan by simulating against likely opponent types; against a proportional allocation this one scores about 47 to 4. If I learned how the field actually plays, I would shift toward the best response to that."

One-line version: "It's Blotto with a penalty, so there's no single optimum; I'd concede castles 1–3, aim for narrow wins on 4–10 weighted by value, avoid round numbers, randomize, and adjust to the opponents I expect."

Part VII — Order Statistics and Geometric Probability

44 Continuous Order Statistics

Problem: A lottery ticket has 10 slots, each hiding an independent Uniform$[0,1]$ number. Your payout is the **maximum** of the ten. What is its fair price (the expected payout)?

Distribution of the Maximum

Write $M = \max(U_1, \dots, U_{10})$ for the payout. To average it we first describe its distribution with two standard objects: the CDF and the density.

The CDF $F_M(x)$. The *cumulative distribution function* is just the probability that M does not exceed a level x, written $F_M(x) = \mathbb{P}(M \le x)$. The maximum is $\le x$ exactly when *every* one of the ten numbers is $\le x$. Each U_i is Uniform$[0,1]$, so $\mathbb{P}(U_i \le x) = x$, and because the ten draws are independent the probabilities multiply:

$$F_M(x) = \mathbb{P}(U_1 \le x) \cdots \mathbb{P}(U_{10} \le x) = x^{10}, \qquad 0 \le x \le 1.$$

The density $f_M(x)$, and why it is $F_M'(x)$. The *probability density function* f_M measures how fast probability accumulates near x: $f_M(x)\,dx$ is the chance that M lands in the tiny interval $[x,\, x+dx]$. That chance is the CDF's increase across the interval,

$$f_M(x)\,dx \approx \mathbb{P}(x < M \le x + dx) = F_M(x+dx) - F_M(x),$$

and dividing by dx is precisely the definition of a derivative. So the density is the derivative of the CDF:

$$f_M(x) = F_M'(x) = \frac{d}{dx}\, x^{10} = 10x^9, \qquad 0 \le x \le 1.$$

(Equivalently $F_M(x) = \int_0^x f_M(t)\,dt$ — the CDF accumulates the density.)

The expectation as $\int x\, f_M(x)\,dx$. An average is a probability-weighted sum of values. Slice $[0,1]$ into slivers $[x,\, x+dx]$; the outcome $M \approx x$ carries probability $f_M(x)\,dx$, so weighting each value x by its probability and summing (integrating) gives

$$\mathbb{E}[M] = \int_0^1 x\, f_M(x)\,dx = \int_0^1 x \cdot 10x^9\,dx = 10\int_0^1 x^{10}\,dx = \frac{10}{11} \approx 0.909.$$

$$\mathbb{E}[\text{max of } 10\ U[0,1]] = \frac{10}{11} \approx \$0.91.$$

The General Result (and the Beta Distribution)

The same idea gives the mean of *any* order statistic, not just the maximum. Sort the n draws as $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$, so $X_{(k)}$ is the k-th smallest.

Density of the k-th smallest. For $X_{(k)}$ to sit right at x, three things must line up among the n draws at once:

- exactly one draw lands in the sliver $[x,\, x+dx]$ — there are n ways to pick which one, each with probability $f(x)\,dx = dx$ (a uniform has density 1);
- exactly $k-1$ of the remaining draws fall *below* x — probability x each;
- the other $n-k$ draws fall *above* x — probability $1-x$ each.

The number of ways to split the draws into these three groups is the multinomial coefficient $\frac{n!}{(k-1)!\,1!\,(n-k)!}$, so the density is

$$f_{(k)}(x) = \frac{n!}{(k-1)!\,(n-k)!}\, x^{\,k-1}(1-x)^{\,n-k}, \qquad 0 \le x \le 1.$$

What "Beta" means. Any density shaped like $x^{a-1}(1-x)^{b-1}$ on $[0,1]$, scaled to integrate to 1, is called the **Beta(a,b) distribution**. The scaling constant is the *Beta function*

$$B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\,dx = \frac{(a-1)!\,(b-1)!}{(a+b-1)!} \quad (\text{integer } a,b),$$

and the Beta(a,b) density $x^{a-1}(1-x)^{b-1}/B(a,b)$ has mean $\frac{a}{a+b}$. Matching exponents, $f_{(k)}$ is exactly Beta$(k,\, n-k+1)$.

The mean. Integrate x against this density; the Beta function evaluates the integral:

$$\mathbb{E}[X_{(k)}] = \frac{n!}{(k-1)!(n-k)!}\int_0^1 x^k(1-x)^{n-k}\,dx = \frac{n!}{(k-1)!(n-k)!}\cdot\frac{k!\,(n-k)!}{(n+1)!} = \frac{k}{n+1}.$$

For n i.i.d. Uniform$[0,1]$ draws, the k-th smallest is Beta$(k,\, n-k+1)$ with mean

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}.$$

So the n order-statistic means are $\frac{1}{n+1}, \frac{2}{n+1}, \ldots, \frac{n}{n+1}$ — equally spaced, cutting $[0,1]$ into $n+1$ equal expected gaps. In particular the maximum ($k=n$) is $\frac{n}{n+1}$ (here $\frac{10}{11}$, matching the direct computation) and the minimum ($k=1$) is $\frac{1}{n+1}$.

Concept Summary and Applications

- **Method:** for a maximum, work with the CDF ($F_M = F^n$); for a minimum, the survival function $(1-(1-F)^n)$; differentiate for the density, or integrate the tail $\mathbb{E}[M] = \int_0^\infty \mathbb{P}(M > x)\,dx$.
- **Uniform mnemonic:** picture dropping n random points on $[0,1]$. On average they slice the interval into $n+1$ gaps of *equal* expected length $\frac{1}{n+1}$, so the k-th smallest point sits at expected position $\frac{k}{n+1}$. This single picture recalls every order-statistic mean with no integration — in this problem the maximum is the 10-th of 10 points, at $\frac{10}{11}$, exactly our answer. (It is the same "$n+1$ equal expected pieces" idea as the broken-stick spacings in Section 48.)
- **Applications:** second-price / highest-bid auctions, extreme-value and tail-risk estimation, and record/max-drawdown statistics.

45 Discrete Order Statistics

Problem: What is the expected value of the **maximum** of two fair six-sided dice?

Use the CDF (Cumulative Distribution Function)

The **CDF** (*cumulative distribution function*) of the maximum is $\mathbb{P}(\max \le k)$, the probability that the max does not exceed k. The maximum is $\le k$ exactly when *both* dice are $\le k$; each die is $\le k$ with probability $k/6$, and the two are independent, so the probabilities multiply:

$$\mathbb{P}(\max \le k) = \left(\frac{k}{6}\right)^2.$$

To get the probability of *exactly* k, subtract consecutive CDF values — the event "max $= k$" is "max $\le k$" with "max $\le k-1$" removed:

$$\mathbb{P}(\max = k) = \mathbb{P}(\max \le k) - \mathbb{P}(\max \le k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

Then, averaging k over its probabilities,

$$\mathbb{E}[\max] = \sum_{k=1}^{6} k \cdot \frac{2k-1}{36} = \frac{1+6+15+28+45+66}{36} = \frac{161}{36} \approx 4.47.$$

$$\mathbb{E}[\max \text{ of two d6}] = \frac{161}{36} \approx 4.47.$$

By the mirror symmetry $k \mapsto 7-k$, $\mathbb{E}[\min] = \frac{91}{36} \approx 2.53$, and as a check $\mathbb{E}[\min] + \mathbb{E}[\max] = \frac{252}{36} = 7 = 2 \cdot 3.5$ (the two always sum to the two dice).

Concept Summary and Applications

- **Method:** for discrete maxima use $\mathbb{P}(\max \le k) = F(k)^n$; the tail sum $\mathbb{E}[X] = \sum_{k \ge 1} \mathbb{P}(X \ge k)$ is often fastest.
- **Symmetry:** for a distribution symmetric about its mean μ, $\mathbb{E}[\min] + \mathbb{E}[\max] = 2\mu$, and range $= \mathbb{E}[\max] - \mathbb{E}[\min]$.
- **Applications:** best-of / worst-of payoffs, range and spread statistics, and tournament or team-formation problems.

Additional Problem 1 — Expected Absolute Difference of Two d6

Problem: Roll two fair six-sided dice, D_1 and D_2. What is the expected value of the absolute difference $|D_1 - D_2|$ between the two rolls?

Why the answer is $\mathbb{E}[\max] - \mathbb{E}[\min]$. For any two numbers the absolute difference equals the larger minus the smaller: $|a-b| = \max(a,b) - \min(a,b)$. Applying this to the two dice and taking expectations (expectation is linear, so it splits across the subtraction),

$$\mathbb{E}\,|D_1 - D_2| = \mathbb{E}[\max] - \mathbb{E}[\min] = \frac{161}{36} - \frac{91}{36} = \frac{70}{36} = \frac{35}{18} \approx 1.94.$$

This is just the *range* of the two dice — the average gap between the higher and lower roll.

Verification by direct counting. We can confirm $\frac{35}{18}$ without the max/min values, by counting how many of the 36 ordered pairs realise each difference d. For $d \geq 1$, $|D_1 - D_2| = d$ occurs in $2(6-d)$ ways — $6-d$ pairs with $D_1 - D_2 = d$ and $6-d$ pairs with $D_2 - D_1 = d$ — while $d = 0$ contributes nothing to the sum. Hence

$$\mathbb{E}\,|D_1 - D_2| = \frac{1}{36}\sum_{d=1}^{5} d \cdot 2(6-d) = \frac{2}{36}\sum_{d=1}^{5} d(6-d) = \frac{2}{36}(6 \cdot 15 - 55) = \frac{2}{36} \cdot 35 = \frac{35}{18}. \checkmark$$

(using $\sum_{d=1}^{5} d = 15$ and $\sum_{d=1}^{5} d^2 = 55$), matching the range computation.

Additional Problem 2 — Absolute Difference of Two d30

Problem: Roll two fair 30-sided dice, each uniform on $\{1, 2, \ldots, 30\}$ and independent. What is the expected value of the absolute difference between the two outcomes?

We solve it for a general n-sided die and then set $n = 30$.

Step 1 — count pairs with a given difference. Let X, Y be the two rolls; there are n^2 equally likely ordered pairs. For a difference $d \geq 1$, $|X - Y| = d$ occurs when $X - Y = d$ (then $Y \in \{1, \ldots, n-d\}$, giving $n-d$ pairs) or $Y - X = d$ (another $n-d$ pairs), so $2(n-d)$ ordered pairs in total. Therefore

$$\mathbb{E}\,|X - Y| = \frac{1}{n^2}\sum_{d=1}^{n-1} d \cdot 2(n-d) = \frac{2}{n^2}\sum_{d=1}^{n-1} d(n-d).$$

Step 2 — evaluate the sum. Write $d(n-d) = n\,d - d^2$ and use the standard sums $\sum_{d=1}^{n-1} d = \frac{(n-1)n}{2}$ and $\sum_{d=1}^{n-1} d^2 = \frac{(n-1)n(2n-1)}{6}$:

$$\sum_{d=1}^{n-1} d(n-d) = n \cdot \frac{(n-1)n}{2} - \frac{(n-1)n(2n-1)}{6} = (n-1)n\Big(\frac{n}{2} - \frac{2n-1}{6}\Big).$$

Put the bracket over 6: $\frac{3n-(2n-1)}{6} = \frac{n+1}{6}$, so the sum is $\frac{(n-1)n(n+1)}{6} = \frac{n(n^2-1)}{6}$.

Step 3 — assemble.

$$\mathbb{E}\,|X - Y| = \frac{2}{n^2} \cdot \frac{n(n^2-1)}{6} = \frac{n^2-1}{3n}.$$

> For two i.i.d. uniforms on $\{1, \ldots, n\}$: $\quad \mathbb{E}\,|X - Y| = \dfrac{n^2-1}{3n}$.

For $n = 30$: $\dfrac{30^2-1}{3 \cdot 30} = \dfrac{899}{90} \approx 9.99$. (The d6 value $\frac{35}{18}$ is the same formula at $n = 6$.)

$\mathbb{E}\,|X-Y| = \dfrac{899}{90} \approx 9.99$ **for two 30-sided dice.**

Additional Problem 3 — Larger of A/B and B/A

Problem: You roll two fair six-sided dice, die A and die B (uniform on $\{1,\dots,6\}$, independent). After seeing both, you may choose to be paid either A/B or B/A dollars, whichever is larger. How much should you be willing to pay to play?

Since you always take the larger ratio, the payoff is

$$X = \max\Big(\frac{A}{B}, \frac{B}{A}\Big) = \frac{\max(A,B)}{\min(A,B)} \geq 1,$$

and the fair price is the expected value $\mathbb{E}[X]$.

Step 1 — organise the 36 outcomes. Of the 36 equally likely ordered pairs (A,B): the 6 ties $A=B$ give $X=1$; and each unordered pair $\{a,b\}$ with $a<b$ appears *twice* (as (a,b) and (b,a)), each time giving payoff b/a. So

$$\mathbb{E}[X] = \frac{1}{36}\Big(6\cdot 1 + 2\sum_{1\leq a<b\leq 6}\frac{b}{a}\Big).$$

Step 2 — the double sum, grouped by the smaller value a.

$$\begin{aligned}
&a=1: \tfrac{2}{1}+\tfrac{3}{1}+\tfrac{4}{1}+\tfrac{5}{1}+\tfrac{6}{1} = 20, && a=2: \tfrac{3}{2}+\tfrac{4}{2}+\tfrac{5}{2}+\tfrac{6}{2} = 9,\\
&a=3: \tfrac{4}{3}+\tfrac{5}{3}+\tfrac{6}{3} = 5, && a=4: \tfrac{5}{4}+\tfrac{6}{4} = \tfrac{11}{4},\\
&a=5: \tfrac{6}{5} = \tfrac{6}{5}.
\end{aligned}$$

Adding over the common denominator 20: $20+9+5+\frac{11}{4}+\frac{6}{5} = \frac{400+180+100+55+24}{20} = \frac{759}{20}$.

Step 3 — take the expectation.

$$\mathbb{E}[X] = \frac{1}{36}\Big(6+2\cdot\frac{759}{20}\Big) = \frac{1}{36}\Big(6+\frac{759}{10}\Big) = \frac{1}{36}\cdot\frac{819}{10} = \frac{819}{360} = \frac{91}{40} = 2.275.$$

$\mathbb{E}\Big[\max\big(\frac{A}{B},\frac{B}{A}\big)\Big] = \dfrac{91}{40} = 2.275$**, so you should pay up to about \$2.28 per play.**

46 Broken Stick Problem

Problem: A stick is broken into 3 pieces at two randomly chosen points. What is the probability that the three pieces form a triangle?

Triangle Inequality Condition

Three segments a, b, c form a triangle iff each is shorter than the sum of the other two. For pieces of a unit stick ($a + b + c = 1$) this reduces to:

$$a < \tfrac{1}{2}, \qquad b < \tfrac{1}{2}, \qquad c < \tfrac{1}{2}.$$

Solution: Two Independent Uniform Break Points

Choose break points $x, y \sim \text{Uniform}(0, 1)$ independently. The three piece lengths are

$$a = \min(x, y), \quad b = |x - y|, \quad c = 1 - \max(x, y).$$

The triangle conditions on the unit-square sample space become:

$$\begin{aligned} \min(x, y) &< \tfrac{1}{2}, \\ |x - y| &< \tfrac{1}{2}, \\ \max(x, y) &> \tfrac{1}{2}. \end{aligned}$$

The favorable region consists of exactly **two triangles** in the unit square (Figure 1):

- **Top-left:** $x \in (0, \tfrac{1}{2}),\ y \in (\tfrac{1}{2},\ x + \tfrac{1}{2})$
- **Bottom-right:** $x \in (\tfrac{1}{2}, 1),\ y \in (x - \tfrac{1}{2},\ \tfrac{1}{2})$

Each triangle has area $\frac{1}{8}$, giving total favorable area $\frac{1}{4}$.

$$P(\textbf{triangle}) = \frac{1}{4}$$

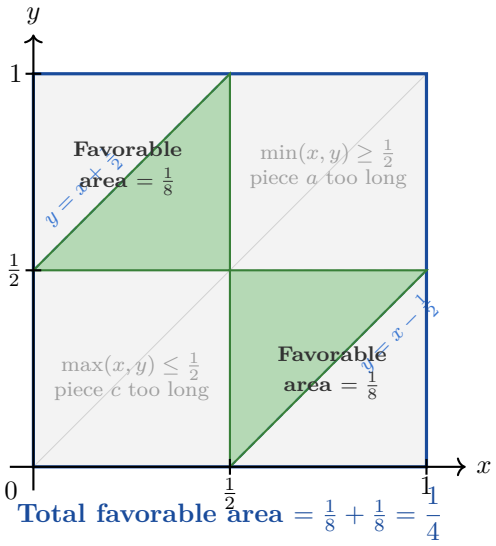


Figure 1: Geometric solution. The sample space is the unit square with break-point coordinates (x, y). **Green triangles** satisfy all three triangle inequalities. The dashed blue lines are the boundaries $|x - y| = \frac{1}{2}$; outside them the middle piece $b = |x - y|$ exceeds $\frac{1}{2}$. The bottom-left and top-right white squares fail because an end piece exceeds $\frac{1}{2}$.

47 Geometric Probability: Random Points on a Circle

When points are dropped uniformly at random on a circle, many questions reduce to one: *do all the points fit inside some half of the circle?* The trick that makes this easy is to stop thinking about "some semicircle" (infinitely many choices) and instead ask *which point starts it* (only n choices). This turns a continuous geometry problem into a sum of n simple, disjoint events.

Representative Problem — All n Points on a Semicircle (Q456)

Problem: You pick n points independently and uniformly at random on the circumference of a circle. What is the probability $P(n)$ that all n points lie within some semicircle? Give a closed form in n.

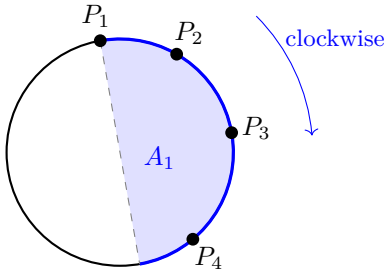
Step 1: Why "some semicircle" is hard, and how to fix it

A semicircle is an arc of half the circumference (180°). There are infinitely many semicircles (one for each starting angle), and we are allowed to choose the semicircle *after* seeing the points. So we cannot simply say "each point lands in the top half with probability $\frac{1}{2}$" — that would compute the probability for one *fixed* semicircle, $(\frac{1}{2})^n$, which is far too small.

Key observation. If the points fit in some semicircle, slide (rotate) that semicircle clockwise until its starting edge hits a point. It still contains all the points, and now it *starts at one of the points.* So:

$$\text{all points fit in some semicircle} \iff \text{for some point } i\text{, all the others lie in the half-circle going clockwise from point } i.$$

Now there are only n candidate semicircles — one starting at each point.



Event A_1: starting at P_1 and going clockwise for half a turn (shaded) covers all the other points.

Step 2: Define the n events

For each point $i = 1, \ldots, n$, let

$$A_i = \{\text{all the other } n-1 \text{ points lie in the half-circle going clockwise from point } i\}.$$

By Step 1, the event we want is $A_1 \cup A_2 \cup \cdots \cup A_n$.

Step 3: Probability of one event

Fix point i wherever it lands. The half-circle clockwise from it is a fixed arc of exactly half the circumference. Each other point is uniform and independent of point i, so it falls in that arc with

probability $\frac{1}{2}$. There are $n-1$ other points, all independent:

$$P(A_i) = \left(\tfrac{1}{2}\right)^{n-1}.$$

Step 4: The events are disjoint

Suppose two of them, A_i and A_j, both happened. A_i says point j is at most half a turn clockwise from point i; A_j says point i is at most half a turn clockwise from point j. Going clockwise from i to j and then from j back to i is exactly one full turn, so both distances must be exactly half a turn — the two points are diametrically opposite. That has probability 0. Hence at most one A_i occurs, and probabilities simply add (no inclusion–exclusion correction is needed):

$$P(n) = P(A_1) + \cdots + P(A_n) = n\left(\tfrac{1}{2}\right)^{n-1}.$$

Intuitively, A_i says "point i is the *first* point (clockwise) of the cluster", and a cluster has only one first point.

Step 5: Sanity checks

n	1	2	3	4	5	6
$P(n) = \frac{n}{2^{n-1}}$	1	1	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{5}{16}$	$\frac{3}{16}$

- $n = 1, 2$: one or two points always fit in a semicircle (two points are at most half a turn apart). ✓
- $n = 3$: $\frac{3}{4}$, which matches the direct calculation in the Additional Problem below. ✓
- A computer simulation (200,000 trials each) gives 0.750, 0.500, 0.312 for $n = 3, 4, 5$. ✓

$$\boxed{P(n) = \frac{n}{2^{n-1}}}$$

— n disjoint choices for the clockwise first point, each with probability $(\frac{1}{2})^{n-1}$.

Concept Summary and Applications

n uniform random points on a circle.

$$P(\text{all in some semicircle}) = \frac{n}{2^{n-1}}, \qquad P(\text{centre inside their convex hull}) = 1 - \frac{n}{2^{n-1}}.$$

Arc of fraction $a \le \frac{1}{2}$ of the circle: $P(\text{all in some arc of length } a) = n\,a^{n-1}$ (same argument; the events stay disjoint as long as $a \le \frac{1}{2}$).
Method: replace "there exists an arc" by "some point *starts* the arc" — n disjoint events of equal probability.

- **Trap:** fixing a semicircle in advance gives $(\frac{1}{2})^n$; the factor n (and the exponent $n-1$) comes from letting the points choose the semicircle.

- **Semicircle ⇔ centre:** the points fail to surround the centre exactly when they all fit in a half-circle — this links the formula to "does the polygon contain the centre?" questions (Additional Problem).
- **Equivalent "gap" view:** all points fit in a semicircle iff the largest gap between neighbouring points is at least half the circle — useful for simulation and for stick-breaking problems.
- **Higher dimensions:** on a sphere, n random points lie in some hemisphere with probability $2^{-(n-1)}\left[\binom{n-1}{0}+\binom{n-1}{1}+\binom{n-1}{2}\right]$ (Wendel's theorem); for 4 points the tetrahedron contains the centre with probability $\frac{1}{8}$.
- **Applications:** stability of random tripods, "is the origin in the convex hull" tests in statistics, clustering of random phases or arrival times on a clock.

Additional Problem 1 — Will the Three-Legged Table Tip Over? (Q338)

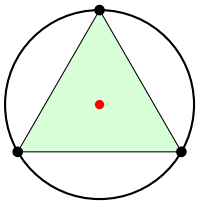
Problem: A round table top has three legs attached at three points chosen independently and uniformly at random on its rim. The table's weight acts at the centre. What is the probability that the table stands without tipping over?

Step 1: Turn "does not tip" into geometry

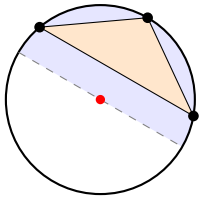
A table resting on legs is stable when its centre of mass lies *over the region the legs enclose* (the triangle formed by the three leg points). If the centre is outside that triangle, the weight creates a turning force about the nearest edge of the triangle and the table tips. So

$$\text{table stands} \iff \text{the centre of the circle lies inside the triangle } P_1P_2P_3.$$

Step 2: Centre inside the triangle ⟺ points not on a semicircle



spread out: centre inside (stands)



all on one semicircle: centre outside (tips)

- If all three points lie on one semicircle, the diameter bounding that semicircle has all three points on one side, so the whole triangle lies on that side and the centre (on the diameter) is not strictly inside it.
- If the points do *not* fit in any semicircle, every side of the triangle cuts off an arc smaller than a semicircle, so the centre lies on the same side of each edge as the opposite vertex — i.e. inside the triangle.

Therefore

$$P(\text{stands}) = 1 - P(\text{all three on some semicircle}).$$

Step 3: Apply the formula with $n = 3$

By the representative problem, $P(3) = \dfrac{3}{2^2} = \dfrac{3}{4}$: each of the 3 points can be the clockwise first point of the cluster, with probability $(\frac{1}{2})^2 = \frac{1}{4}$ each, and these cases are disjoint. Hence

$$P(\text{stands}) = 1 - \tfrac{3}{4} = \tfrac{1}{4}.$$

Step 4: A direct check without the formula

Rotate so P_1 is at angle 0. Let θ be the (shorter) angle between P_1 and P_2; it is uniform on $[0, \pi]$. The triangle contains the centre exactly when P_3 lies in the arc *diametrically opposite* the arc P_1P_2 — the arc between the antipodes of P_1 and P_2, which has length θ. So, given θ,

$$P(\text{stands} \mid \theta) = \frac{\theta}{2\pi}, \qquad P(\text{stands}) = E\Big[\frac{\theta}{2\pi}\Big] = \frac{\pi/2}{2\pi} = \frac{1}{4}. \checkmark$$

Step 5: More legs

With n legs at random rim points, the table stands iff the centre is inside their polygon, i.e. iff the legs are not all on one semicircle:

$$P(\text{stands}) = 1 - \frac{n}{2^{n-1}}: \qquad n=3: \tfrac{1}{4}, \quad n=4: \tfrac{1}{2}, \quad n=5: \tfrac{11}{16}, \quad n=6: \tfrac{13}{16}.$$

More legs make the table safer, but even 4 random legs give only a coin flip.

The table stands iff the centre lies inside the triangle, iff the three legs are *not* all on one semicircle:

$$P(\textbf{stands}) = 1 - \frac{3}{2^2} = \boxed{\tfrac{1}{4}}.$$

48 Expected Length of the Longest Piece

Problem: A unit stick is broken at two independent uniform points. What is the expected length of the **longest** of the three pieces?

Setup

Order the break points as $0 \le U \le V \le 1$; the pieces are

$$X = U, \quad Y = V - U, \quad Z = 1 - V, \qquad M = \max\{X, Y, Z\}.$$

The ordered pair (U, V) is uniform on the triangle $0 < u < v < 1$ (area $\frac{1}{2}$), so (X, Y) is uniform on $\{x > 0,\, y > 0,\, x + y < 1\}$ with **density** 2 (the constant f with $f \cdot \frac{1}{2} = 1$). Hence $\mathbb{P}((X, Y) \in A) = 2\,\text{area}(A)$.

Tail Probability of the Maximum

The longest piece is always at least the average, so $M \ge \frac{1}{3}$ and $\mathbb{P}(M > m) = 1$ for $m < \frac{1}{3}$. For larger m the geometry splits into two regimes.

Case $\frac{1}{3} \le m \le \frac{1}{2}$ — compute $\mathbb{P}(M \le m)$ directly. Here $\{M \le m\}$ means *all three* pieces are $\le m$: $X \le m$, $Y \le m$, and the third piece $Z = 1 - X - Y \le m$, i.e. $X + Y \ge 1 - m$. In the (X, Y) plane these three cuts carve out a small triangle with corners

$$(m, m), \qquad (1{-}2m,\, m), \qquad (m,\, 1{-}2m).$$

Why the side has length $3m - 1$. Take the horizontal leg: it runs at height $y = m$ from the corner $(1{-}2m,\, m)$ across to (m, m), so its length is

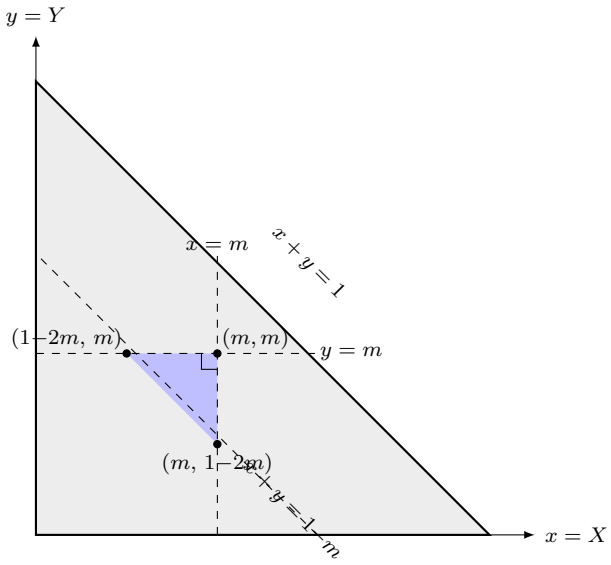
$$m - (1 - 2m) = 3m - 1.$$

By the $x \leftrightarrow y$ symmetry the vertical leg from $(m, 1{-}2m)$ up to (m, m) has the same length $3m - 1$, and the right angle sits at (m, m). So the region is a right isosceles triangle with both legs equal to $3m - 1$ (equivalently $|1 - 3m|$), giving

$$\text{area} = \tfrac{1}{2}(3m - 1)^2.$$

Multiplying by the density 2,

$$\mathbb{P}(M \le m) = 2 \cdot \tfrac{1}{2}(3m - 1)^2 = (3m - 1)^2, \qquad \mathbb{P}(M > m) = 1 - (3m - 1)^2.$$



Drawn at $m = \frac{2}{5}$: the sample space is the large triangle (density 2), and $\{M \le m\}$ is the shaded right-isosceles triangle — right angle at (m, m), legs $3m - 1$; its area $\frac{1}{2}(3m-1)^2$ times 2 gives $\mathbb{P}(M \le m) = (3m-1)^2$.

Case $\frac{1}{2} \le m \le 1$ — compute the tail $\mathbb{P}(M > m)$ directly. Now it is easier to go after the tail. The event $\{M > m\}$ means *at least one* piece exceeds m, i.e. the union $\{X > m\} \cup \{Y > m\} \cup \{Z > m\}$.

Why the factor 3 (disjointness). When $m \ge \frac{1}{2}$, two pieces **cannot both** exceed m: if two of them were each larger than $m \ge \frac{1}{2}$, their lengths alone would already sum to more than 1, which is impossible for pieces of a unit stick. So the three events are **mutually exclusive**, and the probability of their union is simply the sum of the three (no overlaps to subtract):

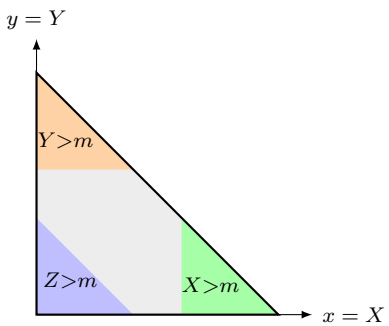
$$\mathbb{P}(M > m) = \mathbb{P}(X > m) + \mathbb{P}(Y > m) + \mathbb{P}(Z > m) = 3\,\mathbb{P}(X > m).$$

Why each term is the square $(1-m)^2$. The event $\{X > m\}$ is the corner triangle $\{x > m,\ y > 0,\ x + y < 1\}$, with corners $(m, 0)$, $(1, 0)$, and $(m,\ 1-m)$. Its horizontal leg (from $x = m$ to $x = 1$ along $y = 0$) has length $1 - m$, and its vertical leg (from $(m, 0)$ up to the hypotenuse $x + y = 1$ at $(m, 1-m)$) also has length $1 - m$. So it is a right triangle with both legs $1 - m$, area $\frac{1}{2}(1-m)^2$; times the density 2,

$$\mathbb{P}(X > m) = 2 \cdot \tfrac{1}{2}(1-m)^2 = (1-m)^2.$$

The **square** is just the area of a right triangle whose two legs both shrink like $1 - m$ as m grows toward 1. Hence

$$\mathbb{P}(M > m) = 3(1-m)^2.$$



Drawn at $m = 0.6$ (so $1 - m = 0.4$): the three events are the three *disjoint* corner triangles, each a right triangle with legs $1 - m$, area $\frac{1}{2}(1-m)^2$ and probability $(1-m)^2$. They do not touch (because $m \geq \frac{1}{2}$), so the tail is their sum $3(1-m)^2$.

Collecting,

$$\mathbb{P}(M > m) = \begin{cases} 1, & 0 \leq m < \frac{1}{3}, \\ 1 - (3m-1)^2, & \frac{1}{3} \leq m \leq \frac{1}{2}, \\ 3(1-m)^2, & \frac{1}{2} \leq m \leq 1. \end{cases}$$

Compute the Expectation

Using $\mathbb{E}[M] = \int_0^1 \mathbb{P}(M > m)\, dm$:

$$\underbrace{\int_0^{1/3} 1\, dm}_{1/3} + \underbrace{\int_{1/3}^{1/2} (6m - 9m^2)\, dm}_{[3m^2 - 3m^3]_{1/3}^{1/2} = 11/72} + \underbrace{\int_{1/2}^{1} 3(1-m)^2\, dm}_{1/8} = \frac{1}{3} + \frac{11}{72} + \frac{1}{8} = \frac{44}{72} = \frac{11}{18}.$$

$$\mathbb{E}[\textbf{longest piece}] = \frac{11}{18} \approx 0.611.$$

The One-Line Shortcut (Uniform Spacings)

Breaking a unit stick at $n-1$ uniform points produces n pieces called **uniform spacings**. Sort them from smallest to largest, $D_{(1)} \leq D_{(2)} \leq \cdots \leq D_{(n)}$. There is a clean formula for the expected value of the j-th smallest,

$$\mathbb{E}[D_{(j)}] = \frac{1}{n} \sum_{i=1}^{j} \frac{1}{n - i + 1}.$$

Apply to $n = 3$. Plug $n = 3$ into the boxed formula. The inner terms $\frac{1}{n-i+1}$ run through $\frac{1}{3}, \frac{1}{2}, 1$, and each partial sum is scaled by $\frac{1}{n} = \frac{1}{3}$:

$$\underbrace{\mathbb{E}[D_{(1)}]}_{\text{shortest}} = \frac{1}{3}\left(\frac{1}{3}\right) = \frac{1}{9}, \qquad \underbrace{\mathbb{E}[D_{(2)}]}_{\text{middle}} = \frac{1}{3}\left(\frac{1}{3} + \frac{1}{2}\right) = \frac{5}{18}, \qquad \underbrace{\mathbb{E}[D_{(3)}]}_{\text{longest}} = \frac{1}{3}\left(\frac{1}{3} + \frac{1}{2} + 1\right) = \frac{11}{18}.$$

Reading the three sums: the shortest keeps only the first term; the middle adds the second; the longest adds the third as well. As a check they must cover the whole stick, and indeed $\frac{1}{9} + \frac{5}{18} + \frac{11}{18} = \frac{2+5+11}{18} = 1$. ✓

The boxed formula itself is derived next; the derivation can be skipped.

Explanation of the One-Line Formula (Can Skip If Too Complex)

Step 1 — spacings are normalised exponentials. A standard and useful fact: if $T_1, \ldots, T_n$ are independent Exponential(1) random variables and $S = T_1 + \cdots + T_n$, then the scaled vector $(\frac{T_1}{S}, \ldots, \frac{T_n}{S})$ has *exactly* the same distribution as the n uniform spacings. Intuitively, dropping n independent "interarrival" times and rescaling so they sum to 1 reproduces the stick breaks. In particular the pieces sorted by size correspond to the T_i sorted by size, and since the average spacing is $\frac{1}{n}$ while the average T_i is 1,

$$\mathbb{E}[D_{(j)}] = \frac{1}{n}\,\mathbb{E}[T_{(j)}],$$

where $T_{(j)}$ is the j-th smallest of the exponentials.

Step 2 — exponential order statistics via memorylessness. Here the magic of the exponential distribution does the work. Wait for the *first* (smallest) of n independent Exp(1) clocks to ring. The minimum of n rate-1 exponentials is itself exponential with rate n, so its mean waiting time is $\frac{1}{n}$. Because the exponential is **memoryless**, once the first clock rings the remaining $n-1$ clocks are "as good as new"; the wait for the *next* ring is the minimum of $n-1$ fresh Exp(1) clocks, mean $\frac{1}{n-1}$. Continuing, the gap between the $(i-1)$-th and i-th rings has mean $\frac{1}{n-i+1}$. Adding these gaps up to the j-th ring,

$$\mathbb{E}[T_{(j)}] = \sum_{i=1}^{j} \frac{1}{n-i+1}.$$

Dividing by n (Step 1) gives the boxed formula. (Substituting $n = 3$ recovers the shortest/middle/longest values found above.)

Concept Summary and Applications

- **Two routes:** geometry (uniform density on the simplex + tail integration) and the spacings formula. The longest piece averages about 61% of the stick — far above the naive $\frac{1}{3}$, because randomness usually produces one dominant piece.
- **Density weighting:** on the (X, Y) simplex, probability = 2× area; forgetting the factor 2 is the classic error.
- **Applications:** load balancing and max-bucket occupancy, spacing statistics for arrivals on an interval, and coverage/gap problems.

Part VIII — Estimation, Logic and Brain Teasers

49 Fermi Estimation

A *Fermi estimate* is a quick, order-of-magnitude answer to a question nobody expects you to know exactly ("how much water is on Earth?"). Interviewers use these to see how you **structure** a problem: break the unknown into pieces you can estimate from everyday facts, multiply them together, state your assumptions, and sanity-check the result. Being within a factor of 2–3 is usually a good answer; the reasoning matters more than the final digit.

Representative Problem — How Many Cubic Metres of Water Are on Earth? (Q388)

Problem: Estimate the number of cubic metres of water on Earth.

Step 1: Decompose

Almost all of Earth's water is in the oceans, and an ocean is roughly a thin layer of water over part of the surface:

$$V \approx (\text{Earth's surface area}) \times (\text{fraction covered by ocean}) \times (\text{average ocean depth}).$$

Each factor can be estimated from facts most people know.

Step 2: Earth's surface area

Earth's radius is about 6,400 km $= 6.4 \times 10^6$ m (memory aid: the equator is about 40,000 km, and $40{,}000/2\pi \approx 6{,}400$). The surface of a sphere is $4\pi R^2$:

$$A = 4\pi(6.4 \times 10^6)^2 \approx 12.6 \times 4.1 \times 10^{13} \approx 5 \times 10^{14} \text{ m}^2.$$

Step 3: Fraction covered by ocean

About 70% of the surface is ocean (a standard fact; a globe is mostly blue):

$$A_{\text{ocean}} \approx 0.7 \times 5 \times 10^{14} = 3.5 \times 10^{14} \text{ m}^2.$$

Step 4: Average depth — the most uncertain factor

The deepest trenches reach about 11 km, but most of the ocean floor (the abyssal plains) lies at 4–5 km, and the shallow continental shelves pull the average down. Take an average depth of about 4 km $= 4 \times 10^3$ m (the measured value is ≈ 3.7 km).

Step 5: Multiply

$$V \approx 3.5 \times 10^{14} \times 4 \times 10^3 = 1.4 \times 10^{18} \text{ m}^3.$$

Step 6: Sanity checks and the rest of the water

- **Thin layer?** 4 km of water on a planet of radius 6,400 km is a layer about $\frac{1}{1600}$ of the radius — the "thin shell" model is fine.
- **Other water:** ice caps, groundwater, lakes and rivers add only a few percent; the atmosphere holds far less. They do not change the order of magnitude.
- **Where is the error?** Surface area and ocean fraction are known well; the depth guess (3–5 km) dominates the uncertainty, giving a range of roughly 1–2 $\times 10^{18}$ m^3. (The accepted value for the oceans is about 1.3×10^{18} m^3.)

$V \approx 4\pi R^2 \times 0.7 \times d \approx 5 \times 10^{14} \times 0.7 \times 4 \times 10^3 \approx \boxed{1.4 \times 10^{18}\ \mathbf{m^3}}$ **(order 10^{18}), dominated by the oceans; the average depth is the main source of uncertainty.**

Concept Summary and Applications

Fermi recipe.

1. **Decompose** the unknown into a product (or sum) of quantities you can estimate: e.g. volume = area × depth, mass = volume × density, flow = stock ÷ lifetime.
2. **Anchor** each factor to a fact you know (Earth's radius 6,400 km, US population 330 million, ...).
3. **Work in powers of 10** and round freely; keep one significant figure.
4. **Cross-check** with an independent route (top-down vs. bottom-up).
5. **State a range**, noting which factor dominates the uncertainty.

Averaging a range: for a quantity between L and H spanning orders of magnitude, use the geometric mean $\sqrt{LH}$.

- **Stock and flow:** for steady-state systems, annual flow ≈ stock/lifetime (planes bought per year, cars sold, piano tuners needed).
- **Errors partly cancel:** independent over- and under-estimates in a product tend to offset, which is why Fermi estimates are often better than any single factor.
- **Say the assumptions out loud:** interviewers grade the structure, the reasonableness of each number, and the sanity checks.
- **Applications:** market sizing, capacity planning, back-of-the-envelope trade sizing, and checking whether a model's output is plausible.

Additional Problem 1 — Mass of the Earth and Weight of Kilimanjaro (Q542, Q611)

Problems: (a) Estimate the mass of the Earth. (b) Estimate the weight of Mount Kilimanjaro.

Both use the same decomposition: mass = volume × density (and, if "weight" means force, weight = mass × g).

(a) Route 1: volume × density

Volume of a sphere of radius $R = 6.4 \times 10^6$ m:

$$V = \tfrac{4}{3}\pi R^3 \approx 4.2 \times (6.4 \times 10^6)^3 \approx 4.2 \times 2.6 \times 10^{20} \approx 1.1 \times 10^{21} \text{ m}^3.$$

Surface rock has density about 3,000 kg/m^3 (three times water), giving 3.3×10^{24} kg. But Earth has an iron core, so the average density is higher — about 5,500 kg/m^3:

$$M \approx 1.1 \times 10^{21} \times 5.5 \times 10^3 \approx 6 \times 10^{24} \text{ kg}.$$

Lesson: using only surface rock underestimates by a factor ≈ 2; knowing that the interior is denser fixes it.

(a) Route 2: from gravity (if you know G)

An object of mass m on the surface feels the force $\frac{GMm}{R^2}$, which is its weight mg. Cancel m:

$$M = \frac{gR^2}{G} = \frac{9.8 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} \approx \frac{3.98 \times 10^{14}}{6.67 \times 10^{-11}} \approx 6.0 \times 10^{24} \text{ kg}.$$

The two routes agree.

(b) Kilimanjaro: model it as a cone

Kilimanjaro is a free-standing volcano whose summit is about 5.9 km above sea level, rising roughly 4–5 km above the surrounding plain, with a base some 30–40 km across. Model it as a cone with height $h \approx 4.5$ km and base radius $r \approx$ 15–20 km:

$$V = \tfrac{1}{3}\pi r^2 h \approx \tfrac{1}{3} \times 3.14 \times (1.5\text{–}2 \times 10^4)^2 \times 4.5 \times 10^3 \approx 1\text{–}2 \times 10^{12} \text{ m}^3.$$

Volcanic rock has density about 2,700 kg/m^3:

$$m \approx (1\text{–}2 \times 10^{12}) \times 2.7 \times 10^3 \approx 3\text{–}5 \times 10^{15} \text{ kg},$$

and its weight (force) is $mg \approx 3\text{–}5 \times 10^{16}$ N. The base radius is the most uncertain input and enters squared, so it dominates the range.

(a) Earth: $\frac{4}{3}\pi R^3 \times 5{,}500$ kg/m^3 $\approx 6 \times 10^{24}$ kg (also gR^2/G). (b) Kilimanjaro: cone of height ≈ 4.5 km and base radius $\approx$ 15–20 km, volume $\approx 10^{12}$ m^3, mass of a few $\times 10^{15}$ kg (weight a few $\times 10^{16}$ N).

Additional Problem 2 — Commercial Planes Bought in the US per Year (Q392)

Problem: Estimate the number of commercial airplanes purchased in the United States each year.

Method 1 — stock ÷ lifetime (bottom line)

A steady fleet must replace about $\frac{\text{fleet size}}{\text{lifetime}}$ planes per year just to stand still, plus any growth.

- **Fleet size:** major, low-cost, regional and cargo airlines operate roughly 7,000–8,000 aircraft; take 7,500.
- **Lifetime:** airliners fly for about 20–25 years.
- **Growth:** about 1–2% a year, i.e. ≈ 100 planes.

$$\frac{7{,}500}{25} + 100 \approx 300 + 100 = 400 \text{ per year.}$$

Method 2 — top-down cross-check

Boeing and Airbus deliver roughly 1,000–1,500 jets a year worldwide; the US is perhaps a quarter of the market: $0.25 \times 1{,}400 \approx 350$.

Method 3 — bottom-up by airline

Four big carriers with ~ 900 planes each replace about $900/25 \approx 36$ each, ≈ 45 with growth: ~ 180. Add mid-size and low-cost carriers (~ 45), cargo airlines (~ 40) and regionals (~ 70): total ~ 340.

Why they agree

All three re-slice the same picture — a $\sim 7{,}500$-plane fleet on a ~ 25-year cycle — from different directions, which is exactly what a good cross-check should do. Purchases are lumpy (airlines order in batches), so any single year varies.

About 400 per year (defensible range 300–500): fleet 7,500 ÷ 25-year life ≈ 300 replacements plus ≈ 100 for growth; cross-checked by global deliveries × US share and an airline-by-airline count.

Additional Problem 3 — Copies of *Infinite Jest* Read Cover to Cover (Q408, part 2)

Problem: Estimate the number of copies of David Foster Wallace's novel *Infinite Jest* (about 1,000 pages, with hundreds of endnotes) that have been read cover to cover.

Step 1: Decompose

$$\#\text{read cover to cover} \approx (\text{copies sold}) \times (\text{fraction of copies finished}).$$

Step 2: Copies sold

A famous, much-discussed literary novel, in print for about three decades with steady sales (and a surge of interest after the author's death). A plausible cumulative figure is 2–4 million; take 3×10^6.

(A sanity bound: far below a global blockbuster's tens of millions, far above an obscure novel's few thousand.)

Step 3: Completion rate

The book is long and difficult, and many copies are bought aspirationally or as gifts. For an ordinary novel perhaps half of buyers finish; here a plausible range is 10–30%; take 20%. Used copies, library copies and lending can mean one copy is read by several people, partly offsetting the low rate.

Step 4: Multiply and give a range

$$3 \times 10^6 \times 0.2 = 6 \times 10^5.$$

Low end: $2 \times 10^6 \times 0.1 = 2 \times 10^5$; high end: $4 \times 10^6 \times 0.3 = 1.2 \times 10^6$. The geometric mean of the ends, $\sqrt{2 \times 10^5 \times 1.2 \times 10^6} \approx 5 \times 10^5$, agrees with the central estimate.

About 600,000 copies (≈ 3 million sold $\times \approx 20\%$ finished); a defensible range is 300,000–1,000,000. The completion rate is the most uncertain factor.

50 Information-Theoretic Guessing: Spending One Bit Well

A yes/no answer carries at most **one bit** of information. These puzzles ask how to spend that single bit. The best question depends entirely on how you are scored, so the first step is always to write the payoff down precisely.

Representative Problem — Guess 100 Coins with One Question (Q372)

Problem: A coin is flipped 100 times. You may ask one yes/no question about the sequence of results, and then you must guess the entire sequence. You earn \$1 for each coin you guess correctly and lose \$1 for each coin you guess wrong. Find a good strategy and compute its expected return.

Step 1 — write the payoff in terms of the number correct. The hidden sequence is $X = (X_1, \dots, X_{100})$, all 2^{100} sequences equally likely. Since $(\#\text{correct}) + (\#\text{wrong}) = 100$,

$$\text{return} = (\#\text{correct}) - (\#\text{wrong}) = 2\,(\#\text{correct}) - 100.$$

Where the factor 2 comes from: the payoff is a $+1/-1$ *spread*, so turning one wrong guess into a right one both gains a dollar and avoids losing one — a swing of \$2. With no information every coin is a fair 50/50 guess, so you expect 50 correct and a return of \$0. The game is about using one bit to push the expected number correct above 50.

Step 2 — the strategy. Let H be the number of heads and $T = 100 - H$ the number of tails. Ask:

"Are there at least 50 heads?"

Guess **all heads** if the answer is yes, **all tails** if no. You then get exactly the majority colour right, so $\#\text{correct} = \max(H, T)$. Because $\max(H, T) = 50 + |H - 50|$,

$$\text{return} = 2\max(H, T) - 100 = 2\,|H - 50| \;\geq 0.$$

The one bit tells you which side is the majority, and you collect twice how far the flips strayed from a perfect 50–50 split. You can never lose money.

> ***The trap — the parity question is worthless here.*** *"Is the number of heads even?" sounds clever, but knowing the total parity leaves* every individual coin *still exactly* 50/50 *(fixing one coin and demanding a given overall parity only constrains the other* 99*). So no coin can be guessed better than even odds and the expected return stays* \$0*. The bit must bias the coins themselves — which the majority question does.*

Step 3 — the distribution of H (binomial). Each flip is heads with probability $\frac{1}{2}$, independently. A particular sequence with exactly k heads has probability $\left(\frac{1}{2}\right)^{100}$, and the number of such sequences is the number of ways to choose which k of the 100 positions are heads, $\binom{100}{k}$. Hence

$$P(H = k) = \binom{100}{k}\left(\frac{1}{2}\right)^{100} = \frac{\binom{100}{k}}{2^{100}}, \qquad k = 0, 1, \dots, 100,$$

i.e. $H \sim \text{Binomial}\left(100, \frac{1}{2}\right)$. Writing H as a sum of 100 independent coin indicators (each with mean $\frac{1}{2}$ and variance $\frac{1}{4}$) gives mean $\mu = 50$ and variance $\sigma^2 = 100 \cdot \frac{1}{4} = 25$, so $\sigma = 5$. The expected return is

$$\mathbb{E}[\text{return}] = 2\,\mathbb{E}|H - 50| = \frac{2}{2^{100}} \sum_{k=0}^{100} \binom{100}{k} |k - 50|.$$

Step 4 — a closed form for the sum. Write $100 = 2N$ with $N = 50$. We show

$$\sum_{k=0}^{2N} \binom{2N}{k} |k - N| = N \binom{2N}{N}.$$

Use symmetry. Since $\binom{2N}{k} = \binom{2N}{2N-k}$, the terms for k and $2N-k$ are equal, and the $k = N$ term is 0. So the sum is twice the upper half:

$$S := \sum_{k=0}^{2N} \binom{2N}{k} |k - N| = 2 \sum_{k=N+1}^{2N} \binom{2N}{k} (k - N), \qquad \frac{S}{2} = \sum_{k=N+1}^{2N} k \binom{2N}{k} - N \sum_{k=N+1}^{2N} \binom{2N}{k}.$$

First piece. The "absorption" identity $k\binom{2N}{k} = 2N\binom{2N-1}{k-1}$ (choosing a team of k with a captain) gives

$$\sum_{k=N+1}^{2N} k \binom{2N}{k} = 2N \sum_{j=N}^{2N-1} \binom{2N-1}{j}.$$

Row $2N-1$ is odd, so its coefficients pair up as mirror images with no middle term, and the upper half is exactly half the total 2^{2N-1}: $\sum_{j=N}^{2N-1} \binom{2N-1}{j} = 2^{2N-2}$. So this piece is $2N \cdot 2^{2N-2} = N\, 2^{2N-1}$.

Second piece. By symmetry the sum above the middle is half of everything except the middle term: $\sum_{k=N+1}^{2N} \binom{2N}{k} = \frac{1}{2}(2^{2N} - \binom{2N}{N})$, so

$$N \sum_{k=N+1}^{2N} \binom{2N}{k} = N\, 2^{2N-1} - \frac{N}{2} \binom{2N}{N}.$$

Subtract. The $N\, 2^{2N-1}$ terms cancel:

$$\frac{S}{2} = \frac{N}{2} \binom{2N}{N} \quad \Longrightarrow \quad S = N \binom{2N}{N}.$$

Therefore

$$\mathbb{E}|H - N| = \frac{N\binom{2N}{N}}{2^{2N}}, \qquad \mathbb{E}[\text{return}] = \frac{2N\binom{2N}{N}}{2^{2N}} = \frac{100\binom{100}{50}}{2^{100}} \approx \$7.96.$$

(Sanity check at $N = 1$, two coins: $\frac{1 \cdot \binom{2}{1}}{4} = \frac{1}{2}$, and indeed $|H - 1|$ is 1 with probability $\frac{1}{2}$.)

Step 5 — the normal approximation and the $\sqrt{2/\pi}$ factor. H is a sum of 100 independent fair bits, so by the central limit theorem it is approximately normal, $H \approx \mathcal{N}(50, 5^2)$. Let $Z = H - 50 \approx \mathcal{N}(0, \sigma^2)$ with density $\varphi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2\sigma^2}$.

Why the integral runs from 0 to ∞. By definition $\mathbb{E}|Z| = \int_{-\infty}^{\infty} |z|\, \varphi(z)\, dz$. The absolute value is awkward, but the density is *symmetric* ($\varphi(-z) = \varphi(z)$) and $|z|$ is symmetric too, so the left half ($z < 0$) contributes exactly the same as the right half ($z > 0$). On the right half $|z| = z$, so

$$\mathbb{E}|Z| = 2 \int_0^{\infty} z\, \varphi(z)\, dz = 2 \int_0^{\infty} \frac{z}{\sigma\sqrt{2\pi}}\, e^{-z^2/2\sigma^2}\, dz.$$

Substitute $u = \frac{z^2}{2\sigma^2}$, so $du = \frac{z}{\sigma^2} dz$, i.e. $z\, dz = \sigma^2 du$:

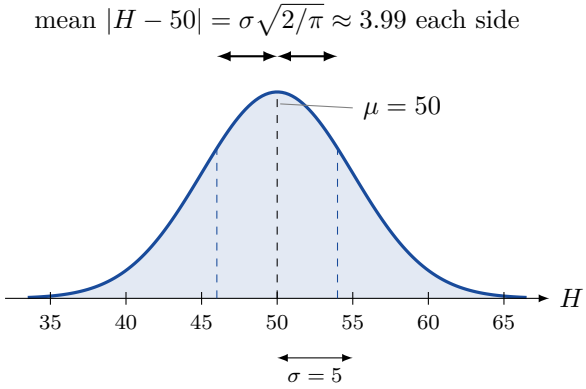
$$\mathbb{E}|Z| = \frac{2}{\sigma\sqrt{2\pi}}\, \sigma^2 \int_0^{\infty} e^{-u}\, du = \frac{2\sigma}{\sqrt{2\pi}} = \sigma\sqrt{\frac{2}{\pi}}.$$

The universal ratio. For *any* normal distribution, the mean absolute deviation is a fixed fraction of the standard deviation:

$$\frac{\mathbb{E}|X-\mu|}{\sigma} = \sqrt{\frac{2}{\pi}} \approx 0.798.$$

So "average distance from the mean" $\approx 0.8\times$ "standard deviation" — a handy mental-math constant.

Here $\mathbb{E}|H-50| \approx 5\sqrt{2/\pi} \approx 3.99$, so the return is about $2 \times 3.99 \approx \$8$ — matching the exact \$7.96 (the tiny gap, 3.979 exact vs. 3.989 normal, is only the discreteness of H). Stirling's formula $\binom{2N}{N}/2^{2N} \approx 1/\sqrt{\pi N}$ shows the exact value $N\binom{2N}{N}/2^{2N} \approx \sqrt{N/\pi}$ is the same expression as $\sigma\sqrt{2/\pi}$ with $\sigma = \sqrt{N/2}$.



The mean absolute deviation (arrows) sits at $\sigma\sqrt{2/\pi} \approx 3.99$, a bit inside one standard deviation $\sigma = 5$. The expected return is twice this.

Ask "Are there at least 50 heads?" and guess every coin as the majority colour. The expected return is

$$\frac{100\binom{100}{50}}{2^{100}} = 2\,\mathbb{E}|H-50| \approx \$7.96$$

(about 53.98 coins correct on average). *Mental shortcut:* $\sigma = 5$ and $\mathbb{E}|H-50| \approx 0.8\sigma = 4$, so the return is about \$8.

Contrast — When Only an Exact Match Counts (Q293, Q355)

Change the scoring: now you win only if you guess *all* 100 *coins* exactly right. A yes/no question splits the 2^{100} equally likely sequences into a "yes" group and a "no" group, and after the answer you can guess just *one* sequence. So at most two sequences (one per answer) can ever win:

$$P(\text{win}) \le \frac{2}{2^{100}} = \frac{1}{2^{99}},$$

and *any* non-trivial question — for example "Is the first flip heads?" — achieves it. One bit can at most **double** your chance of an exact hit. Under this scoring the majority question is no better

than any other; under per-coin scoring it is by far the best. Same puzzle, different payoff, different answer.

Concept Summary and Applications

- **Write the payoff first:** "exact match" and "count of correct pieces" call for completely different questions.
- **One bit at most doubles an exact-match probability**; for per-item scoring, use the bit to *bias many items at once* (the majority question), not to learn a global property that leaves each item 50/50 (parity).
- **Binomial mean absolute deviation:** for $2N$ fair coins, $\mathbb{E}|H - N| = N\binom{2N}{N}/2^{2N} \approx \sqrt{N/\pi}$.
- **Normal rule of thumb:** $\mathbb{E}|X - \mu| = \sigma\sqrt{2/\pi} \approx 0.8\,\sigma$ for any normal.
- **Applications:** designing informative questions/experiments, 20-questions and search strategies, compression and coding (bits of information), and quick estimates of average absolute error.

51 Weighing with a Balance Scale

A two-pan balance does not show a number — it only tells you *which side is heavier*, or that they balance. That makes it a **three-outcome** device (left heavy / balance / right heavy) and, when you may put objects on both pans, a **three-choice** device for each known weight (this pan / other pan / not used). Both problems below come down to counting with powers of 3.

Representative Problem — Which Weights Must Be Known? (Q296)

Problem: You have 40 balls weighing 1 g, 2 g, ..., 40 g (one of each), but you don't know which ball is which. You also have a fair two-pan balance. You are told the weights of a few of the balls in advance, and you will use *those known balls as reference weights*: a ball is "measured" when you find an arrangement in which it balances a combination of known balls (known balls may go on either pan). What is the minimum number of balls whose weights you need to know in advance so that every other ball can be measured?

Step 1 — what one weighing can express. Put the unknown ball on the left pan. Each known ball a_i can go on the *right* pan (it counts $+a_i$), on the *left* pan with the unknown (it counts $-a_i$), or be left off (0). The scale balances when

$$m = \sum_{i=1}^{k} \varepsilon_i \, a_i, \qquad \varepsilon_i \in \{-1, 0, +1\}.$$

So with k known balls there are at most 3^k sign patterns $(\varepsilon_1, \ldots, \varepsilon_k)$.

Step 2 — why only $\frac{3^k-1}{2}$ positive weights. Two adjustments:

- The all-zero pattern (every known ball set aside) nets 0 and measures nothing: $3^k - 1$ patterns remain.
- The rest come in **mirror pairs**: flipping every sign (swapping the two pans) turns a pattern netting $+v$ into one netting $-v$. They describe the same weighing, so each positive weight is counted twice.

Hence at most

$$\frac{3^k - 1}{2}$$

distinct positive weights can be measured. *Example* ($k = 2$, known balls 1 and 3): the 9 patterns give $0, \pm1, \pm2, \pm3, \pm4$ (e.g. $2 = 3 - 1$: unknown + 1 g balances 3 g), i.e. exactly $\frac{9-1}{2} = 4$ weights $\{1, 2, 3, 4\}$.

ball 1	ball 3	net
0	0	0
+	0	+1
−	0	−1
0	+	+3
0	−	−3
+	+	+4
−	−	−4
−	+	+2
+	−	−2

Step 3 — the lower bound. We need all 40 weights, so

$$\frac{3^k-1}{2} \geq 40: \qquad k=3 \Rightarrow \tfrac{27-1}{2}=13<40, \qquad k=4 \Rightarrow \tfrac{81-1}{2}=40.$$

At least 4 known balls are necessary.

Step 4 — four suffice: $1,3,9,27$ (balanced ternary). Every integer from 1 to 40 can be written as

$$m=\varepsilon_0\cdot 1+\varepsilon_1\cdot 3+\varepsilon_2\cdot 9+\varepsilon_3\cdot 27, \qquad \varepsilon_i\in\{-1,0,1\}$$

("balanced ternary" — base 3 with digits $-1,0,1$). For example

$$2=3-1, \quad 5=9-3-1, \quad 8=9-1, \quad 20=27-9+3-1, \quad 40=27+9+3+1.$$

So knowing the 1, 3, 9 and 27 g balls lets you measure every ball.

The minimum is $\boxed{4}$: know the balls of $1,3,9,27$ g. With k reference weights at most $\frac{3^k-1}{2}$ values are measurable, which reaches 40 only at $k=4$, and $1,3,9,27$ achieve every value by balanced ternary.

The answer depends on the rules. *If a ball, once measured, may be* reused *as a new reference, far fewer are needed: knowing* 1 *g and* 3 *g lets you find* 2 *and* 4*, then* $\{1,2,3,4\}$ *reaches everything up to* 10*, and so on — 2 known balls suffice. And if you are told the set is* exactly $\{1,\ldots,40\}$*, you could simply sort the balls by pairwise comparisons and need to know nothing. State your interpretation; the standard (Bachet) answer is* 4*.*

Additional Problem — 12 Coins, One Counterfeit

Problem: You have 12 coins. One is counterfeit and is either *heavier* or *lighter* than the others — you don't know which. Using a balance scale only 3 times, find the fake coin *and* whether it is heavy or light.

Why 3 weighings can be enough (counting). Each weighing has 3 outcomes, so 3 weighings produce at most $3^3=27$ different outcome sequences. There are $12\times 2=24$ possibilities to tell apart (which coin, and heavy or light). Since $24\leq 27$, three weighings are *not ruled out.* (Counting only gives a necessary condition — 14 coins, with 28 possibilities, is impossible — so a concrete scheme is still needed.) The trick is to design weighings whose outcomes split the remaining possibilities as *evenly* as possible; each weighing should leave at most 9 possibilities, then at most 3, then 1.

Why the first weighing is 4 vs 4. Put k coins on each pan. If it tips, the fake is one of the $2k$ coins on the scale, each with a known direction: $2k$ possibilities. If it balances, it is one of the $12-2k$ off-scale coins, heavy or light: $2(12-2k)$ possibilities. Both must be ≤ 9, so $2k\leq 9$ and $24-4k\leq 9$, giving $k=4$ exactly (leaving 8 and 8).

A concrete scheme (label the coins 1–12).

Weighing 1: $\{1,2,3,4\}$ vs $\{5,6,7,8\}$.

- **Balances:** coins 1–8 are genuine; the fake is in $\{9,10,11,12\}$, direction unknown (8 possibilities).

- **Unbalanced**, say the *left* pan is heavier (the right-heavy case is the mirror image): either one of $\{1,2,3,4\}$ is *heavy* or one of $\{5,6,7,8\}$ is *light* (8 possibilities). Coins 9–12 are now known to be genuine.

Balance case (fake in $\{9,10,11,12\}$).

- *Weighing 2:* $\{9,10,11\}$ vs $\{1,2,3\}$ (known genuine).
- If it balances, the fake is 12; *weighing 3:* 12 vs a genuine coin shows whether it is heavy or light.
- If $\{9,10,11\}$ is heavier, one of them is *heavy*; if lighter, one of them is *light* — 3 suspects with a known direction. *Weighing 3:* 9 vs 10; the one tipping the expected way is the fake, and if they balance it is 11.

Imbalance case (left was heavy: $1,2,3,4$ maybe-heavy; $5,6,7,8$ maybe-light).

- *Weighing 2:* $\{1,2,5\}$ vs $\{3,4,6\}$, setting $7,8$ aside. This deliberately *mixes* the groups, so every remaining possibility pushes the scale a predictable way.
- **Left heavy:** the only consistent causes are a heavy 1 or 2 (on the left) or a light 6 (on the right). A light 5 would make the left *lighter*, and a heavy 3 or 4 would make the right heavier. Suspects $\{1\text{H}, 2\text{H}, 6\text{L}\}$. *Weighing 3:* 1 vs 2 — the heavier is the fake; if they balance, 6 is light.
- **Right heavy:** causes are a heavy 3 or 4 or a light 5. *Weighing 3:* 3 vs 4 — the heavier is the fake; if they balance, 5 is light.
- **Balances:** none of 1–6 is fake, so it is 7 or 8, both maybe-light. *Weighing 3:* 7 vs a genuine coin — if 7 is lighter it is the fake; otherwise 8 is the light fake.

Weigh $\{1,2,3,4\}$ vs $\{5,6,7,8\}$, then follow the branches above: every one of the 24 possibilities (coin and direction) is identified in exactly 3 weighings. (Checked by running all 24 cases.)

How far this goes. With 3 weighings the most coins you can handle while also naming the direction is 12; with one extra coin known to be genuine you can handle 13 ($26 \le 27$ possibilities); 14 coins ($28 > 27$) is impossible. In general, w weighings handle $\frac{3^w-3}{2}$ coins, or $\frac{3^w-1}{2}$ with a spare genuine coin.

What a "spare genuine coin" is, and why it helps. *It is an* extra *coin you are told is real, given in addition to the suspects. It can never be the fake; its only use is as a known-good filler on a pan. A balance compares equal numbers of coins, so without a spare every weighing is k suspects vs k suspects. With* 13 *suspects that fails already at the first weighing, which must leave at most* 9 *possibilities for the last two weighings:*

- 4 *vs* 4*: if it tips,* 8 *possibilities remain; if it balances, the* 5 *unweighed coins remain, each heavy or light —* 10 *possibilities (too many).*
- 5 *vs* 5*: if it tips,* 10 *possibilities remain (too many).*

With a spare, weigh 5 *suspects vs* 4 *suspects + the genuine coin. If it tips,* 9 *possibilities remain (the* 9 *suspects on the scale, each with a known direction); if it balances, the* 4 *unweighed coins give* 8*. Both are at most* 9*, so the remaining two weighings can finish — which is why the spare raises the limit from* 12 *to* 13 *coins.*

Concept Summary and Applications

- **A balance is ternary:** each weighing has 3 outcomes, and each reference weight has 3 placements. Count with powers of 3.
- **Information bound:** w weighings distinguish at most 3^w cases; k reference weights measure at most $\frac{3^k-1}{2}$ positive values.
- **Balanced ternary** (digits $-1, 0, 1$) shows $1, 3, 9, 27, \ldots$ achieve the bound.
- **Design for even splits:** choose each weighing so that the three outcomes leave roughly equal numbers of possibilities; mixing suspects across pans is what makes the counterfeit puzzle work.
- **Applications:** optimal search and group testing, coin/weight puzzles, balanced-ternary number systems, and designing tests with few measurements.

52 Rope-Burning Timer Puzzles

Representative Problem — Measure 45 Minutes with Two Ropes (Q302)

Problem: You have two ropes. Each takes exactly 1 hour to burn from one end to the other, but they burn at *non-uniform* rates along their length (some stretches burn fast, others slowly). Using only the ropes and a lighter, how do you measure exactly 45 minutes?

Why the obvious idea fails. You cannot cut a rope in half and expect 30 minutes: because the burn rate varies, half the *length* can take anywhere from almost 0 to almost 60 minutes. The only thing you know for sure is *total* burn time. So the solution must use only that fact.

The key trick — light both ends. A rope lit at *both* ends burns out in exactly **half** its remaining burn time, however uneven it is.

Why. Measure positions along the rope by "burn time from the left end": let $T(x)$ be the time a single flame from the left takes to reach point x, so T runs from 0 at the left end to 60 at the right end. A flame lit at the right end reaches x after $60 - T(x)$ minutes (it has to burn everything the left flame hasn't). The two flames meet at the point where they arrive at the same moment:

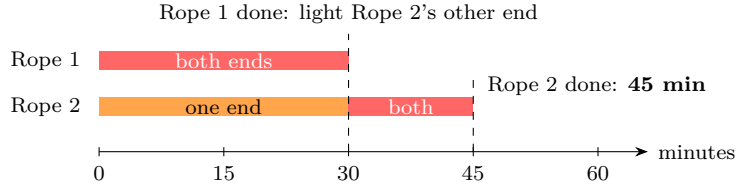
$$T(x) = 60 - T(x) \quad \Longrightarrow \quad T(x) = 30.$$

So they meet after exactly 30 minutes — at an unknown *place* (not necessarily the midpoint), but at a known *time*. The same argument applies to any piece of rope: if the remaining piece would take R minutes from one end, lighting both ends finishes it in $R/2$.

The solution.

1. At time 0, light **Rope 1 at both ends** and **Rope 2 at one end**.
2. Rope 1 burns out after exactly 30 minutes (by the trick above).
3. At that instant, Rope 2 has burned for 30 minutes from one end. You don't know *how much length* is left, but you do know the rest would take exactly $60 - 30 = 30$ more minutes from one end. Now **light the other end of Rope 2**.
4. Burning from both ends, the remaining 30 minutes of rope finishes in $30/2 = 15$ minutes.

When Rope 2 burns out, $30 + 15 = 45$ minutes have passed.



Light Rope 1 at both ends and Rope 2 at one end. When Rope 1 burns out (30 min), light Rope 2's other end. Rope 2 burns out 15 minutes later: exactly 45 minutes in total.

The General Principle

- **Only burn *time* is reliable, never length.** Non-uniform ropes make length useless; reason purely about remaining burn time.
- **Both ends halves the remaining time.** A piece with R minutes left, lit at both ends, lasts $R/2$ — the two flames always meet at the time midpoint.
- **Chain the events.** One rope burning out is a clock tick you can use to light the next end. With two ropes you can measure, for example, 15 min (the last stretch above, from minute 30 to 45), 30, 45, 60, and 90 (Rope 1 from one end, then Rope 2 from both ends).
- **More ropes, finer times.** Each extra rope lets you halve a remaining time again: with three ropes you can reach $52\frac{1}{2}$ minutes ($30 + 15 + 7\frac{1}{2}$). Light Rope 3 at one end at time 0 too; at minute 45 it has 15 minutes left, so lighting its other end then finishes it $7\frac{1}{2}$ minutes later.

Concept Summary and Applications

- **Find the invariant:** when details (the burn profile) are unknown, work only with quantities that are fixed regardless (total burn time).
- **Symmetry trick:** two fronts moving toward each other meet at the midpoint of *time*, not of distance.
- **Applications:** lateral-thinking interview puzzles, measuring with unreliable instruments, and more generally designing procedures that are robust to unknown non-uniformity.

53 Self-Referential Logic Puzzles

A statement is **self-referential** when it talks about itself — for example, "this sentence is false." Such statements can create a **paradox**: if the sentence is true, then it is false; if it is false, then it is true. Interviewers use self-reference to test whether you can step outside the usual "look up the fact" mode and reason carefully about what makes an answer right or wrong.

Representative Problem — Minimise the Correct Answers (Q17)

Problem: You may ask *one* true-or-false question to every American, and you want as *few* people as possible to answer correctly. What question do you ask?

Goal. The smallest possible number of correct answers is **zero**. Can one question *guarantee* zero, no matter how people respond? Only if *every* possible answer is wrong.

Step 1 — why an ordinary question fails. Any factual true/false question ("Is Canberra the capital of Australia?") has a fixed correct answer. Even people who know nothing and simply guess get it right about half the time, so you should expect *at least* around 50% of guessers correct — nowhere near zero. To do better, the correct answer must *depend on what the person says.*

Step 2 — a question that makes every answer wrong. Ask:

"Will you answer *no* to this question?"

(In true/false form: "*True or false: you will answer 'false' to this question.*") Each person answers, and we grade the answer against what they actually did:

- **If they answer "no":** it is now a plain fact that they *did* answer "no," so the truthful answer to "Will you answer no?" is *yes.* They said "no" — **wrong**.
- **If they answer "yes":** they did *not* answer "no," so the truthful answer is *no.* They said "yes" — **wrong**.

Whatever a person says, the correct answer is the opposite. So **every respondent is wrong** — a guaranteed 0 correct, with nothing to hope for or control about how people respond.

Step 3 — why this exact phrasing. The truth of "Will you answer no?" is simply a fact about the person's own response, so each answer has a definite truth value and is definitely wrong: no paradox, nothing undefined.

Compare the tempting phrasing "*Will your answer to this question be wrong?*"

- Answering "true" ("yes, my answer will be wrong") is a genuine *liar paradox*: it is correct exactly when it is wrong, so it has no consistent grade.
- Answering "false" ("no, my answer will not be wrong") is *consistent either way* — it can be graded as correct.

So that version does *not* guarantee the minimum: anyone who answers "false" can be counted correct, and you cannot stop people from doing that. It reaches 0 only in the unrealistic case that *everyone* answers "true." The "Will you answer no?" version closes both loopholes.

Ask "*Will you answer 'no' to this question?*" (equivalently, "*True or false: you will answer 'false' to this question*"). Every possible response is wrong, so the number of people answering correctly is guaranteed to be 0 — the smallest possible.

The Key Idea — Self-Reference That Negates Itself

- **Make the truth depend on the answer.** A question about the respondent's own answer has a correct answer that changes with what they say. That is the only way to beat the $\sim 50\%$ that random guessing gives on a factual question.
- **Make it negate the answer.** "Will you answer *no*?" is built so that each response makes the *opposite* response the true one. Every answer refutes itself.
- **Avoid genuine paradoxes.** A statement like "this sentence is false" (or "will your answer be wrong?") has no consistent truth value on some branch, so it can't be graded reliably. A good self-referential question has a *definite* truth value for every possible answer — here, always "wrong."
- **Check every branch.** For any self-referential puzzle, list each possible response and ask "given that this was the answer, what is actually true?" — exactly as in Step 2.

Concept Summary and Applications

- **Self-reference** lets the correctness of an answer depend on the answer itself; designed as a negation, it forces every response to be wrong.
- **Liar-type statements** ("this is false") are paradoxes with no consistent truth value; well-posed puzzles avoid them.
- **Method:** enumerate the possible answers and evaluate the truth of the question under each one.
- **Applications:** knights-and-knaves and liar puzzles, "what question would you ask the guard?" riddles, the halting problem and Gödel-style arguments in logic and computer science, and interview questions that test careful reasoning over recall.

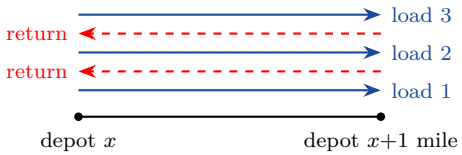
54 Apple Transport Optimization (Relay / Depot Problems)

The Set-Up and the One Idea That Solves It

A truck must move a big pile of apples across a long road. It can carry at most C apples at a time and it "pays" for travel in apples. Because the pile is bigger than one truckload, the truck cannot drive straight through: it must **shuttle** — carry a load forward, drop it at a roadside depot, drive back for the next load, and so on — moving the whole pile forward bit by bit. The question is always: how many apples survive to the destination?

The one idea: cost per mile depends on how many loads are left. Suppose the pile currently needs k truckloads ($k = \lceil \text{pile}/C \rceil$). To advance the *entire* pile by one mile, the truck must make k loaded forward trips and, between them, $k-1$ return trips (it does not return after the last load). How many apples that mile costs depends on whether an *empty* return trip costs anything — which gives two different models:

> **Model A** (loss on *every* mile driven, including empty returns — the classic camel/fuel convention): a mile of progress costs k (forward) $+ (k-1)$ (return) $= \mathbf{2k-1}$ apples.
>
> **Model B** (loss *only while carrying* apples — empty returns are free): a mile of progress costs $\mathbf{k}$ apples.



Moving a 3-load pile ($k = 3$) forward one mile: 3 loaded trips (solid) and 2 empty returns (dashed). Model A charges all 5 legs ($2k-1 = 5$ apples); Model B charges only the 3 loaded legs ($k = 3$ apples).

The strategy (same for both models). Shuttle forward at the current rate until the pile shrinks to *exactly* the next multiple of C — at that moment one whole load disappears and the rate drops ($k \to k-1$). Stopping anywhere else would leave a partial load that still needs a full round trip, wasting apples. So the journey splits into **phases**, one per value of k, and in each phase we lose exactly C apples (except the last phase, which just drives to the end).

Representative Problem 1 — 2000 Apples, Model A (Q43)

Problem: A truck at a depot starts with 2000 apples and can carry at most 1000 at a time. For every mile it travels, it loses 1 apple. What is the maximum number of apples that can be transported across a 1000-mile desert? (Use **Model A**: every mile driven costs an apple, including empty returns.)

Step 1 — the first phase, while the pile is 2000 ($k = 2$). Moving all 2000 apples forward needs two loads, so each mile of progress takes

- one forward trip carrying 1000 apples,

- one return trip (to fetch the rest),
- one forward trip carrying the other 1000 apples.

That is 3 miles of driving per mile of progress, so under Model A the rate is $2k-1=3$ apples per mile. We keep this up until the pile falls to 1000 (one load), i.e. until 1000 apples have been lost:

$$3x = 1000 \implies x = \frac{1000}{3} \approx 333.3 \text{ miles.}$$

So after $\frac{1000}{3}$ miles there are exactly 1000 apples at the depot.

Step 2 — the final phase ($k=1$). Now everything fits in one load, so the truck simply drives the rest of the way at 1 apple per mile. The remaining distance is $1000-\frac{1000}{3}=\frac{2000}{3}$ miles, costing $\frac{2000}{3}$ apples:

$$1000 - \frac{2000}{3} = \frac{1000}{3} = 333\tfrac{1}{3}.$$

Under Model A the truck delivers at most $\frac{1000}{3} = \boxed{333\tfrac{1}{3}}$ apples.

For comparison (Model B): the first phase costs only $k=2$ apples per mile, lasting $1000/2=500$ miles; the last 500 miles cost 1 per mile, so $1000-500=$ **500** apples arrive. The wording "the *truck* loses an apple per mile" is ambiguous — state which model you assume.

Representative Problem 2 — 3000 Apples, Model B (Q266)

Problem: You have 3000 apples in Edinburgh and want to move as many as possible to London, 1000 miles away. The truck holds at most 1000 apples, and *when the truck is carrying apples* it consumes one apple per mile. What is the maximum number you can deliver? (The phrase "when carrying apples" makes this **Model B**: empty return trips are free.)

Step 1 — 3000 apples, three loads ($k=3$). Each mile of progress needs 3 loaded forward trips (each losing 1 apple) and 2 empty returns (free), so the rate is $k=3$ apples per mile. Run until the pile drops to 2000:

$$3000 - 3x = 2000 \implies x = \frac{1000}{3} \approx 333.3 \text{ miles.}$$

Step 2 — 2000 apples, two loads ($k=2$). Now 2 loaded trips per mile: 2 apples per mile. Run until the pile drops to 1000:

$$2000 - 2y = 1000 \implies y = 500 \text{ miles.}$$

Distance covered so far: $\frac{1000}{3}+500=\frac{2500}{3}\approx 833.3$ miles, leaving $1000-\frac{2500}{3}=\frac{500}{3}$ miles to go.

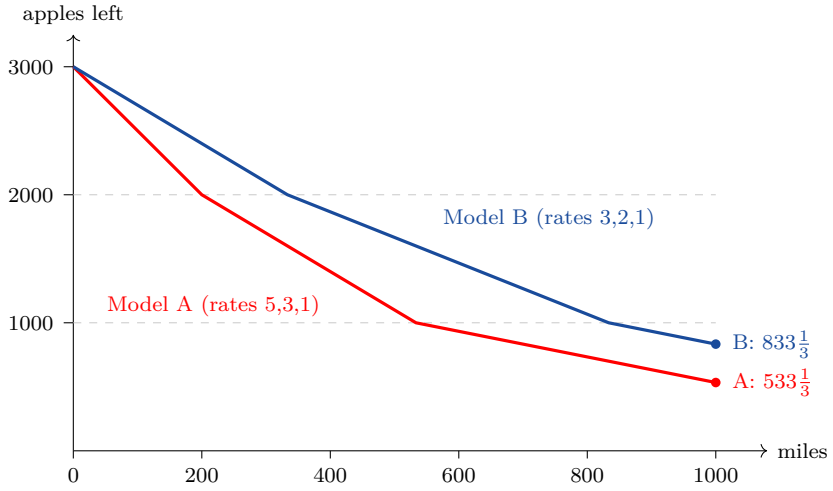
Step 3 — the last 1000 apples, one load ($k=1$). Drive the final $\frac{500}{3}$ miles at 1 apple per mile:

$$1000 - \frac{500}{3} = \frac{2500}{3} = 833\tfrac{1}{3}.$$

Under Model B the truck delivers at most $\frac{2500}{3} = \boxed{833\tfrac{1}{3}}$ apples.

For the same Representative Problem if under Model A. Now empty returns cost too, so the phase rates are $2k-1 = 5, 3, 1$ apples per mile:

$$\begin{aligned}
&\text{phase 1 } (k=3): \ 5 \text{ per mile}, \ 1000/5 = 200 \text{ mi} && \Rightarrow 2000 \text{ left at mile } 200;\\
&\text{phase 2 } (k=2): \ 3 \text{ per mile}, \ 1000/3 \approx 333.3 \text{ mi} && \Rightarrow 1000 \text{ left at mile } 533\tfrac{1}{3};\\
&\text{phase 3 } (k=1): \ 1 \text{ per mile}, \ 1000 - 533\tfrac{1}{3} = 466\tfrac{2}{3} \text{ mi} && \Rightarrow 1000 - 466\tfrac{2}{3} = \mathbf{533\tfrac{1}{3}} \text{ delivered.}
\end{aligned}$$



Apples remaining versus distance for 3000 apples over 1000 miles. Each kink is where the pile hits a multiple of 1000 and one load drops away. Model A (red) burns apples faster because empty returns cost, so it arrives with fewer.

Model A vs. Model B — Key Insights

	Model A	Model B
What costs an apple	every mile driven	only miles driven *loaded*
Empty return trip	costs 1/mile	free
Rate with k loads left	$2k-1$	k
Typical wording	"camel eats a banana per km", fuel	"drops an apple while carrying"
2000 apples, 1000 mi (Q43)	$333\frac{1}{3}$	500
3000 apples, 1000 mi (Q266)	$533\frac{1}{3}$	$833\frac{1}{3}$

- **Read the wording first.** "Consumes an apple *per mile traveled*" (like a camel eating bananas, or a vehicle burning fuel) points to Model A; "loses an apple *while carrying* apples" points to Model B. When in doubt, say which you assume — the answers differ a lot.
- **Phases by load count.** In both models, the pile needs $k = \lceil \text{pile}/C \rceil$ loads and the per-mile cost depends only on k. Advance until the pile is an exact multiple of C; each full phase burns exactly C apples and lasts $C/(\text{rate})$ miles.
- **Model A phases are shorter** ($C/(2k-1)$ miles vs. C/k), because the returns eat into the pile — so Model A always delivers less.

- **Early miles are the expensive ones.** The first phase (most loads) has the highest rate, which is why a big starting pile yields surprisingly few extra apples at the end; extra loads beyond a point are pure waste.
- **Applications:** the jeep/desert-crossing and camel-and-bananas problems, fuel depot planning, supply-chain relay costs, and any "carry capacity + per-distance consumption" optimization.

55 Resource Rotation and Equal Wear

Representative Problem — Tricycle with Two Spare Tyres (Q189)

Problem: You have a tricycle (three tyres on the ground) and two spare tyres, so five tyres in total. You will ride 1000 miles and may swap tyres at stops along the way. (a) If every tyre is to be worn *equally* by the end, how many miles does each tyre spend on the ground? (b) What is the minimum number of stops needed to achieve this?

Label the tyres $T_1, \ldots, T_5$. At every moment exactly 3 of them are on the ground and 2 are resting.

(a) Count the total work, then share it equally. Think of "tyre-miles": one tyre rolling one mile is one tyre-mile of wear. Every mile of the trip uses 3 tyres, so the trip produces

$$3 \times 1000 = 3000 \text{ tyre-miles}$$

of wear in total, *no matter how* you rotate. Spreading it equally over 5 tyres gives each

$$\frac{3000}{5} = 600 \text{ miles on the ground}$$

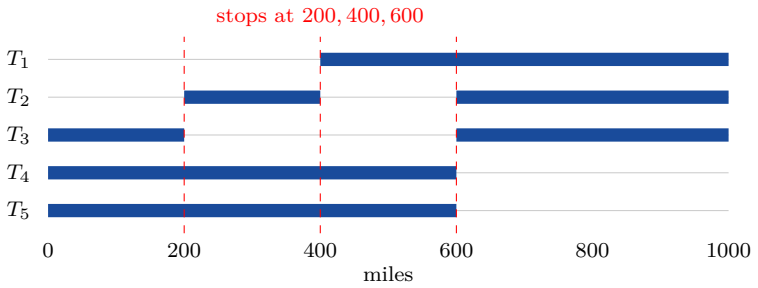
— and therefore $1000 - 600 = 400$ miles resting.

(b) Minimum stops. Split the trip into *segments* between stops; within a segment a fixed pair of tyres rests. Each tyre must rest a total of exactly 400 miles. (Number of stops = number of segments -1.)

Three stops are enough. Group the tyres as a trio $\{T_1, T_2, T_3\}$ and a pair $\{T_4, T_5\}$. Let the trio take turns resting two at a time for 200 miles each, then let the pair rest for the final 400:

miles	resting	on the ground
0–200	T_1, T_2	T_3, T_4, T_5
200–400	T_1, T_3	T_2, T_4, T_5
400–600	T_2, T_3	T_1, T_4, T_5
600–1000	T_4, T_5	T_1, T_2, T_3

Each of T_1, T_2, T_3 rests in two 200-mile segments (400 total); T_4, T_5 rest in the single 400-mile segment. Every tyre is on the ground exactly 600 miles, using four segments — swaps at miles $200, 400, 600$, i.e. 3 **stops**.



Blue = on the ground. Every row has exactly 600 miles of blue, and every vertical slice has exactly 3 blue tyres.

Two stops are impossible. Two stops give three segments with lengths $x+y+z = 1000$, each resting a pair — $3 \times 2 = 6$ "rest slots." Every tyre needs at least one slot (otherwise it rests $0 \neq 400$ miles), and 6 slots over 5 tyres forces one tyre (call it D) to rest twice and the other four to rest once. A tyre that rests in only one segment rests that segment's whole length, so *every segment containing a single-rester must be exactly* 400 *miles.* With D in two segments, the four single-resters fill the remaining slots: one in each of those two segments and two in the third — so all three segments contain a single-rester and must each be 400 miles. Then $x + y + z = 1200 \neq 1000$ (and D would rest 800 miles). Contradiction.

(a) Each tyre is on the ground $\dfrac{3 \times 1000}{5} = \boxed{600}$ miles. (b) The minimum is $\boxed{3}$ stops (at miles $200, 400, 600$ with the schedule above); 2 stops cannot work.

The Key Concept, the Strategy, and the General Formula

Key concept — conserve the total, then divide. However cleverly you rotate, the *total* amount of wear is fixed by the task: with k items in use at every moment over a duration D, the total usage is $k \cdot D$. Equal wear simply means splitting that fixed total evenly over all n items. This "count the total work first" step is the entire trick for part (a) — no schedule is needed.

With n interchangeable items, k in use at all times, over a total duration D:

$$\text{each item is used } \frac{k\,D}{n} \qquad \text{and rests } \frac{(n-k)\,D}{n}.$$

Here $n = 5$, $k = 3$, $D = 1000$: use 600, rest 400.

Strategy — a rotation that always works. Cut the trip into n equal segments of length D/n and let the resting set "slide" around the circle: in segment i, rest items i, $i+1, \ldots, i+(n-k)-1$ (counting cyclically). Each item then rests in exactly $n-k$ of the n segments, i.e. $(n-k)D/n$ in total — equal wear guaranteed, with $n-1$ stops. For the tricycle this is 5 segments of 200 miles and 4 stops (at $200, 400, 600, 800$).

Refinement — fewer stops by grouping. The simple rotation is not always the fewest stops. When the resting requirement can be met by grouping items (as with the trio taking turns in 200-mile blocks while the pair rests together for 400), you can merge segments; the tricycle needs only 3 stops. To prove a stop count is minimal, count the "rest slots" (segments $\times\,(n-k)$) against the number of items and the lengths each must rest, as in the two-stop impossibility argument.

Concept Summary and Applications

- **Total first:** the total usage $k \cdot D$ is fixed by the task; equal wear means each of n items gets kD/n.
- **Cyclic rotation:** n equal segments with a sliding resting set always equalizes wear, using $n-1$ stops.
- **Minimality needs a counting argument:** compare available rest slots with what each item

must receive; grouping can beat the plain rotation.

- **Applications:** tyre rotation, load-balancing servers or workers, shift scheduling so everyone works equal hours, wear-levelling in flash memory, and rotating inventory or equipment.

56 Secure Average Computation

A group wants a *collective* statistic (a total or an average) without anyone revealing their *individual* number. The trick is to hide each number behind random "masks" that cancel out only when everything is added together. This is the simplest example of **secure multi-party computation**.

Representative Problem — Eight Quants' Average Salary (Q294)

Problem: Eight quants from different banks want to know the average salary of the group, but none wants to disclose their own salary to the others. Design a procedure that lets the group compute the average without revealing individual salaries.

Call the salaries $s_1, s_2, \ldots, s_8$. The goal is

$$\frac{s_1 + s_2 + \cdots + s_8}{8}$$

without anyone learning any single s_i. Since the average is just the total divided by 8, it is enough to compute the **total** privately.

Method 1 — the random-offset chain.

1. Quant 1 secretly picks a large random number R and tells no one.
2. She adds her salary and passes $R + s_1$ to Quant 2.
3. Each quant in turn adds their own salary and passes the running total on:

$$R \;\to\; R + s_1 \;\to\; R + s_1 + s_2 \;\to\; \cdots \;\to\; R + \sum_{i=1}^{8} s_i.$$

4. The final total returns to Quant 1, who subtracts her secret R to get $\sum_i s_i$, divides by 8, and announces the average.

Example (salaries in \$k): say $R = 5{,}317$ and the salaries are $200, 150, 300, 180, 250, 220, 170, 130$ (total 1,600). Quant 2 receives 5,517, Quant 3 receives 5,667, ..., and Quant 1 finally receives 6,917; subtracting R gives 1,600, so the average is 200. Every number passed around looks like an arbitrary large number to its receiver.

Method 2 — circulating masks (symmetric). This version has no single "master" holding the secret offset.

1. **Sit in a circle**, labelled $1, 2, \ldots, 8$ (so the neighbour after 8 is 1).
2. **Each quant picks a private random number** r_i.
3. **Pass it on privately:** quant i secretly sends r_i to quant $i + 1$. Now each quant knows their salary s_i, their own r_i, and the r_{i-1} received from the previous person.
4. **Announce a masked value publicly:**

$$a_i = s_i + r_i - r_{i-1} \qquad \text{(indices mod 8)}.$$

5. **Add up the announcements.**

$$\sum_{i=1}^{8} a_i = \sum_{i=1}^{8} s_i + \sum_{i=1}^{8} r_i - \sum_{i=1}^{8} r_{i-1}.$$

The last two sums contain the *same eight numbers* $r_1, \ldots, r_8$ (just listed in a different order), so they cancel:

$$\sum_{i=1}^{8} a_i = \sum_{i=1}^{8} s_i, \qquad \text{average} = \frac{1}{8}\sum_{i=1}^{8} a_i.$$

Each announcement a_i is a salary plus one random number minus another — on its own it says nothing about s_i. Only the grand total is meaningful, because only there do the masks cancel.

Either method gives the exact total, hence the average, while hiding each salary from any single curious participant. Method 1: pass a secret random offset R around the circle, each person adding their salary; the originator subtracts R. Method 2: each quant privately passes a random r_i to the next and publicly announces $s_i + r_i - r_{i-1}$; summing the announcements cancels all masks.

Why the Methods Preserve Privacy

- **Everything a person sees is masked.** In Method 1, everyone except Quant 1 sees only $R+$(some salaries) with R unknown to them, so the number they receive is consistent with *any* set of earlier salaries. Quant 1 knows R but only ever sees the final grand total, never a partial sum. In Method 2, unmasking a_i requires *both* r_i and r_{i-1}, and each of those is known to only one other person.
- **Masks cancel only in the total.** The random numbers are designed so that they disappear exactly when all contributions are combined, and nowhere else.
- **Making the hiding exact.** "Large random number" hides salaries well but not perfectly (a huge received value slightly hints at large salaries). The clean fix is to do all arithmetic modulo a big number M (larger than any possible total) and choose R (or each r_i) uniformly from $0, 1, \ldots, M-1$. Then every value a person sees is *uniformly random*, independent of the salaries — perfect secrecy against a single observer.

The Limits

- **Two neighbours can collude.** In Method 1, the person before quant i knows the total they sent, $R + s_1 + \cdots + s_{i-1}$, and the person after knows the total they received, $R + s_1 + \cdots + s_i$; the difference is exactly s_i. In Method 2, quant $i-1$ chose r_{i-1} and quant $i+1$ received r_i, so together they know both masks and compute $s_i = a_i - r_i + r_{i-1}$. Both methods are safe only against a *single* curious participant.
- **Honest-but-curious only.** The methods assume everyone follows the steps but may try to learn more. A cheater who adds a wrong number corrupts the answer, and nobody can detect it.
- **The answer itself leaks something.** Once the average is announced, each quant can compute the *sum of everyone else's* salaries (total minus their own). With only two people, the average reveals the other person's salary completely — no protocol can prevent what the output itself reveals.

- **Collusion needs a stronger scheme** — additive secret sharing, below.

The Stronger Fix — Additive Secret Sharing

With n participants (here $n = 8$) and salaries $s_1, \dots, s_n$, fix a modulus M larger than any possible total (e.g. $M = 10^9$ dollars). All arithmetic is done "mod M" — keep only the remainder after dividing by M, like a clock that wraps around.

Step 1 — each quant splits their salary into n random shares. Quant i picks $n-1$ numbers $x_{i,1}, \dots, x_{i,n-1}$ uniformly at random from $\{0, 1, \dots, M-1\}$, and sets the last share so that all n shares add up to the salary:

$$x_{i,n} = s_i - \sum_{j=1}^{n-1} x_{i,j} \pmod{M}, \qquad \text{so} \qquad \sum_{j=1}^{n} x_{i,j} \equiv s_i \pmod{M}.$$

Step 2 — hand out the shares. Quant i privately sends share $x_{i,j}$ to quant j (and keeps $x_{i,i}$). After this, quant j holds one share from everybody: the column $x_{1,j}, x_{2,j}, \dots, x_{n,j}$.

Step 3 — each quant announces the sum of what they received.

$$y_j = \sum_{i=1}^{n} x_{i,j} \pmod{M}.$$

Step 4 — add the announcements. Summing over j and swapping the order of summation,

$$\sum_{j=1}^{n} y_j \equiv \sum_{j=1}^{n} \sum_{i=1}^{n} x_{i,j} = \sum_{i=1}^{n} \underbrace{\sum_{j=1}^{n} x_{i,j}}_{\equiv s_i} \equiv \sum_{i=1}^{n} s_i \pmod{M}.$$

Because M exceeds the true total, the remainder *is* the total, and the average is $\frac{1}{n}\sum_j y_j$. Picture an $n \times n$ grid of shares: row i adds up to salary s_i, column j is what quant j receives and announces; the grand total of the grid is the same whether you add rows or columns.

Small example ($n = 3$, $M = 10{,}000$, salaries $200, 150, 300$).

	to 1	to 2	to 3	row sum mod M
Quant 1 ($s_1 = 200$)	4213	7650	8337	$20200 \equiv 200$
Quant 2 ($s_2 = 150$)	9005	1234	9911	$20150 \equiv 150$
Quant 3 ($s_3 = 300$)	5500	6789	8011	$20300 \equiv 300$
announced y_j	8718	5673	6259	

(Each quant chose the first two shares at random; the third makes the row add up to the salary, e.g. $200 - 4213 - 7650 = -11663 \equiv 8337$.) The announcements add to $8718 + 5673 + 6259 = 20650 \equiv 650 = 200 + 150 + 300$, so the average is $650/3 \approx 216.7$ — yet every number anyone saw looks random.

Why it resists collusion. Any $n - 1$ of the n shares of a salary are independent and uniformly random; only *all* n together determine it. So a coalition learns something about quant i's salary only by collecting all of i's shares — including the one i keeps, which leaks only through i's announcement y_i, and extracting it from y_i requires knowing every other share in column i. In short, a coalition of colluders learns nothing beyond the sum of the *non*-colluders' salaries — which

the published total reveals anyway. Only when all $n-1$ others collude is s_i exposed, and at that point the average alone already gives it away.

The price. Method 1 sends n messages; secret sharing sends $n(n-1)$ private shares (56 for eight quants). It still assumes people follow the rules — it protects against curious coalitions, not against a cheater who hands out wrong shares.

Concept Summary and Applications

- **Compute the total, not the parts:** an average only needs the sum, and a sum can be built from masked pieces.
- **Random masks that cancel:** add random numbers that sum to zero across the group (or are removed by their owner), so individual values are hidden and the total survives.
- **Know the threat model:** single curious party vs. colluding groups vs. cheaters; simple masking handles the first, secret sharing the second, verification the third.
- **Applications:** salary and pay-gap surveys, secure voting and polling, privacy-preserving aggregation of data (e.g. federated learning), and auctions or benchmarks among competitors.

57 Clock Hands, Integer Constraints, and Digit-Date Puzzles

Two staples of the mental-math round live here: *clock-hand* problems, solved by comparing the two hands' steady rates, and *"minimum starting number"* problems, solved by chasing a **divisibility** (modular-arithmetic) constraint backward.

Representative Problem 1 — When the Hands Next Align (Q58)

Problem: At 12:00 the hour and minute hands point the same way. Assuming the hands sweep continuously, at what time do they next align?

Compare the two rates. The minute hand sweeps a full circle each hour, and the hour hand one-twelfth of a circle:

$$\text{minute: } 360^\circ/\text{hr}, \qquad \text{hour: } 30^\circ/\text{hr}.$$

So the minute hand *gains* on the hour hand at the relative rate

$$360^\circ - 30^\circ = 330^\circ \text{ per hour}.$$

Starting aligned, the hands align again the instant the minute hand has gained a whole lap of 360° on the hour hand:

$$t = \frac{360^\circ}{330^\circ/\text{hr}} = \frac{12}{11}\text{ hr} = \frac{12}{11}\cdot 60 = \frac{720}{11} = 65\tfrac{5}{11}\text{ min}.$$

That is $65\frac{5}{11}$ minutes after 12:00, i.e. $1{:}05\frac{5}{11}$ (about 1:05:27).

The hands next align at $1{:}05\frac{5}{11}$ ($= 65\frac{5}{11}$ minutes after 12:00). In general they align every $\frac{12}{11}$ hour — 11 times in each 12-hour span.

General formula — the angle at any time (Q564). The same two rates give the angle at any time $h{:}m$. The minute hand moves 6° per minute, so it sits at $6m$ degrees past 12; the hour hand moves 30° per hour $= 0.5^\circ$ per minute, so it sits at $30h + 0.5m$. Their difference is

$$\boldsymbol{\theta = |\,30h - 5.5m\,|^\circ}, \qquad \textbf{angle between the hands} = \min(\theta,\ 360^\circ - \theta)$$

with h on a 12-hour clock ($12{:}m$ counts as $h = 0$, 15:20 as $h = 3$).

Classic slip: forgetting that the hour hand creeps 0.5° per minute — at 2:30 it is halfway between 2 and 3, not on the 2.

time	$30h - 5.5m$	angle
2:30	$60 - 165 = -105$	105°
9:20	$270 - 110 = 160$	160°
9:40	$270 - 220 = 50$	50°
10:02	$300 - 11 = 289 \ \rightarrow\ 360 - 289$	71°
11:40	$330 - 220 = 110$	110°

Setting $\theta = 0$ recovers the alignment times above: $30h = 5.5m$, i.e. the hands coincide every $\frac{360}{5.5} = \frac{720}{11}$ minutes.

Representative Problem 2 — Minimum Starting Passengers (Q79)

Problem: Suppose there are n people on a bus. At each of three stops, $\frac{2}{3}$ of the people on the bus get off and 7 people get on. What is the minimum number of people that were initially on the bus before the first stop, given that the number of people on the bus remains an integer after each stop?

Let the initial number of people on the bus be n. At each stop $\frac{2}{3}$ of the people get off, so $\frac{1}{3}$ remain, and then 7 people get on. If the number of people after each stop must remain an integer, then at each stage the number of people **before** the stop must be divisible by 3.

Let the numbers of people just before the first, second, and third stops be $n,\ n_1,\ n_2$, and after the third stop be n_3. Then

$$n_1 = \frac{n}{3} + 7, \qquad n_2 = \frac{n_1}{3} + 7, \qquad n_3 = \frac{n_2}{3} + 7.$$

We seek the smallest positive integer n such that n, n_1, and n_2 are all divisible by 3 before the corresponding stop.

Work backward. Since n_2 must be divisible by 3 and $n_2 = \frac{n_1}{3} + 7$,

$$\frac{n_1}{3} + 7 \equiv 0 \pmod 3.$$

Because $7 \equiv 1 \pmod 3$, this gives $\frac{n_1}{3} \equiv -1 \equiv 2 \pmod 3$. The smallest such value is $\frac{n_1}{3} = 2$, i.e. $n_1 = 6$. But then

$$6 = \frac{n}{3} + 7 \implies \frac{n}{3} = -1,$$

which is impossible. So take the next value, $\frac{n_1}{3} = 5$, i.e. $n_1 = 15$. Then

$$15 = \frac{n}{3} + 7 \implies \frac{n}{3} = 8 \implies n = 24.$$

The same via a closed form. Composing the recurrence three times expresses the final count n_3 directly in terms of the start n:

$$n_3 = \frac{n}{27} + 7\left(1 + \tfrac{1}{3} + \tfrac{1}{9}\right) = \frac{n}{27} + \frac{91}{9},$$

which rearranges to

$$\boxed{n = 27\,n_3 - 273}.$$

A valid start needs $n > 0$, so $27\,n_3 > 273$, i.e. $n_3 > 10.11$; the smallest integer final count is therefore $n_3 = 11$, giving $n = 27 \cdot 11 - 273 = 24$. (Taking $n_3 = 10$ would force $n = -3 < 0$, which is impossible.)

Check.

- First stop: from 24, $\frac{2}{3} \cdot 24 = 16$ get off, leaving 8; then 7 get on, so 15 remain.

- Second stop: from 15, $\frac{2}{3} \cdot 15 = 10$ get off, leaving 5; then 7 get on, so 12 remain.
- Third stop: from 12, $\frac{2}{3} \cdot 12 = 8$ get off, leaving 4; then 7 get on, so 11 remain.

All counts are integers.

The minimum starting number is $\boxed{24}$ (giving $24 \to 15 \to 12 \to 11$).

Representative Problem 3 — Next Date with All Distinct Digits (Q99)

Problem: Starting from today's date, written `DD/MM/YYYY`, what is the next date in the future whose eight digits are all different?

Distinctness depends only on the *set* of digits, so the `DD/MM/YYYY` vs `YYYY-MM-DD` layout does not matter. Two facts pin the search: a two-digit *month* has tens digit 0 (Jan–Sep) or 1 (Oct–Dec), and a two-digit *day* has tens digit in $\{0, 1, 2, 3\}$. Remarkably the answer is over three centuries away, because whole centuries near us are impossible.

No date in 2000–2099**.** A year $20YY$ always contains a 2 and a 0. Every month then collides: Jan–Sep are $0X$ (the tens 0 clashes with the year's 0); Oct = 10 has a 0; Nov = 11 repeats a 1; Dec = 12 has a 2. *No month fits at all.*

No date in 2100–2199**.** A year $21YY$ contains 2 and 1, so Oct–Dec (which all use a 1) are out, forcing a Jan–Sep month $0X$ — which needs the digit 0. The day's tens digit must come from $\{0, 1, 2, 3\}$, but 0 (month), 1 and 2 (year) are taken, forcing tens digit 3, i.e. day 30 or 31 — whose units 0, 1 are already used. *No day fits.*

No date in 2200–2299**.** The year "$22YY$" already repeats the 2.

The 2300s pin everything down. A year $23YZ$ owns 2 and 3; Oct–Dec are blocked, so the month is Jan–Sep with tens digit **0**. The day's tens digit lies in $\{0, 1, 2, 3\}$, but 0 (month), 2 and 3 (year) are used, so it must be **1** (day 10–19). Thus $\{0, 1, 2, 3\}$ are forced into fixed roles (month-tens 0, day-tens 1, year-lead 2, 3), and the four remaining slots — the year's last two digits, the month's unit, and the day's unit — must be drawn *distinct* from $\{4, 5, 6, 7, 8, 9\}$.

Minimise the date (year first, then month, then day):

- **Year** $23YZ$ with $Y, Z \in \{4, \ldots, 9\}$: smallest is 2345.
- **Month** $0m$ with m from the leftovers $\{6, 7, 8, 9\}$: smallest is 06 (June).
- **Day** $1d$ with d from $\{7, 8, 9\}$: smallest is 17.

The next all-distinct-digit date is $\boxed{17/06/2345}$ — digits $1, 7, 0, 6, 2, 3, 4, 5$ (everything except 8 and 9). No date from 2000 through 2344 qualifies, so the answer does not depend on the exact start date.

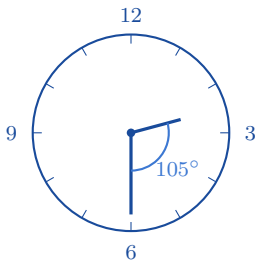
Additional Problem 1 — Angle Between the Hands (Q35)

Problem: What is the angle between the hour and minute hands at 2:30?

Place each hand in degrees, measured clockwise from the 12.

- **Minute hand:** 30 minutes $\times 6^\circ$ per minute (a full 360° in 60 minutes) $= 180^\circ$ — pointing straight down at the 6.
- **Hour hand:** it moves 30° per hour *plus* a bit for the minutes, 0.5° per minute. At 2:30 it is at $2 \times 30^\circ + 30 \times 0.5^\circ = 60^\circ + 15^\circ = 75^\circ$ (halfway between the 2 and the 3).

The angle between them is $|180^\circ - 75^\circ| = 105^\circ$.



At 2:30 the hands are 105° apart. In general the angle at $h{:}m$ is $|30h - 5.5m|$ degrees (take 360° minus this if it exceeds 180°).

Additional Problem 2 — Next Date with at Most One Repeated Digit (Q329)

Problem: Relax the previous puzzle: starting from today (`DD/MM/YYYY`), find the next date whose eight digits repeat *at most once* — that is, at most one digit value may appear twice, and every other digit is distinct (so at least seven of the eight digits are different).

Even with this looser rule, the near future is surprisingly barren, for the same tens-digit reasons as Q99. Recall a month's tens digit is 0 or 1, and a near-term year $20YZ/203Z$ carries the digits 2 and 0.

The 2020s are out. A year $202Z$ already contains a *pair* of 2s — that single repeat uses up the whole allowance — plus a 0. Any month then adds a *second* repeat: Jan–Sep ($0X$) and Oct = 10 repeat the 0; Nov = 11 repeats a 1; Dec = 12 makes a third 2. So no date in the 2020s works.

2030–2033 are out too. The years $2030, 2032, 2033$ each repeat a digit inside the year itself ($00, 22, 33$), using the allowance; the forced month-0 then collides again. The year $2031 = \{2, 0, 3, 1\}$ has distinct digits, but now the four tens-slots values $\{0, 1, 2, 3\}$ are all occupied (month-tens 0, and the day-tens must be one of $0, 1, 2, 3$, all taken) — so the day forces a second repeat. None qualifies.

2034 frees a digit. The year $2034 = \{2, 0, 3, 4\}$ has four distinct digits, which frees the digit 1. Now the one allowed repeat is the pair of 0s (the year's 0 plus a Jan–Sep month's leading 0), the day's tens digit can be 1, and the remaining slots take leftover digits. Minimising month then day: month 05 (May — months 01–04 either strand the day or reuse $2, 3, 4$), and day 16, giving

$$16/05/2034, \qquad \text{digits } 1, 6, 0, 5, 2, 0, 3, 4 \ \text{ (only the 0 repeats).}$$

A day-by-day scan from today confirms nothing earlier qualifies.

The next date with at most one repeated digit is $\boxed{16/05/2034}$ (the single repeat is the pair of 0s). Every year from the 2020s through 2033 is impossible.

Concept Summary and Applications

- **Clock hands are a rate race:** the minute hand gains 330°/hr on the hour hand, so they align every $\frac{12}{11}$ hr; the instantaneous angle is $|30h - 5.5m|$ degrees (30° per hour-mark, 5.5° of net gain per minute).
- **"Minimum integer" problems are divisibility problems:** write the step as a recurrence, demand each intermediate value be a whole number, and work *backward* using congruences (mod) the divisor.
- **Watch for impossible early roots:** the smallest congruence solution can force a negative count; step to the next valid one.
- **Digit-date puzzles:** treat a date as its eight-digit multiset and use the tens-digit constraints (month tens $\in \{0, 1\}$, day tens $\in \{0, 1, 2, 3\}$) to eliminate whole centuries fast; then minimise year, then month, then day.
- **Applications:** calendar/day-of-week arithmetic, gear and orbital alignment (LCM of periods), cycle lengths, and any "keep it a whole number" constraint.

Part IX — Number Theory and Combinatorial Puzzles

58 Prime Factorisation: Divisor Counts, LCM and Divisibility

Every whole number $n \geq 2$ can be written in exactly one way as a product of primes,

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} \qquad (\text{e.g. } 360 = 2^3 \cdot 3^2 \cdot 5),$$

(the *Fundamental Theorem of Arithmetic*). Once a number is written this way, questions about divisors, multiples, common multiples, divisibility and trailing zeros all become simple statements about the *exponents* $e_1, \ldots, e_k$.

Representative Problem — Smallest Number with Exactly 28 Divisors (Q609, part 2)

Problem: What is the smallest positive integer that has exactly 28 positive divisors?

Step 1: How to count divisors

Take $n = p_1^{e_1} \cdots p_k^{e_k}$. A number d divides n exactly when its own factorisation uses only these primes with exponents no larger than in n:

$$d = p_1^{f_1} \cdots p_k^{f_k}, \qquad 0 \leq f_i \leq e_i.$$

Each exponent f_i can be chosen in $e_i + 1$ ways $(0, 1, \ldots, e_i)$, independently, so

$$\boxed{d(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1).}$$

Example: $12 = 2^2 \cdot 3$ has $(2+1)(1+1) = 6$ divisors: $1, 2, 3, 4, 6, 12$.

Step 2: Which exponent patterns give 28 divisors?

We need $(e_1+1)(e_2+1)\cdots = 28$ with every factor ≥ 2. Write 28 as a product of factors ≥ 2 (order does not matter):

$$28, \qquad 14 \cdot 2, \qquad 7 \cdot 4, \qquad 7 \cdot 2 \cdot 2.$$

Subtracting 1 from each factor gives the possible exponent patterns:

$$(27), \qquad (13, 1), \qquad (6, 3), \qquad (6, 1, 1).$$

Step 3: Make each pattern as small as possible

For a given pattern, use the smallest primes $2, 3, 5, \ldots$, and give the *largest exponent to the smallest prime*. (Swapping exponents so that a bigger prime gets a bigger power can only increase n: e.g. $2^6 \cdot 3^3 = 1728 < 2^3 \cdot 3^6 = 5832$.)

28 as a product	exponents	smallest number	value
28	27	2^{27}	134,217,728
$14 \cdot 2$	$13, 1$	$2^{13} \cdot 3$	24,576
$7 \cdot 4$	$6, 3$	$2^6 \cdot 3^3$	1,728
$7 \cdot 2 \cdot 2$	$6, 1, 1$	$2^6 \cdot 3 \cdot 5$	**960**

Step 4: Compare and check

The smallest is $960 = 2^6 \cdot 3 \cdot 5$, with $d(960) = (6+1)(1+1)(1+1) = 28$. *Intuition:* splitting the divisor count into more, smaller factors lets you use more distinct small primes with small exponents, which is cheaper than raising 2 or 3 to high powers. (A computer search confirms no number below 960 has 28 divisors.)

$d(n) = \prod(e_i + 1)$**;** $28 = 7 \cdot 2 \cdot 2$ **gives exponents** $(6, 1, 1)$**, so the smallest number is** $2^6 \cdot 3 \cdot 5 = \boxed{960}$ **(beating** $2^6 \cdot 3^3 = 1728$**,** $2^{13} \cdot 3 = 24576$ **and** 2^{27}**).**

Concept Summary and Applications

For $n = \prod p_i^{e_i}$ and $m = \prod p_i^{f_i}$ (using exponent 0 for missing primes):

- **Divisor count:** $d(n) = \prod(e_i + 1)$. Smallest n with D divisors: factor D, subtract 1, give the largest exponent to the smallest prime, compare the candidates.
- **GCD / LCM:** $\gcd = \prod p_i^{\min(e_i, f_i)}$, $\quad \text{lcm} = \prod p_i^{\max(e_i, f_i)}$.
- **Divisibility by a composite:** split it into prime-power pieces and test each (e.g. $225 = 9 \cdot 25$).
- **Trailing zeros:** $\#\text{zeros}(N) = \min\big(v_2(N), v_5(N)\big)$, where $v_p(N)$ is the exponent of p in N.

- **Factor first:** almost every "divisors / multiples / zeros" question becomes easy once everything is written as prime powers.
- **Exponents add when you multiply:** $v_p(ab) = v_p(a) + v_p(b)$ — the key to products and factorials.
- **Divisibility vs. digit product:** here the *number* must be divisible by something (factor the divisor). In digit-product puzzles the *digits* must multiply to a target (factor the target into digits 2–9). Both start by factorising, but they factor different things.
- **Applications:** periodic events lining up (LCM), simplifying fractions (GCD), counting zeros of $n!$, cryptography (RSA relies on factoring being hard).

Additional Problem 1 — Factor, Then Rearrange the Digits (Q335)

Problem: Write 300,000 as a product of primes. Then arrange all the digits of these prime factors to form the smallest possible number.

Step 1: Factorise

$$300{,}000 = 3 \times 100{,}000 = 3 \times 10^5 = 3 \times (2 \cdot 5)^5 = 2^5 \cdot 3 \cdot 5^5.$$

Written out with multiplicity, the prime factors are $2, 2, 2, 2, 2,\ 3,\ 5, 5, 5, 5, 5$ — eleven single digits.

Step 2: Smallest arrangement

All eleven digits must be used, so every arrangement has 11 digits. Among numbers with the same number of digits, the smaller one has the smaller digit in the first position where they differ, so put the digits in *increasing* order:

$$22222355555.$$

(If the question meant only the *distinct* primes $2, 3, 5$, the answer would be 235; state the reading. The usual reading counts each prime factor with its multiplicity.)

$300{,}000 = 2^5 \cdot 3 \cdot 5^5$; the digits $2, 2, 2, 2, 2, 3, 5, 5, 5, 5, 5$ in increasing order give $\boxed{22222355555}$.

Additional Problem 2 — When Do Three Planets Line Up Again? (Q505)

Problem: Three planets orbit a sun with periods 60, 84 and 140 years. Today all three are aligned with the sun. When will they next be aligned again?

Reading 1 (the standard interview answer): all back at today's positions — LCM

Planet i is back where it started after any whole number of its periods, i.e. at multiples of T_i. All three are back simultaneously at common multiples of $60, 84, 140$; the first is the least common multiple. Factorise:

$$60 = 2^2 \cdot 3 \cdot 5, \qquad 84 = 2^2 \cdot 3 \cdot 7, \qquad 140 = 2^2 \cdot 5 \cdot 7.$$

The LCM takes the highest power of each prime that appears:

$$\operatorname{lcm}(60, 84, 140) = 2^2 \cdot 3 \cdot 5 \cdot 7 = 420 \text{ years}.$$

(Check: $420 = 7 \cdot 60 = 5 \cdot 84 = 3 \cdot 140$.)

Reading 2: aligned anywhere — they line up sooner

"Aligned with the sun" does not require returning to *today's* line. Measure each planet's position as a fraction of a full turn: after t years, planet i has turned t/T_i turns. Two planets are on the same ray from the sun when the difference of their turns is a whole number:

$$\frac{t}{60} - \frac{t}{84} = t \cdot \frac{7-5}{420} = \frac{t}{210}, \qquad \frac{t}{84} - \frac{t}{140} = t \cdot \frac{5-3}{420} = \frac{t}{210}.$$

Both are whole numbers first at $t = \mathbf{210}$ years. Indeed at $t = 210$ the planets have made 3.5, 2.5 and 1.5 turns — all three are on the *opposite* side of the sun from today, but on one straight ray. If "aligned" also allows planets on opposite sides of the sun (all on one straight line through the sun), the differences need only be multiples of $\frac{1}{2}$, which first happens at $t = 105$ years (1.75, 1.25, 0.75 turns: positions 270°, 90°, 270°).

Back in today's exact positions: $\operatorname{lcm}(60, 84, 140) = 2^2 \cdot 3 \cdot 5 \cdot 7 = \boxed{420 \text{ years}}$ (the expected interview answer). If any common ray counts, they line up after 210 years; if opposite sides of the sun also count, after 105 years. State the reading you use.

Additional Problem 3 — Smallest 0/1 Number Divisible by 225 (Q657)

Problem: What is the smallest positive number written only with the digits 0 and 1 that is divisible by 225?

Step 1: Split the divisor into prime-power pieces

$225 = 3^2 \cdot 5^2 = 9 \cdot 25$, and 9 and 25 share no prime, so a number is divisible by 225 exactly when it is divisible by both 9 and 25.

Step 2: Divisibility by 25 — look at the last two digits

Since 100 is a multiple of 25, only the last two digits matter; they must be 00, 25, 50 or 75. With only 0s and 1s available, the number must end in **00**.

Step 3: Divisibility by 9 — the digit sum

Since $10 \equiv 1 \pmod 9$, every number has the same remainder mod 9 as its digit sum. With digits 0 and 1, the digit sum is the number of 1s, so we need a multiple of 9 ones — at least **9**.

Step 4: Assemble the smallest

We need at least 9 ones plus the final 00: at least 11 digits. With exactly 11 digits (9 ones and 2 zeros, the zeros at the end) there is only one choice:

$$11111111100.$$

Any other valid number has more digits and is therefore larger. (A computer search over all 0/1 numbers confirms it.)

$225 = 9 \cdot 25$**: end in** 00 **(for** 25**) and use nine** 1**s (for** 9**):** $\boxed{11111111100}$**.**

Additional Problem 4 — Product of Two 10-Digit Numbers Ending in Five Zeros (Q4)

Problem: Two 10-digit numbers are chosen independently and uniformly at random. What is the probability that their product ends in exactly five zeros?

Step 1: Trailing zeros come from pairs 2×5

A trailing zero is a factor $10 = 2 \cdot 5$. Let $v_p(N)$ be the exponent of the prime p in N (e.g. $v_2(40) = 3$, $v_5(40) = 1$, so 40 has one trailing zero). Then

$$\#\text{zeros}(N) = \min\big(v_2(N),\, v_5(N)\big).$$

For the product ab the exponents add: $V_2 = v_2(a) + v_2(b)$ and $V_5 = v_5(a) + v_5(b)$. We need $\min(V_2, V_5) = 5$.

Step 2: How often is a random number divisible by 2^k **or** 5^j**?**

A fraction $\frac{1}{2^k}$ of all integers are divisible by 2^k. So

$$P(v_2 = k) = \frac{1}{2^k} - \frac{1}{2^{k+1}} = \frac{1}{2^{k+1}} \quad (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \dots), \qquad P(v_5 = j) = \frac{1}{5^j} - \frac{1}{5^{j+1}} = \frac{4}{5^{j+1}} \quad (\tfrac{4}{5}, \tfrac{4}{25}, \dots).$$

These are exact for practical purposes, because there are 9×10^9 ten-digit numbers — a huge multiple of any small power like 2^6 or 5^6. The 2-part and the 5-part of a random number are independent (divisibility by 2^k and by 5^j are independent events, since $\frac{1}{2^k 5^j} = \frac{1}{2^k} \cdot \frac{1}{5^j}$).

Step 3: Distribution of the product's exponents

To get $V_2 = n$, split $n = k + (n-k)$ between the two numbers:

$$P(V_2 = n) = \sum_{k=0}^{n} \frac{1}{2^{k+1}} \cdot \frac{1}{2^{n-k+1}} = \frac{n+1}{2^{n+2}}, \qquad P(V_5 = n) = \sum_{k=0}^{n} \frac{4}{5^{k+1}} \cdot \frac{4}{5^{n-k+1}} = \frac{16(n+1)}{5^{n+2}}$$

(every term in each sum is the same, and there are $n+1$ terms).

Step 4: Tail probabilities by hand

Powers of 2: $P(V_2 = 0, 1, 2, 3, 4) = \frac{1}{4}, \frac{2}{8}, \frac{3}{16}, \frac{4}{32}, \frac{5}{64} = \frac{16+16+12+8+5}{64} = \frac{57}{64}$, so

$$P(V_2 \geq 5) = \tfrac{7}{64}, \qquad P(V_2 \geq 6) = \tfrac{7}{64} - P(V_2 = 5) = \tfrac{7}{64} - \tfrac{6}{128} = \tfrac{1}{16}.$$

Powers of 5: $P(V_5 = 0, \dots, 4) = \frac{10000+4000+1200+320+80}{15625} = \frac{15600}{15625}$, so

$$P(V_5 \geq 5) = \tfrac{25}{15625} = \tfrac{1}{625}, \qquad P(V_5 \geq 6) = \tfrac{1}{625} - \tfrac{96}{78125} = \tfrac{29}{78125}.$$

Step 5: Combine

"min = 5" means both exponents are ≥ 5, but not both ≥ 6:

$$P(\min = 5) = P(V_2 \geq 5)P(V_5 \geq 5) - P(V_2 \geq 6)P(V_5 \geq 6) = \frac{7}{64} \cdot \frac{1}{625} - \frac{1}{16} \cdot \frac{29}{78125} = \frac{875 - 116}{5{,}000{,}000} = \frac{759}{5{,}000{,}000}.$$

Step 6: Why the 2s matter

It is tempting to count only factors of 5 ("there are always enough 2s"), giving $P(V_5 = 5) = \frac{96}{78125} \approx 0.0012$. But $P(V_2 \geq 5) = \frac{7}{64} \approx 11\%$: two random numbers usually do *not* supply five 2s. The correct answer is about 8 times smaller.

P**(exactly five zeros)** $= \dfrac{7}{64} \cdot \dfrac{1}{625} - \dfrac{1}{16} \cdot \dfrac{29}{78125} = \boxed{\dfrac{759}{5{,}000{,}000}} \approx 0.000152$ **(confirmed by exact calculation and simulation). Both the 2s and the 5s constrain the answer.**

59 Number Theory: Digit Counting

Representative Problem — How Many Digits (Q21)

Problem: How many digits does 99^{10} have (in base 10)?

The key fact. A positive integer N with exactly d digits sits between the smallest and next-smallest powers of ten:

$$10^{d-1} \le N < 10^{d} \qquad \text{(e.g. a 3-digit number satisfies } 100 \le N < 1000\text{)}.$$

Taking $\log_{10}$ of all three parts (the log is increasing, so inequalities are preserved),

$$d-1 \le \log_{10} N < d \implies d-1 = \lfloor \log_{10} N \rfloor,$$

so the digit count is

$$d = \lfloor \log_{10} N \rfloor + 1.$$

The idea: $\log_{10} N$ measures "how many powers of ten fit into N," and its integer part is one less than the number of digits.

Apply it. Using $\log_{10}(ab) = \log_{10} a + \log_{10} b$ to pull the exponent out,

$$\log_{10}(99^{10}) = 10 \log_{10} 99.$$

Since $99 = 100 \cdot 0.99$,

$$\log_{10} 99 = \log_{10} 100 + \log_{10} 0.99 = 2 + \log_{10} 0.99 \approx 2 - 0.00436 = 1.99564,$$

so $10 \log_{10} 99 \approx 19.956$. Therefore

$$d = \lfloor 19.956 \rfloor + 1 = 19 + 1 = 20.$$

99^{10} has $\lfloor 10 \log_{10} 99 \rfloor + 1 = \boxed{20}$ digits.

Additional Problem 1 — How Often a Digit Appears (Q29)

Problem: Among all the integers from 1 to 1,000,000, how many times does the digit 2 appear in total (counting every occurrence, so 22 counts twice)?

The padding trick. Write every number from 0 to 999,999 as a *six-digit string*, padding with leading zeros ($0 = 000000$, $37 = 000037$, etc.). There are exactly $10^6 = 1{,}000{,}000$ such strings, and this padding does not change how many 2s appear (leading zeros are never 2s).

Now look at *one* digit position — say the hundreds place. As the number runs over all 10^6 strings, that single position cycles through $0, 1, 2, \ldots, 9$ equally often, so each digit value shows up

$$\frac{10^6}{10} = 100{,}000$$

times in that position. The same holds for every one of the 6 positions, so the digit 2 appears

$$6 \times 100{,}000 = 600{,}000$$

times in total across 0 to 999,999. Finally adjust the range: 0 contributes no 2, and 1,000,000 (digits $1, 0, 0, 0, 0, 0, 0$) contributes no 2, so the count for 1 to 1,000,000 is unchanged.

The digit 2 appears $6 \times 100{,}000 = \boxed{600{,}000}$ times.

(General rule: over 0 to $10^k - 1$, written as k-digit strings, each digit value appears $k \cdot 10^{k-1}$ times.)

Additional Problem 2 — Whether a Digit Appears (Q75)

Problem: A whole number is chosen uniformly at random from 1 to 100. What is the probability that it does *not* contain the digit 7?

Count the "avoiders" position by position. Again write the numbers as two-digit strings 00 to 99. For a string to contain no 7, each of its two digits may be any of $\{0, 1, 2, 3, 4, 5, 6, 8, 9\}$ — that is 9 allowed values per position. Since the positions are independent choices,

$$\#\{\text{two-digit strings with no 7}\} = 9 \times 9 = 81.$$

These are the numbers 00 through 99 avoiding 7. Drop 00 (it is not in 1–100), leaving 80 such numbers in 1–99; then add 100 (its digits $1, 0, 0$ contain no 7), giving 81 avoiders in 1–100. Hence

$$P(\text{no digit } 7) = \frac{81}{100} = 0.81.$$

P(no digit 7 in 1–100) $= \dfrac{81}{100} = 81\%$ (so P(contains a 7) $= 19\%$).

(General rule: among k-digit strings, a fraction $(9/10)^k$ avoid any one chosen digit; its complement gives the probability the digit *does* appear.)

Additional Problem 3 — Sum of All Digits over a Range (Q195)

Problem: Add up the digits of *every* integer from 1 to 1,000,000. (This is the sum of the *digits*, not of the numbers — e.g. from 1 to 10 the numbers sum to 55 but the digits sum to 46.) What is the total?

Same padding idea, now summing instead of counting. Write every number from 0 to 999,999 as a six-digit string with leading zeros; the padding adds only 0-digits, so it does not change any digit sum. Fix one position: as the number runs over all 10^6 strings, that position shows each value $0, 1, \ldots, 9$ exactly 10^5 times, so the digits in that one position add up to

$$\underbrace{(0 + 1 + 2 + \cdots + 9)}_{=45} \times 10^5 = 45 \times 10^5.$$

With 6 independent positions, the grand total over 0 to 999,999 is

$$6 \times 45 \times 10^5 = 270 \times 10^5 = 27{,}000{,}000.$$

Since 0 adds nothing, this is also the total for 1 to 999,999. Finally add the digit sum of 1,000,000 itself, which is 1:

$$27{,}000{,}000 + 1 = 27{,}000{,}001.$$

The digits of all numbers from 1 to 1,000,000 sum to $6 \times 45 \times 10^5 + 1 = \boxed{27{,}000{,}001}$.

(General rule: over 0 to $10^k - 1$, the digits sum to $k \times 45 \times 10^{k-1}$ — each of the k positions contributes the average digit 4.5 across 10^k numbers.)

Concept Summary and Applications

- **Number of digits:** N has $\lfloor \log_{10} N \rfloor + 1$ digits, because $10^{d-1} \le N < 10^d$; pull out exponents with $\log_{10}(a^b) = b \log_{10} a$ to size huge powers quickly.
- **How often a digit appears:** pad numbers to a fixed length; each position is uniform over 0–9, so a digit shows up $k \cdot 10^{k-1}$ times over 0 to $10^k - 1$.
- **Whether a digit appears:** count the strings that *avoid* it (9 choices per position, fraction $(9/10)^k$), then take the complement.
- **Sum of the digits over a range:** each padded position averages the digit 4.5, so over 0 to $10^k - 1$ the digits sum to $k \times 45 \times 10^{k-1}$ (the same position-by-position idea, summed rather than counted).
- **Applications:** estimating the magnitude of large numbers, log-scale sizing and Fermi estimates, combinatorial digit counts, and quick market-making on "probability a number looks like…" questions.

60 Optimal Partition: Maximise the Product (Integer and Continuous)

Representative Problem — Split Coins to Maximise the Product (Q148)

Problem: You have x coins. Into how many stacks, and with how many coins in each stack, should you divide them so that the *product* of the stack sizes is as large as possible?

Warm-up example ($x = 10$). A few ways to split 10:

10 (product 10), 5+5 (25), 3+3+3+1 (27), 3+3+4 (36), 3+3+2+2 (36), 2+2+2+2+2 (32).

The best is 36 — mostly 3s. The rest of the solution explains why.

Step 1 — never use a stack of 1. A stack of 1 multiplies the product by 1, i.e. contributes nothing, yet uses up a coin. Merging that coin into any other stack ($k \to k+1$) strictly increases the product. So every stack has at least 2 coins.

Step 2 — never use a stack of 4 or more. For $n \geq 4$, splitting it as $n = 2 + (n-2)$ gives

$$2(n-2) - n = n - 4 \ \geq 0,$$

so the split is never worse: equal at $n = 4$ ($2 \cdot 2 = 4$) and strictly better for $n \geq 5$ (e.g. $5 \to 2 \cdot 3 = 6 > 5$, $6 \to 2 \cdot 4 = 8 > 6$). So any stack of 4 or more can be broken up without losing anything. Hence the optimal stacks can be taken to be only 2s and 3s.

Step 3 — prefer 3 over 2. Compare how much each size multiplies the product *per coin*: a stack of 3 multiplies by 3 using 3 coins, i.e. by $3^{1/3}$ per coin; a stack of 2 gives $2^{1/2}$ per coin. Raise both to the 6th power to compare: $(3^{1/3})^6 = 9 > 8 = (2^{1/2})^6$. So $3^{1/3} > 2^{1/2}$ and 3s win. In particular never keep three 2s: $2+2+2 = 3+3$ but $2^3 = 8 < 9 = 3^2$ — so at most *two* 2s ever remain.

Rule: use as many 3s as possible, absorbing the remainder with 2s:

$x \equiv 0 \pmod 3$:	all 3s		product $3^{x/3}$
$x \equiv 2 \pmod 3$:	$\frac{x-2}{3}$ threes and one 2		product $2 \cdot 3^{(x-2)/3}$
$x \equiv 1 \pmod 3$:	$\frac{x-4}{3}$ threes and two 2s		product $4 \cdot 3^{(x-4)/3}$

(For $x \equiv 1$, never leave a lone 1: turn one $3+1$ into $2+2$, since $4 > 3$.)

Integer examples. $x = 10 \equiv 1$: $3+3+2+2$, product 36 (beats $3+3+3+1 = 27$). $x = 8 \equiv 2$: $3+3+2$, product 18. $x = 9 \equiv 0$: $3+3+3$, product 27. The full table of best products:

x	2	3	4	5	6	7	8	9	10	11	12
best	2	3	4	6	9	12	18	27	**36**	54	81

Use about $x/3$ stacks, almost all of size 3, with one 2 (if $x \equiv 2$) or two 2s (if $x \equiv 1$). E.g. 10 coins $\to 3+3+2+2$, maximum product 36.

Extending to the Continuous Case — Why e Is Optimal

The continuous problem. Now let the pieces be any positive real numbers: split a total S into parts of *any* size to maximise their product. (Think of cutting a rope of length S into pieces and multiplying the lengths.)

Step 1 — for a fixed number of parts, make them equal. By the AM–GM inequality (explained in the next subsection), n parts with sum S have product at most $(S/n)^n$, reached when every part equals S/n. So only the *size* $t = S/n$ of the equal parts remains to be chosen.

Step 2 — choose the best part size (the derivation from Q148). With parts of size t there are S/t of them, so the product is

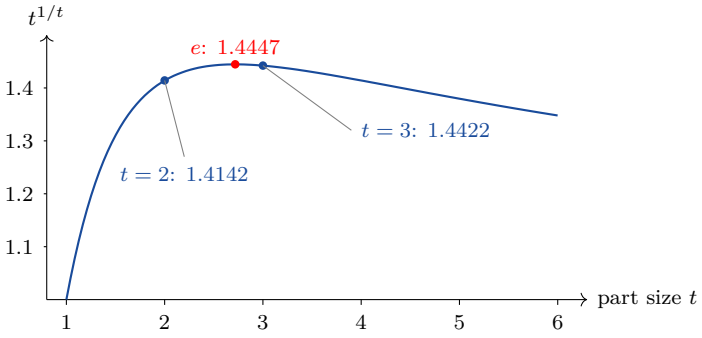
$$t^{S/t} = \left(t^{1/t}\right)^S.$$

Since S is fixed, we maximise $f(t) = t^{1/t}$. Take logarithms, $g(t) = \ln f(t) = \dfrac{\ln t}{t}$, and differentiate with the quotient rule:

$$g'(t) = \frac{\frac{1}{t}\cdot t - \ln t \cdot 1}{t^2} = \frac{1 - \ln t}{t^2}.$$

Because $t^2 > 0$, the sign of g' is the sign of $1 - \ln t$: positive for $t < e$ and negative for $t > e$. So g (and hence f) *rises up to* $t = e$ *and falls after*, and the unique maximiser is

$$1 - \ln t = 0 \implies \boxed{t = e \approx 2.718}.$$



The "growth per unit" $t^{1/t}$ peaks at $t = e$. Among whole numbers, 3 sits closest to the peak (1.4422 vs. 1.4142 for 2) — which is exactly why the integer rule is "use 3s."

Step 3 — respect that the number of parts is a whole number. The ideal $n = S/e$ is usually not an integer, so compare the two nearest whole numbers of parts, $\lfloor S/e \rfloor$ and $\lceil S/e \rceil$, each with equal parts. For $S = 10$: $S/e \approx 3.68$, and

$$n = 3: \ \left(\tfrac{10}{3}\right)^3 \approx 37.04, \qquad n = 4: \ 2.5^4 = 39.0625, \qquad n = 5: \ 2^5 = 32.$$

So the continuous optimum is four pieces of 2.5, product 39.06 — larger than the integer answer 36, because the pieces are now free to sit closer to e.

Continuous case: cut into $n \approx S/e$ *equal* pieces, each close to $e \approx 2.718$ (pick whichever of $\lfloor S/e \rfloor, \lceil S/e \rceil$ gives the larger $(S/n)^n$). Integer case: the nearest usable size to e is 3, hence "as many 3s as possible."

The Connection to Arithmetic and Geometric Means

For positive numbers $a_1, \ldots, a_n$, the **arithmetic mean** (AM) is $\frac{a_1+\cdots+a_n}{n}$ and the **geometric mean** (GM) is $\sqrt[n]{a_1 a_2 \cdots a_n}$. The **AM–GM inequality** says

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \;\geq\; \sqrt[n]{a_1 a_2 \cdots a_n}, \qquad \text{with equality iff } a_1 = a_2 = \cdots = a_n.$$

Why it is true for two numbers. For $a, b > 0$,

$$\Big(\frac{a+b}{2}\Big)^2 - ab = \frac{a^2 + 2ab + b^2 - 4ab}{4} = \frac{(a-b)^2}{4} \;\geq 0,$$

so $\frac{a+b}{2} \geq \sqrt{ab}$, with equality only when $a = b$. (Picture a rectangle with fixed perimeter: its area ab is largest when it is a square.)

How it drives the partition problem. Raise AM–GM to the n-th power: if the parts add up to S, then

$$a_1 a_2 \cdots a_n \;\leq\; \Big(\frac{S}{n}\Big)^n.$$

So, with the *sum fixed*, the *product* is capped by the equal-parts value and reaches it only when all parts are equal. This gives two lessons:

- **Equalise the parts.** Any two unequal parts can be moved toward each other (keeping their sum) and the product goes up — e.g. $5+1$ (5) $\to 3+3$ (9). In the integer case, this is why the optimal stacks differ by at most 1 (only 2s and 3s together).
- **AM–GM fixes the shape, calculus fixes the size.** AM–GM says "make them equal" for a *given* number of parts; it does not say how many parts. The $t^{1/t}$ calculation answers that: equal parts of size e. Integers force the size to 3.

Concept Summary and Applications

- **Integer rule:** split into as many 3s as possible, using one or two 2s for the remainder; never 1s, never parts ≥ 4.
- **Continuous rule:** equal parts of size $\approx e$; the number of parts is $\lfloor S/e \rfloor$ or $\lceil S/e \rceil$, whichever gives the larger $(S/n)^n$.
- **Why e:** maximising the product per unit, $t^{1/t}$, gives $\frac{d}{dt}\frac{\ln t}{t} = \frac{1-\ln t}{t^2} = 0 \Rightarrow t = e$.
- **AM–GM:** for a fixed sum, the product is maximised by equal parts — the principle behind "balance your resources."
- **Applications:** maximising volumes/areas with fixed perimeter or budget, splitting work or capital into balanced chunks, radix economy (base 3 is the most "efficient" number base), and the related urn/ball allocation puzzles.

61 Fibonacci and Tiling Recurrences

Representative Problem — Ten Light Bulbs, No Two Adjacent (Q63)

Problem: Ten light bulbs stand in a row, each either on or off, but *no two adjacent bulbs may be on at the same time*. How many valid on/off patterns are there?

Set up a recurrence by looking at the last bulb. Let a_n be the number of valid patterns for a row of n bulbs. The trick that unlocks these problems is to condition on the *last* bulb and reduce to smaller rows:

- **Last bulb off.** Then the first $n-1$ bulbs may be any valid pattern — no new restriction. That contributes a_{n-1} patterns.
- **Last bulb on.** Then its neighbour, bulb $n-1$, is forced *off* (no two adjacent on); the first $n-2$ bulbs may be any valid pattern. That contributes a_{n-2} patterns.

These two cases are mutually exclusive and cover everything, so

$$a_n = a_{n-1} + a_{n-2} \qquad \text{(the \textbf{Fibonacci recurrence})}.$$

Pin down the starting values by hand. For one bulb, the patterns are `0` and `1`, so $a_1 = 2$. For two bulbs, `00`, `01`, `10` are valid (only `11` is banned), so $a_2 = 3$. Now climb the recurrence:

n	1	2	3	4	5	6	7	8	9	10
a_n	2	3	5	8	13	21	34	55	89	**144**

Each entry is the sum of the two before it ($5 = 3 + 2$, $8 = 5 + 3$, …, $144 = 89 + 55$).

There are $a_{10} = \boxed{144}$ valid patterns. (These are the Fibonacci numbers: $a_n = F_{n+2}$, so $a_{10} = F_{12} = 144$.)

Additional Problem — Climbing a Staircase

Problem: A staircase has n steps. On each move you may climb either **one** step or **two** steps. How many distinct sequences of moves take you from the bottom to the top?

Same idea — condition on the last move. Let $f(n)$ be the number of ways to reach step n. Your final move landed you there either from step $n-1$ (a 1-step) or from step $n-2$ (a 2-step), and these two possibilities are distinct and exhaustive:

$$f(n) = f(n-1) + f(n-2)$$

— the Fibonacci recurrence again. Only the starting values differ from the light-bulb problem: there is exactly $f(1) = 1$ way to climb one step (a single 1-step) and $f(2) = 2$ ways to climb two

(1+1 or 2). Building up,

n	1	2	3	4	5	6	7	8	9	10
$f(n)$	1	2	3	5	8	13	21	34	55	89

so, for example, a 5-step staircase has $f(5) = 8$ ways and a 10-step staircase has $f(10) = 89$.

The number of ways to climb n steps (by 1s and 2s) is the Fibonacci number $f(n) = f(n-1) + f(n-2)$ with $f(1) = 1, f(2) = 2$; here $f(n) = F_{n+1}$, e.g. $f(10) = 89$.

Concept Summary and Applications

- **The signature:** whenever a valid configuration is built one item at a time and each item's choice depends only on the *immediately previous* one — "no two adjacent," "take 1 or 2," "no `11`" — the count obeys $a_n = a_{n-1} + a_{n-2}$, the Fibonacci recurrence.
- **Derivation recipe:** condition on the last element; split into the cases it allows; each case leaves a smaller instance of the *same* problem, giving the recurrence directly.
- **Only the seeds differ:** light bulbs ($a_1 = 2, a_2 = 3$), staircases ($f_1 = 1, f_2 = 2$), 1×2 domino tilings of a $1 \times n$ strip, and binary strings with no `11` are all the *same* recurrence, landing on different offsets of the Fibonacci sequence $1, 1, 2, 3, 5, 8, 13, \ldots$.
- **Applications:** tiling and packing counts, restricted binary strings and codes, path-counting on a line, and any "build it up step by step with a local constraint" combinatorics problem.

62 Parity of Sums of Primes: 2 Is the Only Even Prime

Questions like "pick some primes at random — is their sum odd or even?" look like they require adding up many cases. They don't. Two facts settle everything: **(1)** the parity (odd/even) of a sum depends only on how many odd terms it contains, and **(2)** every prime except 2 is odd. So the whole question becomes: *is the number* 2 *in the selection or not?*

Representative Problem — Six of the First Twelve Primes (Q262)

Problem: You choose 6 different numbers at random from the first 12 prime numbers. What is the probability that their sum is odd?

Step 1: List the primes and sort them by parity

The first 12 primes are

$$2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 17,\ 19,\ 23,\ 29,\ 31,\ 37.$$

A prime has no divisors other than 1 and itself. Every even number greater than 2 is divisible by 2, so it is not prime. Hence 2 **is the only even prime**; here there is 1 even prime and 11 odd ones.

Step 2: Parity rules for sums

- Adding an **even** number never changes parity (odd + even = odd, even + even = even).
- Adding an **odd** number flips parity. So a sum of k odd numbers is odd when k is odd and even when k is even (e.g. $3+5=8$ even, $3+5+7=15$ odd).

Therefore the parity of the sum is decided by **how many odd numbers** are chosen.

Step 3: Only two kinds of selection

Since there is only one even prime, a selection of 6 primes either

- **does not contain 2:** all 6 are odd; 6 odd terms $\Rightarrow$ the sum is **even**;
- **contains 2:** the other 5 are odd; 5 odd terms give an odd subtotal, and adding 2 keeps it odd $\Rightarrow$ the sum is **odd**.

So *the sum is odd exactly when* 2 *is chosen.* Example: $2+3+5+7+11+13=41$ (odd), while $3+5+7+11+13+17=56$ (even).

Step 4: Count

All $\binom{12}{6}=924$ selections are equally likely. Selections containing 2: fix 2, then choose the other 5 from the 11 odd primes, $\binom{11}{5}=462$ ways. So

$$P(\text{sum odd}) = \frac{\binom{11}{5}}{\binom{12}{6}} = \frac{462}{924} = \frac{1}{2}.$$

Step 5: A faster way — $P(2$ is chosen$) = r/n$

Choosing r of n items at random, every item has the same chance of being chosen (symmetry), and the expected number of chosen items is r. So each item — in particular 2 — is chosen with probability $\frac{r}{n}$. Check: $\binom{n-1}{r-1}/\binom{n}{r} = \frac{r}{n}$. Here $\frac{6}{12}=\frac{1}{2}$, with no binomial coefficients at all.

The sum is odd exactly when the only even prime, 2, is among the six chosen, so $P = \frac{\binom{11}{5}}{\binom{12}{6}} = \frac{6}{12} = \boxed{\frac{1}{2}}$.

Concept Summary and Applications

Choose r distinct primes from the first n primes (1 even prime, $n-1$ odd):

$$P(2 \text{ chosen}) = \frac{\binom{n-1}{r-1}}{\binom{n}{r}} = \frac{r}{n}.$$

- 2 **not** chosen: r odd terms $\Rightarrow$ sum has the parity of r.
- 2 chosen: $r-1$ odd terms $\Rightarrow$ sum has the parity of $r-1$.

So: r even $\Rightarrow P(\text{sum odd}) = \frac{r}{n}$; r odd $\Rightarrow P(\text{sum even}) = \frac{r}{n}$.

- **Reduce to parity:** replace each number by "odd" or "even"; only the count of odd terms matters.
- **One special element:** when exactly one item differs (here the prime 2), the event is just "is that item chosen?", with probability r/n by symmetry.
- **Watch the sampling model:** "different primes" (without replacement) and "independent picks" (with replacement) give different answers — see Additional Problem 3.
- **Applications:** parity arguments in puzzles (can a sum equal a given number?), checking invariants, and any "pick r of n, does item X appear?" question.

Additional Problem 1 — Even or Odd Sums from Different Sets of Primes (Q331, Q379, Q567)

Problems: Numbers are chosen at random without replacement.

(a) (Q331) 3 numbers from the first 9 primes. Probability that the sum is *even*?
(b) (Q379) 4 numbers from the first 15 primes. Probability that the sum is *odd*?
(c) (Q567) 4 numbers from the first 20 primes. Probability that the sum is *even*?

In each case the list contains exactly one even prime (2) and the rest are odd. Decide which parity "2 chosen" produces, then use $P(2 \text{ chosen}) = r/n$.

(a) 3 of the first 9 primes $(2, 3, 5, 7, 11, 13, 17, 19, 23)$

- Without 2: 3 odd terms $\Rightarrow$ odd sum.
- With 2: 2 odd terms $+2 \Rightarrow$ even sum.

Even sum $\iff$ 2 chosen:

$$P = \frac{\binom{8}{2}}{\binom{9}{3}} = \frac{28}{84} = \frac{1}{3} = \frac{3}{9}.$$

(b) 4 of the first 15 primes

Without 2: 4 odd terms $\Rightarrow$ even. With 2: 3 odd terms $+2 \Rightarrow$ odd. Odd sum $\iff$ 2 chosen:

$$P = \frac{\binom{14}{3}}{\binom{15}{4}} = \frac{364}{1365} = \frac{4}{15}.$$

(c) 4 of the first 20 primes

Same parity pattern as (b), but now we want an *even* sum, which happens exactly when 2 is *not* chosen:

$$P = 1 - \frac{4}{20} = \frac{4}{5} \qquad \left(\text{check: } \frac{\binom{19}{4}}{\binom{20}{4}} = \frac{3876}{4845} = \frac{4}{5}\right).$$

(a) $\frac{1}{3}$ (even $\iff$ 2 chosen) (b) $\frac{4}{15}$ (odd $\iff$ 2 chosen) (c) $\frac{4}{5}$ (even $\iff$ 2 not chosen).

Additional Problem 2 — The Sum of the First Eight Primes (Q588)

Problem: What is the probability that the sum of the first eight prime numbers is odd?

There is nothing random here — the set is fixed: $2, 3, 5, 7, 11, 13, 17, 19$. It contains 7 odd primes (an odd count, so their sum is odd) plus the even 2, which does not change parity. So the sum is odd; indeed $2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 = 77$. The "probability" is 1. (Interviewers include such questions to check that you notice when nothing is random.)

The sum is 77, always odd: probability 1.

Additional Problem 3 — Two Primes Between 1 and 20: Two Sampling Models (Q487)

Problem: Two prime numbers between 1 and 20 are picked at random. What is the probability that their sum is even? Answer (a) if the two primes must be different, and (b) if they are picked independently (repeats allowed).

The primes are $2, 3, 5, 7, 11, 13, 17, 19$: 8 primes, 7 odd. A sum of two numbers is even when both are odd or both are even; "both even" means both are 2.

(a) Two different primes

There are $\binom{8}{2} = 28$ equally likely pairs, and $(2, 2)$ is impossible. Even sum $\iff$ both odd: $\binom{7}{2} = 21$ pairs.

$$P = \frac{21}{28} = \frac{3}{4} \qquad \text{(equivalently } 1 - P(2 \text{ chosen}) = 1 - \tfrac{2}{8}\text{)}.$$

(b) Independent picks with repeats

Each pick is 2 with probability $\frac{1}{8}$ and odd with probability $\frac{7}{8}$, independently:

$$P = P(\text{both odd}) + P(\text{both } 2) = \left(\frac{7}{8}\right)^2 + \left(\frac{1}{8}\right)^2 = \frac{49+1}{64} = \frac{25}{32}.$$

Common mistake: adding the case $(2,2)$ to the 21 odd pairs and dividing by 28 (giving $\frac{22}{28}$). That mixes the models: $\binom{8}{2} = 28$ counts only pairs of *different* primes, where $(2,2)$ cannot occur. Pick one model and count consistently within it.

(a) Different primes: $\frac{3}{4}$. (b) Independent picks: $\frac{25}{32}$.

63 The Frobenius (Chicken McNugget) Formula

What the Frobenius Formula Says

Suppose you have coins (or stamps, or McNugget boxes) in a few fixed positive denominations and an unlimited supply of each. Using non-negative whole numbers of each coin, which total amounts can you make? If the denominations share no common factor, then *every* large enough amount is makeable, but some small amounts are not. The **Frobenius number** is the **largest amount that cannot be made**. (It is also called the **Chicken McNugget** problem: with boxes of 6, 9, and 20 nuggets, the largest un-orderable number is 43.)

For **two** coprime denominations m and n (i.e. $\gcd(m, n) = 1$), the largest integer that *cannot* be written as $mx + ny$ with $x, y \geq 0$ is

$$g(m, n) = mn - m - n,$$

and the number of non-makeable amounts is $\frac{(m-1)(n-1)}{2}$.

Two things are worth stressing. First, the **coprimality condition is essential**: if $\gcd > 1$, only multiples of that common factor are ever reachable, so there is *no* largest un-makeable number (infinitely many amounts fail). Second, for **three or more** denominations there is no simple closed formula in general — but a denomination that is itself makeable from the others is *redundant* and can be dropped, sometimes collapsing the problem back to the two-coin formula.

Representative Problem — Coins of 3, 8, and 20

Problem: With an unlimited supply of coins worth 3, 8, and 20, what is the largest positive integer that cannot be formed as a non-negative combination $3a + 8b + 20c$?

Step 1 — a largest un-makeable number exists. A finite Frobenius number exists exactly when the denominations are collectively coprime. Here $\gcd(3, 8, 20) = 1$, so all sufficiently large integers are makeable and there is a well-defined largest exception to find.

Step 2 — drop the redundant 20. Notice 20 is itself makeable from the others:

$$20 = 4 \cdot 3 + 1 \cdot 8.$$

So any total that uses a 20-coin could instead use four 3s and one 8 — the 20 adds nothing new. The amounts makeable from $\{3, 8, 20\}$ are exactly those makeable from $\{3, 8\}$, and the problem reduces to the two-coin case $3a + 8b$.

Step 3 — the two-coin formula, and why it holds. For coprime m, n the answer is $g(m, n) = mn - m - n$. Here is the reasoning, which also shows *why* coprimality is needed. Sort every non-negative integer by its remainder modulo m, and study each remainder class separately.

1. **An m-coin keeps you in the same class.** Adding m does not change a number's remainder mod m. So m-coins let you climb *within* a remainder class in steps of m, but never switch classes.

2. **Each class has a single threshold.** Within one class, once some value T is makeable, so are $T+m, T+2m, \ldots$ (just add m-coins), while everything in the class below T is not. The largest un-makeable number in that class is therefore $T-m$.
3. **Only the n-coins choose the class.** Since m-coins do not affect the remainder, the class of $mx+ny$ is decided by ny. To reach a class you must use just enough n-coins to land on its remainder; the cheapest way uses the smallest count k with $kn \equiv r \pmod m$, costing kn.
4. **Coprimality reaches every class exactly once.** Because $\gcd(m,n)=1$, the multiples $0, n, 2n, \ldots, (m-1)n$ hit all m remainders, each exactly once — so every class gets a finite threshold. (If gcd >1 this fails: some classes are never reached, and no finite Frobenius number exists.)
5. **Worst class wins.** The class reached with k n-coins has threshold kn, hence largest gap $kn-m$. The biggest gap over all classes is at the largest $k=m-1$:

$$g(m,n) = (m-1)n - m = mn - m - n.$$

For $m=3$, $n=8$, list the multiples of 8 and their remainders mod 3:

k	$8k$	$8k \bmod 3$	largest unreachable $= 8k-3$
0	0	0	–
1	8	2	5
2	16	1	**13**

The class $\equiv 1 \pmod 3$ is the bottleneck — it needs two 8-coins (16) before it is reachable — so $16-3=13$ is the largest unreachable value:

$$g(3,8) = 3\cdot 8 - 3 - 8 = 24 - 11 = 13.$$

Step 4 — verify. 13 is not $3a+8b$: with $b=0$, 13 is not a multiple of 3; with $b=1$, $13-8=5$ is not a multiple of 3; with $b\geq 2$, $8b>13$. And every integer from 14 up is makeable:

$$14 = 8+3+3, \quad 15 = 3\cdot 5, \quad 16 = 8+8, \quad 17 = 8+3+3+3,$$

after which adding 3s covers all larger integers. The complete list of un-makeable amounts is $1, 2, 4, 5, 7, 10, 13$ (seven of them, matching $\frac{(3-1)(8-1)}{2} = 7$).

The largest integer not formable from 3, 8, 20 is $\boxed{13}$ (the 20 is redundant since $20 = 4\cdot 3 + 8$, and $g(3,8) = 3\cdot 8 - 3 - 8 = 13$).

Additional Problem — The Chicken McNugget Numbers $(6, 9, 20)$

Problem: Nuggets come in boxes of 6, 9, and 20. What is the largest number of nuggets you *cannot* order exactly?

Here no coin is redundant (none of $6, 9, 20$ is makeable from the other two) and there is no closed formula, so we exploit the structure. The key observation: **6 and 9 are both multiples of 3**, so only the 20-boxes can change the total's remainder modulo 3. Handle each remainder class separately.

What 6 and 9 alone make. Since $6a + 9b = 3(2a + 3b)$ and $2a + 3b$ takes every value ≥ 2 (plus 0, missing only 1), the reachable totals are

$$\{0, 6, 9, 12, 15, 18, \dots\} = 0 \text{ and every multiple of 3 that is } \geq 6.$$

So the *only* multiple of 3 you cannot make from 6s and 9s is 3 itself.

Split by remainder mod 3 (note $20 \equiv 2,\ 40 \equiv 1,\ 60 \equiv 0 \pmod 3$):

- **Totals** $\equiv 0 \pmod 3$**:** use no 20-boxes; reachable set is $\{0, 6, 9, 12, \dots\}$, so the largest unreachable value is just **3**.
- **Totals** $\equiv 2 \pmod 3$**:** you need at least one 20 (as $20 \equiv 2$). Reachable $= 20 + \{0, 6, 9, \dots\} = \{20, 26, 29, 32, \dots\}$. The value $20 + 3 = 23$ falls in the "missing 3" gap and is unreachable, while $20, 26, 29, \dots$ are fine — largest unreachable = **23**.
- **Totals** $\equiv 1 \pmod 3$**:** you need at least *two* 20s (since $40 \equiv 1$; one 20 gives the wrong remainder). Reachable $= 40 + \{0, 6, 9, \dots\} = \{40, 46, 49, \dots\}$, and $40 + 3 = 43$ is the unreachable one — largest unreachable = **43**.

Taking the worst class, the Frobenius number is $\max(3, 23, 43) = 43$. Every total from 44 up is orderable ($44 = 6 + 9 + 9 + 20$, $45 = 9 \cdot 5$, $46 = 6 + 20 + 20$, $47 = 9 + 9 + 9 + 20$, $48 = 6 \cdot 8$, then add 6s).

Cross-check (Apéry-set method). For any set of denominations, find the smallest orderable amount in each residue class modulo one coin, then subtract that coin. Modulo 6, the smallest orderable amounts are

$$r = 0\colon 0, \quad r = 1\colon 49, \quad r = 2\colon 20, \quad r = 3\colon 9, \quad r = 4\colon 40, \quad r = 5\colon 29,$$

and the Frobenius number is the largest of these minus 6: $49 - 6 = 43$. ✓

The largest un-orderable number of nuggets from boxes of $6, 9, 20$ is $\boxed{43}$ (the $\equiv 1 \bmod 3$ class needs two 20-boxes, and $43 = 40 + 3$ lands on the one gap 6s and 9s cannot fill).

Concept Summary and Applications

- **Two-coin formula:** for coprime m, n, the largest un-makeable amount is $mn - m - n$, with $\frac{(m-1)(n-1)}{2}$ amounts un-makeable in total.
- **Check coprimality first:** a finite answer exists only when the denominations are collectively coprime; otherwise infinitely many amounts fail.
- **Reduce redundant coins:** drop any denomination expressible from the others; with luck the problem falls back to two coins (no closed form exists for three or more in general).
- **Applications:** postage/stamp and coin-change problems, the McNugget puzzle, packaging and bundling questions, and modular-reachability arguments in number theory.

64 Pairwise-Coprime Sets: Spending Each Prime Once

A set of whole numbers is *pairwise coprime* if no two of them share a common factor greater than 1. Questions about such sets look like number puzzles, but they reduce to a simple resource-allocation idea: **every prime is a resource that can be used by at most one number in the set.** Once you see this, finding the largest sum, the largest size, or any other optimum becomes a matter of deciding how to "spend" the primes.

Representative Problem — Maximum Sum of a Pairwise-Coprime Set (Q437)

Problem: Choose a set of numbers from $\{2, 3, \ldots, 30\}$ such that no two of them share a common factor greater than 1. Which set has the largest possible sum? Using the same rule, what can you say about the numbers 2 to 1000?

Step 1: Translate "coprime" into a statement about primes

Two numbers have a common factor greater than 1 exactly when some prime divides both of them (any common factor > 1 has a prime factor, and that prime divides both numbers). So

$$\text{the set is pairwise coprime} \iff \text{no prime divides two different chosen numbers.}$$

Think of each chosen number as "using up" the primes that divide it: $28 = 2^2 \cdot 7$ uses the primes 2 and 7; $27 = 3^3$ uses only 3; 11 uses only 11. The rule says each prime may be used *at most once*.

Step 2: The obvious answer — all primes — and why it is not optimal

The primes up to 30 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$. They are pairwise coprime, and their sum is

$$2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 + 29 = 129.$$

But this wastes the small primes. The prime 3 contributes only 3 to the sum, whereas the same prime could be "spent" on $27 = 3^3$, which uses no other prime and contributes 27. That single swap already gains 24. The task is to spend each prime on the largest number it can buy.

Step 3: Large primes stay as themselves

For $p \in \{11, 13, 17, 19, 23, 29\}$, the only multiples of p in range are p itself and $22 = 2 \cdot 11$, $26 = 2 \cdot 13$. Using 22 or 26 would also use up the prime 2, which is valuable elsewhere (Step 5 checks this). So these six primes are taken as themselves:

$$11 + 13 + 17 + 19 + 23 + 29 = 112.$$

Step 4: What can the small primes $2, 3, 5, 7$ buy?

List, for each group of small primes, the largest number up to 30 built *only* from that group (using every prime in the group):

primes used	best number	primes used	best number
$\{2\}$	$16 = 2^4$	$\{2, 7\}$	$28 = 2^2 \cdot 7$
$\{3\}$	$27 = 3^3$	$\{2, 3\}$	$24 = 2^3 \cdot 3$
$\{5\}$	$25 = 5^2$	$\{2, 5\}$	$20 = 2^2 \cdot 5$
$\{7\}$	7	$\{2, 3, 5\}$	$30 = 2 \cdot 3 \cdot 5$

($7^2 = 49$ is too big, so 7 alone can only buy 7.)

Step 5: Split $\{2, 3, 5, 7\}$ into groups to maximise the total

- each prime separately: $16 + 27 + 25 + 7 = 75$;
- pair 2 with 7: $\mathbf{28} + \mathbf{27} + \mathbf{25} = \mathbf{80}$ (the best);
- $30 + 7 = 37$; $24 + 25 + 7 = 56$; $20 + 27 + 7 = 54$ — all worse;
- using 22 (or 26) also spends 11 (or 13): e.g. $22 + 27 + 25 + 7 = 81$, but it loses the 11, so it is worth 81 against $80 + 11 = 91$.

Pairing 2 with 7 wins because 7 alone can only buy 7 ($7^2 = 49$ is too big) and 2 alone buys 16, a total of 23; together they buy 28, a gain of 5, while 3 and 5 are already at their best (27, 25).

Step 6: The optimal set

$$\{11, 13, 17, 19, 23, 25, 27, 28, 29\}, \qquad 112 + 80 = 192.$$

An exhaustive computer search over all subsets confirms that 192 is the maximum. (If 1 is allowed, as in Q390/Q444, add it — 1 is coprime to everything — for 193.)

Step 7: Scaling up to 2–1000

Because $32^2 = 1024 > 1000$, every number up to 1000 has *at most one* prime factor larger than 31 (two such factors would multiply to more than 1000). So the large primes $37, \ldots, 997$ are never in competition with one another; only the 11 small primes $2, 3, \ldots, 31$ need to be assigned. Each small prime is spent either

- on its own largest power, e.g. $961 = 31^2$, $841 = 29^2$; or
- on a product with a large prime landing close to 1000, e.g. $976 = 16 \cdot 61$, $981 = 9 \cdot 109$, $925 = 25 \cdot 37$, $973 = 7 \cdot 139$, $979 = 11 \cdot 89$, $949 = 13 \cdot 73$, $901 = 17 \cdot 53$, $893 = 19 \cdot 47$, $989 = 23 \cdot 43$ (the large prime in each product is then not used again).

The remaining 148 primes are taken as themselves. A dynamic program over subsets of the small primes (exact) gives 159 numbers with maximum sum 85,684, compared with 76,127 for all 168 primes.

2–30: $\{11, 13, 17, 19, 23, 25, 27, 28, 29\}$, maximum sum $\boxed{192}$ (all primes give only 129).
2–1000: maximum sum 85,684.

Concept Summary and Applications

Pairwise coprime $\iff$ each prime is used by at most one chosen number. Choosing a set means assigning primes to numbers.

- **Maximum sum:** spend small primes on the *largest* number they can buy (high prime powers $27 = 3^3$, $25 = 5^2$, or products such as $28 = 2^2 \cdot 7$); large primes with no room for multiples stay as themselves.
- **Maximum count:** every number needs at least one prime of its own, so

$$\max |S| = \pi(N) \quad (\text{number of primes} \leq N),$$

  achieved by spending each prime on its own *prime power*.
- **Size bound trick:** if $N < (\text{next prime after } p)^2$, every number $\leq N$ has at most one prime factor above p — only primes $\leq \sqrt{N}$ need careful assignment.

- **Same constraint, different objectives:** sum rewards concentrating primes into big numbers; count rewards spreading them out one per number. The two goals give different optimal sets (192 with 9 numbers vs. 10 numbers with sum at most 187 — see below).
- **Exhaustive check is cheap:** only the small primes interact, so you can enumerate the ways to group them (here, groups of $\{2, 3, 5, 7\}$).
- **Applications:** interview "choose numbers with no common factor" puzzles (Q390, Q437, Q444), scheduling/assignment problems where each resource can serve one task, and the Chinese Remainder Theorem (moduli must be pairwise coprime).

Additional Problem — The Largest Pairwise-Coprime Set (Opposite Goal)

Problem: From the integers $\{2, 3, \ldots, 30\}$, choose *as many numbers as possible* such that no two share a common factor greater than 1.

(a) What is the largest possible number of elements?
(b) How many different sets achieve this maximum size?
(c) Among these largest sets, what are the smallest and largest possible sums? Why can a largest set not reach the maximum sum 192 of the representative problem?
(d) What is the maximum size for $\{2, \ldots, 1000\}$?

(a) Upper bound: one prime per number

Every number $n \geq 2$ has at least one prime factor. Pick one prime factor of each chosen number (say its smallest). Two different chosen numbers cannot get the same prime, because that prime would then divide both of them, contradicting pairwise coprimality. So the chosen numbers are matched to *different* primes up to 30, and there are only 10 such primes $(2, 3, 5, 7, 11, 13, 17, 19, 23, 29)$. Hence

$$|S| \leq 10.$$

Achieving it: the 10 primes themselves are pairwise coprime. So the maximum size is $\boxed{10}$.

(b) Every largest set uses one prime per number exactly

In a 10-element set the 10 numbers must use all 10 primes, and they cannot use more than 10 in total. If one number used two primes (e.g. 28 uses 2 and 7), some other number would have

no prime left. So **each number uses exactly one prime**, i.e. each number is a *prime power* $p^k \le 30$. The choices are:

prime	allowed powers ≤ 30	number of choices
2	2, 4, 8, 16	4
3	3, 9, 27	3
5	5, 25	2
$7, 11, 13, 17, 19, 23, 29$	only the prime itself	1 each

The choices are independent, so the number of largest sets is

$$4 \times 3 \times 2 \times 1^7 = \boxed{24}.$$

(c) Range of sums among the largest sets

- **Smallest:** take the lowest power of each prime — the primes themselves: sum 129.
- **Largest:** take the highest power of each prime: $16, 27, 25$ and $7, 11, 13, 17, 19, 23, 29$:

$$16 + 27 + 25 + (7 + 11 + 13 + 17 + 19 + 23 + 29) = 68 + 119 = \boxed{187}.$$

So $129 \le \text{sum} \le 187$ for every 10-element set.

Why not 192**?** The maximum-sum set $\{11, 13, 17, 19, 23, 25, 27, 28, 29\}$ uses $28 = 2^2 \cdot 7$, which spends two primes on one number. That is good for the sum ($28 > 16 + 7$) but costs one element: it has only 9 numbers. A largest set is forced to keep 2 and 7 apart, giving $16 + 7 = 23$ instead of 28, so it loses 5: $192 - 5 = 187$. This is the trade-off between the two goals.

(d) Scaling up to 1000

The same argument works for any N: at most one number per prime, and all primes work. So

$$\max |S| = \pi(1000) = \boxed{168}.$$

(a) 10 (one number per prime). (b) 24 sets — each number must be a prime power. (c) Sums range from 129 (all primes) to 187 ($16, 27, 25$ plus the primes 7–29); 192 needs $28 = 2^2 \cdot 7$, which spends two primes and leaves only 9 numbers. (d) $\pi(1000) = 168$. (Checked by exhaustive computer search.)

65 Recovering Five Numbers from Their Pairwise Sums

Representative Problem — Five Pumpkins (Q284)

Problem: There are 5 pumpkins. You are told the total weight of every pair of pumpkins — all $\binom{5}{2} = 10$ pairwise sums — but not which pair each sum came from. Determine the weight of each pumpkin.

The crux: the sums are unlabelled. If you knew which pair produced each sum, this would be an easy system of linear equations. You don't — you only have a list of ten numbers. (A formula like "pumpkin $i = \frac{S_i - T/4}{3}$" where S_i is the total of the four sums containing pumpkin i is true but useless, because you cannot tell which four sums those are.) The trick is to **sort** the sums and read the information hiding in the extremes.

Setup. Name the weights in increasing order, $a \le b \le c \le d \le e$, and sort the ten pairwise sums: $s_1 \le s_2 \le \cdots \le s_{10}$.

Step 1 — the grand total, by counting. Each pumpkin is paired with each of the other 4 pumpkins, so it appears in exactly 4 of the 10 sums. Adding all ten sums therefore counts every weight 4 times:

$$s_1 + s_2 + \cdots + s_{10} = 4\,(a+b+c+d+e) \quad \Longrightarrow \quad T := a+b+c+d+e = \tfrac{1}{4}\sum_{i=1}^{10} s_i.$$

Step 2 — four sums are identified for free. The extremes can only come from specific pairs:

- $s_1 = a + b$: the two lightest pumpkins give the smallest sum.
- $s_2 = a + c$: every other sum except $a + b$ is at least $a + c$ (for example $a + d \ge a + c$ and $b + c \ge a + c$), so $a + c$ is next smallest.
- $s_{10} = d + e$: the two heaviest give the largest sum.
- $s_9 = c + e$: symmetrically, every other sum except $d + e$ is at most $c + e$.

(The middle sums $s_3, \ldots, s_8$ are ambiguous — e.g. $a + d$ and $b + c$ can come in either order — so we avoid them.)

Step 3 — solve, starting from the middle pumpkin. The sums $s_1 = a + b$ and $s_{10} = d + e$ together contain every pumpkin except c, so

$$c = T - (a+b) - (d+e) = T - s_1 - s_{10}.$$

Now peel off the rest using the identified sums:

$$a = s_2 - c, \qquad b = s_1 - a, \qquad e = s_9 - c, \qquad d = s_{10} - e.$$

Worked example. Suppose the ten pairwise sums are $\{3, 4, 5, 5, 6, 6, 7, 7, 8, 9\}$.

- Total: $\sum s_i = 60$, so $T = \frac{60}{4} = 15$.
- Middle pumpkin: $c = 15 - s_1 - s_{10} = 15 - 3 - 9 = 3$.

- Then $a = s_2 - c = 4 - 3 = 1,\ b = s_1 - a = 3 - 1 = 2,\ e = s_9 - c = 8 - 3 = 5,\ d = s_{10} - e = 9 - 5 = 4.$

Check: the ten pairwise sums of $\{1, 2, 3, 4, 5\}$ are $3, 4, 5, 5, 6, 6, 7, 7, 8, 9$ — exactly the given list.

Sort the ten sums $s_1 \le \cdots \le s_{10}$. Compute the total $T = \frac{1}{4}\sum s_i$, then $c = T - s_1 - s_{10}$, $a = s_2 - c$, $b = s_1 - a$, $e = s_9 - c$, $d = s_{10} - e$. In the example the weights are $\boxed{1, 2, 3, 4, 5}$.

Why It Works, and a Caveat

- **Counting gives the total.** With n numbers, each appears in $n-1$ pairwise sums, so the total is $\frac{1}{n-1}\sum$(pairwise sums).
- **Sorting gives anchors.** The two smallest and two largest sums must be $a+b$, $a+c$, $c+e$, $d+e$ no matter what the weights are. Together with the total, these four equations pin down all five weights, with the middle weight c as the linchpin. (Checked by exact recovery on many random weight sets.)
- **Caveat — the answer is not always unique for other n.** For 5 numbers the list of pairwise sums determines the numbers. But for some sizes it does not: with 4 numbers, $\{1, 4, 6, 7\}$ and $\{2, 3, 5, 8\}$ both have pairwise sums $\{5, 7, 8, 10, 11, 13\}$. (A classical result: non-uniqueness can only happen when n is a power of 2.)

Concept Summary and Applications

- **Double counting:** summing all the given quantities and counting how often each unknown appears turns a messy list into one clean equation.
- **Extremes are informative:** when data are unlabelled, sorted extremes (the smallest and largest values) are often the only unambiguous pieces — anchor on them.
- **Solve from the linchpin:** find one unknown directly (here c), then unwind the rest.
- **Applications:** reconstruction puzzles ("weigh in pairs"), recovering positions from pairwise distances (the turnpike problem), and checksum-style reasoning in coding and data validation.

66 Rational Numbers: Fractions in an Interval

Representative Problem — Which Denominators Fit (Q305)

Problem: Two positive integers a and b satisfy $\dfrac{a}{b} \in [0.48, 0.52]$. For which values of b is this possible — i.e. for which b does *some* positive integer a make a/b land in that interval?

Step 1 — turn the decimals into exact fractions. Since $0.48 = \frac{12}{25}$ and $0.52 = \frac{13}{25}$, the condition is $\frac{12}{25} \le \frac{a}{b} \le \frac{13}{25}$. Multiplying through by $b > 0$ turns it into a statement about a:

$$\frac{12b}{25} \le a \le \frac{13b}{25}.$$

So b works exactly when the interval $\left[\frac{12b}{25}, \frac{13b}{25}\right]$ contains at least one whole number a.

Step 2 — long intervals always contain an integer. The interval has length

$$\frac{13b}{25} - \frac{12b}{25} = \frac{b}{25}.$$

Any interval of length ≥ 1 must contain an integer, so as soon as $\frac{b}{25} \ge 1$, i.e. $b \ge 25$, the value b automatically works. That leaves only $1 \le b \le 24$ to examine by hand.

Step 3 — the small cases, and a clean pattern. For $1 \le b \le 24$ the interval is shorter than 1, so it may or may not catch an integer. The interval is centred at $\frac{12.5b}{25} = \frac{b}{2}$ with half-width $\frac{b}{50}$. Two cases:

- b **even:** take $a = \frac{b}{2}$, giving $\frac{a}{b} = \frac{1}{2} = 0.5$, which is inside $[0.48, 0.52]$. So *every* even b works — in particular $b = 2, 4, 6, \ldots, 24$.
- b **odd:** the centre $\frac{b}{2}$ is a half-integer, so the nearest integer sits a distance $\frac{1}{2}$ away. The interval only reaches it when its half-width $\frac{b}{50} \ge \frac{1}{2}$, i.e. $b \ge 25$. So no odd $b \le 24$ works, as the intervals $\left[\frac{12b}{25}, \frac{13b}{25}\right]$ for $b = 1, 3, 5, \ldots$ confirm — $[0.48, 0.52]$, $[1.44, 1.56]$, $[2.4, 2.6]$, … each falls strictly between two integers and catches none.

b works iff b is even or $b \ge 25$:

$$b \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24\} \cup \{25, 26, 27, \ldots\}.$$

The smallest is $b = 2$ (with $a = 1$, giving exactly $\frac{1}{2} = 0.5$).

The Bigger Picture — Farey Sequences and the Stern–Brocot Tree

The problem is really asking: *what is the simplest fraction (smallest denominator) sitting inside a given interval?* Two classical structures answer exactly this, and both are worth knowing in layman's terms.

Farey sequences. The *Farey sequence* F_n is simply the list of all fractions between 0 and 1 whose denominator is at most n, written in lowest terms and in increasing order. The first few are

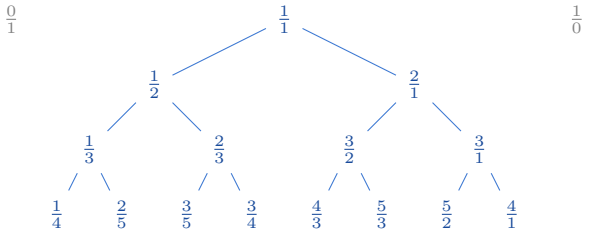
$$\begin{aligned} F_1 &: \ \tfrac{0}{1},\ \tfrac{1}{1}. \\ F_2 &: \ \tfrac{0}{1},\ \tfrac{1}{2},\ \tfrac{1}{1}. \\ F_3 &: \ \tfrac{0}{1},\ \tfrac{1}{3},\ \tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{1}{1}. \\ F_4 &: \ \tfrac{0}{1},\ \tfrac{1}{4},\ \tfrac{1}{3},\ \tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{3}{4},\ \tfrac{1}{1}. \end{aligned}$$

(Going from F_3 to F_4 only adds $\frac{1}{4}$ and $\frac{3}{4}$, since $\frac{2}{4}$ reduces to the already-listed $\frac{1}{2}$.) As you raise n, new fractions squeeze in between the old ones. The magic is *how* they appear: the first fraction to show up between two neighbours $\frac{p}{q}$ and $\frac{r}{s}$ is their **mediant**

$$\frac{p+r}{q+s} \qquad \text{(add the tops, add the bottoms),}$$

which always lies strictly between them and, remarkably, is the fraction with the *smallest possible denominator* in that gap. (Neighbours in a Farey sequence also satisfy the tidy identity $ps - qr = \pm 1$.) Note the mediant is *not* the same as averaging: the mediant of $\frac{0}{1}$ and $\frac{1}{1}$ is $\frac{1}{2}$, not $\frac{1}{1}$-ish nonsense — it is a genuinely different, and very useful, operation.

The Stern–Brocot tree. Start with the two "endpoints" $\frac{0}{1}$ and $\frac{1}{0}$ and repeatedly insert mediants. This grows a binary tree that contains *every* positive fraction in lowest terms exactly once, ordered left-to-right by size. To find the simplest fraction in an interval, you **binary-search** the tree: form the mediant of your current bounds; if it lands in the target interval you are done, otherwise it tells you to go left (mediant too big) or right (too small), and you repeat. Because each step is a mediant, the first fraction you land inside the interval is guaranteed to have the smallest denominator.



The Stern–Brocot tree. The root $\frac{1}{1}$ is the mediant of the boundary values $\frac{0}{1}$ and $\frac{1}{0}$; every node is the mediant of its nearest left and right ancestors, and each fraction in lowest terms appears exactly once.

Back to the problem. Run this search on $[0.48, 0.52]$ starting from $\frac{0}{1}$ and $\frac{1}{1}$: their mediant is $\frac{0+1}{1+1} = \frac{1}{2} = 0.5$, which is already inside the interval. So $\frac{1}{2}$ is the simplest fraction in $[0.48, 0.52]$ — denominator 2 — which is exactly why $b = 2$ is the smallest denominator that works, and why every even b (a scaled-up copy of $\frac{1}{2}$) works too.

Concept Summary and Applications

- **Existence test:** a fraction a/b lands in $[\ell, u]$ for some integer a iff the interval $[\ell b,\ ub]$ contains an integer; if its length $(u - \ell)b \geq 1$ this is automatic.
- **Mediant, not average:** the mediant $\frac{p+r}{q+s}$ of two fractions lies between them and is the smallest-denominator fraction in the gap — the engine behind Farey sequences and the Stern–Brocot tree.

- **Simplest fraction in an interval:** binary-search the Stern–Brocot tree with mediants; the first one inside the interval has the least denominator.
- **Applications:** best rational approximations (gear ratios, calendar and timing fractions), continued fractions, reducing/deriving tolerances, and clock/alignment problems.

67 Equal-Sum Partitions: The Broken Clock

An *equal-sum partition* splits a collection of numbers into groups with the same total. In puzzles the groups usually must also satisfy a *shape* rule — here, they must be pieces of a physically broken clock. Two ideas solve almost every problem of this type: **(1)** the target for each group is forced, total ÷ number of groups; **(2)** if the obvious shape cannot work, ask exactly which shapes the physical setting allows.

Representative Problem — The Broken Clock (Q237)

Problem: A clock face with the numbers $1, 2, \ldots, 12$ falls and breaks into three pieces. The numbers on each piece have the same sum. Which numbers are on each piece?

Step 1: The target is forced

The sum of all twelve numbers is

$$1 + 2 + \cdots + 12 = \frac{12 \cdot 13}{2} = 78.$$

Three pieces with equal sums must each total $78/3 = \mathbf{26}$. (If the total were not divisible by the number of pieces, the task would be impossible immediately.)

Step 2: What shapes can a piece have?

Model each crack as a straight line across the face. Two cracks that cross inside the clock would make four pieces, so for three pieces the two cracks do not cross. Then the face is cut into

- two **caps** (the part on the far side of each crack) — a cap holds a run of *consecutive* numbers around the rim;
- one **band** between the cracks — it holds the two remaining runs, one on each side of the dial.

Step 3: Find runs of consecutive numbers summing to 26

Going round the rim (and wrapping past 12 to 1), the only runs that total 26 are

$$\{5, 6, 7, 8\}\ (5 + 6 + 7 + 8 = 26) \qquad \text{and} \qquad \{11, 12, 1, 2\}\ (11 + 12 + 1 + 2 = 26).$$

Check by length of the run (a run of length L summing to 26 needs an average of $26/L$):

- $L \le 2$: at most $11 + 12 = 23$ — too small.
- $L = 3$ (average ≈ 8.7): 7+8+9 = 24, 8+9+10 = 27, 9+10+11 = 30; wrapping runs such as 11+12+1 = 24 are no better — none equals 26.
- $L = 4$ (average 6.5): $\{5, 6, 7, 8\}$ and, wrapping round, $\{11, 12, 1, 2\}$.
- $L = 5$ (average 5.2): $3+\cdots+7 = 25$ and $4+\cdots+8 = 30$; wrapping round, 12+1+2+3+4 = 22 and 11+12+1+2+3 = 29 — none.
- $L = 6$: going round the dial, the 6-runs sum to $21, 27, 33, 39, 45, 51, 57$ (each shift changes the sum by a multiple of 6) — all odd, never 26.
- $L \ge 7$: the other $12 - L \le 5$ numbers would have to sum to $78 - 26 = 52$, but five numbers total at most 8+9+10+11+12 = 50.

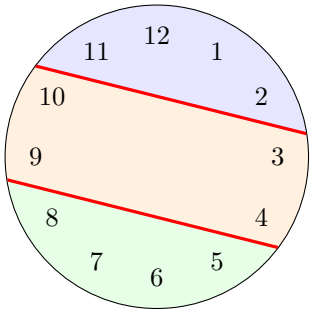
So the **two caps must be** $\{11, 12, 1, 2\}$ **and** $\{5, 6, 7, 8\}$, and the band is what is left:

$$\{3, 4, 9, 10\}, \qquad 3 + 4 + 9 + 10 = 26 \checkmark$$

Step 4: Why three "slices of pie" fail

If all three pieces were runs of consecutive numbers (wedges), we would need three disjoint 26-runs. There are only two, and the leftover $\{3, 4\}$ and $\{9, 10\}$ sit on opposite sides of the dial. The trick is that a *band* can collect numbers from both sides.

Step 5: Draw the cracks



top: $11 + 12 + 1 + 2 = 26$

band: $10 + 9 + 3 + 4 = 26$

bottom: $8 + 7 + 6 + 5 = 26$

The cracks must tilt slightly (lower on the right): 2 and 10 sit at the same height, so a level crack could not keep 2 in the top piece while putting 10 in the band; likewise 4 and 8 for the lower crack.

Step 6: Uniqueness

With straight cracks, the caps must be 26-runs, and Step 3 shows there are only two. So this is the *only* way to break the clock into three equal-sum pieces.

$\{11, 12, 1, 2\}$, $\{3, 4, 9, 10\}$, $\{5, 6, 7, 8\}$ — each sums to $78/3 = 26$. Two nearly horizontal cracks (slightly lower on the right) make a top piece, a middle band and a bottom piece. (Same question: Q510.)

Concept Summary and Applications

Equal-sum partition into k groups: target $= \dfrac{\text{total}}{k}$ (must be a whole number).
Broken clock with straight cracks: non-crossing cracks cut caps (runs of consecutive numbers) and bands (two runs, one on each side). Find the runs that hit the target; the caps must be among them.

- 3 pieces: $\{11, 12, 1, 2\}, \{3, 4, 9, 10\}, \{5, 6, 7, 8\}$ (sum 26).
- Useful structure: opposite-side pairs across a band (12+1, 11+2, 10+3, ...) all sum to 13.

- **Start with the total:** it fixes the target and can prove impossibility at once.
- **List the building blocks:** enumerate the few runs (or subsets) that hit the target; the answer is assembled from them.

- **Question the shape:** when the obvious shape (wedges) fails, check what the physical rules really allow.
- **Applications:** load balancing (splitting tasks into equal workloads), fair division, and the "partition problem" in computer science.

Additional Problem 1 — Equal Sums of Digits (Q237, second reading)

Problem: Read "sum of the digits" literally: 10 counts as $1+0=1$, 11 as 2, 12 as 3. Can the clock break (with straight cracks) into three pieces whose *digit* sums are equal?

Step 1: Target

Digit sum of the whole face: $(1+2+\cdots+9)+1+2+3=45+6=51$, so each piece would need $51/3=\mathbf{17}$.

Step 2: On paper, groupings exist

For example

$$\{1,5,8,12\}:\ 1{+}5{+}8{+}3=17, \qquad \{2,6,7,11\}:\ 2{+}6{+}7{+}2=17, \qquad \{3,4,9,10\}:\ 3{+}4{+}9{+}1=17.$$

Step 3: But not as pieces of the clock

As in the representative problem, two straight cracks produce two caps, and each cap is a run of consecutive numbers that must have digit sum 17. Listing the runs: digit sums around the rim are $1,2,3,4,5,6,7,8,9,1,2,3$, and the only consecutive run totalling 17 is $\{8,9\}$ (digit sum $8+9$). For example, the run $7,8$ gives 15 and $7,8,9$ gives 24; the run $9,10,11,12,1$ gives $9+1+2+3+1=16$ and adding 2 jumps to 18. With only one possible cap, there cannot be two caps.

No. Each piece would need digit sum 17. Such groupings exist on paper (e.g. $\{1,5,8,12\}$, $\{2,6,7,11\}$, $\{3,4,9,10\}$), but only one run of consecutive numbers, $\{8,9\}$, has digit sum 17, so no break with straight cracks works.

Additional Problem 2 — Two, Four or Six Pieces?

Problem: The clock breaks with straight cracks into (a) 2, (b) 4, (c) 6 pieces with equal number sums. Which are possible, and what are the pieces?

(a) Two pieces: one crack, target 39

One crack splits the rim into two runs, each summing to $78/2=39$. Checking all runs (as in the representative problem), the only ones summing to 39 are the two complementary 6-runs

$$\{4,5,6,7,8,9\}=39, \qquad \{10,11,12,1,2,3\}=39.$$

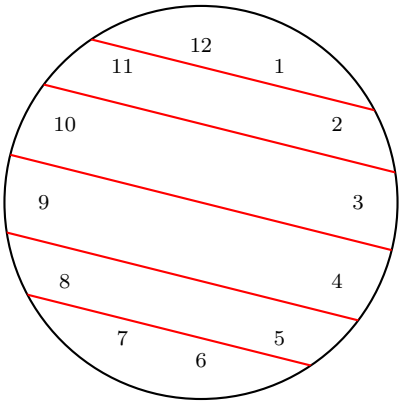
A nearly vertical crack between 3|4 and 9|10 does it (the only solution).

(b) Four pieces: impossible

$78/4 = 19.5$ is not a whole number, but every piece has a whole-number sum. Impossible — no geometry needed.

(c) Six pieces: target 13

$78/6 = 13$. The numbers pair up as $12+1 = 11+2 = 10+3 = 9+4 = 8+5 = 7+6 = 13$. In each pair the two numbers sit at the same height on opposite sides of the dial (except $12, 1$ and $6, 7$, which are neighbours). Five parallel cracks, tilted slightly lower on the right as before, cut the face into six bands, each holding one pair:



Bands from top to bottom: $\{12,1\}, \{11,2\}, \{10,3\}, \{9,4\}, \{8,5\}, \{7,6\}$ — each sums to 13.

(a) $\{4,\ldots,9\}$ and $\{10,11,12,1,2,3\}$ (sum 39). (b) Impossible, since $78/4$ is not an integer. (c) Six parallel bands $\{12,1\}, \{11,2\}, \{10,3\}, \{9,4\}, \{8,5\}, \{7,6\}$, each summing to 13.

Part X — Geometry, Regions and Graphs

68 Circle Regions from Chords (Euler's Formula)

Background: Euler's Formula in 3D (Convex Polyhedra)

The most famous form of Euler's formula is about solid shapes. Take any convex polyhedron — a cube, a pyramid, a soccer-ball-shaped solid — and count its corners (**vertices** V), its **edges** E, and its flat **faces** F. Whichever convex polyhedron you pick, these three counts always satisfy

$$V - E + F = 2.$$

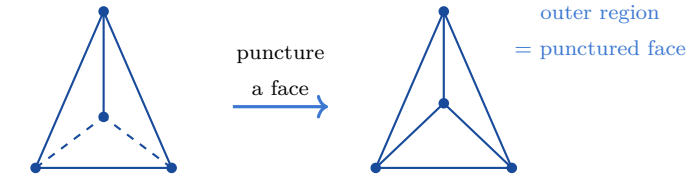
Check three familiar solids:

$$\text{cube } 8 - 12 + 6 = 2, \qquad \text{tetrahedron } 4 - 6 + 4 = 2, \qquad \text{octahedron } 6 - 12 + 8 = 2.$$

It is genuinely striking that the *same* fixed number 2 falls out of solids with wildly different vertex, edge, and face counts.

From Polyhedra to the Plane (2D Planar Graphs)

The identical law governs flat drawings, and a single picture bridges the two. Take a convex polyhedron, puncture one face, and stretch that hole open until the whole surface flattens into the plane — like cutting one face off a balloon and pulling its rim wide. No vertex, edge, or face is created or destroyed, so V, E, F are unchanged. The one face you punctured is now stretched around the *outside* of the drawing: it becomes the flat picture's single unbounded **outer face**.

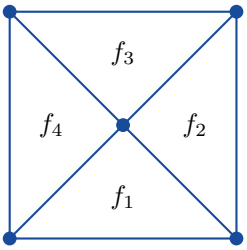


So a **planar graph** — dots (vertices) joined by non-crossing edges into one connected piece — obeys the very same identity, as long as we count *all* faces including the outer region:

$$V - E + F = 2 \qquad \text{(connected planar graph, } F \text{ includes the outer face).}$$

Simplest check — a triangle. Three corners, three sides, and two faces (the inside and the outside): $V = 3$, $E = 3$, $F = 2$, so $3 - 3 + 2 = 2$. ✓

A slightly richer example — a square with both diagonals. Add the two diagonals of a square. In a planar graph edges may not cross, so the point where the diagonals meet becomes a *new vertex* in the middle. Now count:



Vertices: 4 corners $+1$ centre $= 5$. Edges: 4 sides $+4$ half-diagonals (each diagonal is split in two by the centre) $= 8$. Faces: 4 inner triangles $+1$ outer $= 5$. Indeed $V - E + F = 5 - 8 + 5 = 2$. ✓ (This is exactly the $n = 4$ chord picture we return to below.)

The common value 2 is the **Euler characteristic of the sphere**, a topological invariant — a polyhedron's surface and a connected planar graph are both the sphere in disguise, which is why the count never depends on the exact drawing.

The Problem

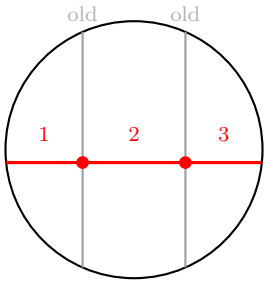
Problem: Place n points on a circle and draw all $\binom{n}{2}$ chords, in general position (no three chords meeting at one interior point, which maximises regions). Into how many regions is the disk divided?

Incremental Count — Why Each Chord and Each Crossing Adds a Region

Start with 1 region (the empty disk) and add the chords one at a time. Track how many *new* regions each chord creates as you sweep along it.

A chord with no crossings adds exactly one region. It enters the disk, runs through a single existing region, and slices that region into two — a net gain of one.

Each crossing the chord makes adds one more region. Every time the new chord crosses a chord already drawn, it passes from one region into the next, and that passage cuts an extra region in two. So a chord that crosses c existing chords is itself chopped into $c + 1$ little segments, and each segment splits a region: it adds $c + 1$ regions in all $= 1$ (for the chord) $+ c$ (one per crossing).



Here the new (red) chord crosses 2 old chords, so it is cut into 3 segments and creates 3 new regions

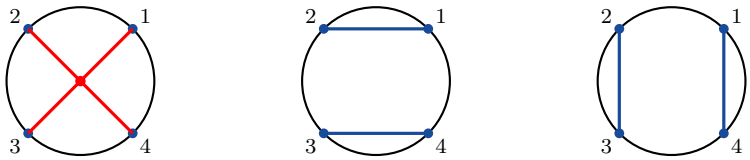
$= 1 + 2$. Summing over all chords, the total count is

$$\underbrace{1}_{\text{empty disk}} + \underbrace{(\text{number of chords})}_{\text{one region per chord}} + \underbrace{(\text{number of crossings})}_{\text{one region per crossing}}.$$

There are $\binom{n}{2}$ chords (one per pair of points), so all that remains is to count the interior crossings.

Why Exactly $\binom{n}{4}$ Crossings

The clean way to count crossings is to match each one with a set of 4 boundary points. Pick any 4 of the n points; they form a convex quadrilateral. There are three ways to pair up 4 points into two chords, but only *one* of them makes the chords cross inside — the pair of **diagonals**. The other two pairings join adjacent points ("opposite sides") and stay apart:



This is a **bijection**: every crossing comes from a unique set of 4 endpoints (two crossing chords have 4 distinct ends — if they shared one, they would meet on the boundary, not inside), and every set of 4 points yields exactly one crossing. So the number of interior crossings equals the number of ways to choose 4 points, $\binom{n}{4}$.

$$R(n) = 1 + \binom{n}{2} + \binom{n}{4}.$$

Numeric examples. $n = 5$: $\binom{5}{2} = 10$ chords and $\binom{5}{4} = 5$ crossings, so $R = 1 + 10 + 5 = 16$. $n = 6$: $\binom{6}{2} = 15$ chords and $\binom{6}{4} = 15$ crossings, so $R = 1 + 15 + 15 = 31$.

Warning: the Answer Is *Not* 2^{n-1}

The values for $n = 1, 2, 3, 4, 5$ are $1, 2, 4, 8, 16$ — exactly the powers of two, which tempts everyone to guess $R(n) = 2^{n-1}$. It is **wrong**. The doubling breaks at $n = 6$:

n	1	2	3	4	5	**6**	7
$R(n) = 1 + \binom{n}{2} + \binom{n}{4}$	1	2	4	8	16	**31**	57
2^{n-1} (the trap)	1	2	4	8	16	**32**	64

Five data points agreeing with 2^{n-1} prove nothing: at $n = 6$ the true count is **31**, *not* 32. *This is the textbook warning against extrapolating a pattern from small cases — the polynomial $1 + \binom{n}{2} + \binom{n}{4}$ just* happens *to coincide with 2^{n-1} for $n \le 5$.*

Verifying with Euler's Formula

The incremental count is convincing but informal; Euler's formula pins it down rigorously. Treat the whole figure — boundary points, crossings, arcs, and chord-segments — as one connected planar graph and apply $V - E + F = 2$.

Vertices are the n points on the circle plus the $\binom{n}{4}$ interior crossings:

$$V = n + \binom{n}{4}.$$

Edges are the n boundary arcs between adjacent points, plus the chord pieces. Each of the $\binom{n}{2}$ chords is cut into segments by the crossings on it; since every crossing lies on exactly two chords, it creates 2 extra chord pieces, giving

$$E = n + \binom{n}{2} + 2\binom{n}{4}.$$

Solving Euler's formula for the face count:

$$F = 2 - V + E = 2 + \binom{n}{2} + \binom{n}{4}.$$

Finally — and this is the key last step — **discard the outer face**. Euler's F counts *every* region including the unbounded one outside the circle, but the problem asks only for regions *inside* the disk. Removing that single outer region gives

$$R(n) = F - 1 = 1 + \binom{n}{2} + \binom{n}{4},$$

matching the incremental count exactly.

Concept Summary and Applications

- **Two-tool method:** count regions either *incrementally* (initial region + one per edge + one per new crossing) or *rigorously* via $V - E + F = 2$ (then subtract the outer face); they must agree.
- **General position maximises** regions: forcing three chords to be concurrent merges crossings and *loses* regions.
- **Applications:** planar subdivision and arrangement counting in computational geometry, face counts in graph drawings, and the classic "don't trust $1, 2, 4, 8, \ldots$" cautionary tale about pattern-matching.

69 Painted Cube and 3D Geometry

The whole family of "painted cube" problems rests on one classification. Paint the outside of an $m \times m \times m$ cube and slice it into m^3 unit cubes; a unit cube gets painted only on the faces that lay on the surface, and *where* it sat determines how many painted faces it has:

Position	Painted faces	Count	
Corner	3	8	(a cube has 8 corners)
Edge (non-corner)	2	$12(m-2)$	(12 edges, $m-2$ each)
Face (non-edge)	1	$6(m-2)^2$	(6 faces, $(m-2)^2$ each)
Interior	0	$(m-2)^3$	(the hidden inner block)

Everything else is bookkeeping on these four groups.

Representative Problem — The $3 \times 3 \times 3$ Cube (Q61)

Problem: You dip a cube in paint and let it dry, then cut it into 27 unit sub-cubes and drop them in a bag. You pick one at random and roll it on the table. What is the probability that *no paint shows on top*?

Step 1 — classify the 27 sub-cubes. With $m = 3$ the table above gives

$$\underbrace{8}_{3\text{ faces}}\ \text{(corners)},\quad \underbrace{12(1)=12}_{2\text{ faces}}\ \text{(edges)},\quad \underbrace{6(1)^2=6}_{1\text{ face}}\ \text{(face-centres)},\quad \underbrace{1^3=1}_{0\text{ faces}}\ \text{(the centre)}.$$

(Check: $8+12+6+1=27$.)

Step 2 — probability of a blank top for each type. After a random roll, each of the sub-cube's 6 faces is equally likely to land on top, so the chance the top is unpainted is $\dfrac{6-(\text{painted faces})}{6}$:

$$0 \text{ painted} \to 1, \qquad 1 \to \tfrac{5}{6}, \qquad 2 \to \tfrac{4}{6}, \qquad 3 \to \tfrac{3}{6}.$$

Step 3 — average over the bag. Weight each type by how many sub-cubes it has and divide by 27:

$$P = \frac{1}{27}\left(1\cdot 1 + 6\cdot\tfrac{5}{6} + 12\cdot\tfrac{4}{6} + 8\cdot\tfrac{3}{6}\right) = \frac{1}{27}(1+5+8+4) = \frac{18}{27} = \frac{2}{3}.$$

$$P(\textbf{no paint on top}) = \frac{18}{27} = \frac{2}{3}.$$

(If instead you just asked for a sub-cube with *no painted face at all*, that is only the single centre cube: probability $\frac{1}{27}$.)

Additional Problem 1 — The $4 \times 4 \times 4$ Cube (Q62)

Problem: Paint the outside of a $4 \times 4 \times 4$ cube and cut it into 64 unit cubes. (1) How many unit cubes have at least one painted face? (2) If you pick one of the 64 at random and roll it, what is the probability the top face is painted?

Part (1) — count by the complement. The unpainted cubes are exactly the hidden interior block, of size $(4-2)^3 = 2^3 = 8$. So the number with *at least one* painted face is

$$64 - 8 = 56.$$

Part (2) — classify, then average. With $m = 4$ the table gives

$$\text{corners } 8 \text{ (3 faces)}, \quad \text{edges } 12(2) = 24 \text{ (2)}, \quad \text{faces } 6(2)^2 = 24 \text{ (1)}, \quad \text{interior } 8 \text{ (0)}.$$

(Check: $8 + 24 + 24 + 8 = 64$.) After a random roll the chance the top *is* painted is $\frac{\text{painted faces}}{6}$, so

$$P = \frac{1}{64}\left(8 \cdot \tfrac{3}{6} + 24 \cdot \tfrac{2}{6} + 24 \cdot \tfrac{1}{6} + 8 \cdot 0\right) = \frac{1}{64} \cdot \frac{24 + 48 + 24}{6} = \frac{1}{64} \cdot \frac{96}{6} = \frac{16}{64} = \frac{1}{4}.$$

56 of the 64 cubes have at least one painted face; and P(top face painted) $= \frac{16}{64} = \frac{1}{4}$.

A quick sanity check for part (2): the total painted area is 6 faces of side 4, i.e. $6 \cdot 16 = 96$ unit faces, spread over $64 \cdot 6 = 384$ unit faces in all; a uniformly random unit face is painted with probability $96/384 = \frac{1}{4}$ — matching the roll calculation, since rolling makes every unit face equally likely to be on top.

Additional Problem 2 — Red *Edges* Instead of Faces (Q407)

Problem: A grey $3 \times 3 \times 3$ cube is painted red on all its *edges* (only the 12 edge lines of the big cube, not the faces). It is cut into 27 unit cubes; one is chosen uniformly at random and placed randomly on a table. What is the probability that *no red edge is visible*?

Step 1 — classify by red *edges* (not faces). A unit cube gets a red edge only where it touches one of the big cube's 12 edges:

Position	Count	Red edges	
Corner	8	3	(the 3 edges meeting at the big cube's corner)
Edge-middle	12	1	(lies along one big edge)
Face-centre	6	0	(touches no big edge)
Centre	1	0	

(Check: $8 + 12 + 6 + 1 = 27$.) Compare the face-painting table: the edge-middle cubes now carry 1 red edge (not 2 painted faces), and the face-centres are clean.

Step 2 — which edges are hidden? A cube has 6 faces and 12 edges, and each edge borders exactly 2 faces. On the table, one face (equally likely any of the 6) is on the bottom. The natural convention is that the 4 edges of the bottom face, which lie on the table, are hidden; the other 8 are visible. So no red shows iff *every* red edge lies on the bottom face.

Step 3 — probability for each type.

- 0 **red edges (7 cubes):** always clean, probability 1.

- 1 **red edge (12 cubes):** that edge belongs to exactly 2 of the 6 faces; it is hidden iff one of those 2 is on the bottom: $\frac{2}{6} = \frac{1}{3}$.
- 3 **red edges at a corner (8 cubes):** no single face contains all 3 (each face through that corner holds only 2 of them), so at least one always shows: probability 0.

Step 4 — average over the bag.

$$P = \frac{1}{27}\left(7 \cdot 1 + 12 \cdot \tfrac{1}{3} + 8 \cdot 0\right) = \frac{7+4}{27} = \frac{11}{27} \approx 40.7\%.$$

$P(\textbf{no red edge visible}) = \frac{11}{27}$ **(edges lying on the table count as hidden).**

How to read "visible". Unlike faces, an edge is shared by two faces, so the answer depends on the convention:

Convention	Per-type probabilities (0 / 1 / 3 red edges)	Answer
Edges touching the table are hidden (used above)	$1,\ \frac{1}{3},\ 0$	$\frac{11}{27}$
Strict: viewed from sides and top, a bottom edge still shows as the lower border of a side face	$1,\ 0,\ 0$	$\frac{7}{27}$
Top view only: just the 4 edges of the top face	$1,\ \frac{2}{3},\ \frac{1}{2}$	$\frac{19}{27}$

For the top view: a single red edge lies on the top face with probability $\frac{2}{6}$, so it is hidden with probability $\frac{2}{3}$. For a corner cube, if the top face contains that corner (probability $\frac{3}{6}$) two red edges show; otherwise the top face contains none of the three, so it is clean with probability $\frac{1}{2}$. Then

$$P = \frac{7 + 12 \cdot \frac{2}{3} + 8 \cdot \frac{1}{2}}{27} = \frac{7+8+4}{27} = \frac{19}{27}.$$

In an interview, state the convention first; the intended answer is $\frac{11}{27}$.

Takeaway. Same recipe as face painting: *classify by position, find the per-type probability, average over the bag.* Only the per-type rule changes: for faces it is a face-count ratio ($\frac{\text{painted faces}}{6}$); for edges you must ask which *faces* put an edge out of sight (an edge belongs to 2 faces).

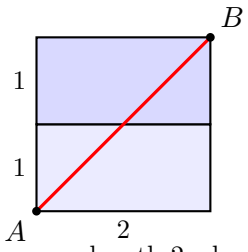
Additional Problem 3 — Shortest and Longest Paths on the Surface of a Box (Q408, Part 3)

Problem: (a) What is the shortest path between opposite corners of a $2 \times 1 \times 1$ rectangular box, if the path must stay on the *surface* of the box? (b) What is the maximum surface distance between *any* two points of the box?

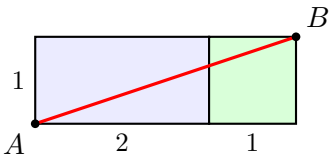
Key idea — unfold the box. An ant walking on the box cannot cut through the inside, so the straight 3D diagonal ($\sqrt{2^2 + 1^2 + 1^2} = \sqrt{6}$) is not allowed. The trick is to *unfold* (flatten) the faces

the path crosses into one plane, like opening a cardboard box. Unfolding does not change lengths on the faces, and in the plane the shortest route between two points is a straight line. So: *list the possible unfoldings, draw a straight line in each, and take the shortest.*

Step 1 — which faces can a corner-to-corner path cross? Call the starting corner A and the opposite corner B. Each face touches either A or B (three faces each), so a shortest path leaves A on one face and arrives at B on an adjacent face, crossing the single edge they share. There are three edge directions to cross (length 2, 1, or 1), giving three unfoldings. When two adjacent faces are laid flat they form one rectangle: one side is the length of the shared edge, the other is the sum of the two remaining dimensions, and A, B sit at opposite ends of its diagonal.



cross a length-2 edge
$2 \times (1{+}1)$: $\sqrt{8} = 2\sqrt{2}$



cross a length-1 edge
$(2{+}1) \times 1$: $\sqrt{10}$

Step 2 — compare the three lengths.

Path crosses an edge of length	Unfolded rectangle	Straight-line length
2	$2 \times (1+1)$	$\sqrt{2^2+2^2} = 2\sqrt{2} \approx 2.828$
1	$1 \times (2+1)$	$\sqrt{1^2+3^2} = \sqrt{10} \approx 3.162$
1 (other)	$1 \times (1+2)$	$\sqrt{10} \approx 3.162$

(a) Shortest surface path between opposite corners $= 2\sqrt{2} \approx 2.83$: cross two of the long 2×1 faces, which unfold into a 2×2 square, and go along its diagonal.

General rule. For an $a \times b \times c$ box with $a \geq b \geq c$, the three candidates are $\sqrt{a^2+(b+c)^2}$, $\sqrt{b^2+(a+c)^2}$, $\sqrt{c^2+(a+b)^2}$. Expanding, each equals $\sqrt{a^2+b^2+c^2+2\cdot(\text{product of the two summed sides})}$, so the smallest is the one whose summed sides are the two *shortest*:

$$d_{\min}(\text{opposite corners}) = \sqrt{a^2+(b+c)^2}$$

— pair the **longest** side with the **sum of the other two** (the cross term $2bc$ is the smallest of $2bc$, $2ac$, $2ab$).

Part (b) — is the farthest pair the opposite corners? It is tempting to say yes, but it is **false** for this box.

Counterexample: the centres of the two 1×1 end faces. Any surface path between them must

1. walk from the centre of the first end face to its edge: at least $\frac{1}{2}$ (all of this happens on the end face, so it makes no progress along the long axis);
2. cross the long faces from one end to the other: at least 2 (the change in the long coordinate);

3. walk from the edge of the second end face to its centre: at least $\frac{1}{2}$.

So their distance is at least $\frac{1}{2}+2+\frac{1}{2}=3$, and the straight route over one long face achieves exactly 3. Since $3>2\sqrt{2}\approx 2.83$, the opposite corners are *not* the farthest pair.

Slightly farther still. Put coordinates $(y,z)\in[0,1]^2$ on each end face and move the two points a little off-centre in opposite directions along a diagonal:

$$P=(a,\,1-a),\qquad Q=(1-a,\,a),\qquad a=\tfrac{3-\sqrt{3}}{2}\approx 0.634.$$

Unfolding over a long face, the straight segment has horizontal run $(1-a)+2+a=3$ and vertical offset $2a-1=2-\sqrt{3}$, so

$$\sqrt{3^2+(2-\sqrt{3})^2}=\sqrt{16-4\sqrt{3}}=2\sqrt{4-\sqrt{3}}\approx 3.012,$$

and for this pair every other route (over a different long face, or across two or three long faces) is equally long or longer. A numerical search over all pairs of end-face points finds nothing larger.

Why $a=\frac{3-\sqrt{3}}{2}$? The surface distance is the *shortest* of all routes, and moving the points off-centre pulls two kinds of route in opposite directions:

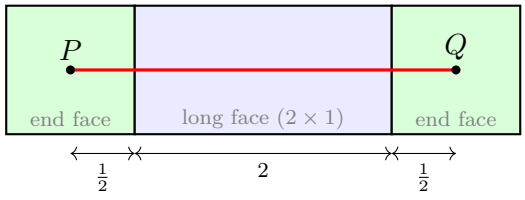
- **Over one long face** (all four long faces give the same length for this pair): run 3, offset $2a-1$, length $\sqrt{9+(2a-1)^2}$ — this *grows* as a moves away from $\frac{1}{2}$.
- **Around a long edge** (end face → long face → adjacent long face → end face): unfolded, the run is $(1-a)+2+(1-a)=4-2a$ and the offset is $a+a=2a$, length $\sqrt{(4-2a)^2+(2a)^2}$ — this *shrinks* as a grows, because the points get closer to that shared edge.

The distance is the smaller of the two, so it is largest where they are equal:

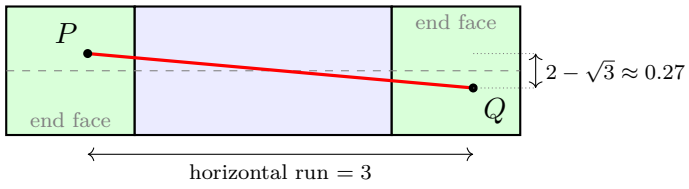
$$9+(2a-1)^2=(4-2a)^2+4a^2 \iff 10-4a+4a^2=16-16a+8a^2 \iff 2a^2-6a+3=0 \iff a=\frac{6-\sqrt{12}}{4}=\frac{3-\sqrt{3}}{2}$$

(the other root exceeds 1). Beyond this point the corner route becomes the shorter one and the distance starts to fall.

(i) End-face centres: distance 3



(ii) Off-centre pair: distance $2\sqrt{4-\sqrt{3}}\approx 3.012$



Each panel shows the box unfolded along one long face (blue) with the two square end faces (green) folded down flat on either side. In (i) the path is $\frac{1}{2}+2+\frac{1}{2}=3$. In (ii) the points are shifted off-centre in opposite

directions (by $a - \frac{1}{2} = \frac{2-\sqrt{3}}{2} \approx 0.134$ each), so the straight line is slightly tilted and longer; the shift is chosen so that the routes over the other long faces become no shorter than this one.

(b) The maximum surface distance is about 3.01 (at least 3, from the end-face centres) — larger than the $2\sqrt{2} \approx 2.83$ between opposite corners.

Takeaway. "Unfold, then draw a straight line" turns any "shortest path on a surface" problem into plane geometry; just remember to check *every* unfolding. And do not assume that the farthest points on a solid are its corners: on a long box, points in the middle of the small end faces are farther apart.

Concept Summary and Applications

- **Four groups by location:** corners (3 painted, always 8), edges (2, $12(m-2)$), faces (1, $6(m-2)^2$), interior (0, $(m-2)^3$); they sum to m^3.
- **Count the interior by shrinking:** the unpainted block is $(m-2)^3$, so "at least one painted face" $= m^3 - (m-2)^3$.
- **Roll = average over faces:** a random roll makes each of a cube's 6 faces equally likely on top, so "top painted" has probability $\frac{\text{painted faces}}{6}$; averaging over the bag equals (total painted unit-faces)/(total unit-faces).
- **Painted edges (Q407):** classify by red *edges* instead (corners 3, edge-middles 1, others 0); an edge is hidden only if a face containing it is out of sight, so state the visibility convention.
- **Surface paths (Q408):** unfold the faces a path crosses and draw a straight line; opposite corners of an $a \times b \times c$ box ($a \geq b \geq c$) are $\sqrt{a^2 + (b+c)^2}$ apart. The farthest surface points need not be corners ($2 \times 1 \times 1$: corners $2\sqrt{2}$, end-face centres 3).
- **Applications:** surface-area vs. volume scaling, expected-value counting by category, and Bayesian "I see a painted face, which cube is it?" variants (weight cubes by painted-face count).

70 Region Counting in 2-Dimensional and 3-Dimensional Space

Both problems below share one idea: **add the cuts one at a time and count how many new regions each cut creates**. In 2D a new line is chopped into pieces by the lines it crosses; in 3D a new plane is chopped into flat pieces by the planes it crosses — and each piece splits exactly one region in two.

Representative Problem 1 — n Lines Dividing the Plane

Problem: Into how many regions do n straight lines divide the plane, when the lines are in "general position" (no two parallel, no three through a common point, which maximises the count)?

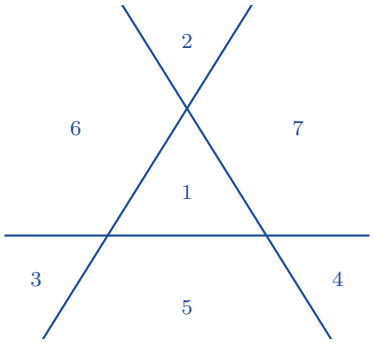
Add one line at a time. Start with the empty plane: 1 region. Now suppose $k-1$ lines are already drawn and you add the k-th. In general position it crosses each of the existing $k-1$ lines at a distinct point, so those $k-1$ crossing points chop the new line into k pieces ($k-2$ interior segments plus 2 infinite rays). *Each piece slices one existing region into two*, so the new line adds exactly k regions. Summing from the first line to the n-th,

$$L(n) = 1 + \underbrace{(1+2+3+\cdots+n)}_{\text{one term per line}} = 1 + \frac{n(n+1)}{2}.$$

Written with binomial coefficients this is the memorable form

$$L(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} = 1 + n + \frac{n(n-1)}{2}.$$

These are the **lazy-caterer** (pancake) numbers $1, 2, 4, 7, 11, 16, 22, \ldots$. The picture below shows $n = 3$: three lines make a central triangle plus six unbounded pieces, $L(3) = 1 + 3 + 3 = 7$.



n lines in general position make $L(n) = 1 + n + \binom{n}{2} = \dfrac{n^2+n+2}{2}$ regions (the lazy-caterer numbers); e.g. $L(3) = 7$, $L(4) = 11$.

Representative Problem 2 — Tetrahedron Planes (Q38)

Problem: Extend the four faces of a tetrahedron into full infinite planes. Into how many regions do these four planes divide three-dimensional space?

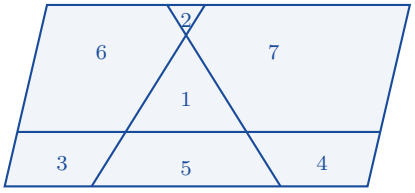
The four face-planes are in general position (no two parallel, no three sharing a line, no four sharing a point). Add them one at a time; the new idea is that the k-th plane is cut by the $k-1$ existing planes into flat pieces, and *each flat piece splits one 3D region in two.* On the new plane those $k-1$ cutting lines are themselves in general position, so they carve it into $L(k-1)$ pieces — exactly the 2D lazy-caterer count from Problem 1. Hence each plane adds $L(k-1)$ regions:

planes so far	0	1	2	3	4
new regions added	1	1	2	4	7
running total	1	2	4	8	**15**

The fourth plane is cut by 3 lines into $L(3) = 7$ pieces (drawn in the right-hand figure below), so it adds 7 regions, giving $1+1+2+4+7 = 15$.



tetrahedron: its 4 face-planes, extended, cut space into 15



the 4th plane: sliced by the other three along 3 lines into $L(3) = 7$ regions (1–7) $\Rightarrow$ adds 7 regions

General formula. The same "add one term per dimension" pattern extends the lazy-caterer numbers by one binomial term:

$$R(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = \frac{n^3 + 5n + 6}{6}.$$

These are the **cake numbers** $1, 2, 4, 8, 15, 26, 42, \ldots$, and $R(4) = 15$ recovers the answer.

Why (recurrence). Add the planes one at a time. The n-th plane is sliced by the $n-1$ existing planes into flat pieces, and *each piece splits one 3D region in two*, so it adds exactly that many regions. On that plane the $n-1$ slicing lines are in general position, so they create

$$L(n-1) = \binom{n-1}{0} + \binom{n-1}{1} + \binom{n-1}{2}$$

pieces — the 2D lazy-caterer count for $n-1$ lines. Hence

$$R(n) = R(n-1) + L(n-1),$$

and summing this telescoping recurrence gives the boxed formula. Our step table $1+1+2+4+7 = 15$ is exactly $R(3) + L(3)$ with $L(3) = 1+3+3 = 7$.

Four planes in general position (the tetrahedron's faces) cut space into $R(4) = 1+4+6+4 = \boxed{15}$ regions.

Concept Summary and Applications

- **One recipe, every dimension:** add each cut and count the pieces it is chopped into; each piece splits one region. This gives "sum the first $d+1$ binomial coefficients" in dimension d:

$$\text{points (1D)}: \ 1+n, \quad \text{lines (2D)}: \ 1+n+\tbinom{n}{2}, \quad \text{planes (3D)}: \ 1+n+\tbinom{n}{2}+\tbinom{n}{3}.$$

- **General position maximises:** making cuts parallel or concurrent merges intersections and *loses* regions.
- **Name-check:** 2D gives the lazy-caterer/pancake numbers, 3D the cake numbers.
- **Applications:** arrangement counting in computational geometry, hyperplane arrangements, and "maximum pieces from n cuts" puzzles.

71 Knight-Move Graphs: Tours, Obstructions and Reachable Squares

A knight moves two squares in one direction and one square in the other: from (x, y) to $(x \pm 1, y \pm 2)$ or $(x \pm 2, y \pm 1)$. Two kinds of questions recur: which squares can a knight reach in a given number of moves, and can a knight visit every square of a board exactly once (a *knight's tour*)? Both are best answered by thinking of the board as a *graph*.

Representative Problem — Squares Reachable in 10 Moves (Q107)

Problem: On an infinite chessboard, how many different squares can a knight be on after exactly 10 moves? Give a 90% confidence interval, and say whether the exact answer is even or odd.

A knight move changes (x, y) by one of the eight vectors $(\pm 1, \pm 2), (\pm 2, \pm 1)$. We want the number of distinct squares reachable in *exactly* 10 moves. The clean way is to pin down the reachable region as an **octagon** and count lattice points by hand.

1. Which squares are reachable

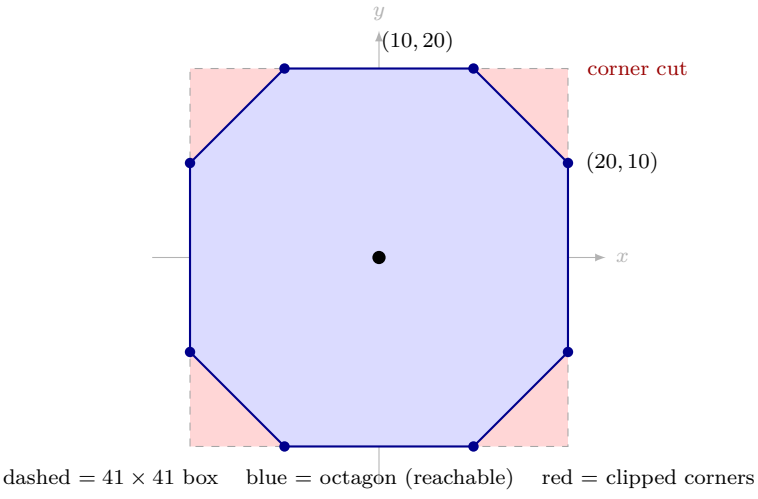
Colour (parity). Every move flips the parity of $x + y$ (it changes by $\pm 1 \pm 2$, always odd). After 10 moves $x + y$ is even — the knight is back on its starting colour. Moreover the knight can always "waste" two moves (step to a neighbour and back), so a square reachable in d moves is also reachable in $d + 2, d + 4, \ldots$. Hence

$$\{\text{reachable in exactly } 10\} = \{\text{same-colour squares reachable in } \leq 10 \text{ moves}\}.$$

Three linear bounds. Each move changes a single coordinate by at most 2, so $|x| \leq 20$ and $|y| \leq 20$. Each move changes $|x| + |y|$ by at most $1 + 2 = 3$, so $|x| + |y| \leq 30$. These three inequalities are **tight** by $n = 10$: the knight can actually reach *every* same-colour lattice point satisfying them. So the reachable set is exactly

$$\{(x, y) : \ |x| \leq 20, \ |y| \leq 20, \ |x| + |y| \leq 30, \ x + y \text{ even}\},$$

which is a 41×41 square with its four corners sliced off along $|x| + |y| = 30$ — an octagon.



2. Counting the octagon by hand: box minus corners

Same-colour squares in the full 41×41 *box.* With $x \in [-20, 20]$ there are 21 even values $(-20, \dots, 20)$ and 20 odd values; likewise for y. Now $x + y$ is even exactly when x, y are *both even* or *both odd*:

$$\underbrace{21 \times 21}_{\text{both even}} + \underbrace{20 \times 20}_{\text{both odd}} = 441 + 400 = \mathbf{841}.$$

Subtract the four clipped corners. Take the top-right corner (near $(20, 20)$): the removed points have $x + y = 31, 32, \dots, 40$, and the diagonal $x + y = s$ holds $41 - s$ points. Only *even* s are the reachable colour, so the same-colour points removed from one corner are

$$s = 32, 34, 36, 38, 40: \quad 9 + 7 + 5 + 3 + 1 = \mathbf{25} \ \ (= 5^2).$$

Four identical corners remove $4 \times 25 = 100$ same-colour squares.

Answer.

$$841 - 100 = \boxed{741}.$$

(For a cross-check, the octagon holds $41^2 - 4 \cdot 55 = 1681 - 220 = 1461$ lattice points of *both* colours, splitting as 741 reachable-colour $+720$ opposite. A common slip is to report 1461 — that forgets the colour restriction and counts squares of the wrong parity too.)

3. A 90% confidence interval

Without a computer you would estimate the count from the octagon's *area*. The distance-d "ring" holds about $28d$ squares, so the number reachable grows like $N(n) \approx 7n^2$, giving $N(10) \approx 7 \cdot 100 = 700$. A candidate would therefore quote something like

$$\textbf{90\% CI:}\ [650,\ 800]$$

(a more conservative band, allowing for the octagon's corner corrections, would be $[600, 850]$). Both comfortably contain the exact value 741.

4. Even or odd?

$21^2 + 20^2$ is (odd) + (even) = odd, and subtracting 100 (even) leaves the parity unchanged, so 741 is **odd**. Equivalently, by the knight's 8-fold symmetry (four reflections, four rotations) every reachable square falls into an orbit of size 4 or 8 except the origin, which is a lone fixed point; the origin *is* reachable (out-and-back five times), so the total is even $+ 1 =$ **odd**.

741 squares (a 90% interval such as $[650, 800]$ from the area estimate $\approx 7n^2 = 700$), and the answer is odd. In general, for $n \geq 5$ moves the count is $7n^2 + 4n + 1$.

Concept Summary and Applications

The board as a graph. A *graph* is a set of dots (*vertices*) joined by lines (*edges*). For knight problems the vertices are squares, and two squares are joined if a knight can move between them. The *degree* of a square is its number of knight moves.

- **Colour flips every move** (the sum $x + y$ changes by an odd number), so knight routes alternate colours.
- **Reachable squares (infinite board):** after n moves the knight is inside the octagon $|x|, |y| \leq 2n$, $|x| + |y| \leq 3n$, on the starting colour if n is even; two moves can be wasted by stepping out and back. Count $\approx 7n^2$ (exactly $7n^2 + 4n + 1$ for $n \geq 5$): quadratic growth, not 8^n.
- **Tours:** a knight's tour is a path through every vertex of the knight graph (see the Additional Problem for paths, cycles and Euler paths).

- **Counting lattice points:** bound the region by simple inequalities, count a big box, and subtract the cut-off corners; check parity with symmetry.
- **Applications:** reachability and distance on grids and networks, route planning with movement constraints, and parity arguments in puzzles.

Additional Problem — No Knight's Tour on a 4×4 Board (Q431)

Problem: On a 4×4 chessboard, can a knight start from some square and visit every other square exactly once (without revisiting any square)?

Background: paths, cycles and Euler paths.

- A **Hamiltonian path** visits every *vertex* exactly once and may end anywhere. For knights this is an **open tour** — exactly what this question asks for.
- A **Hamiltonian cycle** visits every vertex exactly once *and* its last vertex is joined to its first, so the route closes into a loop. For knights this is a **closed tour**. Every Hamiltonian cycle gives a Hamiltonian path (drop one edge), but not conversely: the 5×5 board has open tours but no closed tour.
- An **Euler path** uses every *edge* exactly once (vertices may repeat), as in the Königsberg bridges problem. Euler paths have a simple test: a connected graph has one if and only if it has 0 or 2 vertices of odd degree. Hamiltonian paths have *no* such simple test — deciding whether one exists is computationally hard in general — so you either *construct* one (by search; for knights, Warnsdorff's rule of always moving to the square with the fewest onward moves works well) or *prove impossibility* with a structural argument, as below.

Tools for proving that no tour exists.

- **Colouring / parity.** A closed tour needs equal numbers of each colour (so it is impossible on boards with an odd number of squares); an open tour may have one more square of one colour, starting and ending on that colour. Colouring gives necessary conditions only.
- **Low-degree squares force edges.** A square with only 2 moves must use both unless it is an endpoint of the path.
- **Forced structure $\Rightarrow$ disconnected leftovers.** After placing the forced edges, check whether

the unused squares can still be joined.

Answer: No. There is no knight's tour of a 4×4 board from *any* starting square; the longest possible knight path visits only 15 of the 16 squares (confirmed by exhaustive computer search).

Setup: a knight's tour is a Hamiltonian path

Treat each square as a vertex and join two squares if a knight can move between them. A tour visiting every square exactly once is a path through all 16 vertices (a *Hamiltonian path*). Label squares (row, column) with rows and columns 1–4. Two facts about any such path:

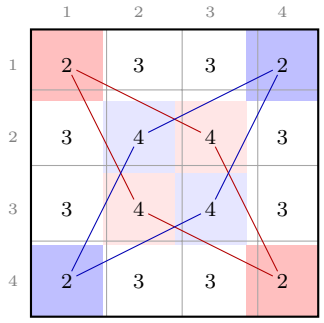
- a square in the *middle* of the path uses exactly two of its knight moves (one in, one out); only the two *endpoints* use one;
- the path can never close up into a loop.

(The usual two-colour argument does not help here: knight moves alternate colours, and 16 squares split 8 black, 8 white, which is consistent with a tour. We need a sharper argument.)

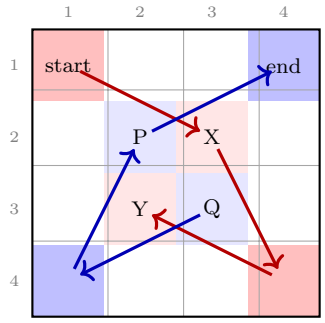
Step 1: count each square's knight moves

Panel (a) shows how many knight moves each square has: the four **corners have only** 2, the other edge squares have 3, and the four centre squares have 4. The corners come in two pairs that share *both* neighbours:

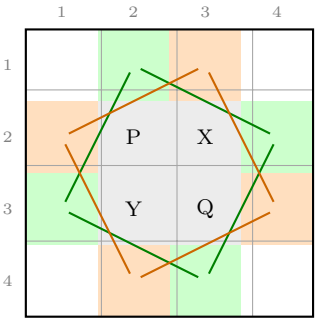
- $(1,1)$ and $(4,4)$ both move only to $X = (2,3)$ and $Y = (3,2)$ (red);
- $(1,4)$ and $(4,1)$ both move only to $P = (2,2)$ and $Q = (3,3)$ (blue).



(a) number of knight moves; corner pairs share neighbours



(b) forced ends of the path (one of the symmetric cases)



(c) the 8 edge squares form two separate 4-square loops

Step 2: each corner pair must contain an endpoint

A corner in the middle of the path must use *both* of its moves. If $(1,1)$ and $(4,4)$ were both in the middle, the path would contain X–$(1,1)$–Y and X–$(4,4)$–Y, which together form the closed loop $X \to (1,1) \to Y \to (4,4) \to X$. A path cannot contain a loop, so at least one of $(1,1)$, $(4,4)$ is an endpoint. The same argument applies to $(1,4)$, $(4,1)$ with P, Q. A path has only two endpoints, so **both endpoints are corners, one from each pair**.

Step 3: the two ends of the path are forced

By symmetry, say $(1,1)$ is an endpoint and $(4,4)$ is in the middle, and $(1,4)$ is an endpoint and $(4,1)$ is in the middle (the other cases are mirror images). The middle corner $(4,4)$ must use both X and Y, and the endpoint $(1,1)$ attaches to one of them, say X. So one end of the path is

$$(1,1) \to X \to (4,4) \to Y \to \cdots$$

Likewise the other end is $\cdots \to Q \to (4,1) \to P \to (1,4)$ (panel (b)). Now X and P are *used up*: each already has both its path neighbours. The remaining stretch must run from Y to Q through the 8 unvisited squares — all the non-corner edge squares.

Step 4: the edge squares cannot all be covered

Among themselves the 8 edge squares form **two separate 4-square loops** (panel (c)):

$$\text{green: } (1,2)\text{–}(3,1)\text{–}(4,3)\text{–}(2,4), \qquad \text{orange: } (1,3)\text{–}(2,1)\text{–}(4,2)\text{–}(3,4).$$

Apart from its two loop neighbours, every edge square's only other knight move goes to a centre square (X, Y, P or Q). Starting from Y, the path enters one loop. To get from that loop to the other one it would have to pass through a centre square, but X and P are used up and Y is already behind us, so the only exit is Q — which leads straight to the end of the path. Hence the four squares of the second loop are never visited, contradicting the requirement that the path covers all 16 squares.

No. There is no knight's tour of a 4×4 board from any starting square; the longest knight path visits only 15 of the 16 squares. The four 2-move corners force a rigid skeleton, and the eight edge squares split into two loops that the remaining path cannot join. (Open tours exist on 3×4, 5×5 and every board with both sides at least 5; the standard 8×8 board even has closed tours.)

72 De Bruijn Sequences

A **de Bruijn sequence** of order n over an alphabet of k symbols is a *cyclic* sequence in which every possible string of length n appears as a block of consecutive symbols **exactly once**. It is the shortest possible sequence that "contains every combination." For binary strings ($k = 2$) of length $n = 5$ there are $2^5 = 32$ possible blocks, so the cyclic sequence has length 32.

Representative Problem — A Length-36 Binary Sequence (Q185)

Problem: Build a sequence of 36 digits, each 0 or 1, that starts with 00000 and in which every block of 5 consecutive digits is *different* from every other. (For example, 00000100001 . . . fails, because the block 00001 appears twice.) How would you construct it?

Step 1 — count the blocks. A sequence of length 36 has

$$36 - 5 + 1 = 32$$

blocks of 5 consecutive digits (starting at positions $1, 2, \ldots, 32$). There are exactly $2^5 = 32$ different binary strings of length 5. So "all 32 blocks distinct" means **every 5-bit string appears exactly once**. That is exactly a de Bruijn sequence of order 5: a cyclic sequence of length 32, written out linearly by repeating its first 4 digits at the end so that the blocks wrapping around are visible:

$$32 + 4 = 36.$$

Step 2 — turn it into a graph problem. Make a directed graph whose *vertices* are the 16 binary strings of length 4. From a vertex $a_1a_2a_3a_4$, draw two arrows: to $a_2a_3a_4\,0$ and to $a_2a_3a_4\,1$ (drop the first digit, append a new one). Each arrow corresponds to the 5-bit string $a_1a_2a_3a_4b$ — so

$$\text{vertices} = \text{4-bit strings}, \qquad \text{arrows} = \text{5-bit strings}.$$

Walking along arrows and writing down the appended digit builds the sequence one digit at a time; the 5-bit block just completed is the arrow just used. A sequence using every 5-bit block exactly once is therefore a walk using *every arrow exactly once* — an **Eulerian cycle**. Every vertex has 2 arrows in and 2 arrows out (and the graph is connected), so by Euler's theorem such a cycle always exists.

Step 3 — one explicit answer. A valid cyclic de Bruijn sequence of order 5 is

$$00000100011001010011101011011111 \qquad \text{(length 32)}.$$

Appending its first four digits 0000 gives a length-36 answer starting with 00000:

$$\boxed{000001000110010100111010110111110000}.$$

Its 32 blocks of length 5 are exactly the 32 binary strings, each once.

Constructing It by Hand — the Greedy "Prefer-1" Rule

You do not need to draw the graph. A simple greedy rule (Martin's algorithm) produces a valid sequence directly:

1. Start with 00000.
2. Look at the *last* 4 *digits*. Try appending a **1**: if the new 5-digit block has not appeared before, keep it.
3. Otherwise append a **0** (if that block is new).
4. Stop when neither choice gives a new block — this happens after exactly 36 digits.

last 4 digits	append	new block	
0000	1	00001	
0001	1	00011	
0011	1	00111	
0111	1	01111	
1111	1	11111	
1111	0	11110	(11111 already used)
1110	1	11101	
		⋮	

Carrying this to the end gives

$$00000111110111001101011000101001000 0,$$

which also starts with 00000 and has all 32 blocks distinct. (De Bruijn sequences are *not* unique; there are $2^{11} = 2048$ different cyclic binary ones of order 5.)

Why the Greedy "Prefer-1" Rule Always Works

It is not obvious that such a short-sighted rule never paints itself into a corner. It does not, provided you **start from all zeros** (from other starting seeds it usually stalls early: from 11111 it covers only 6 of the 32 blocks). This is **Martin's theorem** (1934). The proof uses the graph from Step 2 of the representative problem.

Set-up in graph language. Vertices are the 4-bit strings; the arrow $v\,b$ leaves vertex v and appends bit b. Every vertex has exactly 2 arrows out ($v0$ and $v1$) and 2 arrows in. The rule is a walk that starts at vertex 0000 (the seed 00000 counts as having already used the loop arrow 00000) and, at each vertex v, takes arrow $v1$ if unused, else $v0$ if unused, else stops. Each arrow is a 5-bit block, and the walk never reuses an arrow — so the blocks are automatically distinct. What must be shown is that the walk does not stop until *all* 32 arrows are used.

Fact 1 — the walk can only stop at the start vertex 0000. Suppose the walk is at some vertex $u \neq 0000$. It got there by using an arrow *into* u, and every earlier visit to u also entered and then left. So the number of arrows used into u is one more than the number used out of u. Since u has only 2 arrows in, it has used at most 1 arrow out — so an unused arrow out of u remains, and the walk continues. Hence it can only get stuck back at 0000.

Fact 2 — at the end, every vertex is "balanced." Each visit to a vertex enters once and leaves once, and the walk starts and ends at 0000. So when it stops, every vertex has used *the same number* of arrows in as arrows out.

Fact 3 — "prefer-1" means a used 0-arrow implies a used 1-arrow. The first time the walk leaves any vertex v, the arrow $v1$ is still unused (only a departure from v can use it), so the rule takes $v1$. Therefore, if $v0$ was ever used, $v1$ was used before it. Equivalently: *if v has any unused arrow out, then $v0$ is unused.*

Putting it together. Suppose, for contradiction, that when the walk stops some vertex $v = c_1c_2c_3c_4$ still has an unused arrow out.

1. By Fact 3, the arrow $v0 = c_1c_2c_3c_4\,0$ is unused.
2. That arrow points *into* the vertex $w = c_2c_3c_4\,0$ (drop the first digit, append 0). So w has an unused arrow in, and by Fact 2 (balance) it also has an unused arrow out.
3. Repeat with w: its 0-arrow is unused, so $c_3c_4\,00$ has an unused arrow out; then $c_4\,000$; then 0000.

After four steps every original digit has been pushed out and replaced by 0: the vertex 0000 has an unused arrow out. But by Fact 1 the walk stopped *at* 0000, which means both of its arrows out were used — a contradiction. So no arrow is left unused: the walk uses all $2^5 = 32$ blocks, each exactly once, and the output has $32 + 4 = 36$ digits.

The same argument works for any length n (push in 0s $n - 1$ times) and for any alphabet of k symbols (start from all copies of the smallest symbol and always prefer the largest symbol that gives a new block).

Any binary de Bruijn sequence of order 5, written out with its first 4 digits repeated at the end, works. For example

$$000001000110010100111010110111110000$$

(from an Eulerian cycle) or $000001111101110011010110001010010000$ (greedy "prefer-1"). Each has $36 - 5 + 1 = 32$ distinct blocks of length 5 — all 2^5 binary strings once.

Concept Summary and Applications

- **Counting first:** a sequence of length L has $L - n + 1$ windows of length n; with k^n possible strings, the shortest sequence containing all of them has length $k^n + n - 1$ (cyclically, k^n).
- **Graph view:** vertices = $(n-1)$-strings, arrows = n-strings; a de Bruijn sequence is an Eulerian cycle, which exists because every vertex has equal in- and out-degree.
- **Two constructions:** an Eulerian cycle in the de Bruijn graph, or the greedy "prefer-1" rule from all zeros (Martin's theorem guarantees it never gets stuck early).
- **Applications:** cracking combination/keypad locks with the fewest presses, position encoding (reading any n consecutive marks tells you where you are), test-pattern generation, and DNA sequence assembly.